

TABLE 6-1

Forward and reverse rate constants (k_1 and k_{-1}) and association and dissociation constants (K_a and K_d) for three ligand-antibody interactions

Antibody	Ligand	k_1	k_{-1}	K_a	K_d
Anti-DNP	ϵ -DNP-L-lysine	8×10^7	1	1×10^8	1×10^{-8}
Anti-fluorescein	Fluorescein	4×10^8	5×10^{-3}	1×10^{11}	1×10^{-11}
Anti-bovine serum albumin (BSA)	Dansyl-BSA	3×10^5	2×10^{-3}	1.7×10^8	5.9×10^{-9}

SOURCE: Adapted from H. N. Eisen, 1990, *Immunology*, 3rd ed., Harper & Row Publishers.

other methods are evaluated, it is described here. This procedure uses a dialysis chamber containing two equal compartments separated by a semipermeable membrane. Antibody is placed in one compartment, and a radioactively labeled ligand that is small enough to pass through the semipermeable membrane is placed in the other compartment (Figure 6-2). Suitable ligands include haptens, oligosaccharides, and oligopeptides. In the absence of antibody, ligand added to compartment B will equilibrate on both sides of the membrane (Figure 6-2a). In the presence of antibody, however, part

of the labeled ligand will be bound to the antibody at equilibrium, trapping the ligand on the antibody side of the vessel, whereas unbound ligand will be equally distributed in both compartments. Thus the total concentration of ligand will be greater in the compartment containing antibody (Figure 6-2b). The difference in the ligand concentration in the two compartments represents the concentration of ligand bound to the antibody (i.e., the concentration of Ag-Ab complex). The higher the affinity of the antibody, the more ligand is bound.

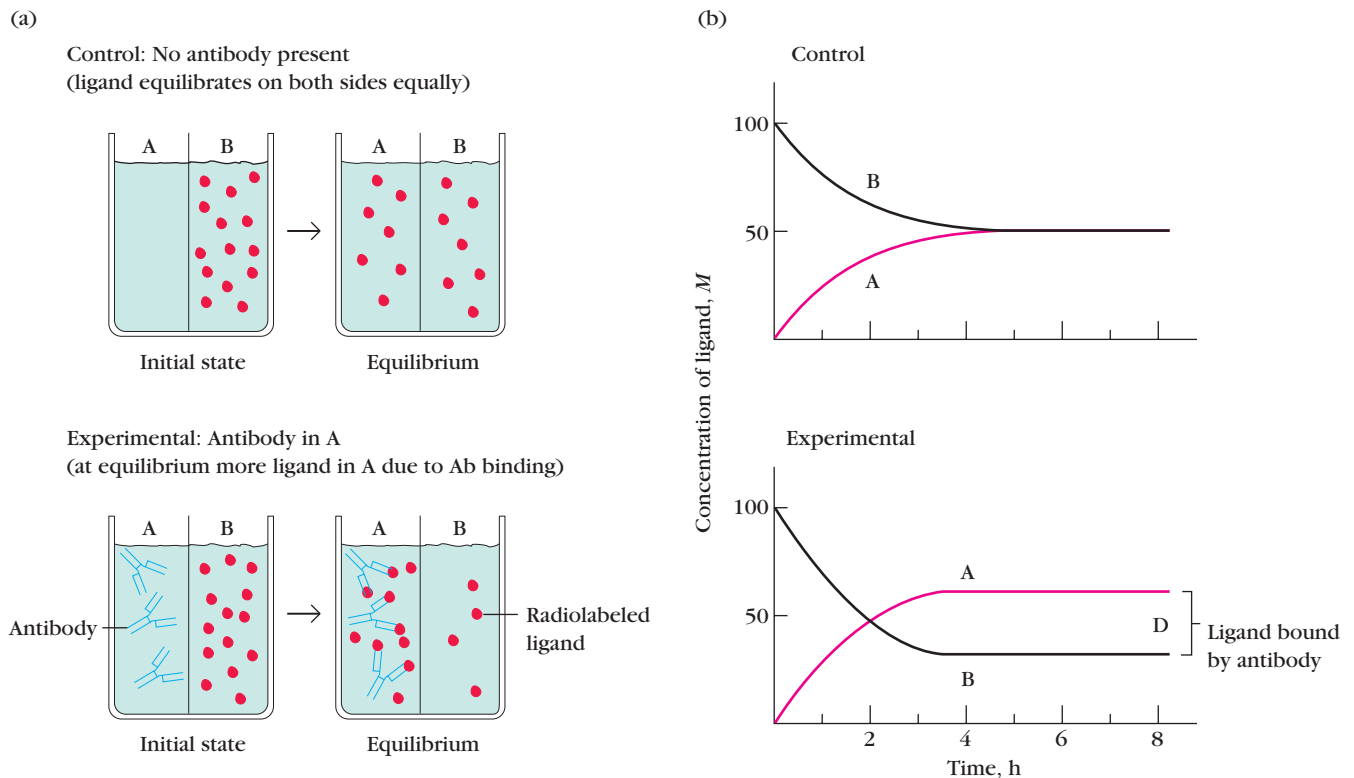


FIGURE 6-2 Determination of antibody affinity by equilibrium dialysis. (a) The dialysis chamber contains two compartments (A and B) separated by a semipermeable membrane. Antibody is added to one compartment and a radiolabeled ligand to another. At equilibrium, the concentration of radioactivity in both compartments is mea-

sured. (b) Plot of concentration of ligand in each compartment with time. At equilibrium, the difference in the concentration of radioactive ligand in the two compartments represents the amount of ligand bound to antibody.

Since the total concentration of antibody in the equilibrium dialysis chamber is known, the equilibrium equation can be rewritten as:

$$K_a = [\text{Ab-Ag}]/[\text{Ab}][\text{Ag}] = \frac{r}{c(n-r)}$$

where r equals the ratio of the concentration of bound ligand to total antibody concentration, c is the concentration of free ligand, and n is the number of binding sites per antibody molecule. This expression can be rearranged to give the **Scatchard equation**:

$$\frac{r}{c} = K_a n - K_a r$$

Values for r and c can be obtained by repeating the equilibrium dialysis with the same concentration of antibody but with different concentrations of ligand. If K_a is a constant, that is, if all the antibodies within the dialysis chamber have the same affinity for the ligand, then a Scatchard plot of r/c versus r will yield a straight line with a slope of $-K_a$ (Figure 6-3a). As the concentration of unbound ligand c increases, r/c approaches 0, and r approaches n , the **valency**, equal to the number of binding sites per antibody molecule.

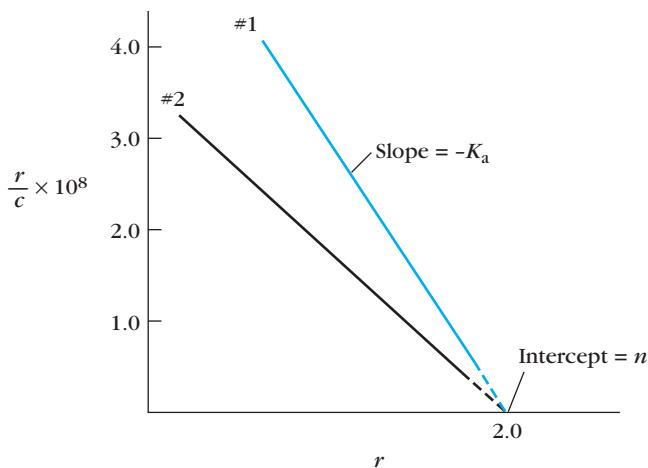
Most antibody preparations are polyclonal, and K_a is therefore not a constant because a heterogeneous mixture of antibodies with a range of affinities is present. A Scatchard plot of heterogeneous antibody yields a curved line whose

slope is constantly changing, reflecting this antibody heterogeneity (Figure 6-3b). With this type of Scatchard plot, it is possible to determine the average affinity constant, K_0 , by determining the value of K_a when half of the antigen-binding sites are filled. This is conveniently done by determining the slope of the curve at the point where half of the antigen binding sites are filled.

Antibody Avidity Incorporates Affinity of Multiple Binding Sites

The affinity at one binding site does not always reflect the true strength of the antibody-antigen interaction. When complex antigens containing multiple, repeating antigenic determinants are mixed with antibodies containing multiple binding sites, the interaction of an antibody molecule with an antigen molecule at one site will increase the probability of reaction between those two molecules at a second site. The strength of such multiple interactions between a multivalent antibody and antigen is called the **avidity**. The avidity of an antibody is a better measure of its binding capacity within biological systems (e.g., the reaction of an antibody with antigenic determinants on a virus or bacterial cell) than the affinity of its individual binding sites. High avidity can compensate for low affinity. For example, secreted pentameric

(a) Homogeneous antibody



(b) Heterogeneous antibody

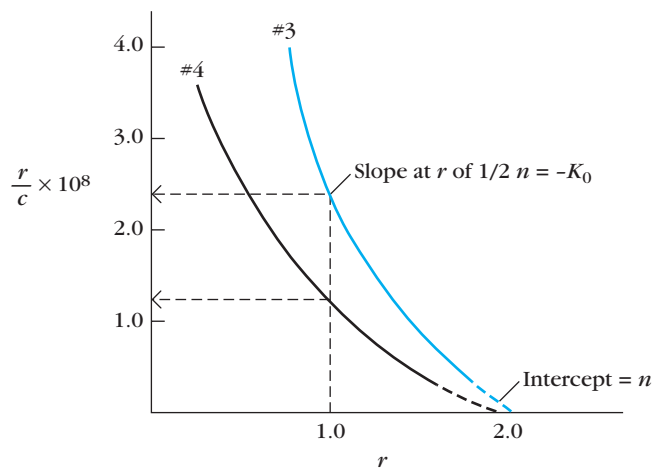


FIGURE 6-3 Scatchard plots are based on repeated equilibrium dialyses with a constant concentration of antibody and varying concentration of ligand. In these plots, r equals moles of bound ligand/mole antibody and c is the concentration of free ligand. From a Scatchard plot, both the equilibrium constant (K_a) and the number of binding sites per antibody molecule (n), or its valency, can be obtained. (a) If all antibodies have the same affinity, then a Scatchard plot yields a straight line with a slope of $-K_a$. The x intercept is n , the valency of the antibody, which is 2 for IgG and other divalent Igs. For IgM, which is pentameric, $n = 10$, and for dimeric IgA, $n = 4$. In this

graph, antibody #1 has a higher affinity than antibody #2. (b) If the antibody preparation is polyclonal and has a range of affinities, a Scatchard plot yields a curved line whose slope is constantly changing. The average affinity constant K_0 can be calculated by determining the value of K_a when half of the binding sites are occupied (i.e., when $r = 1$ in this example). In this graph, antiserum #3 has a higher affinity ($K_0 = 2.4 \times 10^8$) than antiserum #4 ($K_0 = 1.25 \times 10^8$). Note that the curves shown in (a) and (b) are for divalent antibodies such as IgG.

IgM often has a lower affinity than IgG, but the high avidity of IgM, resulting from its higher valence, enables it to bind antigen effectively.

Cross-Reactivity

Although Ag-Ab reactions are highly specific, in some cases antibody elicited by one antigen can cross-react with an unrelated antigen. Such **cross-reactivity** occurs if two different antigens share an identical or very similar epitope. In the latter case, the antibody's affinity for the cross-reacting epitope is usually less than that for the original epitope.

Cross-reactivity is often observed among polysaccharide antigens that contain similar oligosaccharide residues. The **ABO blood-group antigens**, for example, are glycoproteins expressed on red blood cells. Subtle differences in the terminal residues of the sugars attached to these surface proteins distinguish the A and B blood-group antigens. An individual lacking one or both of these antigens will have serum antibodies to the missing antigen(s). The antibodies are induced not by exposure to red blood cell antigens but by exposure to cross-reacting microbial antigens present on common intestinal bacteria. These microbial antigens induce the formation of antibodies in individuals lacking the similar blood-group antigens on their red blood cells. (In individuals possessing these antigens, complementary antibodies would be eliminated during the developmental stage in which antibodies that recognize self epitopes are weeded out.) The blood-group antibodies, although elicited by microbial antigens, will cross-react with similar oligosaccharides on foreign red blood cells, providing the basis for blood typing tests and accounting for the necessity of compatible blood types during blood transfusions. A type A individual has anti-B antibodies; a type B individual has anti-A; and a type O individual thus has anti-A and anti-B (Table 6-2).

A number of viruses and bacteria have antigenic determinants identical or similar to normal host-cell components. In some cases, these microbial antigens have been shown to elicit antibody that cross-reacts with the host-cell components, resulting in a tissue-damaging autoimmune reaction.

The bacterium *Streptococcus pyogenes*, for example, expresses cell-wall proteins called M antigens. Antibodies produced to streptococcal M antigens have been shown to cross-react with several myocardial and skeletal muscle proteins and have been implicated in heart and kidney damage following streptococcal infections. The role of other cross-reacting antigens in the development of autoimmune diseases is discussed in Chapter 20.

Some vaccines also exhibit cross-reactivity. For instance, vaccinia virus, which causes cowpox, expresses cross-reacting epitopes with variola virus, the causative agent of smallpox. This cross-reactivity was the basis of Jenner's method of using vaccinia virus to induce immunity to smallpox, as mentioned in Chapter 1.

Precipitation Reactions

Antibody and soluble antigen interacting in aqueous solution form a lattice that eventually develops into a visible precipitate. Antibodies that aggregate soluble antigens are called **precipitins**. Although formation of the soluble Ag-Ab complex occurs within minutes, formation of the visible precipitate occurs more slowly and often takes a day or two to reach completion.

Formation of an Ag-Ab lattice depends on the valency of both the antibody and antigen:

- The antibody must be bivalent; a precipitate will not form with monovalent Fab fragments.
- The antigen must be either bivalent or polyvalent; that is, it must have at least two copies of the same epitope, or have different epitopes that react with different antibodies present in polyclonal antisera.

Experiments with myoglobin illustrate the requirement that protein antigens be bivalent or polyvalent for a precipitin reaction to occur. Myoglobin precipitates well with specific polyclonal antisera but fails to precipitate with a specific monoclonal antibody because it contains multiple, distinct epitopes but only a single copy of each epitope (Figure 6-4a). Myoglobin thus can form a crosslinked lattice structure with polyclonal antisera but not with monoclonal antisera. The principles that underlie precipitation reactions are presented because they are essential for an understanding of commonly used immunological assays. Although various modifications of the precipitation reaction were at one time the major types of assay used in immunology, they have been largely replaced by methods that are faster and, because they are far more sensitive, require only very small quantities of antigen or antibody. Also, these modern assay methods are not limited to antigen-antibody reactions that produce a precipitate. Table 6-3 presents a comparison of the *sensitivity*, or minimum amount of antibody detectable, by a number of immunoassays.

TABLE 6-2 ABO blood types

Blood type	Antigens on RBCs	Serum antibodies
A	A	Anti-B
B	B	Anti-A
AB	A and B	Neither
O	Neither	Anti-A and anti-B

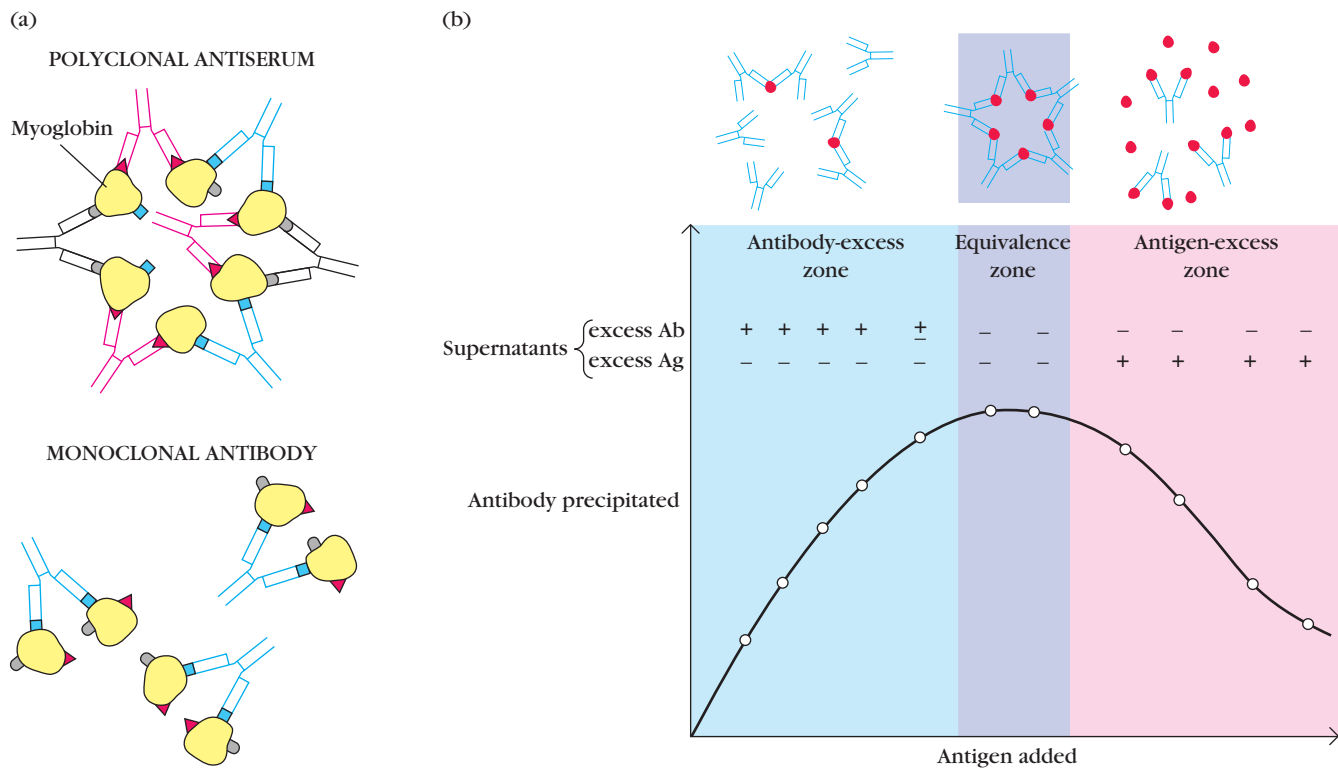


FIGURE 6-4 Precipitation reactions. (a) Polyclonal antibodies can form lattices, or large aggregates, that precipitate out of solution. However, if each antigen molecule contains only a single epitope recognized by a given monoclonal antibody, the antibody can link only two molecules of antigen and no precipitate is formed. (b) A precipitation curve for a system of one antigen and its antibodies. This plot of the amount of antibody precipitated versus increasing antigen concentrations (at constant total antibody) reveals three zones: a

zone of antibody excess, in which precipitation is inhibited and antibody not bound to antigen can be detected in the supernatant; an equivalence zone of maximal precipitation in which antibody and antigen form large insoluble complexes and neither antibody nor antigen can be detected in the supernatant; and a zone of antigen excess in which precipitation is inhibited and antigen not bound to antibody can be detected in the supernatant.

Precipitation Reactions in Fluids Yield a Precipitin Curve

A quantitative precipitation reaction can be performed by placing a constant amount of antibody in a series of tubes and adding increasing amounts of antigen to the tubes. At one time this method was used to measure the amount of antigen or antibody present in a sample of interest. After the precipitate forms, each tube is centrifuged to pellet the precipitate, the supernatant is poured off, and the amount of precipitate is measured. Plotting the amount of precipitate against increasing antigen concentrations yields a precipitin curve. As Figure 6-4b shows, excess of either antibody or antigen interferes with maximal precipitation, which occurs in the so-called **equivalence zone**, within which the ratio of antibody to antigen is optimal. As a large multimolecular lattice is formed at equivalence, the complex increases in size and precipitates out of solution. As shown in Figure 6-4, under conditions of *antibody excess* or *antigen excess*, extensive lattices do not form and precipitation is inhibited. Although the quantitative precipitation reaction is seldom used exper-

imentally today, the principles of antigen excess, antibody excess, and equivalence apply to many Ag-Ab reactions.

Precipitation Reactions in Gels Yield Visible Precipitin Lines

Immune precipitates can form not only in solution but also in an agar matrix. When antigen and antibody diffuse toward one another in agar, or when antibody is incorporated into the agar and antigen diffuses into the antibody-containing matrix, a visible line of precipitation will form. As in a precipitation reaction in fluid, visible precipitation occurs in the region of equivalence, whereas no visible precipitate forms in regions of antibody or antigen excess. Two types of *immunodiffusion reactions* can be used to determine relative concentrations of antibodies or antigens, to compare antigens, or to determine the relative purity of an antigen preparation. They are **radial immunodiffusion** (the Mancini method) and **double immunodiffusion** (the Ouchterlony method); both are carried out in a semisolid medium such as agar. In radial immunodiffusion, an antigen sample is placed in a well and allowed to diffuse into

TABLE 6-3 Sensitivity of various immunoassays

Assay	Sensitivity* (μg antibody/ml)
Precipitation reaction in fluids	20–200
Precipitation reactions in gels	
Mancini radial immunodiffusion	10–50
Ouchterlony double immunodiffusion	20–200
Immunoelectrophoresis	20–200
Rocket electrophoresis	2
Agglutination reactions	
Direct	0.3
Passive agglutination	0.006–0.06
Agglutination inhibition	0.006–0.06
Radioimmunoassay	0.0006–0.006
Enzyme-linked immunosorbent assay (ELISA)	<0.0001–0.01
ELISA using chemiluminescence	<0.0001–0.01 [†]
Immunofluorescence	1.0
Flow cytometry	0.06–0.006

*The sensitivity depends upon the affinity of the antibody as well as the epitope density and distribution.

[†]Note that the sensitivity of chemiluminescence-based ELISA assays can be made to match that of RIA.

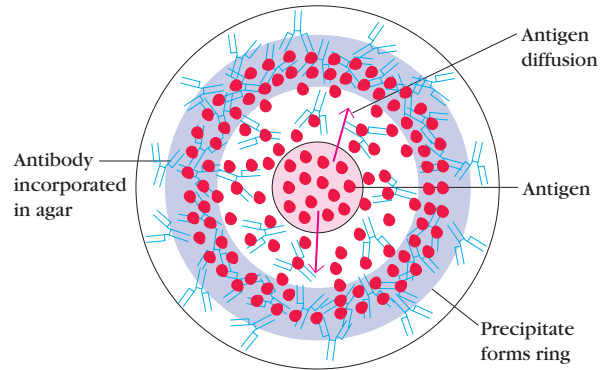
SOURCE: Adapted from N. R. Rose et al., eds., 1997, *Manual of Clinical Laboratory Immunology*, 5th ed., American Society for Microbiology, Washington, D.C.

agar containing a suitable dilution of an antiserum. As the antigen diffuses into the agar, the region of equivalence is established and a ring of precipitation, a precipitin ring, forms around the well (Figure 6-5, upper panel). The area of the precipitin ring is proportional to the concentration of antigen. By comparing the area of the precipitin areas of known concentrations of the antigen, the concentration of the antigen sample can be determined. In the Ouchterlony method, both antigen and antibody diffuse radially from wells toward each other, thereby establishing a concentration gradient. As equivalence is reached, a visible line of precipitation, a precipitin line, forms (Figure 6-5, lower panel).

Immunoelectrophoresis Combines Electrophoresis and Double Immunodiffusion

In **immunoelectrophoresis**, the antigen mixture is first electrophoresed to separate its components by charge. Troughs are then cut into the agar gel parallel to the direction of

RADIAL IMMUNODIFFUSION



DOUBLE IMMUNODIFFUSION

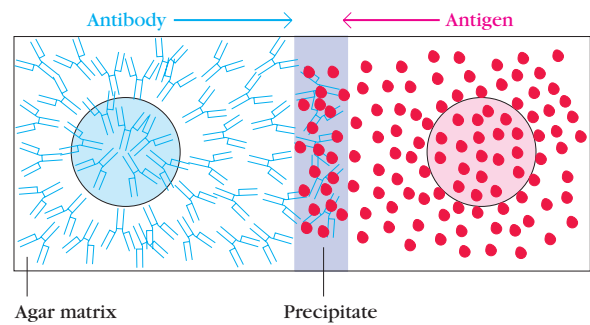


FIGURE 6-5 Diagrammatic representation of radial immunodiffusion (Mancini method) and double immunodiffusion (Ouchterlony method) in a gel. In both cases, large insoluble complexes form in the agar in the zone of equivalence, visible as lines of precipitation (purple regions). Only the antigen (red) diffuses in radial immunodiffusion, whereas both the antibody (blue) and antigen (red) diffuse in double immunodiffusion.

the electric field, and antiserum is added to the troughs. Antibody and antigen then diffuse toward each other and produce lines of precipitation where they meet in appropriate proportions (Figure 6-6a). Immunoelectrophoresis is used in clinical laboratories to detect the presence or absence of proteins in the serum. A sample of serum is electrophoresed, and the individual serum components are identified with antisera specific for a given protein or immunoglobulin class (Figure 6-6b). This technique is useful in determining whether a patient produces abnormally low amounts of one or more isotypes, characteristic of certain immunodeficiency diseases. It can also show whether a patient overproduces some serum protein, such as albumin, immunoglobulin, or transferrin. The immunoelectrophoretic pattern of serum from patients with multiple myeloma, for example, shows a heavy distorted arc caused by the large amount of myeloma protein, which is monoclonal Ig and therefore uniformly charged (Figure 6-6b). Because immunoelectrophoresis is a strictly *qualitative* technique that only detects relatively high antibody concentrations (greater than several hundred $\mu\text{g}/\text{ml}$), its utility is limited to the detection

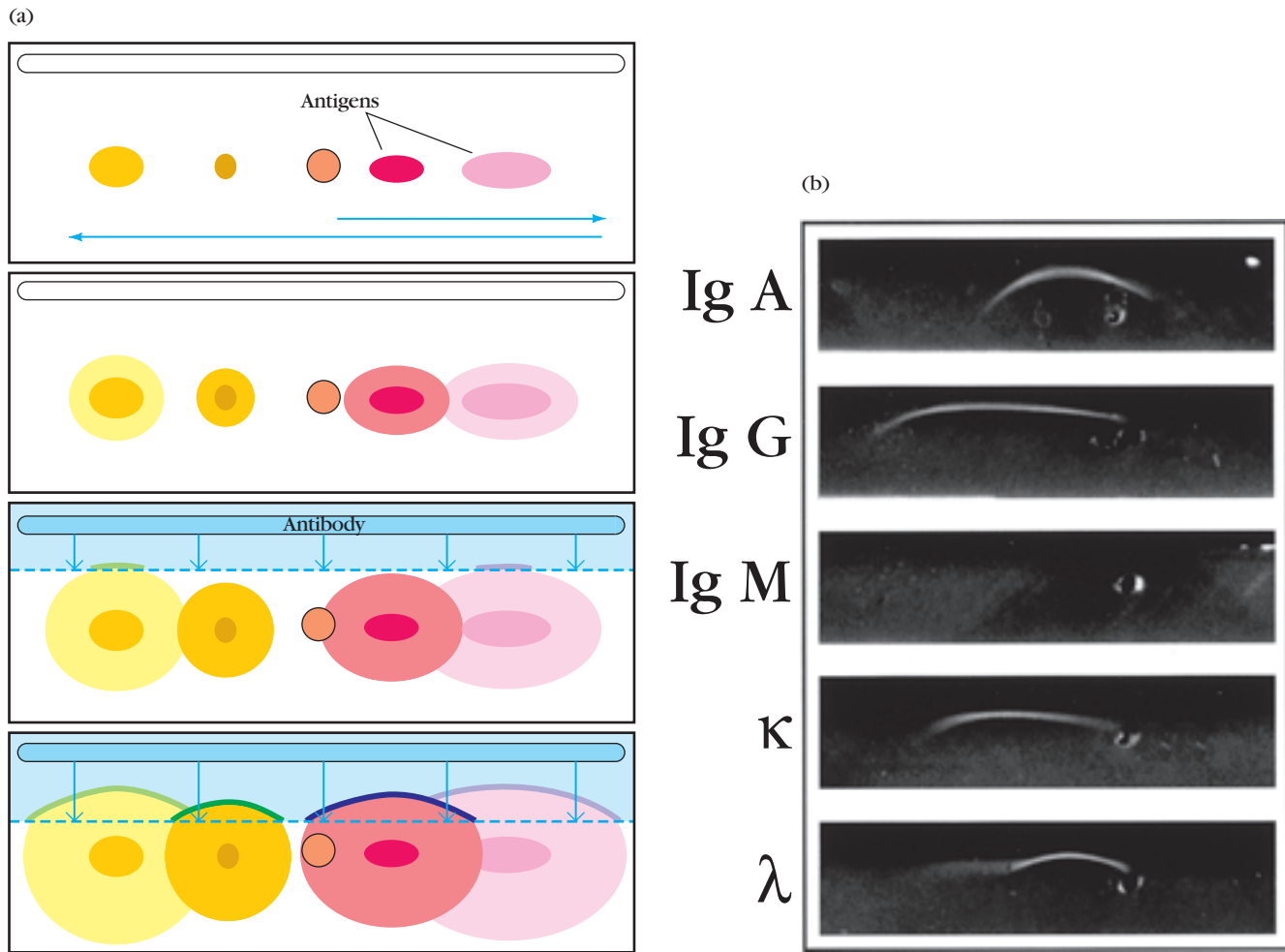


FIGURE 6-6 Immunoelectrophoresis of an antigen mixture. (a) An antigen preparation (orange) is first electrophoresed, which separates the component antigens on the basis of charge. Antiserum (blue) is then added to troughs on one or both sides of the separated antigens and allowed to diffuse; in time, lines of precipitation (colored arcs) form where specific antibody and antigen interact. (b) Immunoelectrophoretic patterns of human serum from a patient with myeloma. The patient produces a large amount of a monoclonal IgG

(λ-light-chain-bearing) antibody. A sample of serum from the patient was placed in the well of the slide and electrophoresed. Then antiserum specific for the indicated antibody class or light chain type was placed in the top trough of each slide. At the concentrations of patient's serum used, only anti-IgG and anti-λ antibodies produced lines of precipitation. [Part (b), Robert A. Kyle and Terry A. Katzman, *Manual of Clinical Immunology*, 1997, N. Rose, ed., ASM Press, Washington, D.C., p. 164.]

of quantitative abnormalities only when the departure from normal is striking, as in immunodeficiency states and immunoproliferative disorders.

A related *quantitative* technique, **rocket electrophoresis**, does permit measurement of antigen levels. In rocket electrophoresis, a negatively charged antigen is electrophoresed in a gel containing antibody. The precipitate formed between antigen and antibody has the shape of a rocket, the height of which is proportional to the concentration of antigen in the well. One limitation of rocket electrophoresis is the need for the antigen to be negatively charged for electrophoretic movement within the agar matrix. Some proteins, immunoglobulins for example,

are not sufficiently charged to be quantitatively analyzed by rocket electrophoresis; nor is it possible to measure the amounts of several antigens in a mixture at the same time.

Agglutination Reactions

The interaction between antibody and a particulate antigen results in visible clumping called **agglutination**. Antibodies that produce such reactions are called **agglutinins**. Agglutination reactions are similar in principle to precipitation reactions; they depend on the crosslinking of polyvalent antigens. Just as

an excess of antibody inhibits precipitation reactions, such excess can also inhibit agglutination reactions; this inhibition is called the **prozone effect**. Because prozone effects can be encountered in many types of immunoassays, understanding the basis of this phenomenon is of general importance.

Several mechanisms can cause the prozone effect. First, at high antibody concentrations, the number of antibody binding sites may greatly exceed the number of epitopes. As a result, most antibodies bind antigen only univalently instead of multivalently. Antibodies that bind univalently cannot crosslink one antigen to another. Prozone effects are readily diagnosed by performing the assay at a variety of antibody (or antigen) concentrations. As one dilutes to an optimum antibody concentration, one sees higher levels of agglutination or whatever parameter is measured in the assay being used. When one is using polyclonal antibodies, the prozone effect can also occur for another reason. The antiserum may contain high concentrations of antibodies that bind to the antigen but do not induce agglutination; these antibodies, called **incomplete antibodies**, are often of the IgG class. At high concentrations of IgG, incomplete antibodies may occupy most of the antigenic sites, thus blocking access by IgM, which is a good agglutinin. This effect is not seen with agglutinating monoclonal antibodies. The lack of agglutinating activity of an incomplete antibody may be due to restricted flexibility in the hinge region, making it difficult for the antibody to assume the required angle for optimal cross-linking of epitopes on two or more particulate antigens. Alternatively, the density of epitope distribution or the location of some epitopes in deep pockets of a particulate antigen may make it difficult for the antibodies specific for these epitopes to agglutinate certain particulate antigens. When feasible, the solution to both of these problems is to try different antibodies that may react with other epitopes of the antigen that do not present these limitations.

Hemagglutination Is Used in Blood Typing

Agglutination reactions (Figure 6-7) are routinely performed to type red blood cells (RBCs). In typing for the ABO

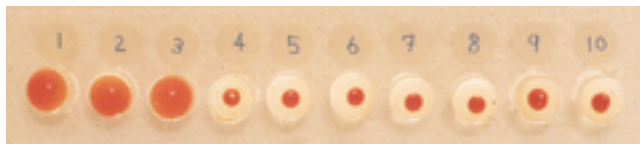


FIGURE 6-7 Demonstration of hemagglutination using antibodies against sheep red blood cells (SRBCs). The control tube (10) contains only SRBCs, which settle into a solid “button.” The experimental tubes 1–9 contain a constant number of SRBCs plus serial two-fold dilutions of anti-SRBC serum. The spread pattern in the experimental series indicates positive hemagglutination through tube 3. [Louisiana State University Medical Center/MIP. Courtesy of Harriet C. W. Thompson.]

antigens, RBCs are mixed on a slide with antisera to the A or B blood-group antigens. If the antigen is present on the cells, they agglutinate, forming a visible clump on the slide. Determination of which antigens are present on donor and recipient RBCs is the basis for matching blood types for transfusions.

Bacterial Agglutination Is Used To Diagnose Infection

A bacterial infection often elicits the production of serum antibodies specific for surface antigens on the bacterial cells. The presence of such antibodies can be detected by bacterial agglutination reactions. Serum from a patient thought to be infected with a given bacterium is serially diluted in an array of tubes to which the bacteria is added. The last tube showing visible agglutination will reflect the serum antibody **titer** of the patient. The agglutinin titer is defined as the reciprocal of the greatest serum dilution that elicits a positive agglutination reaction. For example, if serial twofold dilutions of serum are prepared and if the dilution of 1/640 shows agglutination but the dilution of 1/1280 does not, then the agglutination titer of the patient’s serum is 640. In some cases serum can be diluted up to 1/50,000 and still show agglutination of bacteria.

The agglutinin titer of an antiserum can be used to diagnose a bacterial infection. Patients with typhoid fever, for example, show a significant rise in the agglutination titer to *Salmonella typhi*. Agglutination reactions also provide a way to type bacteria. For instance, different species of the bacterium *Salmonella* can be distinguished by agglutination reactions with a panel of typing antisera.

Passive Agglutination Is Useful with Soluble Antigens

The sensitivity and simplicity of agglutination reactions can be extended to soluble antigens by the technique of passive hemagglutination. In this technique, antigen-coated red blood cells are prepared by mixing a soluble antigen with red blood cells that have been treated with tannic acid or chromium chloride, both of which promote adsorption of the antigen to the surface of the cells. Serum containing antibody is serially diluted into microtiter plate wells, and the antigen-coated red blood cells are then added to each well; agglutination is assessed by the size of the characteristic spread pattern of agglutinated red blood cells on the bottom of the well, like the pattern seen in agglutination reactions (see Figure 6-7).

Over the past several years, there has been a shift away from red blood cells to synthetic particles, such as latex beads, as matrices for agglutination reactions. Once the antigen has been coupled to the latex beads, the preparation can either be used immediately or stored for later use. The use of synthetic beads offers the advantages of consistency,

uniformity, and stability. Furthermore, agglutination reactions employing synthetic beads can be read rapidly, often within 3 to 5 minutes of mixing the beads with the test sample. Whether based on red blood cells or the more convenient and versatile synthetic beads, agglutination reactions are simple to perform, do not require expensive equipment, and can detect small amounts of antibody (concentrations as low as nanograms per milliliter).

In Agglutination Inhibition, Absence of Agglutination Is Diagnostic of Antigen

A modification of the agglutination reaction, called **agglutination inhibition**, provides a highly sensitive assay for small quantities of an antigen. For example, one of the early

types of home pregnancy test kits included latex particles coated with human chorionic gonadotropin (HCG) and antibody to HCG (Figure 6-8). The addition of urine from a pregnant woman, which contained HCG, inhibited agglutination of the latex particles when the anti-HCG antibody was added; thus the absence of agglutination indicated pregnancy.

Agglutination inhibition assays can also be used to determine whether an individual is using certain types of illegal drugs, such as cocaine or heroin. A urine or blood sample is first incubated with antibody specific for the suspected drug. Then red blood cells (or other particles) coated with the drug are added. If the red blood cells are not agglutinated by the antibody, it indicates the sample contained an antigen recognized by the antibody, suggesting that the individual was

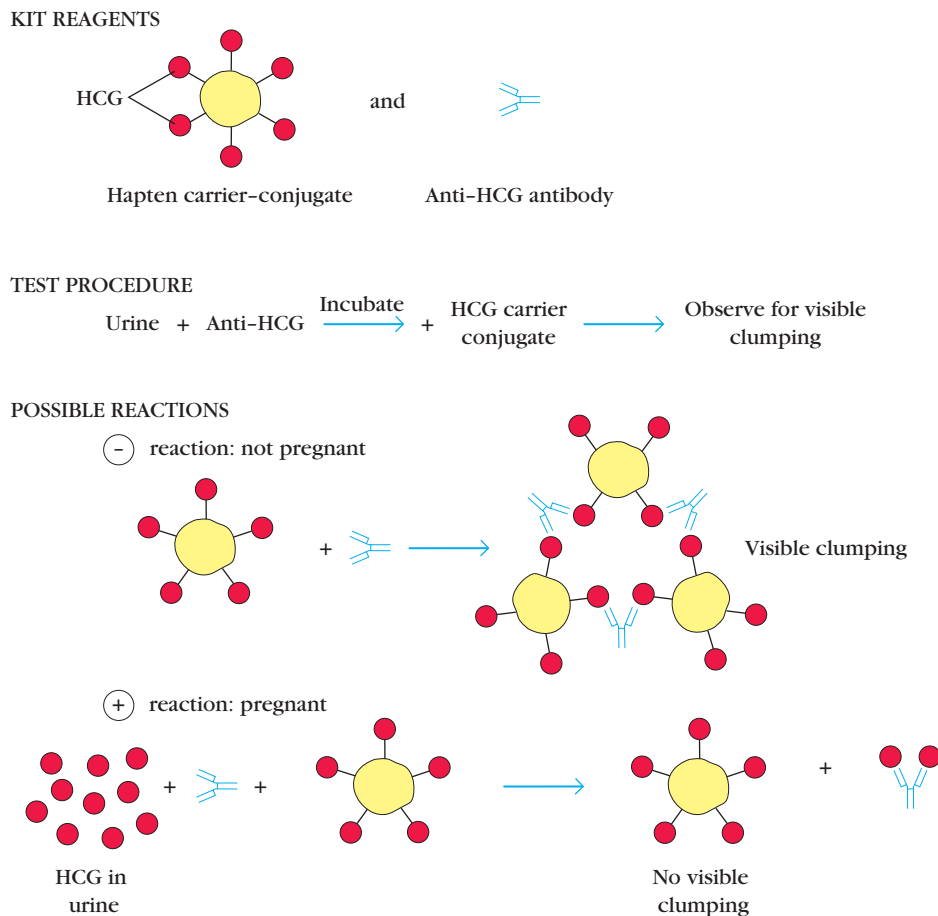


FIGURE 6-8 The original home pregnancy test kit employed hapten inhibition to determine the presence or absence of human chorionic gonadotropin (HCG). The original test kits used the presence or absence of visible clumping to determine whether HCG was present. If a woman was not pregnant, her urine would not contain HCG; in this case, the anti-HCG antibodies and HCG-carrier conjugate in the

kit would react, producing visible clumping. If a woman was pregnant, the HCG in her urine would bind to the anti-HCG antibodies, thus inhibiting the subsequent binding of the antibody to the HCG-carrier conjugate. Because of this inhibition, no visible clumping occurred if a woman was pregnant. The kits currently on the market use ELISA-based assays (see Figure 6-10).

using the illicit drug. One problem with these tests is that some legal drugs have chemical structures similar to those of illicit drugs, and these legal drugs may cross-react with the antibody, giving a false-positive reaction. For this reason a positive reaction must be confirmed by a nonimmunologic method.

Agglutination inhibition assays are widely used in clinical laboratories to determine whether an individual has been exposed to certain types of viruses that cause agglutination of red blood cells. If an individual's serum contains specific antiviral antibodies, then the antibodies will bind to the virus and interfere with hemagglutination by the virus. This technique is commonly used in premarital testing to determine the immune status of women with respect to rubella virus. The reciprocal of the last serum dilution to show inhibition of rubella hemagglutination is the titer of the serum. A titer greater than 10 (1:10 dilution) indicates that a woman is immune to rubella, whereas a titer of less than 10 is indicative of a lack of immunity and the need for immunization with the rubella vaccine.

Radioimmunoassay

One of the most sensitive techniques for detecting antigen or antibody is **radioimmunoassay (RIA)**. The technique was first developed in 1960 by two endocrinologists, S. A. Berson and Rosalyn Yalow, to determine levels of insulin–anti-insulin complexes in diabetics. Although their technique encountered some skepticism, it soon proved its value for measuring hormones, serum proteins, drugs, and vitamins at concentrations of 0.001 *micrograms* per milliliter or less. In 1977, some years after Berson's death, the significance of the technique was acknowledged by the award of a Nobel Prize to Yalow.

The principle of RIA involves competitive binding of radiolabeled antigen and unlabeled antigen to a high-affinity antibody. The labeled antigen is mixed with antibody at a concentration that saturates the antigen-binding sites of the antibody. Then test samples of unlabeled antigen of unknown concentration are added in progressively larger amounts. The antibody does not distinguish labeled from unlabeled antigen, so the two kinds of antigen compete for available binding sites on the antibody. As the concentration of unlabeled antigen increases, more labeled antigen will be displaced from the binding sites. The decrease in the amount of radiolabeled antigen bound to specific antibody in the presence of the test sample is measured in order to determine the amount of antigen present in the test sample.

The antigen is generally labeled with a gamma-emitting isotope such as ^{125}I , but beta-emitting isotopes such as tritium (^3H) are also routinely used as labels. The radiolabeled antigen is part of the assay mixture; the test sample may be a complex mixture, such as serum or other body fluids, that contains the unlabeled antigen. The first step in

setting up an RIA is to determine the amount of antibody needed to bind 50%–70% of a fixed quantity of radioactive antigen (Ag^*) in the assay mixture. This ratio of antibody to Ag^* is chosen to ensure that the number of epitopes presented by the labeled antigen always exceeds the total number of antibody binding sites. Consequently, unlabeled antigen added to the sample mixture will compete with radiolabeled antigen for the limited supply of antibody. Even a small amount of unlabeled antigen added to the assay mixture of labeled antigen and antibody will cause a decrease in the amount of radioactive antigen bound, and this decrease will be proportional to the amount of unlabeled antigen added. To determine the amount of labeled antigen bound, the Ag-Ab complex is precipitated to separate it from free antigen (antigen not bound to Ab), and the radioactivity in the precipitate is measured. A standard curve can be generated using unlabeled antigen samples of known concentration (in place of the test sample), and from this plot the amount of antigen in the test mixture may be precisely determined.

Several methods have been developed for separating the bound antigen from the free antigen in RIA. One method involves precipitating the Ag-Ab complex with a secondary anti-isotype antiserum. For example, if the Ag-Ab complex contains rabbit IgG antibody, then goat anti-rabbit IgG will bind to the rabbit IgG and precipitate the complex. Another method makes use of the fact that protein A of *Staphylococcus aureus* has high affinity for IgG. If the Ag-Ab complex contains an IgG antibody, the complex can be precipitated by mixing with formalin-killed *S. aureus*. After removal of the complex by either of these methods, the amount of free labeled antigen remaining in the supernatant can be measured in a radiation counter; subtracting this value from the total amount of labeled antigen added yields the amount of labeled antigen bound.

Various solid-phase RIAs have been developed that make it easier to separate the Ag-Ab complex from the unbound antigen. In some cases, the antibody is covalently cross-linked to Sepharose beads. The amount of radiolabeled antigen bound to the beads can be measured after the beads have been centrifuged and washed. Alternatively, the antibody can be immobilized on polystyrene or polyvinylchloride wells and the amount of free labeled antigen in the supernatant can be determined in a radiation counter. In another approach, the antibody is immobilized on the walls of microtiter wells and the amount of bound antigen determined. Because the procedure requires only small amounts of sample and can be conducted in small 96-well microtiter plates (slightly larger than a 3×5 card), this procedure is well suited for determining the concentration of a particular antigen in large numbers of samples. For example, a microtiter RIA has been widely used to screen for the presence of the hepatitis B virus (Figure 6-9). RIA screening of donor blood has sharply reduced the incidence of hepatitis B infections in recipients of blood transfusions.

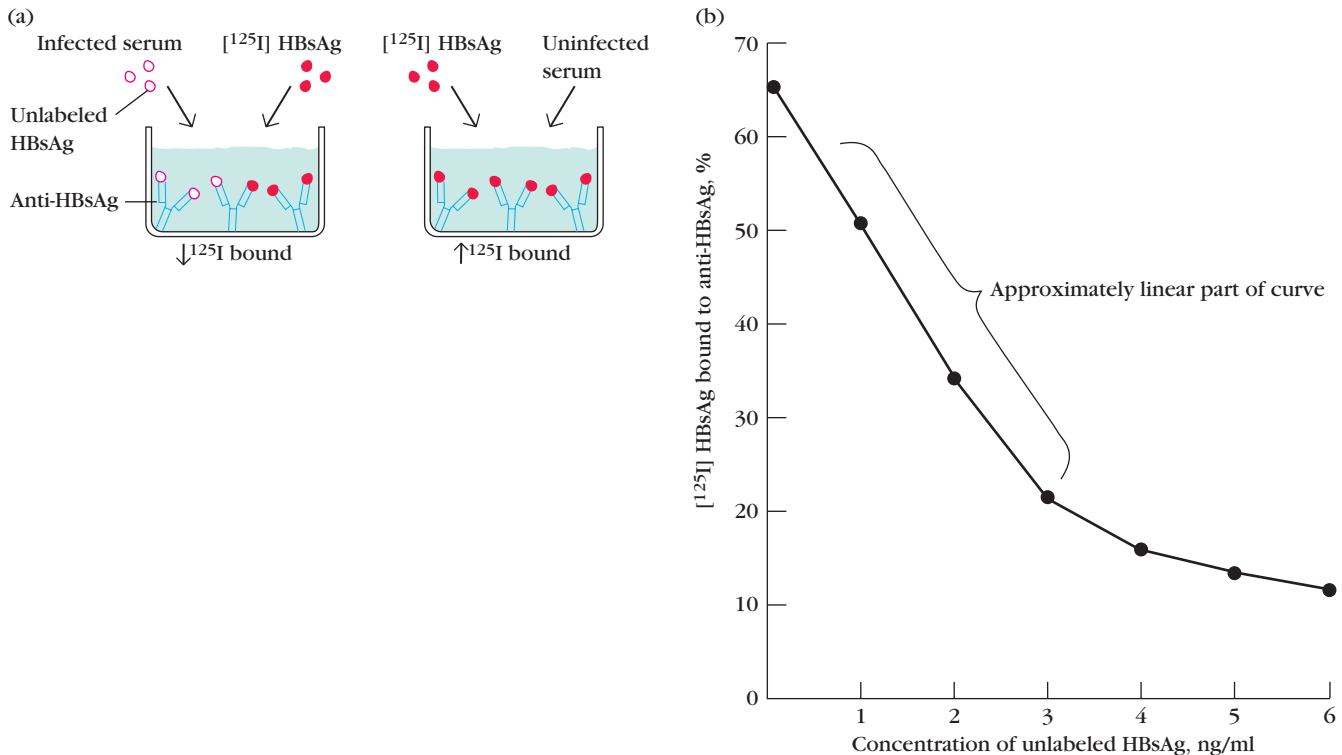


FIGURE 6-9 A solid-phase radioimmunoassay (RIA) to detect hepatitis B virus in blood samples. (a) Microtiter wells are coated with a constant amount of antibody specific for HBsAg, the surface antigen on hepatitis B virions. A serum sample and [¹²⁵I]HBsAg are then added. After incubation, the supernatant is removed and the radioactivity of the antigen-antibody complexes is measured. If the sample is infected, the amount of label bound will be less than

in controls with uninfected serum. (b) A standard curve is obtained by adding increasing concentrations of unlabeled HBsAg to a fixed quantity of [¹²⁵I]HBsAg and specific antibody. From the plot of the percentage of labeled antigen bound versus the concentration of unlabeled antigen, the concentration of HBsAg in unknown serum samples can be determined by using the linear part of the curve.

Enzyme-Linked Immunosorbent Assay

Enzyme-linked immunosorbent assay, commonly known as **ELISA** (or EIA), is similar in principle to RIA but depends on an enzyme rather than a radioactive label. An enzyme conjugated with an antibody reacts with a colorless substrate to generate a colored reaction product. Such a substrate is called a **chromogenic substrate**. A number of enzymes have been employed for ELISA, including alkaline phosphatase, horseradish peroxidase, and β -galactosidase. These assays approach the sensitivity of RIAs and have the advantage of being safer and less costly.

There Are Numerous Variants of ELISA

A number of variations of ELISA have been developed, allowing qualitative detection or quantitative measurement of either antigen or antibody. Each type of ELISA can be used qualitatively to detect the presence of antibody or antigen. Alternatively, a standard curve based on known

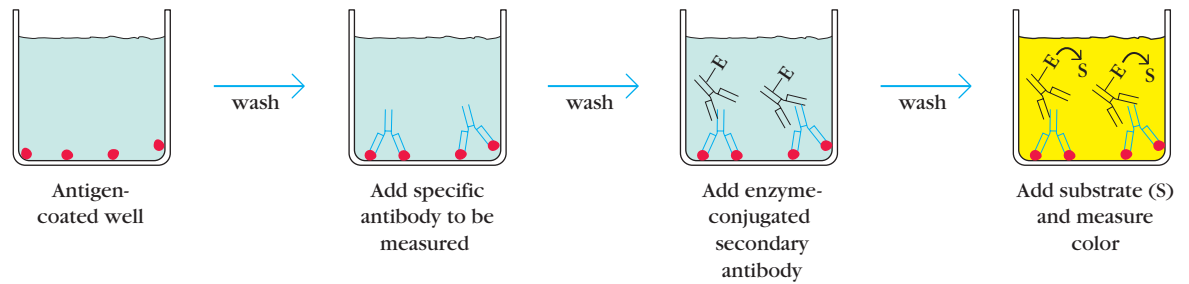
concentrations of antibody or antigen is prepared, from which the unknown concentration of a sample can be determined.

INDIRECT ELISA

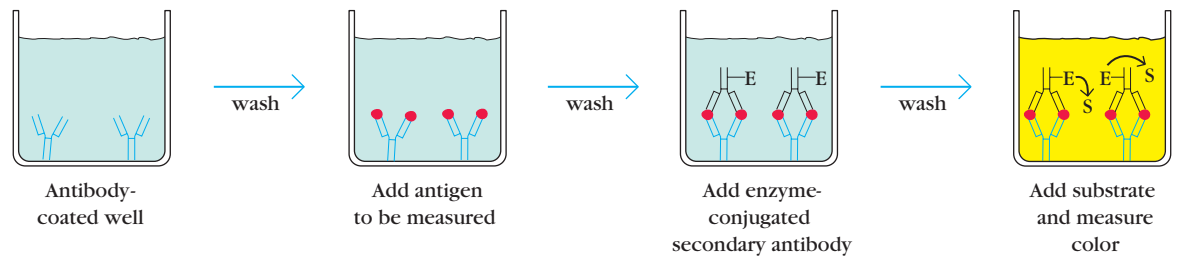
Antibody can be detected or quantitatively determined with an indirect ELISA (Figure 6-10a). Serum or some other sample containing primary antibody (Ab_1) is added to an antigen-coated microtiter well and allowed to react with the antigen attached to the well. After any free Ab_1 is washed away, the presence of antibody bound to the antigen is detected by adding an enzyme-conjugated secondary anti-isotype antibody (Ab_2), which binds to the primary antibody. Any free Ab_2 then is washed away, and a substrate for the enzyme is added. The amount of colored reaction product that forms is measured by specialized spectrophotometric plate readers, which can measure the absorbance of all of the wells of a 96-well plate in seconds.

Indirect ELISA is the method of choice to detect the presence of serum antibodies against human immunodeficiency virus (HIV), the causative agent of AIDS. In this assay, recombinant envelope and core proteins of HIV are adsorbed

(a) Indirect ELISA



(b) Sandwich ELISA



(c) Competitive ELISA

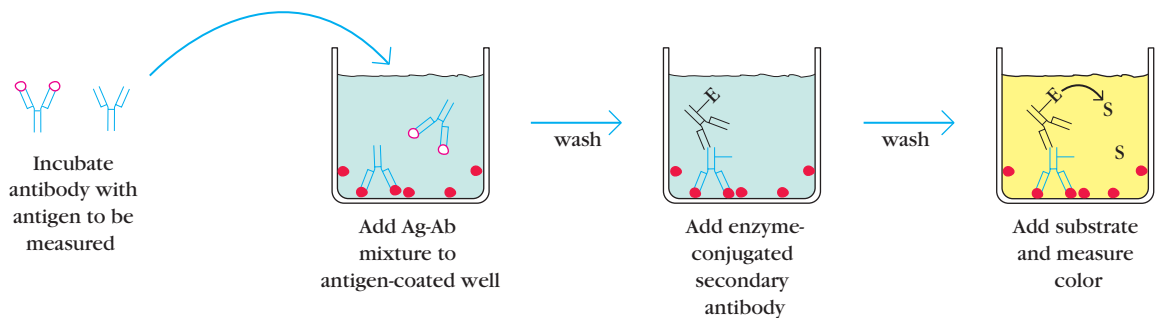


FIGURE 6-10 Variations in the enzyme-linked immunosorbent assay (ELISA) technique allow determination of antibody or antigen. Each assay can be used qualitatively, or quantitatively by comparison with standard curves prepared with known concentrations of antibody or antigen. Antibody can be determined with an indirect ELISA

(a), whereas antigen can be determined with a sandwich ELISA (b) or competitive ELISA (c). In the competitive ELISA, which is an inhibition-type assay, the concentration of antigen is inversely proportional to the color produced.

as solid-phase antigens to microtiter wells. Individuals infected with HIV will produce serum antibodies to epitopes on these viral proteins. Generally, serum antibodies to HIV can be detected by indirect ELISA within 6 weeks of infection.

SANDWICH ELISA

Antigen can be detected or measured by a sandwich ELISA (Figure 6-10b). In this technique, the antibody (rather than the antigen) is immobilized on a microtiter well. A sample containing antigen is added and allowed to react with the immobilized antibody. After the well is washed, a second enzyme-linked antibody specific for a different epitope on the antigen is added and allowed to react with the bound antigen. After any free second antibody is removed by washing,

substrate is added, and the colored reaction product is measured.

COMPETITIVE ELISA

Another variation for measuring amounts of antigen is competitive ELISA (Figure 6-10c). In this technique, antibody is first incubated in solution with a sample containing antigen. The antigen-antibody mixture is then added to an antigen-coated microtiter well. The more antigen present in the sample, the less free antibody will be available to bind to the antigen-coated well. Addition of an enzyme-conjugated secondary antibody (Ab_2) specific for the isotype of the primary antibody can be used to determine the amount of primary antibody bound to the well as in an indirect ELISA. In the

competitive assay, however, the higher the concentration of antigen in the original sample, the lower the absorbance.

CHEMILUMINESCENCE

Measurement of light produced by chemiluminescence during certain chemical reactions provides a convenient and highly sensitive alternative to absorbance measurements in ELISA assays. In versions of the ELISA using chemiluminescence, a luxogenic (light-generating) substrate takes the place of the chromogenic substrate in conventional ELISA reactions. For example, oxidation of the compound luminol by H_2O_2 and the enzyme horseradish peroxidase (HRP) produces light:



The advantage of chemiluminescence assays over chromogenic ones is enhanced sensitivity. In general, the detection limit can be increased at least ten-fold by switching from a chromogenic to a luxogenic substrate, and with the addition of enhancing agents, more than 200-fold. In fact, under ideal conditions, as little as 5×10^{-18} moles (5 attomoles) of target antigen have been detected.

ELISPOT ASSAY

A modification of the ELISA assay called the ELISPOT assay allows the quantitative determination of the number of cells in a population that are producing antibodies specific for a given antigen or an antigen for which one has a specific antibody (Figure 6-11). In this approach, the plates are coated with the antigen (capture antigen) recognized by the antibody of interest or with the antibody (capture antibody) specific for the antigen whose production is being assayed. A suspension of the cell population under investigation is then added to the coated plates and incubated. The cells settle onto the surface of the plate, and secreted molecules reactive with the capture molecules are bound by the capture molecules in the vicinity of the secreting cells, producing a ring of antigen-antibody complexes around each cell that is producing the molecule of interest. The plate is then washed and an enzyme-linked antibody specific for the secreted antigen or specific for the species (e.g., goat anti-rabbit) of the secreted antibody is added and allowed to bind. Subsequent development of the assay by addition of a suitable chromogenic or chemiluminescence-producing substrate reveals the position of each antibody- or antigen-producing cell as a point of color or light.

Western Blotting

Identification of a specific protein in a complex mixture of proteins can be accomplished by a technique known as **Western blotting**, named for its similarity to Southern blotting,

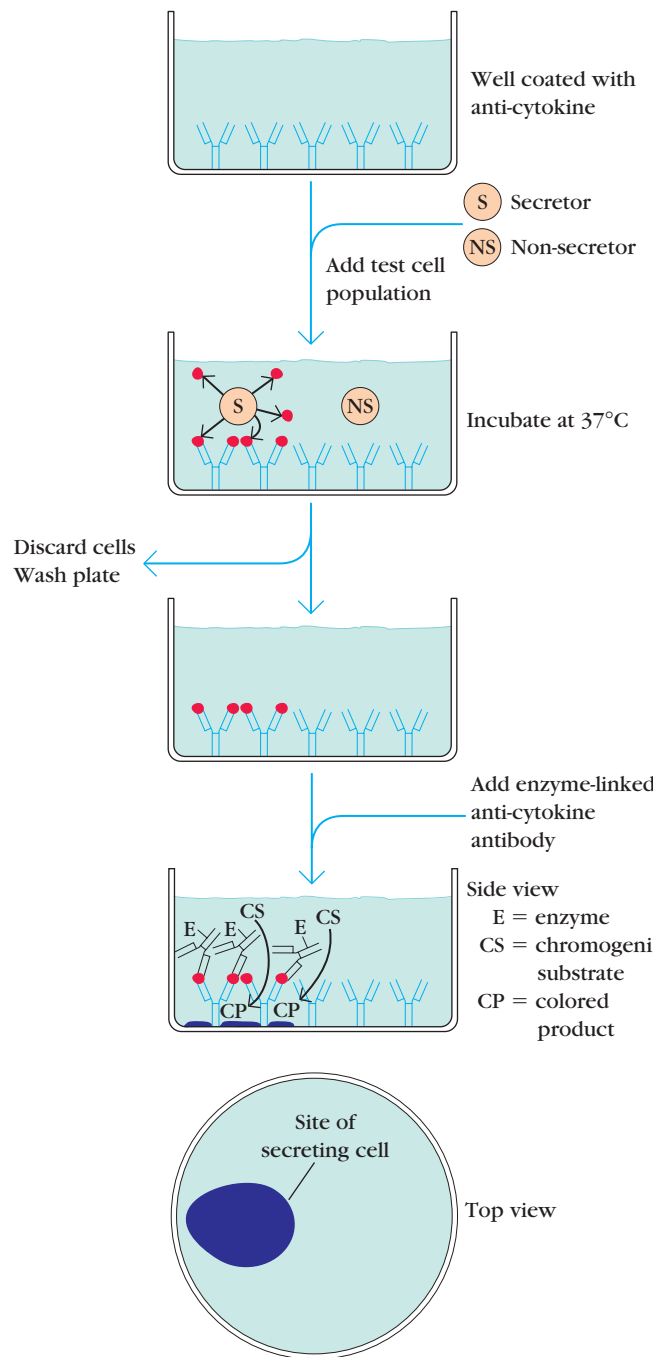


FIGURE 6-11 In the ELISPOT assay, a well is coated with antibody against the antigen of interest, a cytokine in this example, and then a suspension of a cell population thought to contain some members synthesizing and secreting the cytokine are layered onto the bottom of the well and incubated. Most of the cytokine molecules secreted by a particular cell react with nearby well-bound antibodies. After the incubation period, the well is washed and an enzyme-labeled anti-cytokine antibody is added. After washing away unbound antibody, a chromogenic substrate that forms an insoluble colored product is added. The colored product (purple) precipitates and forms a spot only on the areas of the well where cytokine-secreting cells had been deposited. By counting the number of colored spots, it is possible to determine how many cytokine-secreting cells were present in the added cell suspension.

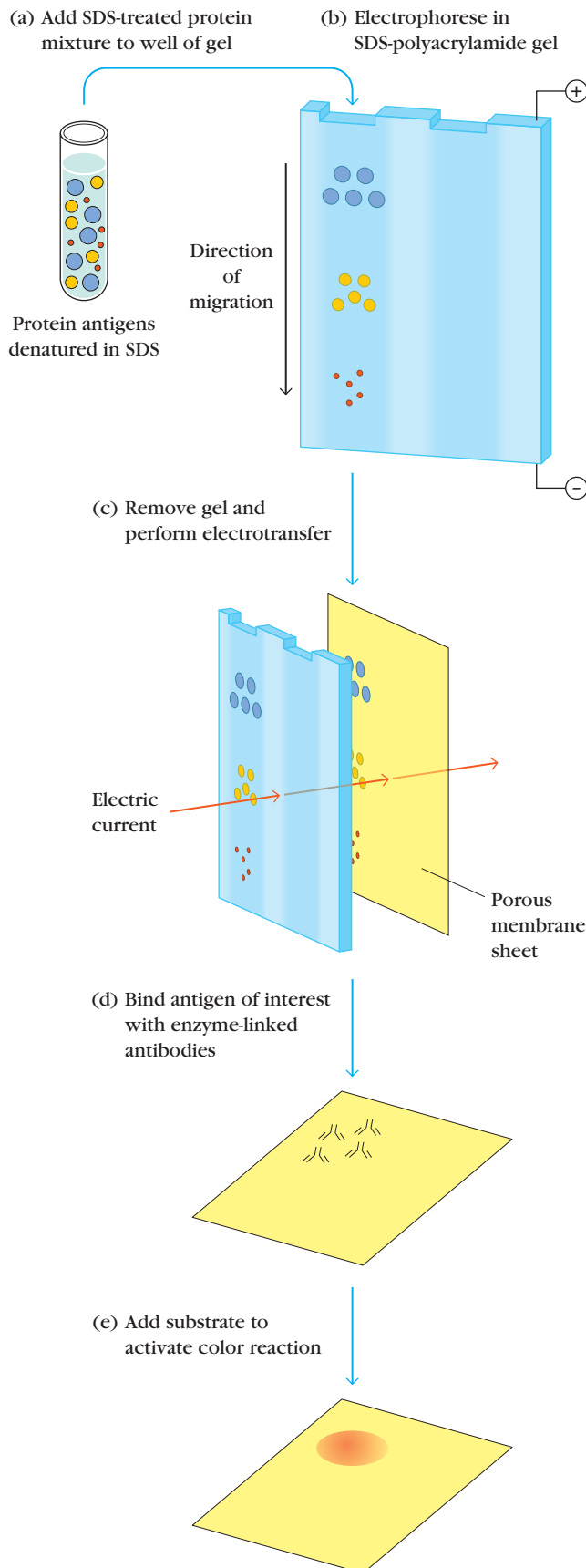


FIGURE 6-12 In Western blotting, a protein mixture is (a) treated with SDS, a strong denaturing detergent, (b) then separated by electrophoresis in an SDS polyacrylamide gel (SDS-PAGE) which separates the components according to their molecular weight; lower molecular weight components migrate farther than higher molecular weight ones. (c) The gel is removed from the apparatus and applied to a protein-binding sheet of nitrocellulose or nylon and the proteins in the gel are transferred to the sheet by the passage of an electric current. (d) Addition of enzyme-linked antibodies detects the antigen of interest, and (e) the position of the antibodies is visualized by means of an ELISA reaction that generates a highly colored insoluble product that is deposited at the site of the reaction. Alternatively, a chemiluminescent ELISA can be used to generate light that is readily detected by exposure of the blot to a piece of photographic film.

which detects DNA fragments, and Northern blotting, which detects mRNAs. In Western blotting, a protein mixture is electrophoretically separated on an **SDS-polyacrylamide gel (SDS-PAGE)**, a slab gel infused with sodium dodecyl sulfate (SDS), a dissociating agent (Figure 6-12). The protein bands are transferred to a nylon membrane by electrophoresis and the individual protein bands are identified by flooding the nitrocellulose membrane with radiolabeled or enzyme-linked polyclonal or monoclonal antibody specific for the protein of interest. The Ag-Ab complexes that form on the band containing the protein recognized by the antibody can be visualized in a variety of ways. If the protein of interest was bound by a radioactive antibody, its position on the blot can be determined by exposing the membrane to a sheet of x-ray film, a procedure called autoradiography. However, the most generally used detection procedures employ enzyme-linked antibodies against the protein. After binding of the enzyme-antibody conjugate, addition of a chromogenic substrate that produces a highly colored and insoluble product causes the appearance of a colored band at the site of the target antigen. The site of the protein of interest can be determined with much higher sensitivity if a chemiluminescent compound along with suitable enhancing agents is used to produce light at the antigen site.

Western blotting can also identify a specific antibody in a mixture. In this case, known antigens of well-defined molecular weight are separated by SDS-PAGE and blotted onto nitrocellulose. The separated bands of known antigens are then probed with the sample suspected of containing antibody specific for one or more of these antigens. Reaction of an antibody with a band is detected by using either radiolabeled or enzyme-linked secondary antibody that is specific for the species of the antibodies in the test sample. The most widely used application of this procedure is in confirmatory testing for HIV, where Western blotting is used to determine whether the patient has antibodies that react with one or more viral proteins.

Immunoprecipitation

The immunoprecipitation technique has the advantage of allowing the isolation of the antigen of interest for further analysis. It also provides a sensitive assay for the presence of a particular antigen in a given cell or tissue type. An extract produced by disruption of cells or tissues is mixed with an antibody against the antigen of interest in order to form an antigen-antibody complex that will precipitate. However, if the antigen concentration is low (often the case in cell and tissue extracts), the assembly of antigen-antibody complexes into precipitates can take hours, even days, and it is difficult to isolate the small amount of immunoprecipitate that forms.

Fortunately, there are a number of ways to avoid these limitations. One is to attach the antibody to a solid support, such as a synthetic bead, which allows the antigen-antibody complex to be collected by centrifugation. Another is to add a secondary antibody specific for the primary antibody to bind the antigen-antibody complexes. If the secondary antibody is attached to a bead, the immune complexes can be collected by centrifugation. A particularly ingenious version of this procedure involves the coupling of the secondary antibody to magnetic beads. After the secondary antibody binds to the primary antibody, immunoprecipitates are collected by placing a magnet against the side of the tube (Figure 6-13).

When used in conjunction with biosynthetic radioisotope labeling, immunoprecipitation can also be used to determine

whether a particular antigen is actually synthesized by a cell or tissue. Radiolabeling of proteins synthesized by cells of interest can be done by growing the cells in cell-culture medium containing one or more radiolabeled amino acids. Generally, the amino acids used for this application are those most resistant to metabolic modification, such as leucine, cysteine, or methionine. After growth in the radioactive medium, the cells are lysed and subjected to a primary antibody specific for the antigen of interest. The Ag-Ab complex is collected by immunoprecipitation, washed free of unincorporated radiolabeled amino acid and other impurities, and then analyzed. The complex can be counted in a scintillation counter to obtain a quantitative determination of the amount of the protein synthesized. Further analysis often involves disruption of the complex, usually by use of SDS and heat, so that the identity of the immunoprecipitated antigen can be confirmed by checking that its molecular weight is that expected for the antigen of interest. This is done by separation of the disrupted complex by SDS-PAGE and subsequent autoradiography to determine the position of the radiolabeled antigen on the gel.

Immunofluorescence

In 1944, Albert Coons showed that antibodies could be labeled with molecules that have the property of fluorescence. Fluorescent molecules absorb light of one wavelength

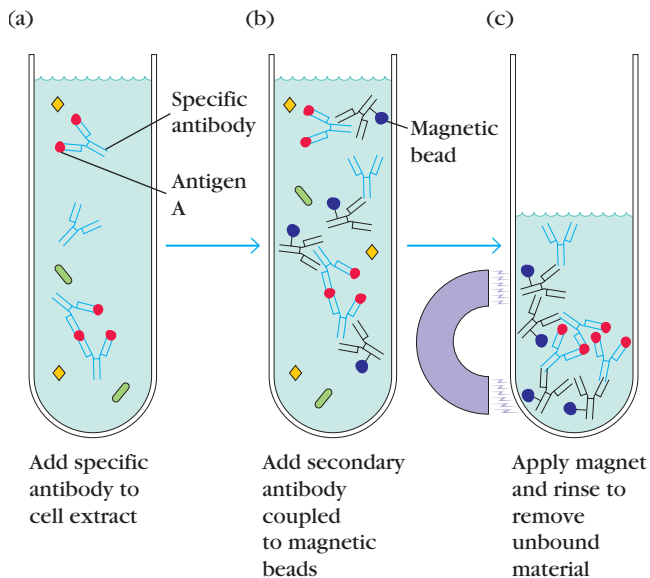
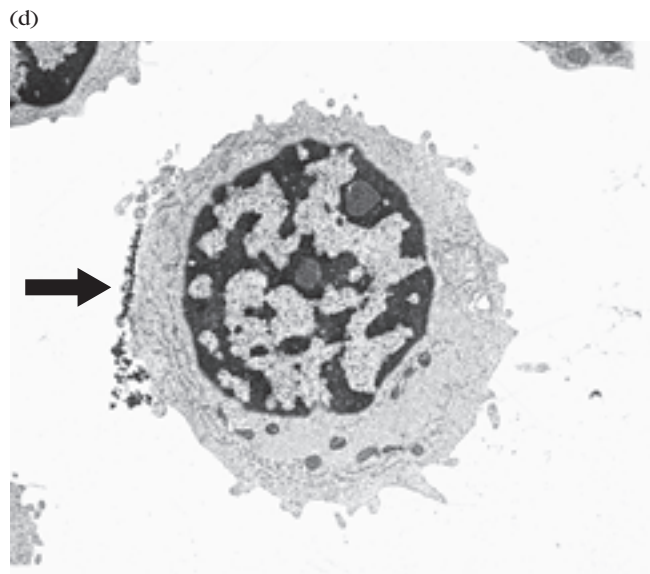


FIGURE 6-13 Immunoprecipitates can be collected using magnetic beads coupled to a secondary antibody. (a) Treatment of a cell extract containing antigen A (red) with a mouse anti-A antibody (blue) results in the formation of antigen-antibody complexes. (b) Addition of magnetic beads to which a rabbit anti-mouse antibody is linked binds the antigen-antibody complexes (and any unreacted mouse Ig). (c) Placing a magnet against the side of the tube



allows the rapid collection of the antigen-antibody complexes. After rinsing to remove any unbound material, the antigen-antibody complexes can be dissociated and the antigen studied. (d) An electron micrograph showing a cell with magnetic beads attached to its surface via antibodies. [Part (d), P. Groscurth, *Institute of Anatomy, University of Zurich-Irchel*.]

(excitation) and emit light of another wavelength (emission). If antibody molecules are tagged with a fluorescent dye, or **fluorochrome**, immune complexes containing these fluorescently labeled antibodies (FA) can be detected by colored light emission when excited by light of the appropriate wavelength. Antibody molecules bound to antigens in cells or tissue sections can similarly be visualized. The emitted light can be viewed with a fluorescence microscope, which is equipped with a UV light source. In this technique, known as **immunofluorescence**, fluorescent compounds such as fluorescein and rhodamine are in common use, but other highly fluorescent substances are also routinely used, such as phycoerythrin, an intensely colored and highly fluorescent pigment obtained from algae. These molecules can be conjugated to the Fc region of an antibody molecule without affecting the specificity of the antibody. Each of the fluorochromes below absorbs light at one wavelength and emits light at a longer wavelength:

- **Fluorescein**, an organic dye that is the most widely used label for immunofluorescence procedures, absorbs blue light (490 nm) and emits an intense yellow-green fluorescence (517 nm).
- **Rhodamine**, another organic dye, absorbs in the yellow-green range (515 nm) and emits a deep red

fluorescence (546 nm). Because it emits fluorescence at a longer wavelength than fluorescein, it can be used in two-color immunofluorescence assays. An antibody specific to one determinant is labeled with fluorescein, and an antibody recognizing a different antigen is labeled with rhodamine. The location of the fluorescein-tagged antibody will be visible by its yellow-green color, easy to distinguish from the red color emitted where the rhodamine-tagged antibody has bound. By conjugating fluorescein to one antibody and rhodamine to another antibody, one can, for example, visualize simultaneously two different cell-membrane antigens on the same cell.

- **Phycoerythrin** is an efficient absorber of light (~30-fold greater than fluorescein) and a brilliant emitter of red fluorescence, stimulating its wide use as a label for immunofluorescence.

Fluorescent-antibody staining of cell membrane molecules or tissue sections can be direct or indirect (Figure 6-14). In **direct staining**, the specific antibody (the primary antibody) is directly conjugated with fluorescein; in **indirect staining**, the primary antibody is unlabeled and is detected with an additional fluorochrome-labeled reagent. A number of reagents have been developed for indirect staining.

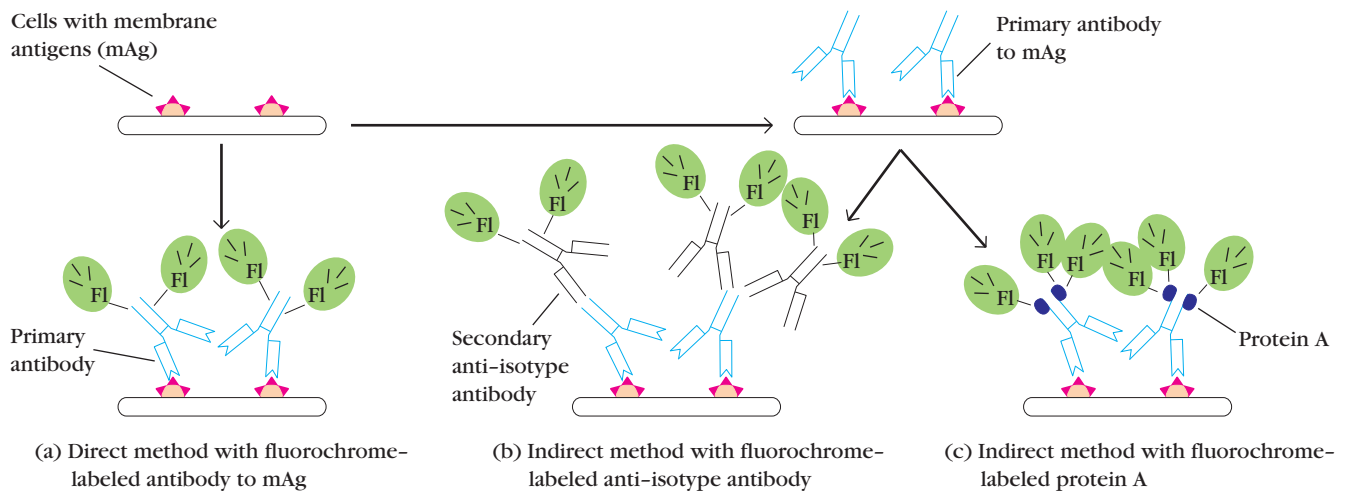
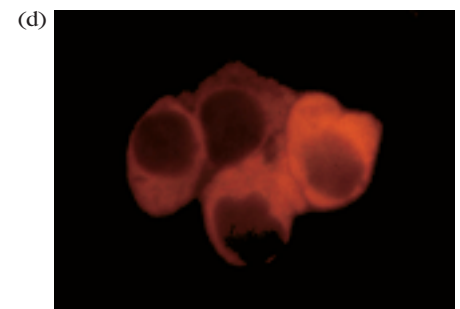


FIGURE 6-14 Direct and indirect immunofluorescence staining of membrane antigen (mAg). Cells are affixed to a microscope slide. In the direct method (a), cells are stained with anti-mAg antibody that is labeled with a fluorochrome (FI). In the indirect methods (b and c), cells are first incubated with unlabeled anti-mAg antibody and then stained with a fluorochrome-labeled secondary reagent that binds to the primary antibody. Cells are viewed under a fluorescence microscope to see if they have been stained. (d) In this micrograph, antibody molecules bearing μ heavy chains are detected by indirect staining of cells with rhodamine-conjugated second antibody. [Part(d), H. A. Schreuder et al., 1997, *Nature* **386**:196, courtesy H. Schreuder, Hoechst Marion Roussel.]



The most common is a fluorochrome-labeled secondary antibody raised in one species against antibodies of another species, such as fluorescein-labeled goat anti-mouse immunoglobulin.

Indirect immunofluorescence staining has two advantages over direct staining. First, the primary antibody does not need to be conjugated with a fluorochrome. Because the supply of primary antibody is often a limiting factor, indirect methods avoid the loss of antibody that usually occurs during the conjugation reaction. Second, indirect methods increase the sensitivity of staining because multiple molecules of the fluorochrome reagent bind to each primary antibody molecule, increasing the amount of light emitted at the location of each primary antibody molecule.

Immunofluorescence has been applied to identify a number of subpopulations of lymphocytes, notably the CD4⁺ and CD8⁺ T-cell subpopulations. The technique is also suitable for identifying bacterial species, detecting Ag-Ab complexes in autoimmune disease, detecting complement components in tissues, and localizing hormones and other cellular products stained in situ. Indeed, a major application of the fluorescent-antibody technique is the localization of antigens in tissue sections or in subcellular compartments. Because it can be used to map the actual location of target antigens, fluorescence microscopy is a powerful tool for relating the molecular architecture of tissues and organs to their overall gross anatomy.

Flow Cytometry and Fluorescence

The fluorescent antibody techniques described are extremely valuable qualitative tools, but they do not give quantitative data. This shortcoming was remedied by development of the flow cytometer, which was designed to automate the analysis and separation of cells stained with fluorescent antibody. The flow cytometer uses a laser beam and light detector to count single intact cells in suspension (Figure 6-15). Every time a cell passes the laser beam, light is deflected from the detector, and this interruption of the laser signal is recorded. Those cells having a fluorescently tagged antibody bound to their cell surface antigens are excited by the laser and emit light that is recorded by a second detector system located at a right angle to the laser beam. The simplest form of the instrument counts each cell as it passes the laser beam and records the level of fluorescence the cell emits; an attached computer generates plots of the number of cells as the ordinate and their fluorescence intensity as the abscissa. More sophisticated versions of the instrument are capable of sorting populations of cells into different containers according to their fluorescence profile. Use of the instrument to determine which and how many members of a cell population bind fluorescently labeled antibodies is called analysis; use of the instrument to place cells having different patterns of reactivity into different containers is called cell sorting.

The flow cytometer has multiple applications to clinical and research problems. A common clinical use is to determine the kind and number of white blood cells in blood samples. By treating appropriately processed blood samples with a fluorescently labeled antibody and performing flow cytometric analysis, one can obtain the following information:

- How many cells express the target antigen as an absolute number and also as a percentage of cells passing the beam. For example, if one uses a fluorescent antibody specific for an antigen present on all T cells, it would be possible to determine the percentage of T cells in the total white blood cell population. Then, using the cell-sorting capabilities of the flow cytometer, it would be possible to isolate the T-cell fraction of the leukocyte population.
- The distribution of cells in a sample population according to antigen densities as determined by fluorescence intensity. It is thus possible to obtain a measure of the distribution of antigen density within the population of cells that possess the antigen. This is a powerful feature of the instrument, since the same type of cell may express different levels of antigen depending upon its developmental or physiological state.
- The size of cells. This information is derived from analysis of the light-scattering properties of members of the cell population under examination.

Flow cytometry also makes it possible to analyze cell populations that have been labeled with two or even three different fluorescent antibodies. For example, if a blood sample is reacted with a fluorescein-tagged antibody specific for T cells, and also with a phycoerythrin-tagged antibody specific for B cells, the percentages of B and T cells may be determined simultaneously with a single analysis. Numerous variations of such “two-color” analyses are routinely carried out, and “three-color” experiments are common. Aided by appropriate software, highly sophisticated versions of the flow cytometer can even perform “five-color” analyses.

Flow cytometry now occupies a key position in immunology and cell biology, and it has become an indispensable clinical tool as well. In many medical centers, the flow cytometer is one of the essential tools for the detection and classification of leukemias (see the Clinical Focus). The choice of treatment for leukemia depends heavily on the cell types involved, making precise identification of the neoplastic cells an essential part of clinical practice. Likewise, the rapid measurement of T-cell subpopulations, an important prognostic indicator in AIDS, is routinely done by flow-cytometric analysis. In this procedure, labeled monoclonal antibodies against the major T-cell subtypes bearing the CD4 and CD8 antigens are used to determine their ratios in the patient's blood. When the number of CD4 T cells falls below a certain level, the patient is at high risk for opportunistic infections.

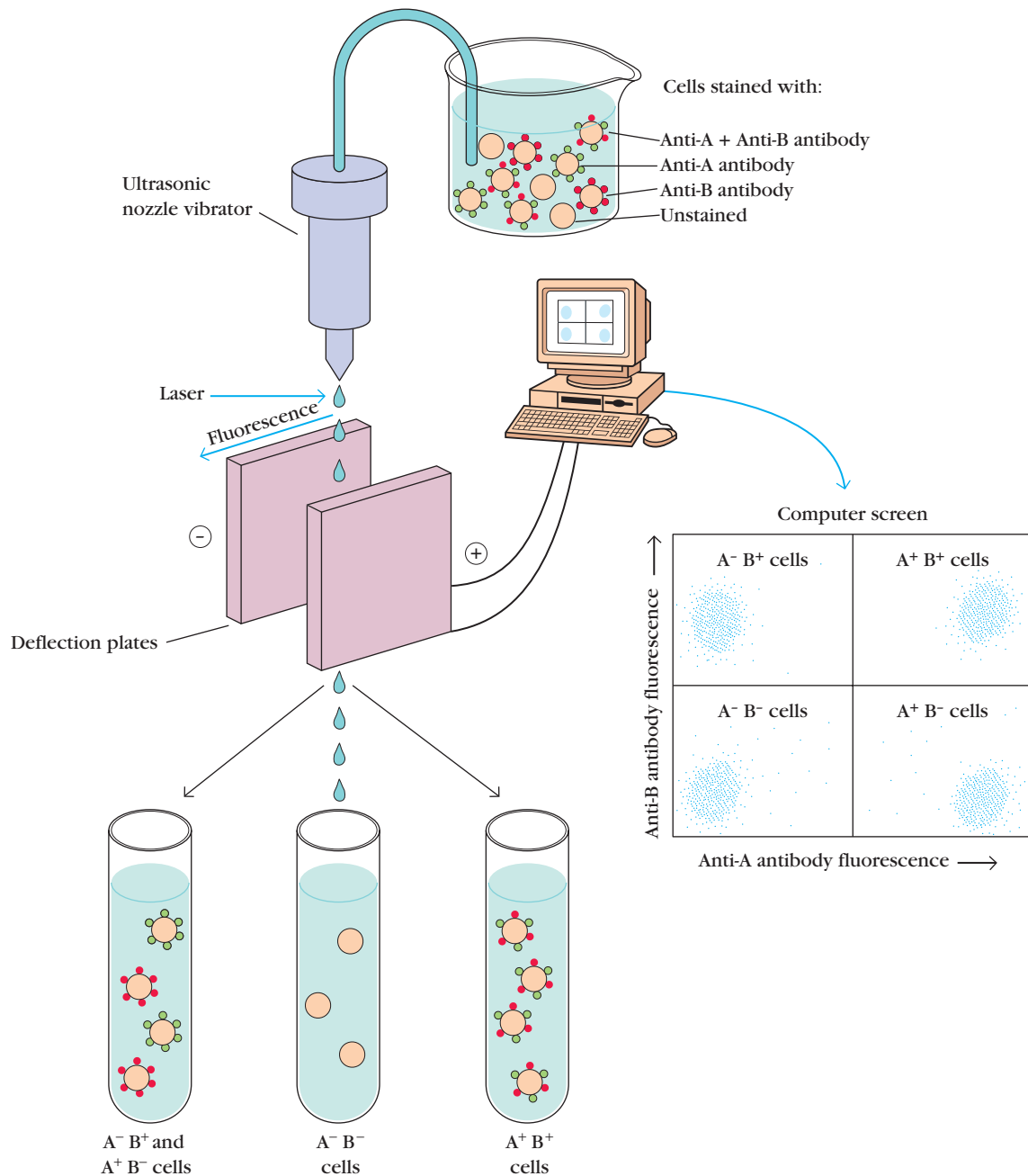


FIGURE 6-15 Separation of fluorochrome-labeled cells with the flow cytometer. In the example shown, a mixed cell population is stained with two antibodies, one specific for surface antigen A and the other specific for surface antigen B. The anti-A antibodies are labeled with fluorescein (green) and the anti-B antibodies with rhodamine (red). The stained cells are loaded into the sample chamber of the cytometer. The cells are expelled, one at a time, from a small vibrating nozzle that generates microdroplets, each containing no more than a single cell. As it leaves the nozzle, each droplet receives a small electrical charge, and the computer that controls the flow cytometer can detect exactly when a drop generated by the nozzle passes through the beam of laser light that excites the fluorochrome. The intensity of the fluorescence emitted by each droplet that contains a cell is monitored by a detector and displayed on a computer screen. Because the computer tracks the position of each droplet, it is possible to determine when a partic-

ular droplet will arrive between the deflection plates. By applying a momentary charge to the deflection plates when a droplet is passing between them, it is possible to deflect the path of a particular droplet into one or another collecting vessel. This allows the sorting of a population of cells into subpopulations having different profiles of surface markers.

In the computer display, each dot represents a cell. Cells that fall into the lower left-hand panel have background levels of fluorescence and are judged not to have reacted with either antibody anti-A or anti-B. Those that appear in the upper left panel reacted with anti-B but not anti-A, and those in the lower right panel reacted with anti-A but not anti-B. The upper right panel contains cells that react with both anti-A and anti-B. In the example shown here, the A⁻B⁻—and the A⁺B⁺—subpopulations have each been sorted into a separate tube. Staining with anti-A and anti-B fluorescent antibodies allows four subpopulations to be distinguished: A⁻B⁻, A⁺B⁺, A⁻B⁺, and A⁺B⁻.



CLINICAL FOCUS

Flow Cytometry and Leukemia Typing

Leukemia is the unchecked proliferation of an abnormal clone of hematopoietic cells. Typically, leukemic cells respond poorly or inappropriately to regulatory signals, display aberrant patterns of differentiation, or even fail to differentiate. Furthermore, they sometimes suppress the growth of normal lymphoid and myeloid cells. Leukemia can arise at any maturational stage of any one of the hematopoietic lineages. Lymphocytic leukemias display many characteristics of cells of the lymphoid lineage; another broad group, myelogenous leukemias, have attributes of members of the myeloid lineage. Aside from lineage, many leukemias can be classified as acute or chronic. Some examples are acute lymphocytic leukemia (ALL), the most common childhood leukemia; acute myelogenous leukemia (AML), found more often in

adults than in children; and chronic lymphocytic leukemia (CLL), which is rarely seen in children but is the most common form of adult leukemia in the Western world. A fourth type, chronic myelogenous leukemia (CML), occurs much more often in older adults than in children.

The diagnosis of leukemia is made on the basis of two findings. One is the detection of abnormal cells in the bloodstream, and the other is observation of abnormal cells in the bone marrow. Clinical experience has shown that designing the most appropriate therapy for the patient requires knowing which type of leukemia is present. In this regard, two of the important questions are: (1) What is the lineage of the abnormal cells and (2) What is their maturational stage? A variety of approaches, including cytologic examination of cell morphology and staining characteristics, immuno-

phenotyping, and, in some cases, an analysis of gene rearrangements, are useful in answering these questions. One of the most powerful of these approaches is immunophenotyping, the determination of the profile of selected cell-surface markers displayed by the leukemic cell. Although leukemia-specific antigens have not yet been found, profiles of expressed surface antigens often can establish cell lineage, and they are frequently helpful in determining the maturational stages present in leukemic cell populations. For example, an abnormal cell that displays surface immunoglobulin would be assigned to the B-cell lineage and its maturational stage would be that of a mature B cell. On the other hand, a cell that had cytoplasmic μ heavy chains, but no surface immunoglobulin, would be a B-lineage leukemic cell but at the maturational stage of a pre-B cell. The most efficient and precise technology for immunophenotyping uses flow cytometry and monoclonal antibodies. The availability of monoclonal antibodies specific for each of the scores of antigens found on various types and subtypes of hematopoietic cells has made it possible to identify patterns of antigen

Alternatives to Antigen-Antibody Reactions

As a defense against host antibodies, some bacteria have evolved the ability to make proteins that bind to the Fc region of IgG molecules with high affinity ($K_a \sim 10^8$). One such molecule, known as **protein A**, is found in the cell walls of some strains of *Staphylococcus aureus*, and another, **protein G**, appears in the walls of group C and G *Streptococcus*. By cloning the genes for protein A and protein G and generating a hybrid of both, one can make a recombinant protein, known as **protein A/G**, that combines some of the best features of both. These molecules are useful because they bind IgG from many different species. Thus they can be labeled with fluorochromes, radioactivity, or biotin and used to detect IgG molecules in the antigen-antibody complexes formed during ELISA, RIA, or such fluorescence-based assays as flow cytometry or fluorescence microscopy. These bacterial IgG-binding proteins can also be used to make affinity columns for the isolation of IgG.

Egg whites contain a protein called *avidin* that binds biotin, a vitamin that is essential for fat synthesis. Avidin is believed to have evolved as a defense against marauding rodents that rob nests and eat the stolen eggs. The binding between avidin and biotin is extremely specific and of much higher affinity ($K_a \sim 10^{15}$) than any known antigen-antibody reaction. A bacterial protein called **streptavidin**, made by *streptomyces avidinii*, has similarly high affinity and specificity. The extraordinary affinity and exquisite specificity of the interaction of these proteins with biotin is widely used in many immunological procedures. The primary or secondary antibody is labeled with biotin and allowed to react with the target antigen, and the unbound antibody is then washed away. Subsequently, streptavidin or avidin conjugated with an enzyme, fluorochrome, or radioactive label is used to detect the bound antibody.

Immunoelectron Microscopy

The fine specificity of antibodies has made them powerful tools for visualizing specific intracellular tissue components

expression that are typical of cell lineages, maturational stages, and a number of different types of leukemia. Most cancer centers are equipped with flow cytometers that are capable of performing and interpreting the multiparameter analyses necessary to provide useful pro-

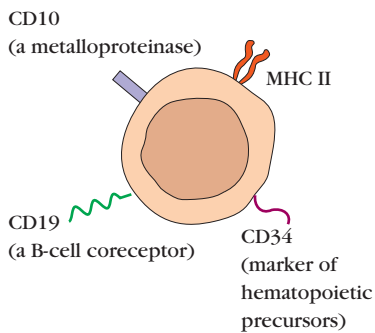
files of surface markers on tumor cell populations. Flow cytometric determination of immuno-phenotypes allows:

- Confirmation of diagnosis
- Diagnosis when no clear judgment can be made based on morphology or

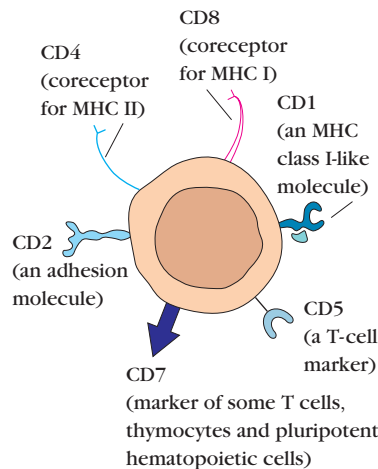
patterns of cytochemical staining

- Identification of aberrant antigen profiles that can help identify the return of leukemia during remission
- Improved prediction of the course of the disease

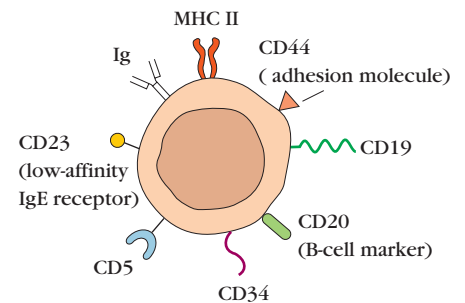
An ALL of the pre-B lineage
(the most commonly occurring ALL)



ALL of the T lineage



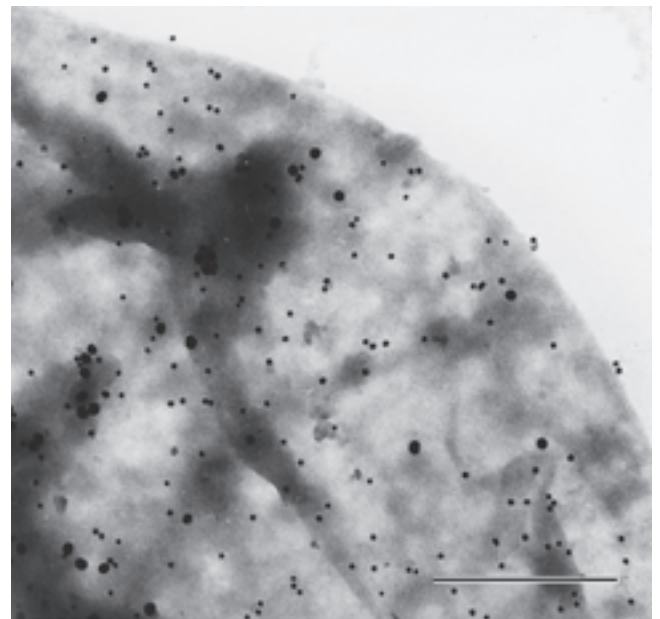
A B-lineage CLL



Distribution of selected markers on some leukemic cell types. Shown are typical surface antigen profiles found on many, but not all, ALLs and CLLs.

by **immunoelectron microscopy**. In this technique, an electron-dense label is either conjugated to the Fc portion of a specific antibody for direct staining or conjugated to an anti-immunoglobulin reagent for indirect staining. A number of electron-dense labels have been employed, including *ferritin* and *colloidal gold*. Because the electron-dense label absorbs electrons, it can be visualized with the electron microscope as small black dots. In the case of immunogold labeling, different antibodies can be conjugated with gold particles of different sizes, allowing identification of several antigens within a cell by the different sizes of the electron-dense gold particles attached to the antibodies (Figure 6-16).

FIGURE 6-16 An immunoelectronmicrograph of the surface of a B-cell lymphoma was stained with two antibodies: one against class II MHC molecules labeled with 30-nm gold particles, and another against MHC class I molecules labeled with 15-nm gold particles. The density of class I molecules exceeds that of class II on this cell. Bar = 500 nm. [From A. Jenei et al., 1997, PNAS **94**:7269–7274; courtesy of A. Jenei and S. Damjanovich, University Medical School of Debrecen, Hungary.]



SUMMARY

- Antigen-antibody interactions depend on four types of noncovalent interactions: hydrogen bonds, ionic bonds, hydrophobic interactions, and van der Waals interactions.
- The affinity constant, which can be determined by Scatchard analysis, provides a quantitative measure of the strength of the interaction between an epitope of the antigen and a single binding site of an antibody. The avidity reflects the overall strength of the interactions between a multivalent antibody molecule and a multivalent antigen molecule at multiple sites.
- The interaction of a soluble antigen and precipitating antibody in a liquid or gel medium forms an Ag-Ab precipitate. Electrophoresis can be combined with precipitation in gels in a technique called immunoelectrophoresis.
- The interaction between a particulate antigen and agglutinating antibody (agglutinin) produces visible clumping, or agglutination that forms the basis of simple, rapid, and sensitive immunoassays.
- Radioimmunoassay (RIA) is a highly sensitive and quantitative procedure that utilizes radioactively labeled antigen or antibody.
- The enzyme-linked immunosorbent assay (ELISA) depends on an enzyme-substrate reaction that generates a colored reaction product. ELISA assays that employ chemiluminescence instead of a chromogenic reaction are the most sensitive immunoassays available.
- In Western blotting, a protein mixture is separated by electrophoresis; then the protein bands are electrophoretically transferred onto nitrocellulose and identified with labeled antibody or labeled antigen.
- Fluorescence microscopy using antibodies labeled with fluorescent molecules can be used to visualize antigen on or within cells.
- Flow cytometry provides an unusually powerful technology for the quantitative analysis and sorting of cell populations labeled with one or more fluorescent antibodies.

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USEFUL WEB SITES

<http://pathlabsofark.com/flowcyttests.html>

Explore the Pathology Laboratories of Arkansas to see what kinds of samples are taken from patients and what markers are used to evaluate lymphocyte populations by flow cytometry.

<http://jcsmr.anu.edu.au/facslab/AFCCG/standards.html>

At the highly informative Australian Flow Cytometry Group Web site, one can find a carefully detailed and illustrated guide to the interpretation of flow cytometric analyses of clinical samples.

<http://www.kpl.com>

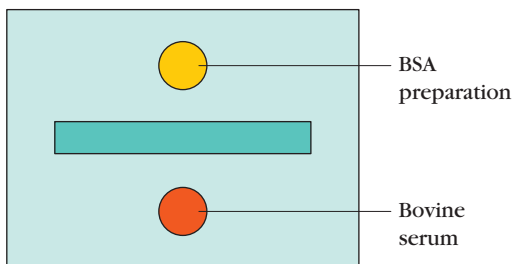
The Kirkegaard & Perry Laboratories Web site contains a sub-site, <http://www.kpl.com/support/immun/pds/50datasht/54-12-10.html>, which allows one to follow a step-by-step procedure for using a chemiluminescent substrate in a sensitive immunoassay.

Study Questions

CLINICAL FOCUS QUESTION Flow-cytometric analysis for the detection and measurement of subpopulations of leukocytes, including those of leukemia, is usually performed using monoclonal antibodies. Why is this the case?

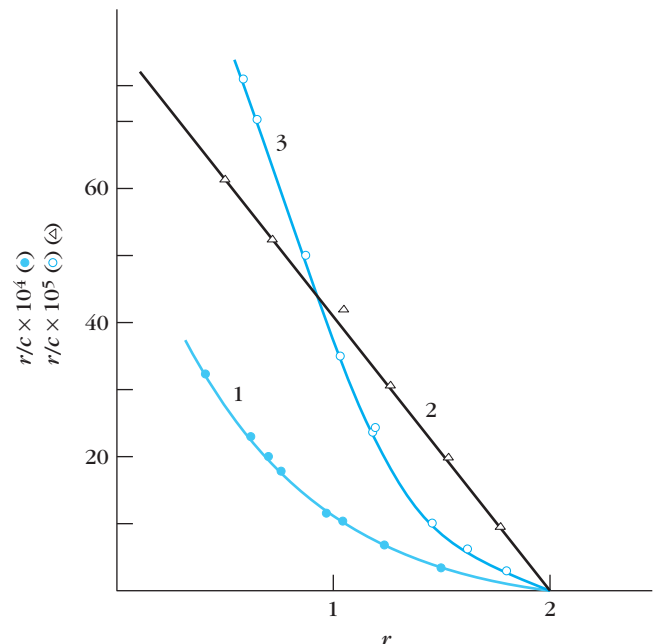
1. Indicate whether each of the following statements is true or false. If you think a statement is false, explain why.
 - a. Indirect immunofluorescence is a more sensitive technique than direct immunofluorescence.
 - b. Most antigens induce a polyclonal response.
 - c. A papain digest of anti-SRBC antibodies can agglutinate sheep red blood cells (SRBCs).
 - d. A pepsin digest of anti-SRBC antibodies can agglutinate SRBCs.
 - e. Indirect immunofluorescence can be performed using a Fab fragment as the primary, nonlabeled antibody.
 - f. For precipitation to occur, both antigen and antibody must be multivalent.
 - g. Analysis of a cell population by flow cytometry can simultaneously provide information on both the size distribution and antigen profile of cell populations containing several different cell types.

- h. ELISA tests using chemiluminescence are more sensitive than chromogenic ones and precipitation tests are more sensitive than agglutination tests.
 - i. Western blotting and immunoprecipitation assays are useful quantitative assays for measuring the levels of proteins in cells or tissues.
 - j. Assume antibody A and antibody B both react with an epitope C. Furthermore, assume that antibody A has a K_a 5 times greater than that of antibody B. The strength of the monovalent reaction of antibody A with epitope C will always be greater than the avidity of antibody B for an antigen with multiple copies of epitope C.
2. You have obtained a preparation of purified bovine serum albumin (BSA) from normal bovine serum. To determine whether any other serum proteins remain in this preparation of BSA, you decide to use immunoelectrophoresis.
- a. What antigen would you use to prepare the antiserum needed to detect impurities in the BSA preparation?
 - b. Assuming that the BSA preparation is pure, draw the immunoelectrophoretic pattern you would expect if the assay was performed with bovine serum in a well above a trough containing the antiserum you prepared in (a) and the BSA sample in a well below the trough as shown below:



3. The labels from four bottles (A, B, C, and D) of hapten-carrier conjugates were accidentally removed. However, it was known that each bottle contained either 1) hapten 1-carrier 1 (H1-C1), 2) hapten 1-carrier 2 (H1-C2), 3) hapten 2-carrier 1 (H2-C1), or 4) hapten 2-carrier 2 (H2-C2). Carrier 1 has a molecular weight of 60,000 daltons and carrier 2 has a molecular weight of over 120,000 daltons. Assume you have an anti-H1 antibody and an anti-H-2 antibody and a molecular-weight marker that is 100,000 daltons. Use Western blotting to determine the contents of each bottle and show the Western blots you would expect from 1, 2, 3, and 4. Your answer should also tell which antibody or combination of antibodies was used to obtain each blot.
4. The concentration of a small amount (250 nanograms/ml) of hapten can be determined by which of the following assays: (a) ELISA (chromogenic), (b) Ouchterlony method, (c) RIA, (d) fluorescence microscopy, (e) flow cytometry, (f) immunoprecipitation, (g) immunoelectron microscopy, (h) ELISPOT assay, (i) chemiluminescent ELISA.
5. You have a myeloma protein, X, whose isotype is unknown and several other myeloma proteins of all known isotypes (e.g., IgG, IgM, IgA, and IgE).

- a. How could you produce isotype-specific antibodies that could be used to determine the isotype of myeloma protein, X?
 - b. How could you use this anti-isotype antibody to measure the level of myeloma protein X in normal serum?
6. For each antigen or antibody listed below, indicate an appropriate assay method and the necessary test reagents. Keep in mind the sensitivity of the assay and the expected concentration of each protein.
- a. IgG in serum
 - b. Insulin in serum
 - c. IgE in serum
 - d. Complement component C3 on glomerular basement membrane
 - e. Anti-A antibodies to blood-group antigen A in serum
 - f. Horsemeat contamination of hamburger
 - g. Syphilis spirochete in a smear from a chancre
7. Which of the following does *not* participate in the formation of antigen-antibody complexes?
- a. Hydrophobic bonds
 - b. Covalent bonds
 - c. Electrostatic interactions
 - d. Hydrogen bonds
 - e. Van der Waals forces
8. Explain the difference between antibody affinity and antibody avidity. Which of these properties of an antibody better reflects its ability to contribute to the humoral immune response to invading bacteria?
9. You want to develop a sensitive immunoassay for a hormone that occurs in the blood at concentrations near 10^{-7} M. You are offered a choice of three different antisera whose affinities for the hormone have been determined by equilibrium dialysis. The results are shown in the Scatchard plots.



- a. What is the value of K_0 for each antiserum?
 - b. What is the valence of each of the antibodies?
 - c. Which of the antisera might be a monoclonal antibody?
 - d. Which of the antisera would you use for your assay? Why?
10. In preparing a demonstration for her immunology class, an instructor purified IgG antibodies to sheep red blood cells (SRBCs) and digested some of the antibodies into Fab, Fc, and $F(ab)_2$ fragments. She placed each preparation in a separate tube, labeled the tubes with a water-soluble marker, and left them in an ice bucket. When the instructor returned for her class period, she discovered that the labels had smeared and were unreadable. Determined to salvage the demonstration, she relabeled the tubes 1, 2, 3, and 4 and proceeded. Based on the test results described below, indicate which preparation was contained in each tube and explain how you identified the contents.
- a. The preparation in tube 1 agglutinated SRBCs but did not lyse them in the presence of complement.
 - b. The preparation in tube 2 did not agglutinate SRBCs or lyse them in the presence of complement. However, when this preparation was added to SRBCs before the addition of whole anti-SRBC, it prevented agglutination of the cells by the whole anti-SRBC antiserum.
 - c. The preparation in tube 3 agglutinated SRBCs and also lysed the cells in the presence of complement.
 - d. The preparation in tube 4 did not agglutinate or lyse SRBCs and did not inhibit agglutination of SRBCs by whole anti-SRBC antiserum.
11. You are given two solutions, one containing protein X and the other containing antibody to protein X. When you add 1 ml of anti-X to 1 ml of protein X, a precipitate forms. But when you dilute the antibody solution 100-fold and then mix 1 ml of the diluted anti-X with 1 ml of protein X, no precipitate forms.
- a. Explain why no precipitate formed with the diluted antibody.
 - b. Which species (protein X or anti-X) would likely be present in the supernatant of the antibody-antigen mixture in each case?
12. Consider equation 1 and derive the form of the Scatchard equation that appears in equation 2.
1. $S + L = SL$
 2. $B/F = K_a([S]_t - B)$
- Where: S = antibody binding sites; [S] = molar concentration of antibody binding sites; L = ligand (monovalent antigen); [L] = molar concentration of ligand; SL = site-ligand complex; [SL] = molar concentration of site ligand complex; B is substituted for [SL] and F for [L]. Hint: It will be helpful to begin by writing the law of mass action for the reaction shown in equation 1.

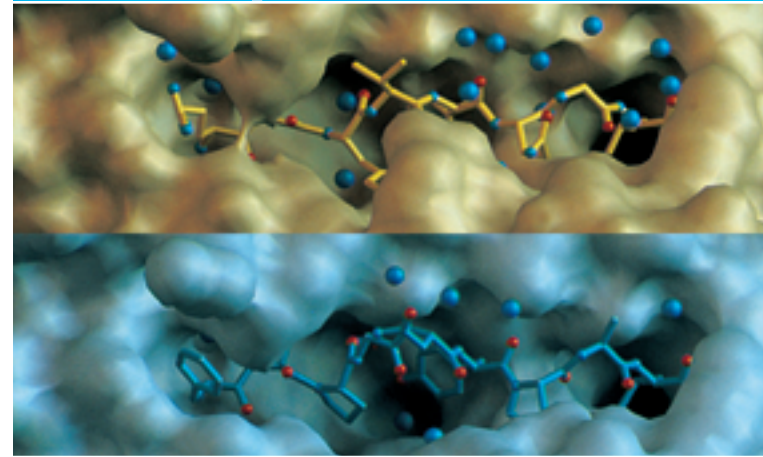
chapter 7

Major Histocompatibility Complex

EVERY MAMMALIAN SPECIES STUDIED TO DATE possesses a tightly linked cluster of genes, the **major histocompatibility complex (MHC)**, whose products play roles in intercellular recognition and in discrimination between self and nonself. The MHC participates in the development of both humoral and cell-mediated immune responses. While antibodies may react with antigens alone, most T cells recognize antigen only when it is combined with an MHC molecule. Furthermore, because MHC molecules act as antigen-presenting structures, the particular set of MHC molecules expressed by an individual influences the repertoire of antigens to which that individual's T_H and T_C cells can respond. For this reason, the MHC partly determines the response of an individual to antigens of infectious organisms, and it has therefore been implicated in the susceptibility to disease and in the development of autoimmunity. The recent understanding that natural killer cells express receptors for MHC class I antigens and the fact that the receptor–MHC interaction may lead to inhibition or activation expands the known role of this gene family (see Chapter 14). The present chapter examines the organization and inheritance of MHC genes, the structure of the MHC molecules, and the central function that these molecules play in producing an immune response.

General Organization and Inheritance of the MHC

The concept that the rejection of foreign tissue is the result of an immune response to cell-surface molecules, now called **histocompatibility antigens**, originated from the work of Peter Gorer in the mid-1930s. Gorer was using inbred strains of mice to identify blood-group antigens. In the course of these studies, he identified four groups of genes, designated I through IV, that encoded blood-cell antigens. Work carried out in the 1940s and 1950s by Gorer and George Snell established that antigens encoded by the genes in the group designated II took part in the rejection of transplanted tumors and other tissue. Snell called these genes “histocompatibility



Presentation of Vesicular Stomatitis Virus Peptide (top) and Sendai Virus Nucleoprotein Peptide by Mouse MHC Class I Molecule H-2K^b

- General Organization and Inheritance of the MHC
- MHC Molecules and Genes
- Detailed Genomic Map of MHC Genes
- Cellular Distribution of MHC Molecules
- Regulation of MHC Expression
- MHC and Immune Responsiveness
- MHC and Disease Susceptibility

genes”; their current designation as histocompatibility-2 (H-2) genes was in reference to Gorer’s group II blood-group antigens. Although Gorer died before his contributions were recognized fully, Snell was awarded the Nobel prize in 1980 for this work.

The MHC Encodes Three Major Classes of Molecules

The major histocompatibility complex is a collection of genes arrayed within a long continuous stretch of DNA on chromosome 6 in humans and on chromosome 17 in mice. The MHC is referred to as the **HLA complex** in humans and as the **H-2 complex** in mice. Although the arrangement of genes is somewhat different, in both cases the MHC genes are organized into regions encoding three classes of molecules (Figure 7-1):

- **Class I MHC genes** encode glycoproteins expressed on the surface of nearly all nucleated cells; the major function of the class I gene products is presentation of peptide antigens to T_C cells.



VISUALIZING CONCEPTS

Mouse H-2 complex

Complex	H-2						
MHC class	I	II		III		I	
Region	K	IA	IE	S		D	
Gene products	H-2K	IA $\alpha\beta$	IE $\alpha\beta$	C' proteins	TNF- α TNF- β	H-2D	H-2L

Human HLA complex

Complex	HLA							
MHC class	II			III		I		
Region	DP	DQ	DR	C4, C2, BF		B	C	A
Gene products	DP $\alpha\beta$	DQ $\alpha\beta$	DR $\alpha\beta$	C' proteins	TNF- α TNF- β	HLA-B	HLA-C	HLA-A

FIGURE 7-1 Simplified organization of the major histocompatibility complex (MHC) in the mouse and human. The MHC is referred to as the H-2 complex in mice and as the HLA complex in humans. In both species the MHC is organized into a number of regions encoding class I (pink), class II (blue), and class III

(green) gene products. The class I and class II gene products shown in this figure are considered to be the classical MHC molecules. The class III gene products include complement (C') proteins and the tumor necrosis factors (TNF- α and TNF- β).

- **Class II MHC genes** encode glycoproteins expressed primarily on antigen-presenting cells (macrophages, dendritic cells, and B cells), where they present processed antigenic peptides to T_H cells.
- **Class III MHC genes** encode, in addition to other products, various secreted proteins that have immune functions, including components of the complement system and molecules involved in inflammation.

Class I MHC molecules encoded by the K and D regions in mice and by the A, B, and C loci in humans were the first discovered, and they are expressed in the widest range of cell types. These are referred to as *classical class I molecules*. Additional genes or groups of genes within the H-2 or HLA complexes also encode class I molecules; these genes are designated *nonclassical class I genes*. Expression of the nonclassical gene products is limited to certain specific cell types. Although functions are not known for all of these gene products, some may have highly specialized roles in immunity. For example, the expression of the class I HLA-G molecules on cytotrophoblasts at the fetal-maternal interface has been implicated in protection of the fetus from being recognized as foreign (this may occur when paternal

antigens begin to appear) and from being rejected by maternal T_C cells.

The two chains of the class II MHC molecules are encoded by the IA and IE regions in mice and by the DP, DQ, and DR regions in humans. The terminology is somewhat confusing, since the D region in mice encodes class I MHC molecules, whereas the D region (DR, DQ, DP) in humans refers to genes encoding class II MHC molecules! Fortunately, the designation D for the general chromosomal location encoding the human class II molecules is seldom used today; the sequence of the entire MHC region is available so the more imprecise reference to region is seldom necessary. As with the class I loci, additional class II molecules encoded within this region have specialized functions in the immune process.

The class I and class II MHC molecules have common structural features and both have roles in antigen processing. By contrast, the class III MHC region, which is flanked by the class I and II regions, encodes molecules that are critical to immune function but have little in common with class I or II molecules. Class III products include the complement components C4, C2, BF (see Chapter 13), and inflammatory cytokines, including tumor necrosis factor (TNF) and heat-shock proteins (see Chapter 12).

Allelic Forms of MHC Genes Are Inherited in Linked Groups Called Haplotypes

As described in more detail later, the loci constituting the MHC are highly **polymorphic**; that is, many alternative forms of the gene, or **alleles**, exist at each locus among the population. The genes of the MHC loci lie close together; for example, the recombination frequency within the H-2 complex (i.e., the frequency of chromosome crossover events during mitosis, indicative of the distance between given gene segments) is only 0.5%—crossover occurs only once in every 200 mitotic cycles. For this reason, most individuals inherit the alleles encoded by these closely linked loci as two sets, one from each parent. Each set of alleles is referred to as a **haplotype**. An individual inherits one haplotype from the mother and one haplotype from the father. In outbred populations, the offspring are generally heterozygous at many loci and will express both maternal and paternal MHC alleles. The alleles are *codominantly expressed*; that is, both maternal and paternal gene products are expressed in the same cells. If mice are inbred (that is, have identical alleles at all loci), each H-2 locus will be homozygous because the maternal and paternal haplotypes are identical, and all offspring therefore express identical haplotypes.

Certain inbred mouse strains have been designated as prototype strains, and the MHC haplotype expressed by these strains is designated by an arbitrary italic superscript (e.g., H-2^a, H-2^b). These designations refer to the entire set of inherited H-2 alleles within a strain without having to list each allele individually (Table 7-1). Different inbred strains may have the same set of alleles, that is the same MHC haplotype, as the prototype strain. For example, the CBA, AKR, and C3H strains all have the same MHC haplotype (H-2^k). The three strains differ, however, in genes outside the H-2 complex.

If two mice from inbred strains having different MHC haplotypes are bred to one another, the F₁ generation inherits haplotypes from both parental strains and therefore ex-

presses both parental alleles at each MHC locus. For example, if an H-2^b strain is crossed with an H-2^k, then the F₁ inherits both parental sets of alleles and is said to be H-2^{b/k} (Figure 7-2a). Because such an F₁ expresses the MHC proteins of both parental strains on its cells, it is histocompatible with both strains and able to accept grafts from either parental strain (see example in Figure 7-2b). However, neither of the inbred parental strains can accept a graft from the F₁ mice because half of the MHC molecules will be foreign to the parent.

The inheritance of HLA haplotypes from heterozygous human parents is illustrated in Figure 7-2c. In an outbred population, each individual is generally heterozygous at each locus. The human HLA complex is highly polymorphic and multiple alleles of each class I and class II gene exist. However, as with mice, the human MHC loci are closely linked and usually inherited as a haplotype. When the father and mother have different haplotypes, as in the example shown (Figure 7-2c) there is a one-in-four chance that siblings will inherit the same paternal and maternal haplotypes and therefore be histocompatible with each other; none of the offspring will be histocompatible with the parents.

Although the rate of recombination by crossover is low within the HLA, it still contributes significantly to the diversity of the loci in human populations. Genetic recombination generates new allelic combinations (Figure 7-2d), and the high number of intervening generations since the appearance of humans as a species has allowed extensive recombination, so that it is rare for any two unrelated individuals to have identical sets of HLA genes.

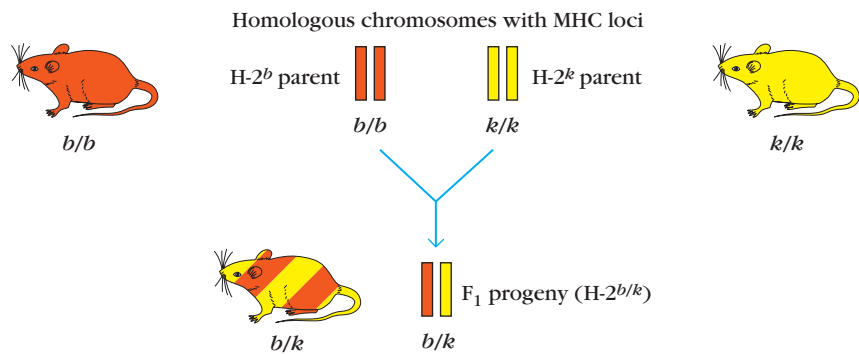
MHC Congenic Mouse Strains Are Identical at All Loci Except the MHC

Detailed analysis of the H-2 complex in mice was made possible by the development of congenic mouse strains. Inbred mouse strains are **syngeneic** or identical at all genetic loci. Two strains are **congenic** if they are genetically identical

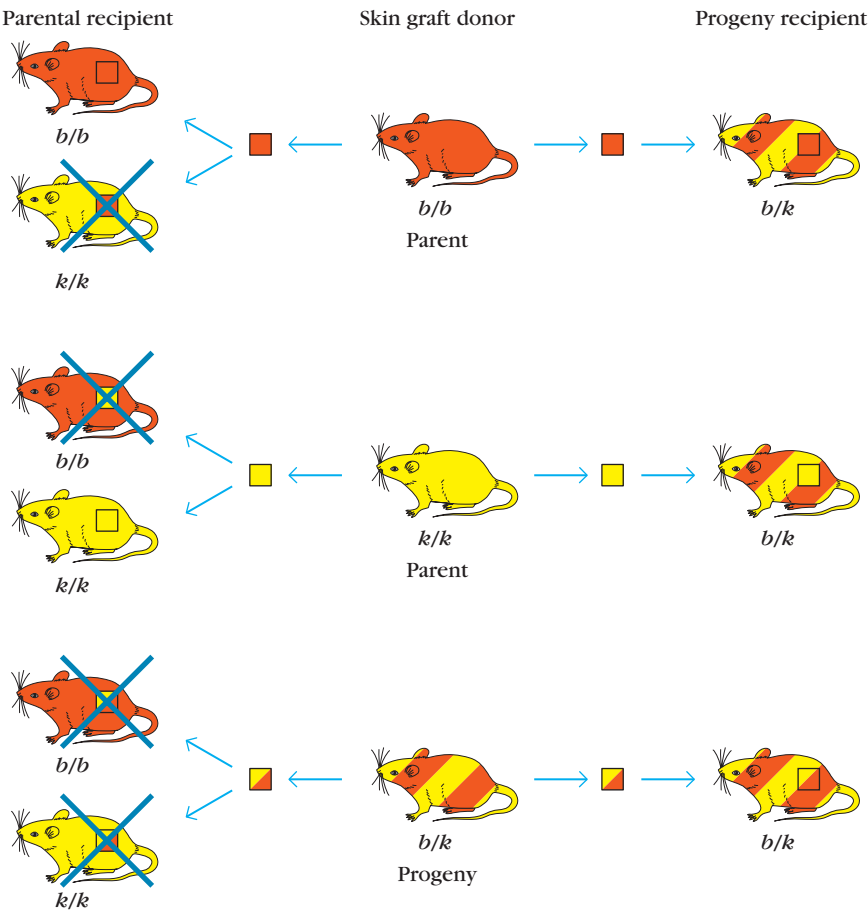
TABLE 7-1 H-2 Haplotypes of some mouse strains

Prototype strain	Other strains with the same haplotype	Haplotype	H-2 ALLELES				
			<i>K</i>	<i>IA</i>	<i>IE</i>	<i>S</i>	<i>D</i>
CBA	AKR, C3H, B10.BR, C57BR	<i>k</i>	<i>k</i>	<i>k</i>	<i>k</i>	<i>k</i>	<i>k</i>
DBA/2	BALB/c, NZB, SEA, YBR	<i>d</i>	<i>d</i>	<i>d</i>	<i>d</i>	<i>d</i>	<i>d</i>
C57BL/10 (B10)	C57BL/6, C57L, C3H.SW, LP, 129	<i>b</i>	<i>b</i>	<i>b</i>	<i>b</i>	<i>b</i>	<i>b</i>
A	A/He, A/Sn, A/Wy, B10.A	<i>a</i>	<i>k</i>	<i>k</i>	<i>k</i>	<i>d</i>	<i>d</i>
A.SW	B10.S, SJL	<i>s</i>	<i>s</i>	<i>s</i>	<i>s</i>	<i>s</i>	<i>s</i>
A.TL		<i>t1</i>	<i>s</i>	<i>k</i>	<i>k</i>	<i>k</i>	<i>d</i>
DBA/1	STOLI, B10.Q, BDP	<i>q</i>	<i>q</i>	<i>q</i>	<i>q</i>	<i>q</i>	<i>q</i>

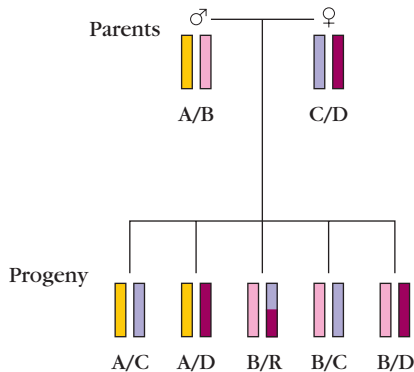
(a) Mating of inbred mouse strains with different MHC haplotypes



(b) Skin transplantation between inbred mouse strains with same or different MHC haplotypes



(c) Inheritance of HLA haplotypes in a typical human family



(d) A new haplotype (R) arises from recombination of maternal haplotypes

	HLA Alleles					
	A	B	C	DR	DQ	DP
A	1	7	w3	2	1	1
B	2	8	w2	3	2	2
C	3	44	w4	4	1	3
D	11	35	w1	7	3	4
R	3	44	w4	7	3	4

FIGURE 7-2 (a) Illustration of inheritance of MHC haplotypes in inbred mouse strains. The letters b/b designate a mouse homozygous for the H-2^b MHC haplotype, k/k homozygous for the H-2^k haplotype, and b/k a heterozygote. Because the MHC loci are closely linked and inherited as a set, the MHC haplotype of F₁ progeny from the mating of two different inbred strains can be predicted easily. (b) Acceptance or rejection of skin grafts is controlled by the MHC type of the inbred mice. The progeny of the cross between two inbred strains with different MHC haplotypes (H-2^b and H-2^k) will express both haplotypes (H-2^{b/k}) and will accept grafts from either parent and from one another. Neither parent strain will accept grafts from the offspring. (c) Inheritance of HLA haplotypes in a hypothetical human family. In humans, the paternal HLA haplotypes are arbitrarily designated A and B, maternal C and D. Because humans are an outbred species and there are many alleles at each HLA locus, the alleles comprising the haplotypes must be determined by typing parents and progeny. (d) The genes that make up each parental haplotype in the hypothetical family in (c) are shown along with a new haplotype that arose from recombination (R) of maternal haplotypes.

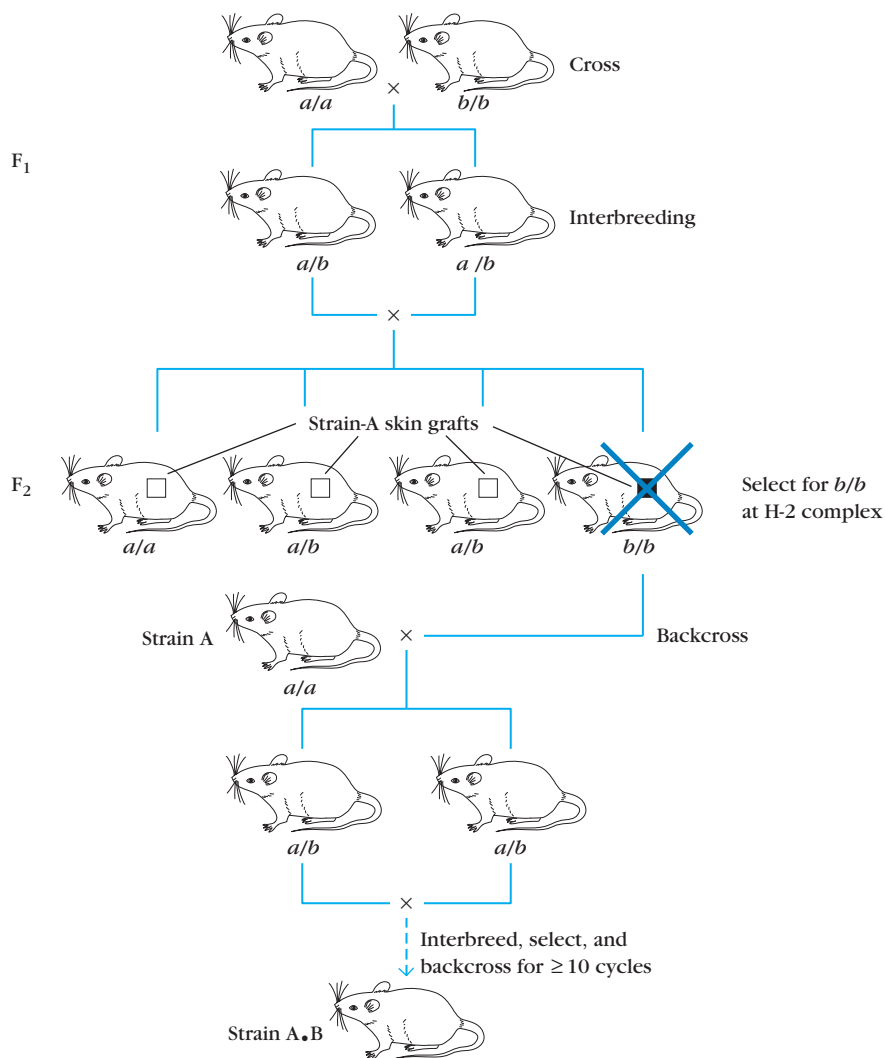


FIGURE 7-3 Production of congenic mouse strain A.B, which has the genetic background of parental strain A but the H-2 complex of strain B. Crossing inbred strain A (H-2^a) with strain B (H-2^b) generates F₁ progeny that are heterozygous (*a/b*) at all H-2 loci. The F₁ progeny are interbred to produce an F₂ generation, which includes *a/a*, *a/b*, and *b/b* individuals. The F₂ progeny homozygous for the B-strain H-2 complex are selected by their ability to reject a skin graft from strain A; any progeny that accept an A-strain graft are eliminated from future breeding. The selected *b/b* homozygous mice are then backcrossed to strain A; the resulting progeny are again interbred and their offspring are again selected for *b/b* homozygosity at the H-2 complex. This process of backcrossing to strain A, intercrossing, and selection for ability to reject an A-strain graft is repeated for at least 12 generations. In this way A-strain homozygosity is restored at all loci except the H-2 locus, which is homozygous for the B strain.

except at a single genetic locus or region. Any phenotypic differences that can be detected between congenic strains are related to the genetic region that distinguishes the strains. Congenic strains that are identical with each other except at the MHC can be produced by a series of crosses, backcrosses, and selections. Figure 7-3 outlines the steps by which the H-2 complex of homozygous strain B can be introduced into the background genes of homozygous strain A to generate a congenic strain, denoted A.B. The first letter in a congenic strain designation refers to the strain providing the genetic background and the second letter to the strain providing the genetically different MHC region. Thus, strain A.B will be genetically identical to strain A except for the MHC locus or loci contributed by strain B.

During production of congenic mouse strains, a crossover event sometimes occurs within the H-2 complex, yielding a recombinant strain that differs from the parental strains or the congenic strain at one or a few loci within the H-2 complex. Figure 7-4 depicts haplotypes present in several **recombinant congenic** strains that were obtained during pro-

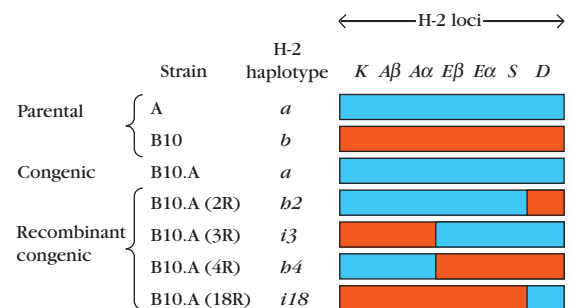


FIGURE 7-4 Examples of recombinant congenic mouse strains generated during production of the B10.A strain from parental strain B10 (H-2^b) and parental strain A (H-2^a). Crossover events within the H-2 complex produce recombinant strains, which have *a*-haplotype alleles (blue) at some H-2 loci and *b*-haplotype alleles (orange) at other loci.

duction of a B10.A congenic strain. Such recombinant strains have been extremely useful in analyzing the MHC because they permit comparisons of functional differences

between strains that differ in only a few genes within the MHC. Furthermore, the generation of new H-2 haplotypes under the experimental conditions of congenic strain development provides an excellent illustration of the means by which the MHC continues to maintain heterogeneity even in populations with limited diversity.

MHC Molecules and Genes

Class I and class II MHC molecules are membrane-bound glycoproteins that are closely related in both structure and function. Both class I and class II MHC molecules have been isolated and purified and the three-dimensional structures of their extracellular domains have been determined by x-ray crystallography. Both types of membrane glycoproteins function as highly specialized antigen-presenting molecules that form unusually stable complexes with antigenic peptides, displaying them on the cell surface for recognition by T cells. In contrast, class III MHC molecules are a group of unrelated proteins that do not share structural similarity and common function with class I and II molecules. The class III molecules will be examined in more detail in later chapters.

Class I Molecules Have a Glycoprotein Heavy Chain and a Small Protein Light Chain

Class I MHC molecules contain a 45-kilodalton (kDa) α chain associated noncovalently with a 12-kDa β_2 -microglobulin molecule (see Figure 7-5). The α chain is a transmembrane glycoprotein encoded by polymorphic genes within the A, B, and C regions of the human HLA complex and within the K and D/L regions of the mouse H-2 complex (see Figure 7-1). β_2 -Microglobulin is a protein encoded by a highly conserved gene located on a different chromosome. Association of the α chain with β_2 -microglobulin is required for expression of class I molecules on cell membranes. The α chain is anchored in the plasma membrane by its hydrophobic transmembrane segment and hydrophilic cytoplasmic tail.

Structural analyses have revealed that the α chain of class I MHC molecules is organized into three external domains (α_1 , α_2 , and α_3), each containing approximately 90 amino acids; a transmembrane domain of about 25 hydrophobic amino acids followed by a short stretch of charged (hydrophilic) amino acids; and a cytoplasmic anchor segment of 30 amino acids. The β_2 -microglobulin is similar in size and organization to the α_3 domain; it does not contain a transmembrane region and is noncovalently bound to the class I glycoprotein. Sequence data reveal homology between the α_3

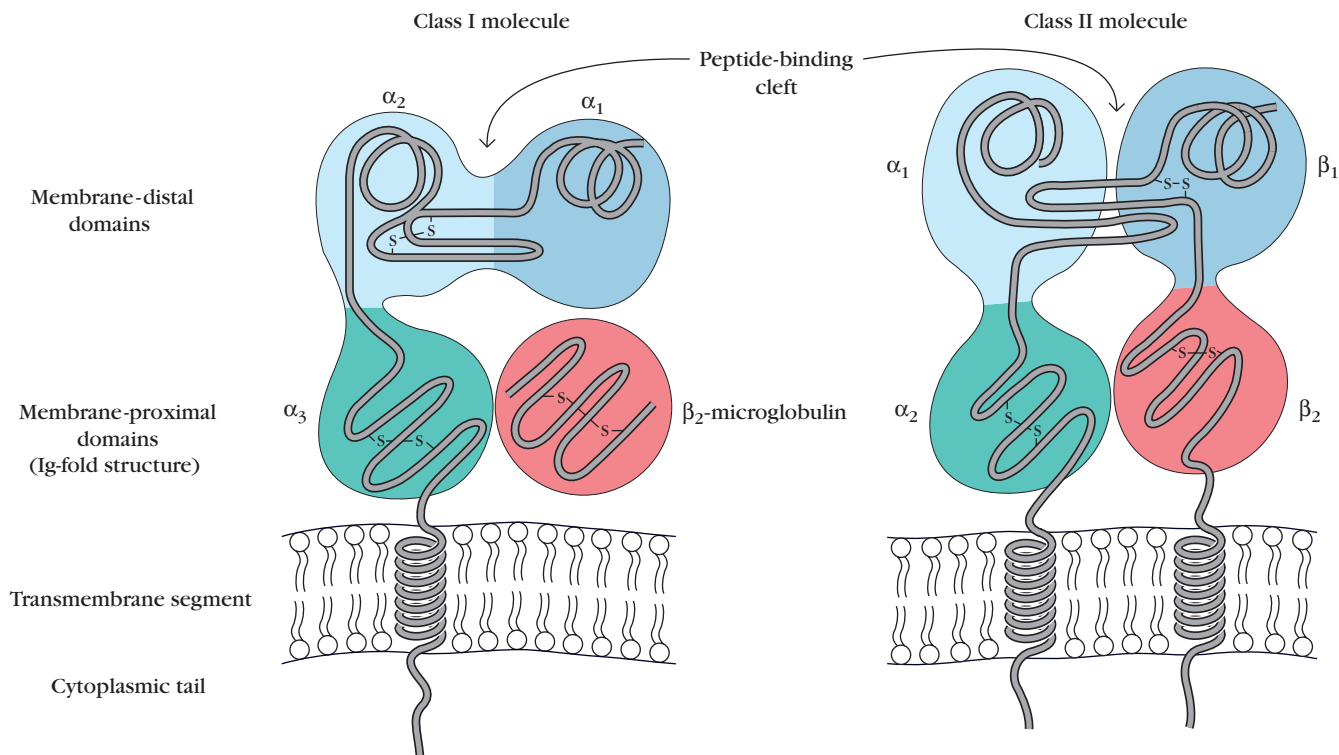


FIGURE 7-5 Schematic diagrams of a class I and a class II MHC molecule showing the external domains, transmembrane segment, and cytoplasmic tail. The peptide-binding cleft is formed by the membrane-distal domains in both class I and class II molecules. The

membrane-proximal domains possess the basic immunoglobulin fold structure; thus, class I and class II MHC molecules are classified as members of the immunoglobulin superfamily.

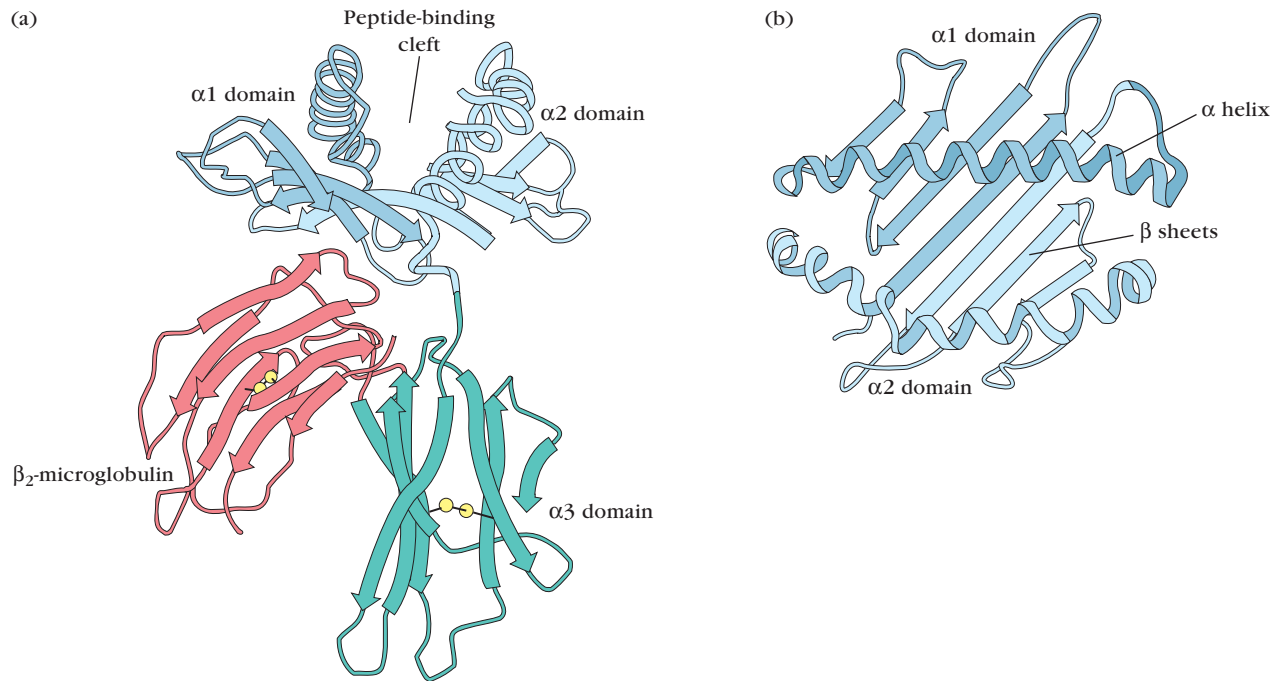


FIGURE 7-6 Representations of the three-dimensional structure of the external domains of a human class I MHC molecule based on x-ray crystallographic analysis. (a) Side view in which the β strands are depicted as thick arrows and the α helices as spiral ribbons. Disulfide bonds are shown as two interconnected spheres. The $\alpha 1$ and $\alpha 2$ domains interact to form the peptide-binding cleft. Note the im-

munoglobulin-fold structure of the $\alpha 3$ domain and β_2 -microglobulin. (b) The $\alpha 1$ and $\alpha 2$ domains as viewed from the top, showing the peptide-binding cleft consisting of a base of antiparallel β strands and sides of α helices. This cleft in class I molecules can accommodate peptides containing 8–10 residues.

domain, β_2 -microglobulin, and the constant-region domains in immunoglobulins. The enzyme papain cleaves the α chain just 13 residues proximal to its transmembrane domain, releasing the extracellular portion of the molecule, consisting of $\alpha 1$, $\alpha 2$, $\alpha 3$, and β_2 -microglobulin. Purification and crystallization of the extracellular portion revealed two pairs of interacting domains: a membrane-distal pair made up of the $\alpha 1$ and $\alpha 2$ domains and a membrane-proximal pair composed of the $\alpha 3$ domain and β_2 -microglobulin (Figure 7-6a).

The $\alpha 1$ and $\alpha 2$ domains interact to form a platform of eight antiparallel β strands spanned by two long α -helical regions. The structure forms a deep groove, or cleft, approximately $25 \text{ \AA} \times 10 \text{ \AA} \times 11 \text{ \AA}$, with the long α helices as sides and the β strands of the β sheet as the bottom (Figure 7-6b). This *peptide-binding cleft* is located on the top surface of the class I MHC molecule, and it is large enough to bind a peptide of 8–10 amino acids. The great surprise in the x-ray crystallographic analysis of class I molecules was the finding of small peptides in the cleft that had cocrystallized with the protein. These peptides are, in fact, processed antigen and self-peptides bound to the $\alpha 1$ and $\alpha 2$ domains in this deep groove.

The $\alpha 3$ domain and β_2 -microglobulin are organized into two β pleated sheets each formed by antiparallel β strands of amino acids. As described in Chapter 4, this structure, known as the immunoglobulin fold, is characteristic of immunoglobulin domains. Because of this structural similarity,

which is not surprising given the considerable sequence similarity with the immunoglobulin constant regions, class I MHC molecules and β_2 -microglobulin are classified as members of the immunoglobulin superfamily (see Figure 4-20). The $\alpha 3$ domain appears to be highly conserved among class I MHC molecules and contains a sequence that interacts with the CD8 membrane molecule present on T_C cells.

β_2 -Microglobulin interacts extensively with the $\alpha 3$ domain and also interacts with amino acids of the $\alpha 1$ and $\alpha 2$ domains. The interaction of β_2 -microglobulin and a peptide with a class I α chain is essential for the class I molecule to reach its fully folded conformation. As described in detail in Chapter 8, assembly of class I molecules is believed to occur by the initial interaction of β_2 -microglobulin with the folding class I α chain. This metastable “empty” dimer is then stabilized by the binding of an appropriate peptide to form the native trimeric class I structure consisting of the class I α chain, β_2 -microglobulin, and a peptide. This complete molecular complex is ultimately transported to the cell surface.

In the absence of β_2 -microglobulin, the class I MHC α chain is not expressed on the cell membrane. This is illustrated by Daudi tumor cells, which are unable to synthesize β_2 -microglobulin. These tumor cells produce class I MHC α chains, but do not express them on the membrane. However, if Daudi cells are transfected with a functional gene encoding β_2 -microglobulin, class I molecules appear on the membrane.

Class II Molecules Have Two Nonidentical Glycoprotein Chains

Class II MHC molecules contain two different polypeptide chains, a 33-kDa α chain and a 28-kDa β chain, which associate by noncovalent interactions (see Figure 7-5b). Like class I α chains, class II MHC molecules are membrane-bound glycoproteins that contain external domains, a transmembrane segment, and a cytoplasmic anchor segment. Each chain in a class II molecule contains two external domains: $\alpha 1$ and $\alpha 2$ domains in one chain and $\beta 1$ and $\beta 2$ domains in the other. The membrane-proximal $\alpha 2$ and $\beta 2$ domains, like the membrane-proximal $\alpha 3/\beta 2$ -microglobulin domains of class I MHC molecules, bear sequence similarity to the immunoglobulin-fold structure; for this reason, class II MHC molecules also are classified in the immunoglobulin superfamily. The membrane-distal portion of a class II molecule is composed of the $\alpha 1$ and $\beta 1$ domains and forms the antigen-binding cleft for processed antigen.

X-ray crystallographic analysis reveals the similarity of class II and class I molecules, strikingly apparent when the molecules are superimposed (Figure 7-7). The peptide-binding cleft of HLA-DR1, like that in class I molecules, is composed of a floor of eight antiparallel β strands and sides of antiparallel α helices. However, the class II molecule lacks the conserved residues that bind to the terminal residues of short peptides and forms instead an open pocket; class I presents more of a socket, class II an open-ended groove. These functional consequences of these differences in fine structure will be explored below.

An unexpected difference between crystallized class I and class II molecules was observed for human DR1 in that the

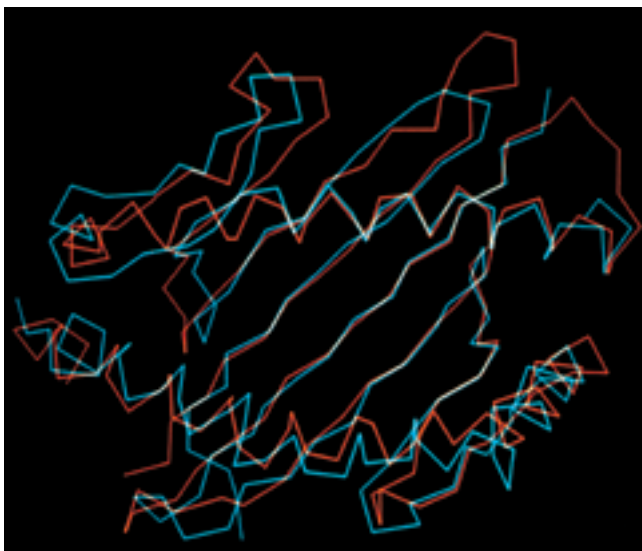


FIGURE 7-7 The membrane-distal, peptide-binding cleft of a human class II MHC molecule, HLA-DR1 (blue), superimposed over the corresponding regions of a human class I MHC molecule, HLA-A2 (red). [From J. H. Brown *et al.*, 1993, *Nature* **364**:33.]

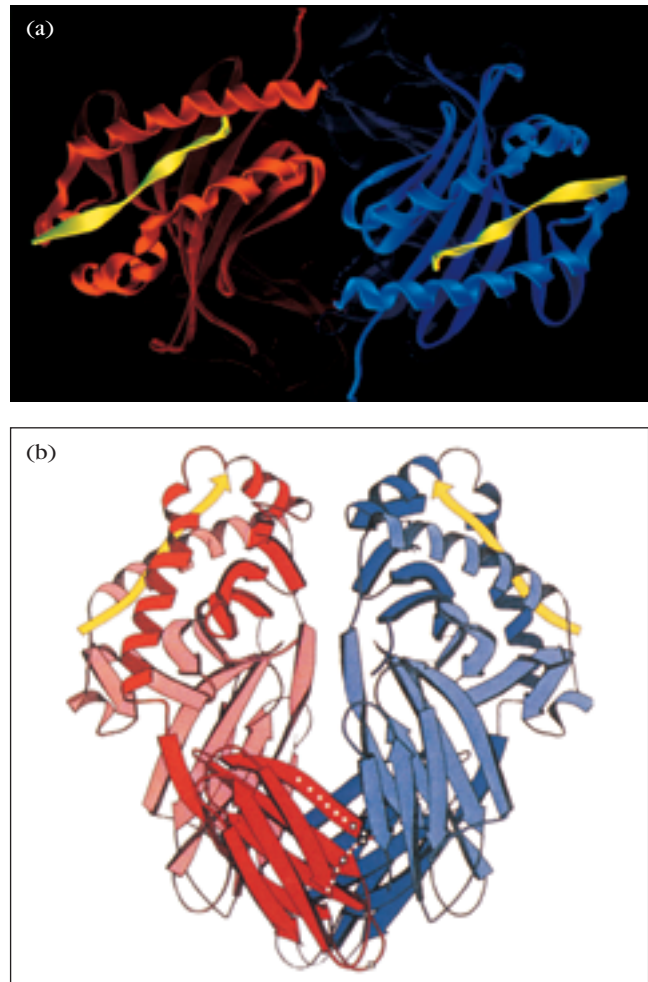


FIGURE 7-8 Antigen-binding cleft of dimeric class II DR1 molecule in (a) top view and (b) side view. This molecule crystallized as a dimer of the $\alpha\beta$ heterodimer. The crystallized dimer is shown with one DR1 molecule in red and the other DR1 molecule in blue. The bound peptides are yellow. The two peptide-binding clefts in the dimeric molecule face in opposite directions. [From J. H. Brown *et al.*, 1993, *Nature* **364**:33.]

latter occurred as a dimer of $\alpha\beta$ heterodimers, a “dimer of dimers” (Figure 7-8). The dimer is oriented so that the two peptide-binding clefts face in opposite directions. While it has not yet been determined whether this dimeric form exists *in vivo*, the presence of CD4 binding sites on opposite sides of the class II molecule suggests that it does. These two sites on the $\alpha 2$ and $\beta 2$ domains are adjacent in the dimer form and a CD4 molecule binding to them may stabilize class II dimers.

The Exon/Intron Arrangement of Class I and II Genes Reflects Their Domain Structure

Separate exons encode each region of the class I and II proteins (Figure 7-9). Each of the mouse and human class I genes has a 5' leader exon encoding a short signal peptide

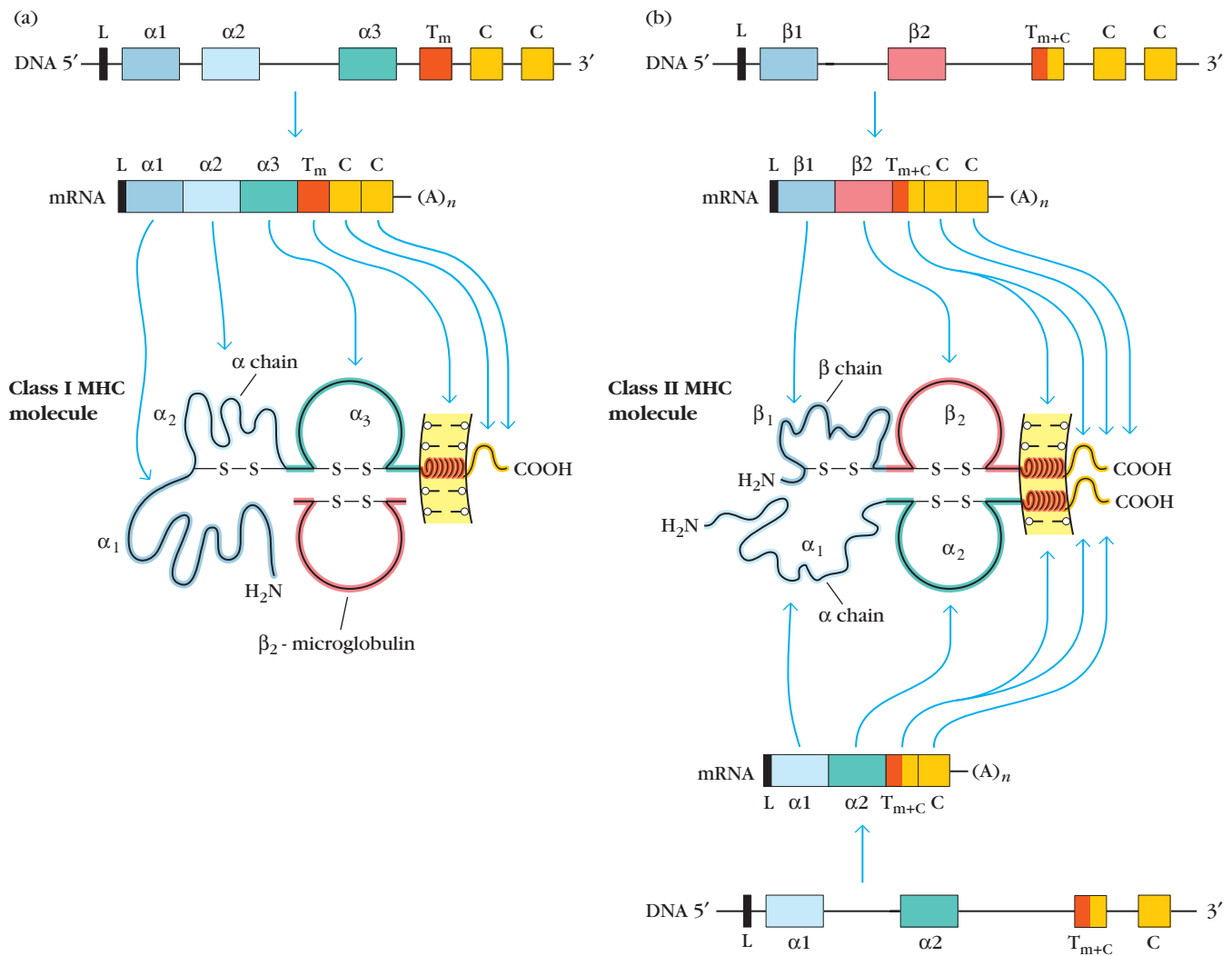


FIGURE 7-9 Schematic diagram of (a) class I and (b) class II MHC genes, mRNA transcripts, and protein molecules. There is correspondence between exons and the domains in the gene products; note that the mRNA transcripts are spliced to remove the intron sequences. Each exon, with the exception of the leader (L) exon, en-

codes a separate domain of the MHC molecule. The leader peptides are removed in a post-translational reaction before the molecules are expressed on the cell surface. The gene encoding β_2 -microglobulin is located on a different chromosome. T_m = transmembrane; C = cytoplasmic.

followed by five or six exons encoding the α chain of the class I molecule (see Figure 7-9a). The signal peptide serves to facilitate insertion of the α chain into the endoplasmic reticulum and is removed by proteolytic enzymes in the endoplasmic reticulum after translation is completed. The next three exons encode the extracellular $\alpha 1$, $\alpha 2$, and $\alpha 3$ domains, and the following downstream exon encodes the transmembrane (T_m) region; finally, one or two 3'-terminal exons encode the cytoplasmic domains (C).

Like class I MHC genes, the class II genes are organized into a series of exons and introns mirroring the domain structure of the α and β chains (see Figure 7-9b). Both the α and the β genes encoding mouse and human class II MHC molecules have a leader exon, an $\alpha 1$ or $\beta 1$ exon, an $\alpha 2$ or $\beta 2$ exon, a transmembrane exon, and one or more cytoplasmic exons.

Class I and II Molecules Exhibit Polymorphism in the Region That Binds to Peptides

Several hundred different allelic variants of class I and II MHC molecules have been identified in humans. Any one individual, however, expresses only a small number of these molecules—up to 6 different class I molecules and up to 12 different class II molecules. Yet this limited number of MHC molecules must be able to present an enormous array of different antigenic peptides to T cells, permitting the immune system to respond specifically to a wide variety of antigenic challenges. Thus, peptide binding by class I and II molecules does not exhibit the fine specificity characteristic of antigen binding by antibodies and T-cell receptors. Instead, a given MHC molecule can bind

TABLE 7-2 Peptide binding by class I and class II MHC molecules

	Class I molecules	Class II molecules
Peptide-binding domain	$\alpha 1/\alpha 2$	$\alpha 1/\beta 1$
Nature of peptide-binding cleft	Closed at both ends	Open at both ends
General size of bound peptides	8–10 amino acids	13–18 amino acids
Peptide motifs involved in binding to MHC molecule	Anchor residues at both ends of peptide; generally hydrophobic carboxyl-terminal anchor	Anchor residues distributed along the length of the peptide
Nature of bound peptide	Extended structure in which both ends interact with MHC cleft but middle arches up away from MHC molecule	Extended structure that is held at a constant elevation above the floor of MHC cleft

numerous different peptides, and some peptides can bind to several different MHC molecules. Because of this broad specificity, the binding between a peptide and an MHC molecule is often referred to as “promiscuous.”

Given the similarities in the structure of the peptide-binding cleft in class I and II MHC molecules, it is not surprising that they exhibit some common peptide-binding features (Table 7-2). In both types of MHC molecules, peptide ligands are held in a largely extended conformation that runs the length of the cleft. The peptide-binding cleft in class I molecules is blocked at both ends, whereas the cleft is open in class II molecules (Figure 7-10). As a result of this difference, class I molecules bind peptides that typically contain 8–10 amino acid residues, while the open groove of class II molecules accommodates slightly longer peptides of 13–18 amino acids. Another difference, explained in more detail below, is that class I binding requires that the peptide contain specific amino acid residues near the N and C termini; there is no such requirement for class II peptide binding.

The peptide–MHC molecule association is very stable ($K_d \sim 10^{-6}$) under physiologic conditions; thus, most of

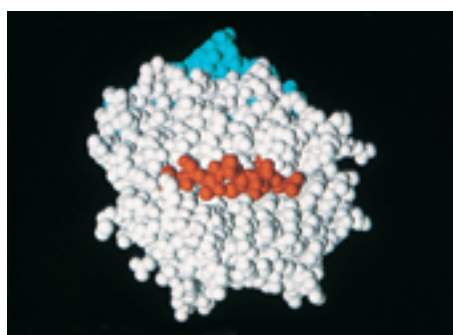
the MHC molecules expressed on the membrane of a cell will be associated with a peptide of self or nonself origin.

CLASS I MHC–PEPTIDE INTERACTION

Class I MHC molecules bind peptides and present them to $CD8^+$ T cells. In general, these peptides are derived from endogenous intracellular proteins that are digested in the cytosol. The peptides are then transported from the cytosol into the cisternae of the endoplasmic reticulum, where they interact with class I MHC molecules. This process, known as the cytosolic or endogenous processing pathway, is discussed in detail in the next chapter.

Each type of class I MHC molecule (K, D, and L in mice or A, B, and C in humans) binds a unique set of peptides. In addition, each allelic variant of a class I MHC molecule (e.g., $H-2K^k$ and $H-2K^d$) also binds a distinct set of peptides. Because a single nucleated cell expresses about 10^5 copies of each class I molecule, many different peptides will be expressed simultaneously on the surface of a nucleated cell by class I MHC molecules.

(a) Class I MHC



(b) Class II MHC

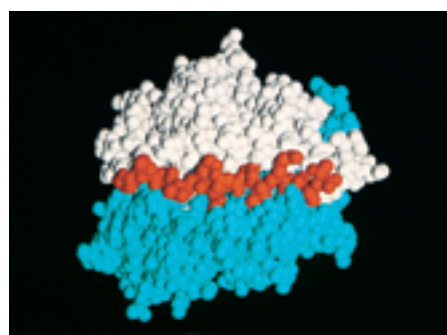


FIGURE 7-10 MHC class I and class II molecules with bound peptides. (a) Space-filling model of human class I molecule HLA-A2 (white) with peptide (red) from HIV reverse transcriptase (amino acid residues 309–317) in the binding groove. β_2 -microglobulin is shown in blue. Residues above the peptide are from the $\alpha 1$ domain,

those below from $\alpha 2$. (b) Space-filling model of human class II molecule HLA-DR1 with the $DR\alpha$ chain shown in white and the $DR\beta$ chain in blue. The peptide (red) in the binding groove is from influenza hemagglutinin (amino acid residues 306–318). [From D. A. Vignali and J. Strominger, 1994, *The Immunologist* 2:112.]

In a critical study of peptide binding by MHC molecules, peptides bound by two allelic variants of a class I MHC molecule were released chemically and analyzed by HPLC mass spectrometry. More than 2000 distinct peptides were found among the peptide ligands released from these two class I MHC molecules. Since there are approximately 10^5 copies of each class I allelic variant per cell, it is estimated that each of the 2000 distinct peptides is presented with a frequency of 100–4000 copies per cell. Evidence suggests that as few as 100 peptide-MHC complexes are sufficient to target a cell for recognition and lysis by a cytotoxic T lymphocyte with a receptor specific for this target structure.

The bound peptides isolated from different class I molecules have been found to have two distinguishing features: they are eight to ten amino acids in length, most commonly nine, and they contain specific amino acid residues that appear to be essential for binding to a particular MHC molecule. Binding studies have shown that nonameric peptides bind to class I molecules with a 100- to 1000-fold higher affinity than do peptides that are either longer or shorter, suggesting that this peptide length is most compatible with the closed-ended peptide-binding cleft in class I molecules. The ability of an individual class I MHC molecule to bind to a diverse spectrum of peptides is due to the presence of the same or similar amino acid residues at several defined positions along the peptides (Figure 7-11). Because these amino acid residues anchor the peptide into the groove of the MHC molecule, they are called *anchor residues*. The side chains of the anchor residues in the peptide are complementary with surface features of the binding cleft of the class I MHC molecule. The amino acid residues lining the binding sites vary among different class I allelic variants and

determine the identity of the anchor residues that can interact with the molecule.

All peptides examined to date that bind to class I molecules contain a carboxyl-terminal anchor. These anchors are generally hydrophobic residues (e.g., leucine, isoleucine), although a few charged amino acids have been reported. Besides the anchor residue found at the carboxyl terminus, another anchor is often found at the second or second and third positions at the amino-terminal end of the peptide (see Figure 7-11). In general, any peptide of correct length that contains the same or similar anchor residues will bind to the same class I MHC molecule. The discovery of conserved anchor residues in peptides that bind to various class I MHC molecules may permit prediction of which peptides in a complex antigen will bind to a particular MHC molecule, based on the presence or absence of these motifs.

X-ray crystallographic analyses of peptide–class I MHC complexes have revealed how the peptide-binding cleft in a given MHC molecule can interact stably with a broad spectrum of different peptides. The anchor residues at both ends of the peptide are buried within the binding cleft, thereby holding the peptide firmly in place (Figure 7-12). As noted already, nonameric peptides are bound preferentially; the main contacts between class I MHC molecules and peptides involve residue 2 at the amino-terminal end and residue 9 at the carboxyl terminus of the nonameric peptide. Between the anchors the peptide arches away from the floor of the cleft in the middle (Figure 7-13), allowing peptides that are slightly longer or shorter to be accommodated. Amino acids that arch away from the MHC molecule are more exposed and presumably can interact more directly with the T-cell receptor.

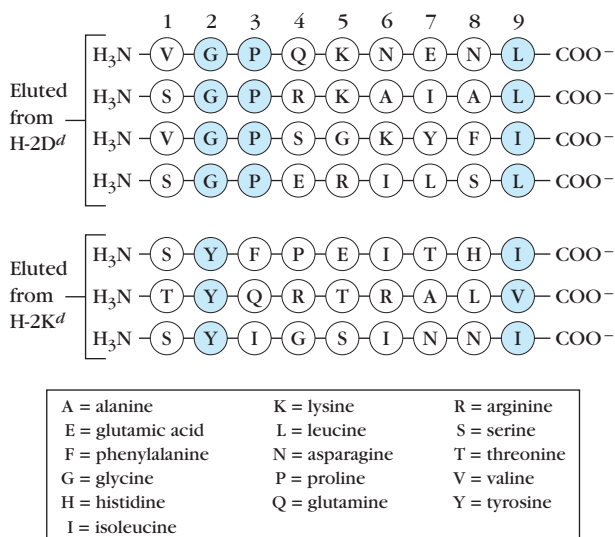


FIGURE 7-11 Examples of anchor residues (blue) in nonameric peptides eluted from two class I MHC molecules. Anchor residues that interact with the class I MHC molecule tend to be hydrophobic amino acids. [Data from V. H. Engelhard, 1994, *Curr. Opin. Immunol.* 6:13.]

CLASS II MHC–PEPTIDE INTERACTION

Class II MHC molecules bind peptides and present these peptides to CD4⁺ T cells. Like class I molecules, molecules of class II can bind a variety of peptides. In general, these peptides are derived from exogenous proteins (either self or nonself), which are degraded within the endocytic processing pathway (see Chapter 8). Most of the peptides associated with class II MHC molecules are derived from membrane-bound proteins or proteins associated with the vesicles of the endocytic processing pathway. The membrane-bound proteins presumably are internalized by phagocytosis or by receptor-mediated endocytosis and enter the endocytic processing pathway at this point. For instance, peptides derived from digestion of membrane-bound class I MHC molecules often are bound to class II MHC molecules.

Peptides recovered from class II MHC–peptide complexes generally contain 13–18 amino acid residues, somewhat longer than the nonameric peptides that most commonly bind to class I molecules. The peptide-binding cleft in class II molecules is open at both ends (see Figure 7-10b), allowing longer peptides to extend beyond the ends, like a long hot dog in a bun. Peptides bound to class II MHC molecules maintain a roughly constant elevation on the

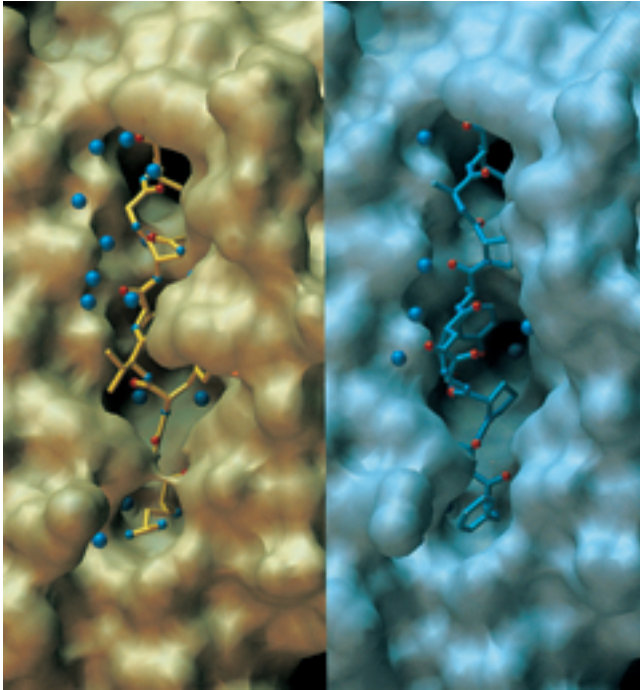


FIGURE 7-12 Model of the solvent-accessible area of class I H-2K^b, depicting the complex formed with a vesicular stomatitis virus (VSV-8) peptide (*left*, yellow backbone) and Sendai virus (SEV-9) nucleoprotein (*right*, blue backbone). Water molecules (blue spheres) interact with the bound peptides. The majority of the surface of both peptides is inaccessible for direct contact with T cells (VSV-8 is 83% buried; SEV-9 is 75% buried). The H-2K^b surface in the two complexes exhibits a small, but potentially significant, conformational variation, especially in the central region of the binding cleft on the right side of the peptides, which corresponds to the α helix in the $\alpha 2$ domain (see Figure 7-6b). [From M. Matsumura *et al.*, 1992, *Science* **257**:927; photographs courtesy of D. H. Fremont, M. Matsumura, M. Pique, and I. A. Watson.]

floor of the binding cleft, another feature that distinguishes peptide binding to class I and class II molecules.

Peptide binding studies and structural data for class II molecules indicate that a central core of 13 amino acids determines the ability of a peptide to bind class II. Longer peptides may be accommodated within the class II cleft, but the binding characteristics are determined by the central 13 residues. The peptides that bind to a particular class II molecule often have internal conserved “motifs,” but unlike class I–binding peptides, they lack conserved anchor residues. Instead, hydrogen bonds between the backbone of the peptide and the class II molecule are distributed throughout the binding site rather than being clustered predominantly at the ends of the site as for class I–bound peptides. Peptides that bind to class II MHC molecules contain an internal sequence comprising 7–10 amino acids that provide the major contact points. Generally, this sequence has an aromatic or hydrophobic residue at the amino terminus and three additional hydrophobic residues in the middle portion and carboxyl-terminal end of the peptide.

In addition, over 30% of the peptides eluted from class II molecules contain a proline residue at position 2 and another cluster of prolines at the carboxyl-terminal end.

Class I and Class II Molecules Exhibit Diversity Within a Species and Multiple Forms Occur in an Individual

An enormous diversity is exhibited by the MHC molecules within a species and within individuals. This variability echoes the diversity of antibodies and T-cell receptors, but the source of diversity for MHC molecules is not the same. Antibodies and T-cell receptors are generated by several somatic processes, including gene rearrangement and somatic mutation of rearranged genes (see Table 5-2). Thus, the generation of T and B cell receptors is dynamic, changing over time within an individual. By contrast, the MHC molecules expressed by an individual are fixed in the genes and do not change over time. The diversity of the MHC within a species stems from polymorphism, the presence of multiple alleles at a given genetic locus within the species. Diversity of MHC molecules in an individual results not only from having different alleles of each gene but also from the presence of duplicated genes with similar or overlapping functions, not unlike the isotypes of immunoglobulins. Because it includes genes with similar, but not identical structure and function (for example, HLA-A, -B, and -C), the MHC may be said to be **polygenic**.

The MHC possesses an extraordinarily large number of different alleles at each locus and is one of the most polymorphic genetic complexes known in higher vertebrates. These alleles differ in their DNA sequences from one individual to another by 5% to 10%. The number of amino acid differences between MHC alleles can be quite significant, with up to 20 amino acid residues contributing to the unique structural nature of each allele. Analysis of human HLA class I genes has revealed, as of early 2002, approximately 240 A alleles, 470 B alleles, and 110 C alleles. In mice, the polymorphism is similarly enormous. The human class II genes are also highly polymorphic and, in some cases, there are different gene numbers in different individuals. The number of HLA-DR beta-chain genes may vary from 2 to 9 in different haplotypes, and approximately 350 alleles of DRB genes have been reported. Interestingly, the DRA chain is highly conserved, with only 2 different alleles reported. Current estimates of actual polymorphism in the human MHC are probably on the low side because the most detailed data were obtained from populations of European descent. The fact that many non-European population groups cannot be typed using the MHC serologic typing reagents available indicates that the worldwide diversity of the MHC genes is far greater. Now that MHC genes can be sequenced directly, it is expected that many additional alleles will be detected.

This enormous polymorphism results in a tremendous diversity of MHC molecules within a species. Using the numbers given above for the allelic forms of human HLA-A, -B,

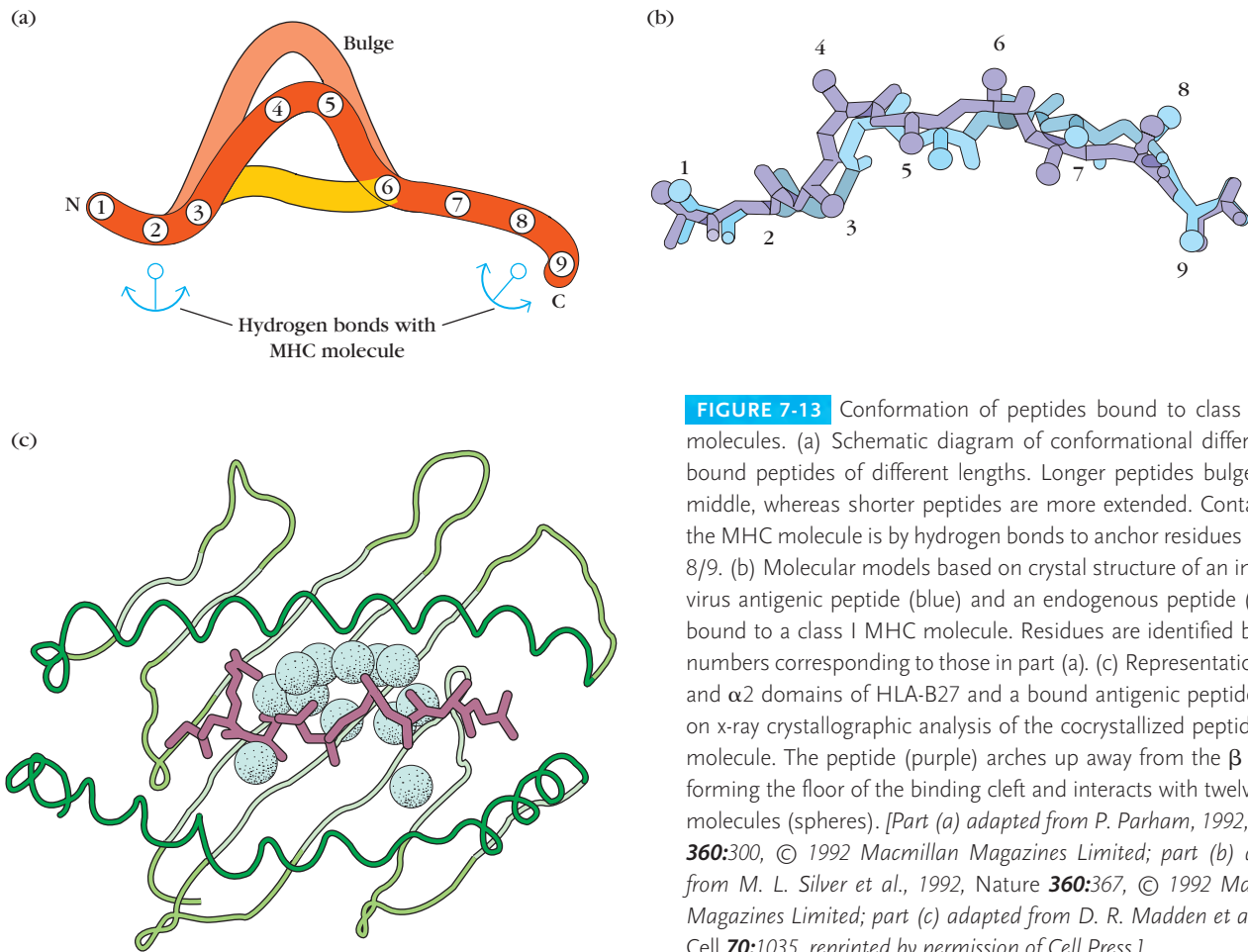


FIGURE 7-13 Conformation of peptides bound to class I MHC molecules. (a) Schematic diagram of conformational difference in bound peptides of different lengths. Longer peptides bulge in the middle, whereas shorter peptides are more extended. Contact with the MHC molecule is by hydrogen bonds to anchor residues 1/2 and 8/9. (b) Molecular models based on crystal structure of an influenza virus antigenic peptide (blue) and an endogenous peptide (purple) bound to a class I MHC molecule. Residues are identified by small numbers corresponding to those in part (a). (c) Representation of $\alpha 1$ and $\alpha 2$ domains of HLA-B27 and a bound antigenic peptide based on x-ray crystallographic analysis of the cocrystallized peptide–HLA molecule. The peptide (purple) arches up away from the β strands forming the floor of the binding cleft and interacts with twelve water molecules (spheres). [Part (a) adapted from P. Parham, 1992, *Nature* **360**:300, © 1992 Macmillan Magazines Limited; part (b) adapted from M. L. Silver et al., 1992, *Nature* **360**:367, © 1992 Macmillan Magazines Limited; part (c) adapted from D. R. Madden et al., 1992, *Cell* **70**:1035, reprinted by permission of Cell Press.]

and -C, we can calculate the theoretical number of combinations that can exist by multiplying $240 \times 470 \times 110$, yielding upwards of 12 million different class I haplotypes possible in the population. If class II loci are considered, the 5 DRB genes B1 through B5 have 304, 1, 35, 11, and 15 alleles respectively, DQA1 and B1 contribute 22 and 49 alleles, respectively and, DPB1 96 alleles; this allows approximately 1.8×10^{11} different class II combinations. Because each haplotype contains both class I and class II genes, the numbers are multiplied to give a total of 2.25×10^{18} possible combinations of these class I and II alleles.

LINKAGE DISEQUILIBRIUM

The calculation of theoretical diversity in the previous paragraph assumes completely random combinations of alleles. The actual diversity is known to be less, because certain allelic combinations occur more frequently in HLA haplotypes than predicted by random combination, a state referred to as *linkage disequilibrium*. Briefly, linkage disequilibrium is the difference between the frequency observed for a particular combination of alleles and that *expected* from the frequencies of the individual alleles. The expected frequency for the combination may be calculated by multiplying the frequencies of

the two alleles. For example, if HLA-A1 occurs in 16% of individuals in a population (frequency = 0.16) and HLA-B8 in 9% of that group (frequency = 0.09) it is expected that about 1.4% of the group should have both alleles ($0.16 \times 0.09 = 0.014$). However, the data show that HLA-A1 and HLA-B8 are found together in 8.8% of individuals studied. This difference is a measure of the linkage disequilibrium between these alleles of class I MHC genes.

Several explanations have been advanced to explain linkage disequilibrium. The simplest is that too few generations have elapsed to allow the number of crossovers necessary to reach equilibrium among the alleles present in founders of the population. The haplotypes that are over-represented in the population today would then reflect the combinations of alleles present in the founders. Alternatively, selective effects could also result in the higher frequency of certain allelic combinations. For example, certain combinations of alleles might produce resistance to certain diseases, causing them to be selected for and over-represented, or they might generate harmful effects, such as susceptibility to autoimmune disorders, and undergo negative selection. A third hypothesis is that crossovers are more frequent in certain DNA sequence regions, and the presence or absence of regions prone to crossover (hotspots) between alleles can dictate the

frequency of allelic association. Data in support of this was found in mouse breeding studies that generated new recombinant H-2 types. The points of crossover in the new MHC haplotypes were not randomly distributed throughout the complex. Instead, the same regions of crossover were found in more than one recombinant haplotype. This suggests that hotspots of recombination do exist that would influence linkage disequilibrium in populations.

Despite linkage disequilibrium, there is still enormous polymorphism in the human MHC, and it remains very difficult to match donor and acceptor MHC types for successful organ transplants. The consequences of this major obstacle to the therapeutic use of transplantation are described in Chapter 21.

FUNCTIONAL RELEVANCE OF MHC POLYMORPHISM

Sequence divergence among alleles of the MHC within a species is very high, as great as the divergence observed for the genes encoding some enzymes across species. Also of interest is that the sequence variation among MHC molecules is not randomly distributed along the entire polypeptide chain but instead is clustered in short stretches, largely within the membrane-distal $\alpha 1$ and $\alpha 2$ domains of class I

molecules (Figure 7-14a). Similar patterns of diversity are observed in the $\alpha 1$ and $\beta 2$ domains of class II molecules.

Progress has been made in locating the polymorphic residues within the three-dimensional structure of the membrane-distal domains in class I and class II MHC molecules and in relating allelic differences to functional differences (Figure 7-14b). For example, of 17 amino acids previously shown to display significant polymorphism in the HLA-A2 molecule, 15 were shown by x-ray crystallographic analysis to be in the peptide-binding cleft of this molecule. The location of so many polymorphic amino acids within the binding site for processed antigen strongly suggests that allelic differences contribute to the observed differences in the ability of MHC molecules to interact with a given antigenic peptide.

Detailed Genomic Map of MHC Genes

The MHC spans some 2000 kb of mouse DNA and some 4000 kb of human DNA. The recently completed human genome sequence shows this region to be densely packed

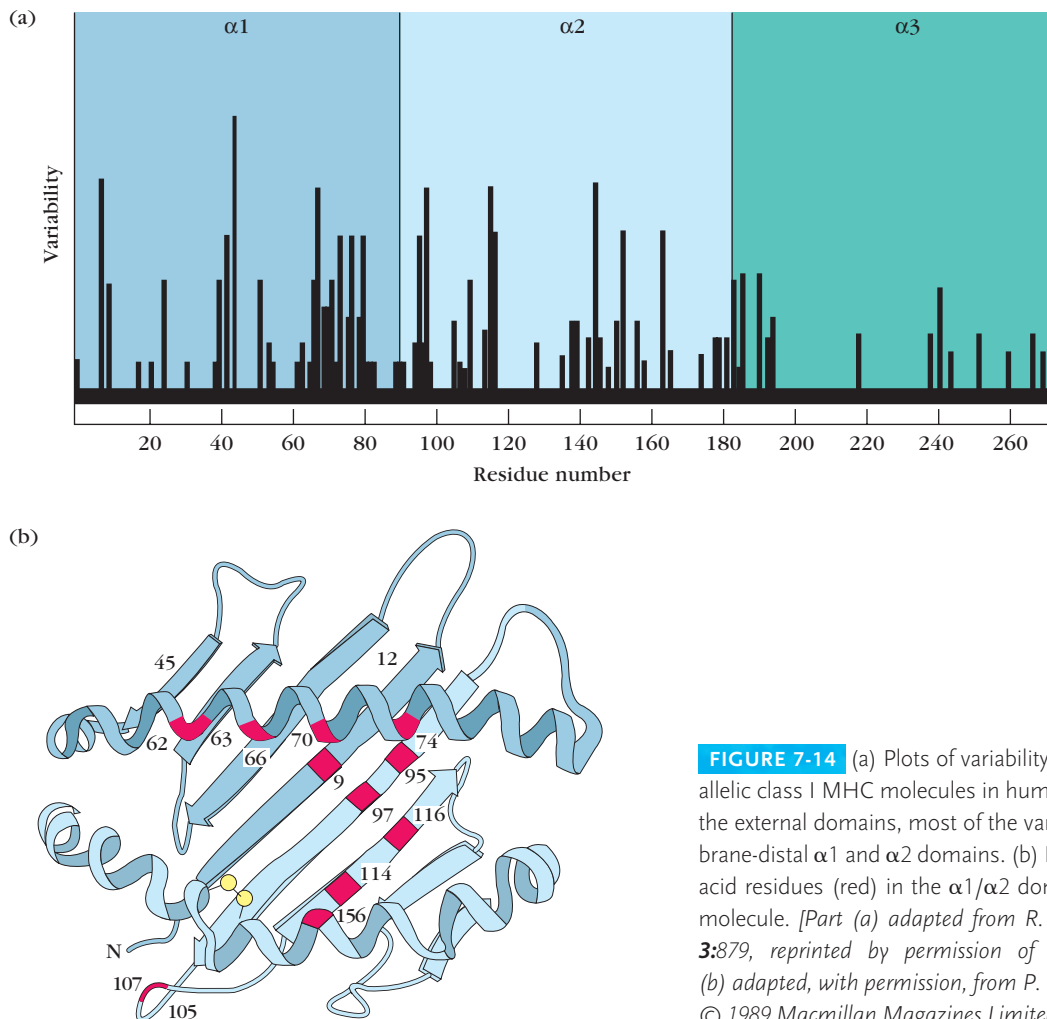
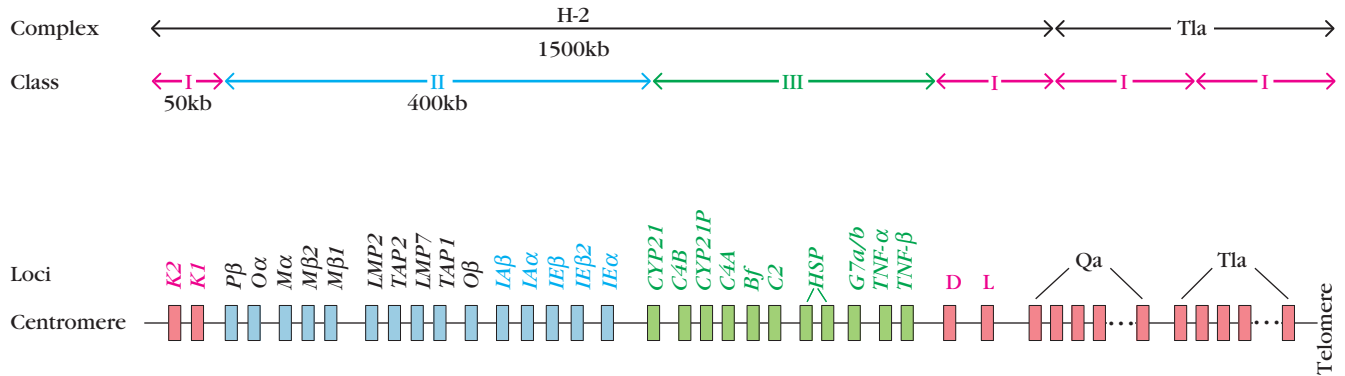
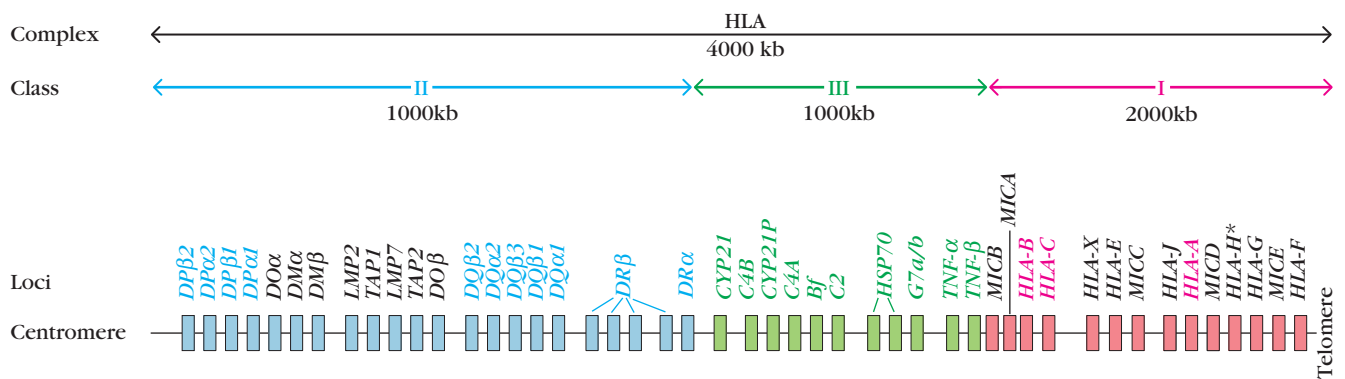


FIGURE 7-14 (a) Plots of variability in the amino acid sequence of allelic class I MHC molecules in humans versus residue position. In the external domains, most of the variable residues are in the membrane-distal $\alpha 1$ and $\alpha 2$ domains. (b) Location of polymorphic amino acid residues (red) in the $\alpha 1/\alpha 2$ domain of a human class I MHC molecule. [Part (a) adapted from R. Sodoyer et al., 1984, EMBO J. 3:879, reprinted by permission of Oxford University Press; part (b) adapted, with permission, from P. Parham, 1989, Nature 342:617, © 1989 Macmillan Magazines Limited.]

MOUSE CHROMOSOME 17



HUMAN CHROMOSOME 6



KEY

Gene	Encoded protein
<i>C2, C4A, C4B, Bf</i>	Complement components
<i>CYP21, CYP21P</i>	Steroid 21-hydroxylase
<i>G7a/b</i>	Valyl-tRNA synthetase
<i>HSP</i>	Heat-shock protein
<i>LMP2, LMP7</i>	Proteasome-like subunits
<i>TAP1, TAP2</i>	Peptide-transporter subunits
<i>TNF-α, TNF-β</i>	Tumor necrosis factors α and β

*Now designated HFE

FIGURE 7-15 Detailed genomic map of the mouse and human MHC, including genes encoding classical and nonclassical MHC molecules. The class I MHC genes are colored red, MHC II genes are colored blue, and genes in MHC III are colored green. Classical class I genes are labeled in red, class II in blue, and the nonclassical MHC genes are labeled in black. The concept of classical and nonclassical does not apply to class III. The functions for certain proteins encoded by the nonclassical class I genes are known. In the mouse, there are nonclassical genes located downstream from Tla that are not shown.

with genes, most of which have known functions. Our current understanding of the genomic organization of mouse and human MHC genes is diagrammed in Figure 7-15.

The Human Class I Region Spans about 2000 kb at the Telomeric End of the HLA Complex

In humans, the class I MHC region is about 2000 kb long and contains approximately 20 genes. In mice, the class I MHC consists of two regions separated by the intervening class II and class III regions. Included within the class I region are the genes encoding the well-characterized classical class I MHC molecules designated HLA-A, HLA-B, and HLA-C in humans and H-2K, H-2D, and H-2L in mice. Many nonclassical class I genes, identified by molecular

mapping, also are present in both the mouse and human MHC. In mice, the nonclassical class I genes are located in three regions (*H-2Q*, *T*, and *M*) downstream from the H-2 complex (*M* is not shown in Figure 7-15). In humans, the nonclassical class I genes include the *HLA-E*, *HLA-F*, *HLA-G*, *HFE*, *HLA-J*, and *HLA-X* loci as well as a recently discovered family of genes called *MIC*, which includes *MICA* through *MICE*. Some of the nonclassical class I MHC genes are pseudogenes and do not encode a protein product, but others, such as *HLA-G* and *HFE*, encode class I-like products with highly specialized functions. The *MIC* family of class I genes has only 15%–30% sequence identity to classical class I, and those designated as *MICA* are highly polymorphic. The *MIC* gene products are expressed at low levels in epithelial cells and are induced by heat or other stimuli that influence heat shock proteins.

The functions of the nonclassical class I MHC molecules remain largely unknown, although a few studies suggest that some of these molecules, like the classical class I MHC molecules, may present peptides to T cells. One intriguing finding is that the mouse molecule encoded by the *H-2M* locus is able to bind a self-peptide derived from a subunit of NADH dehydrogenase, an enzyme encoded by the mitochondrial genome. This particular self-peptide contains an amino-terminal formylated methionine. What is interesting about this finding is that peptides derived from prokaryotic organisms often have formylated amino-terminal methionine residues. This *H-2M*-encoded class I molecule may thus be uniquely suited to present peptides from prokaryotic organisms that are able to grow intracellularly. Such organisms include *Mycobacterium tuberculosis*, *Listeria monocytogenes*, *Brucella abortus*, and *Salmonella typhimurium*.

Up to this point, all description of antigen presentation by class I and class II molecules has been confined to presentation of peptide antigens. As will be seen in the description of antigen presentation (Chapter 8), there are also molecules with structural similarity to class I molecules that present non-peptide antigens, such as glycolipids, to T cells. A major family of such molecules, designated CD1, has been shown to present lipid antigens derived from bacteria. The CD1 molecules are not encoded within the MHC but are located on chromosome 1.

The Class II MHC Genes Are Located at the Centromeric End of HLA

The class II MHC region contains the genes encoding the α and β chains of the classical class II MHC molecules designated HLA-DR, DP, and DQ in humans and H-2IA and -IE in mice. Molecular mapping of the class II MHC has revealed multiple β -chain genes in some regions in both mice and humans, as well as multiple α -chain genes in humans (see Figure 7-15). In the human DR region, for example, there are three or four functional β -chain genes. All of the β -chain gene products can be expressed together with the α -chain gene product in a given cell, thereby increasing the number of different antigen-presenting molecules on the cell. Although the human DR region contains just one α -chain gene, the DP and DQ regions each contains two.

Genes encoding nonclassical class II MHC molecules have also been identified in both humans and mice. In mice, several class II genes ($O\alpha$, $O\beta$, $M\alpha$, and $M\beta$) encode nonclassical MHC molecules that exhibit limited polymorphism and a different pattern of expression than the classical IA and IE class II molecules. In the human class II region, nonclassical genes designated *DM* and *DO* have been identified. The *DM* genes encode a class II-like molecule (HLA-DM) that facilitates the loading of antigenic peptides into the class II MHC molecules. Class II *DO* molecules, which are expressed only in the thymus and mature B cells, have been shown to serve as regulators of class II antigen processing. The functions of HLA-DM and HLA-DO will be described further in Chapter 8.

Human MHC Class III Genes Are Between Class I and II

The class III region of the MHC in humans and mice contains a heterogeneous collection of genes (see Figure 7-15). These genes encode several complement components, two steroid 21-hydroxylases, two heat-shock proteins, and two cytokines (TNF- α and TNF- β). Some of these class III MHC gene products play a role in certain diseases. For example, mutations in the genes encoding 21-hydroxylase have been linked to congenital adrenal hyperplasia. Interestingly, the presence of a linked class III gene cluster is conserved in all species with an MHC region.

Cellular Distribution of MHC Molecules

In general, the classical class I MHC molecules are expressed on most nucleated cells, but the level of expression differs among different cell types. The highest levels of class I molecules are expressed by lymphocytes, where they constitute approximately 1% of the total plasma-membrane proteins, or some 5×10^5 molecules per cell. In contrast, fibroblasts, muscle cells, liver hepatocytes, and neural cells express very low levels of class I MHC molecules. The low level on liver cells may contribute to the considerable success of liver transplants by reducing the likelihood of graft recognition by T_c of the recipient. A few cell types (e.g., neurons and sperm cells at certain stages of differentiation) appear to lack class I MHC molecules altogether.

As noted earlier, any particular MHC molecule can bind many different peptides. Since the MHC alleles are codominantly expressed, a heterozygous individual expresses on its cells the gene products encoded by both alleles at each MHC locus. An F_1 mouse, for example, expresses the K, D, and L from each parent (six different class I MHC molecules) on each of its nucleated cells (Figure 7-16). A similar situation occurs in humans; that is, a heterozygous individual expresses the A, B, and C alleles from each parent (six different class I MHC molecules) on the membrane of each nucleated cell. The expression of so many class I MHC molecules allows each cell to display a large number of peptides in the peptide-binding clefts of its MHC molecules.

In normal, healthy cells, the class I molecules will display self-peptides resulting from normal turnover of self proteins. In cells infected by a virus, viral peptides, as well as self-peptides, will be displayed. A single virus-infected cell should be envisioned as having various class I molecules on its membrane, each displaying different sets of viral peptides. Because of individual allelic differences in the peptide-binding clefts of the class I MHC molecules, different individuals within a species will have the ability to bind different sets of viral peptides.

Unlike class I MHC molecules, class II molecules are expressed constitutively only by antigen-presenting cells, pri-

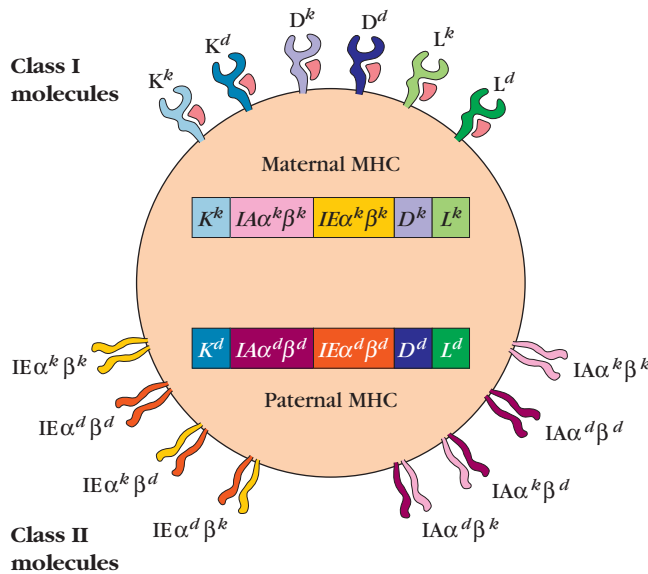


FIGURE 7-16 Diagram illustrating various MHC molecules expressed on antigen-presenting cells of a heterozygous H-2^{k/d} mouse. Both the maternal and paternal MHC genes are expressed. Because the class II molecules are heterodimers, heterologous molecules containing one maternal-derived and one paternal-derived chain are produced. The β_2 -microglobulin component of class I molecules (pink) is encoded by a gene on a separate chromosome and may be derived from either parent.

marily macrophages, dendritic cells, and B cells; thymic epithelial cells and some other cell types can be induced to express class II molecules and to function as antigen-presenting cells under certain conditions and under stimulation of some cytokines (see Chapter 8). Among the various cell types that express class II MHC molecules, marked differences in expression have been observed. In some cases, class II expression depends on the cell's differentiation stage. For example, class II molecules cannot be detected on pre-B cells but are expressed constitutively on the membrane of mature B cells. Similarly, monocytes and macrophages express only low levels of class II molecules until they are activated by interaction with an antigen, after which the level of expression increases significantly.

Because each of the classical class II MHC molecules is composed of two different polypeptide chains, which are encoded by different loci, a heterozygous individual expresses not only the parental class II molecules but also molecules containing α and β chains from different chromosomes. For example, an H-2^k mouse expresses IA^k and IE^k class II molecules; similarly, an H-2^d mouse expresses IA^d and IE^d molecules. The F₁ progeny resulting from crosses of mice with these two haplotypes express four parental class II molecules and four molecules containing one parent's α chain and the other parent's β chain (as shown in Figure 7-16). Since the human MHC contains three classical class II genes (*DP*, *DQ*, and *DR*), a heterozygous individual expresses six parental class II molecules and six molecules containing α and β chain

combinations from either parent. The number of different class II molecules expressed by an individual is increased further by the presence of multiple β -chain genes in mice and humans, and in humans by multiple α -chain genes. The diversity generated by these mechanisms presumably increases the number of different antigenic peptides that can be presented and thus is advantageous to the organism.

Regulation of MHC Expression

Research on the regulatory mechanisms that control the differential expression of MHC genes in different cell types is still in its infancy, but much has been learned. The publication of the complete genomic map of the MHC complex is expected to greatly accelerate the identification and investigation of coding and regulatory sequences, leading to new directions in research on how the system is controlled.

Both class I and class II MHC genes are flanked by 5' promoter sequences, which bind sequence-specific transcription factors. The promoter motifs and transcription factors that bind to these motifs have been identified for a number of MHC genes. Transcriptional regulation of the MHC is mediated by both positive and negative elements. For example, an MHC II transactivator, called *CIITA*, and another transcription factor, called *REFX*, both have been shown to bind to the promoter region of class II MHC genes. Defects in these transcription factors cause one form of *bare lymphocyte syndrome* (see the Clinical Focus box in Chapter 8). Patients with this disorder lack class II MHC molecules on their cells and as a result suffer a severe immunodeficiency due to the central role of class II MHC molecules in T-cell maturation and activation.

The expression of MHC molecules is also regulated by various cytokines. The interferons (alpha, beta, and gamma) and tumor necrosis factor have each been shown to increase expression of class I MHC molecules on cells. Interferon gamma (IFN- γ), for example, appears to induce the formation of a specific transcription factor that binds to the promoter sequence flanking the class I MHC genes. Binding of this transcription factor to the promoter sequence appears to coordinate the up-regulation of transcription of the genes encoding the class I α chain, β_2 -microglobulin, the proteasome subunits (LMP), and the transporter subunits (TAP). IFN- γ also has been shown to induce expression of the class II transactivator (*CIITA*), thereby indirectly increasing expression of class II MHC molecules on a variety of cells, including non-antigen-presenting cells (e.g., skin keratinocytes, intestinal epithelial cells, vascular endothelium, placental cells, and pancreatic beta cells). Other cytokines influence MHC expression only in certain cell types; for example, IL-4 increases expression of class II molecules by resting B cells. Expression of class II molecules by B cells is down-regulated by IFN- γ ; corticosteroids and prostaglandins also decrease expression of class II molecules.

MHC expression is decreased by infection with certain viruses, including human cytomegalovirus (CMV), hepatitis

B virus (HBV), and adenovirus 12 (Ad12). In some cases, reduced expression of class I MHC molecules on cell surfaces is due to decreased levels of a component needed for peptide transport or MHC class I assembly rather than in transcription. In cytomegalovirus infection, for example, a viral protein binds to β_2 -microglobulin, preventing assembly of class I MHC molecules and their transport to the plasma membrane. Adenovirus 12 infection causes a pronounced decrease in transcription of the transporter genes (*TAP1* and *TAP2*). As the next chapter describes, the TAP gene products play an important role in peptide transport from the cytoplasm into the rough endoplasmic reticulum. Blocking of TAP gene expression inhibits peptide transport; as a result, class I MHC molecules cannot assemble with β_2 -microglobulin or be transported to the cell membrane. Decreased expression of class I MHC molecules, by whatever mechanism, is likely to help viruses evade the immune response by reducing the likelihood that virus-infected cells can display MHC–viral peptide complexes and become targets for CTL-mediated destruction.

MHC and Immune Responsiveness

Early studies by B. Benacerraf in which guinea pigs were immunized with simple synthetic antigens were the first to show that the ability of an animal to mount an immune re-

sponse, as measured by the production of serum antibodies, is determined by its MHC haplotype. Later experiments by H. McDevitt, M. Sela, and their colleagues used congenic and recombinant congenic mouse strains to map the control of *immune responsiveness* to class II MHC genes. In early reports, the genes responsible for this phenotype were designated *Ir* or immune response genes, and for this reason mouse class II products are called IA and IE. We now know that the dependence of immune responsiveness on the class II MHC reflects the central role of class II MHC molecules in presenting antigen to T_H cells.

Two explanations have been proposed to account for the variability in immune responsiveness observed among different haplotypes. According to the *determinant-selection model*, different class II MHC molecules differ in their ability to bind processed antigen. According to the *alternative holes-in-the-repertoire model*, T cells bearing receptors that recognize foreign antigens closely resembling self-antigens may be eliminated during thymic processing. Since the T-cell response to an antigen involves a trimolecular complex of the T cell's receptor, an antigenic peptide, and an MHC molecule (see Figure 3-8), both models may be correct. That is, the absence of an MHC molecule that can bind and present a given peptide, or the absence of T-cell receptors that can recognize a given peptide–MHC molecule complex, could result in the absence of immune responsiveness and so account for the observed relationship between

TABLE 7-3 Differential binding of peptides to mouse class II MHC molecules and correlation with MHC restriction

Labeled peptide*	MHC restriction of responders [†]	PERCENTAGE OF LABELED PEPTIDE BOUND TO [‡]			
		IA ^d	IE ^d	IA ^k	IE ^k
Ovalbumin (323–339)	IA ^d	11.8	0.1	0.2	0.1
Influenza hemagglutinin (130–142)	IA ^d	18.9	0.6	7.1	0.3
Hen egg-white lysozyme (46–61)	IA ^k	0.0	0.0	35.2	0.5
Hen egg-white lysozyme (74–86)	IA ^k	2.0	2.3	2.9	1.7
Hen egg-white lysozyme (81–96)	IE ^k	0.4	0.2	0.7	1.1
Myoglobin (132–153)	IE ^d	0.8	6.3	0.5	0.7
Pigeon cytochrome <i>c</i> (88–104)	IE ^k	0.6	1.2	1.7	8.7
λ repressor (12–26) [§]	IA ^d + IE ^k	1.6	8.9	0.3	2.3

*Amino acid residues included in each peptide are indicated by the numbers in parentheses.

[†]Refers to class II molecule (IA or IE) and haplotype associated with a good response to the indicated peptides.

[‡]Binding determined by equilibrium dialysis. Bold-faced values indicate binding was significantly greater ($p < 0.05$) than that of the other three class II molecules tested.

[§]The λ repressor is an exception to the rule that high binding correlates with the MHC restriction of high-responder strains. In this case, the T_H cell specific for the λ peptide–IE^d complex has been deleted; this is an example of the hole-in-the-repertoire mechanism.

SOURCE: Adapted from S. Buus et al., 1987, *Science* 235:1353.

MHC haplotype and immune responsiveness to exogenous antigens.

According to the determinant-selection model, the MHC polymorphism within a species will generate a diversity of binding specificities, and thus different patterns of responsiveness to antigens. If this model is correct, then class II MHC molecules from mouse strains that respond to a particular antigen and those that do not should show differential binding of that antigen. Table 7-3 presents data on the binding of various radiolabeled peptides to class II IA and IE molecules with the H-2^d or H-2^k haplotype. Each of the listed peptides binds significantly to only one of the IA or IE molecules. Furthermore, in all but one case, the haplotype of the class II molecule showing the highest affinity for a particular peptide is the same as the haplotype of responder strains for that peptide, as the determinant-selection model predicts.

The single exception to the general pattern in Table 7-3 (residues 12–26 of the λ repressor protein) gives evidence that the influence on immune responsiveness can also be caused by absence of functional T cells (holes-in-the-repertoire model) capable of recognizing a given antigen–MHC molecule complex. The λ repressor peptide binds best in vitro to IE^d, yet the MHC restriction for response to this pep-

tide is known to be associated not with IE^d but instead with IA^d and IE^k. This suggests that T cells recognizing this repressor peptide in association with IE^d may have been eliminated by negative selection in the thymus, leaving a hole in the T-cell repertoire.

MHC and Disease Susceptibility

Some HLA alleles occur at a much higher frequency in those suffering from certain diseases than in the general population. The diseases associated with particular MHC alleles include autoimmune disorders, certain viral diseases, disorders of the complement system, some neurologic disorders, and several different allergies. The association between HLA alleles and a given disease may be quantified by determining the frequency of the HLA alleles expressed by individuals afflicted with the disease, then comparing these data with the frequency of the same alleles in the general population. Such a comparison allows calculation of **relative risk** (see Table 7-4). A relative risk value of 1 means that the HLA allele is expressed with the same frequency in the patient and general populations, indicating that the allele confers no increased risk for the disease. A relative risk value substantially

TABLE 7-4 Some significant associations of HLA alleles with increased risk for various diseases

Disease	Associated HLA allele	Relative risk*
Ankylosing spondylitis	B27	90
Goodpasture's syndrome	DR2	16
Gluten-sensitive enteropathy	DR3	12
Hereditary hemochromatosis	A3	9.3
	B14	2.3
	A3/B14	90
Insulin-dependent diabetes mellitus	DR4/DR3	20
Multiple sclerosis	DR2	5
Myasthenia gravis	DR3	10
Narcolepsy	DR2	130
Reactive arthritis (<i>Yersinia</i> , <i>Salmonella</i> , <i>Gonococcus</i>)	B27	18
Reiter's syndrome	B27	37
Rheumatoid arthritis	DR4	10
Sjogren's syndrome	Dw3	6
Systemic lupus erythematosus	DR3	5

*Relative risk is calculated by dividing the frequency of the HLA allele in the patient population by the frequency in the general population:

$$RR = \frac{(Ag^+/Ag^-)_{\text{disease}}}{(Ag^+/Ag^-)_{\text{control}}}$$

SOURCE: Data from SAM CD: *A Comprehensive Knowledge Base of Internal Medicine*, D. C. Dale and D. D. Federman, eds., 1997, Scientific American, New York.



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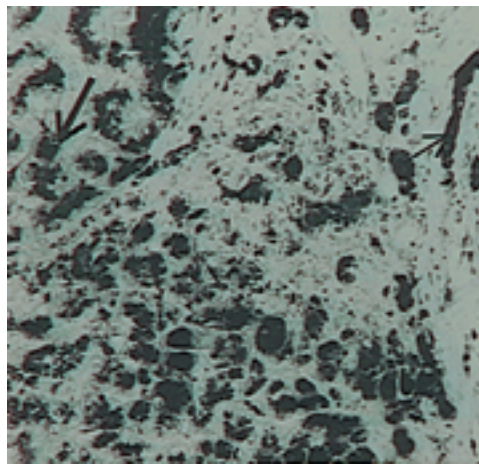
HFE and Hereditary Hemochromatosis

Hereditary hemochromatosis (HH) is a disease in which defective regulation of dietary iron absorption leads to increased levels of iron. HH (which in earlier reports may be referred to as *idiopathic* or *primary* hemochromatosis) is the most common known autosomal recessive genetic disorder in North Americans of European descent, with a frequency of 3–4 cases per 1000 persons. Recent studies show that this disease is associated with a mutation in the nonclassical class I gene *HFE* (formerly designated *HLA-H*), which lies to the telomeric side of *HLA-A*. The association of the *HFE* gene with HH is an example of how potentially lifesaving clinical information can be obtained by studying the connection of HLA genes with disease.

The total iron content of a normal human adult is 3 to 4 grams; the average dietary intake of iron is about 10 to 20

milligrams per day; of this, only 1 to 2 mg is absorbed. The iron balance is maintained by control of its absorption from digested food in the intestinal tract. The primary defect in HH is increased gastrointestinal uptake of iron and, as a result of this, patients with HH may throughout their lives accumulate 15 to 35 grams of

iron instead of the normal 3 to 4 grams. The iron overload results in pathologic accumulation of iron in cells of many organs, including the heart and liver. Although a severe form of HH may result in heart disease in children, the clinical manifestations of the disease are not usually seen until 40 to 50 years of age. Males are affected eight times more frequently than females. Early symptoms of HH are rather nonspecific and include weakness, lethargy, abdominal pain, diabetes, impotence, and severe joint pain. Physical examination of HH sufferers reveals liver damage, skin pigmentation, arthritis, en-



High-magnification iron stain of liver cells from HH patient. The stain confirms the presence of iron in both parenchymal cells (thick arrow) and bile duct cells (thin arrow). This woman with hemochromatosis required removal of 72 units (about 36 liters or 9 gallons) of blood during one and a half years to render her liver free of excess iron. [SAM CD: A Comprehensive Knowledge Base of Internal Medicine, D. C. Dale and D. D. Federman, eds., 1997, *Scientific American*, New York.]

above 1 indicates an association between the HLA allele and the disease. As Table 7-4 shows, individuals with the HLA-B27 allele have a 90 times greater likelihood (relative risk of 90) of developing the autoimmune disease ankylosing spondylitis, an inflammatory disease of vertebral joints characterized by destruction of cartilage, than do individuals with a different HLA-B allele.

The existence of an association between an MHC allele and a disease should not be interpreted to imply that the expression of the allele has caused the disease—the relationship between MHC alleles and development of disease is complex. In the case of ankylosing spondylitis, for example, it has been suggested that because of the close linkage of the *TNF- α* and *TNF- β* genes with the HLA-B locus, these cytokines may be involved in the destruction of cartilage. An association of HLA class I genes with the disease hereditary hemochromatosis is discussed in the Clinical Focus box in this chapter.

When the associations between MHC alleles and disease are weak, reflected by low relative risk values, it is likely that multiple genes influence susceptibility, of which only one is

in the MHC. That these diseases are not inherited by simple Mendelian segregation of MHC alleles can be seen in identical twins; both inherit the MHC risk factor, but it is by no means certain that both will develop the disease. This finding suggests that multiple genetic and environmental factors have roles in the development of disease, especially autoimmune diseases, with the MHC playing an important but not exclusive role. An additional difficulty in associating a particular MHC product with disease is the genetic phenomenon of linkage disequilibrium, which was described above. The fact that some of the class I MHC alleles are in linkage disequilibrium with the class II MHC alleles makes their contribution to disease susceptibility appear more pronounced than it actually is. If, for example, DR4 contributes to risk of a disease, and if it occurs frequently in combination with A3 because of linkage disequilibrium, then A3 would incorrectly appear to be associated with the disease. Improved genomic mapping techniques make it possible to analyze the linkage between the MHC and various diseases more fully and to assess the contributions from other loci.

larged spleen, jaundice, and peripheral edema. If untreated, HH results in hepatic cancer, liver failure, severe diabetes, and heart disease. Exactly how the increase in iron content results in these diseases is not known, but repeated phlebotomy (taking blood) is an effective treatment if the disease is recognized before there is extensive damage to organs. Phlebotomy does not reverse damage already done. Phlebotomy (also called blood-letting) was used as treatment for many conditions in former times; HH may be one of the rare instances in which the treatment had a positive rather than a harmful effect on the patient.

Prior to appearance of the recognized signs of the disease, such as the characteristic skin pigmentation or liver dysfunction, diagnosis is difficult unless for some reason (such as family history of the disease) HH is suspected and specific tests for iron metabolism are performed. A reliable genetic test for HH would allow treatment to commence prior to disease manifestation and irreversible organ damage.

Because it is a common disease, the association of HH with HLA was studied; initially a significant association with the *HLA-A3* allele was found (RR of 9.3). This

association is well documented, but the relatively high frequency of the *HLA-A3* allele (present in 20% of the North American population) makes this an inadequate marker; the majority of individuals with *HLA-A3* will not have HH. Further studies showed a greatly increased relative risk in individuals with the combination of *HLA-A3* and *HLA-B14*; homozygotes for these two alleles carried a relative risk for HH of 90. Detailed studies of several populations in the US and France with high incidence of HH revealed a mutation in the nonclassical HLA class I gene *HFE* in 83%–100% of patients with HH. *HFE*, which lies close to the *HLA-A* locus, was shown in several independent studies to carry a characteristic mutation at position 283 in HH patients, with substitution of a tyrosine residue for the cysteine normally found at this position. The substitution precludes formation of the disulfide link between cysteines in the $\alpha 3$ domain, which is necessary for association of the MHC α chain with β_2 -microglobulin and for expression on the cell surface. *HFE* molecules are normally expressed on the surface of cells in the stomach, intestines, and liver. There is evidence showing that *HFE* plays a role in the abil-

ity of these organs to regulate iron uptake from the circulation. The mechanism by which *HFE* functions involves binding to the transferrin receptor, which reduces the affinity of the receptor for iron-loaded transferrin. This lowers the uptake of iron by the cell. Mutations that interfere with the ability of *HFE* to form a complex with transferrin and its receptor can lead to increased iron absorption and HH.

There are several possible reasons for why this defect continues to be so common in our population. Factors that favor the spread of the defective *HFE* gene would include the fact that it is a recessive trait, so only homozygotes are affected; the gene is silent in carriers. In addition, even in most homozygotes affected with HH, the disease does not manifest itself until later in life and so may have minimal influence on the breeding success of the HH sufferer.

Studies of knockout mice that lack the gene for β_2 -microglobulin demonstrate that MHC class I products on cell surfaces are necessary for the maintenance of normal iron metabolism. These mice, which are unable to express any of their class I molecules on the cell surfaces, suffer from iron overload with disease consequences similar to HH.

A number of hypotheses have been offered to account for the role of the MHC in disease susceptibility. As noted earlier, allelic differences may yield differences in immune responsiveness arising from variation in the ability to present processed antigen or the ability of T cells to recognize presented antigen. Allelic forms of MHC genes may also encode molecules that are recognized as receptors by viruses or bacterial toxins. As will be explained in Chapter 16, the genetic analysis of disease must consider the possibility that genes at multiple loci may be involved and that complex interactions among them may be needed to trigger disease.

Some evidence suggests that a reduction in MHC polymorphism within a species may predispose that species to infectious disease. Cheetahs and certain other wild cats, such as Florida panthers, that have been shown to be highly susceptible to viral disease have very limited MHC polymorphism. It is postulated that the present cheetah population (Figure 7-17) arose from a limited breeding stock, causing a loss of MHC diversity. The increased susceptibility of cheetahs to various viral diseases may result from a reduction in



FIGURE 7-17 Cheetah female with two nearly full grown cubs. Polymorphism in MHC genes of the cheetah is very limited, presumably because of a bottleneck in breeding that occurred in the not too distant past. It is assumed that all cheetahs alive today are descendants of a very small breeding pool. [Photograph taken in the Okavango Delta, Botswana, by T. J. Kindt.]

the number of different MHC molecules available to the species as a whole and a corresponding limitation on the range of processed antigens with which these MHC molecules can interact. Thus, the high level of MHC polymorphism that has been observed in various species may provide the advantage of a broad range of antigen-presenting MHC molecules. Although some individuals within a species probably will not be able to develop an immune response to any given pathogen and therefore will be susceptible to infection by it, extreme polymorphism ensures that at least some members of a species will be able to respond and will be resistant. In this way, MHC diversity appears to protect a species from a wide range of infectious diseases.

SUMMARY

- The major histocompatibility complex (MHC) comprises a stretch of tightly linked genes that encode proteins associated with intercellular recognition and antigen presentation to T lymphocytes.
- A group of linked MHC genes is generally inherited as a unit from parents; these linked groups are called haplotypes.
- MHC genes are polymorphic in that there are large numbers of alleles for each gene, and they are polygenic in that there are a number of different MHC genes.
- Class I MHC molecules consist of a large glycoprotein chain with 3 extracellular domains and a transmembrane segment, and β_2 -microglobulin, a protein with a single domain.
- Class II MHC molecules are composed of two noncovalently associated glycoproteins, the α and β chain, encoded by separate MHC genes.
- X-ray crystallographic analyses reveal peptide-binding clefts in the membrane-distal regions of both class I and class II MHC molecules.
- Both class I and class II MHC molecules present antigen to T cells. Class I molecules present processed endogenous antigen to CD8 T cells. Class II molecules present processed exogenous antigen to CD4 T cells.
- Certain conserved motifs in peptides influence their ability to interact with the membrane-distal regions of class I and class II MHC molecules.
- Class I molecules are expressed on most nucleated cells; class II antigens are restricted to B cells, macrophages, and dendritic cells.
- The class III region of the MHC encodes molecules that include a diverse group of proteins that play no role in antigen presentation.
- Detailed maps of the human and mouse MHC reveal the presence of genes involved in antigen processing, including proteasomes and transporters.
- Studies with mouse strains have shown that MHC haplotype influences immune responsiveness and the ability to present antigen.
- Increased susceptibility to a number of diseases, predominantly, but not exclusively, of an autoimmune nature, has been linked to certain MHC alleles.

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- Yung, Y. C., et al. 2000. The human and mouse class III region: a parade of 21 genes at the centromeric segment. *Immunol. Today* **21**:320.



USEFUL WEB SITES

- <http://www.bioscience.org/knockout/b2micr.gl.htm>
for beta-2 microglobulin KO
- <http://www.bioscience.org/knockout/mhci.htm>
for MHC class I KO

<http://www.bioscience.org/knockout/mhcii.htm>
for KO of an MHC class II chain

<http://www.bioscience.org/knockout/mhc2inva.htm>
for KO of the invariant chain

This series of destinations in the Bioscience Web site provides updated information on studies of the consequences of targeted disruption of MHC molecules and other component molecules including β_2 microglobulin and the class II invariant chain.

<http://www.bshi.org.uk/>

British Society for Histocompatibility and Immunogenetics home page contains information on tissue typing, transplantation, and links to worldwide sites concerned with MHC.

<http://www.ebi.ac.uk/imgt/hla/>

The International ImMunoGeneTics (IMGT) database section contains links concerned with HLA gene structure and genetics. It also contains up-to-date listings and sequences for all HLA alleles officially recognized by the World Health Organization HLA nomenclature committee.

Study Questions

CLINICAL FOCUS QUESTION Almost 90% of Caucasians homozygous for a mutation in position 283 of the HFE gene have clinical signs of hemochromatosis. The fact that 10% of those with the mutation are not affected causes a critic of the work to state that the HFE is not involved with HH. She contends that this association is just a result of linkage disequilibrium. How would you answer her? Can you design an experiment to shed further light on this association?

- Indicate whether each of the following statements is true or false. If you think a statement is false, explain why.
 - A monoclonal antibody specific for β_2 -microglobulin can be used to detect both class I MHC K and D molecules on the surface of cells.
 - Antigen-presenting cells express both class I and class II MHC molecules on their membranes.
 - Class III MHC genes encode membrane-bound proteins.
 - In outbred populations, an individual is more likely to be histocompatible with one of its parents than with its siblings.
 - Class II MHC molecules typically bind to longer peptides than do class I molecules.
 - All cells express class I MHC molecules.
 - The majority of the peptides displayed by class I and class II MHC molecules on cells are derived from self-proteins.
- You wish to produce a syngeneic and a congenic mouse strain. Indicate whether each of the following characteristics applies to production of syngeneic (S), congenic (C), or both (S and C) mice.
 - Requires the greatest number of generations
 - Requires backcrosses
 - Yields mice that are genetically identical
 - Requires selection for homozygosity

- Requires sibling crosses
- Can be started with outbred mice
- Yields progeny that are genetically identical to the parent except for a single genetic region

- You have generated a congenic A.B mouse strain that has been selected for its MHC haplotype. The haplotype of strain A was *a/a* and of strain B was *b/b*.
 - Which strain provides the genetic background of this mouse?
 - Which strain provides the haplotype of the MHC of this mouse?
 - To produce this congenic strain, the F₁ progeny are always backcrossed to which strain?
 - Why was backcrossing to one of the parents performed?
 - Why was interbreeding of the F₁ and F₂ progeny performed?
 - Why was selection necessary and what kind of selection was performed?
- You cross a BALB/c (H-2^d) mouse with a CBA (H-2^k) mouse. What MHC molecules will the F₁ progeny express on (a) its liver cells and (b) its macrophages?
- To carry out studies on the structure and function of the class I MHC molecule K^b and the class II MHC molecule IA^b, you decide to transfect the genes encoding these proteins into a mouse fibroblast cell line (L cell) derived from the C3H strain (H-2^k). L cells do not normally function as antigen-presenting cells. In the following table, indicate which of the listed MHC molecules will (+) or will not (−) be expressed on the membrane of the transfected L cells.

Transfected gene	MHC molecules expressed on the membrane of the transfected L cells					
	D ^k	D ^b	K ^k	K ^b	IA ^k	IA ^b
None						
K ^b						
IA ^a ^b						
IA ^b ^b						
IA ^a ^b and IA ^b ^b						

- The SJL mouse strain, which has the H-2^k haplotype, has a deletion of the *IE α* locus.
 - List the classical MHC molecules that are expressed on the membrane of macrophages from SJL mice.
 - If the class II *IE α* and *IE β* genes from an H-2^s strain are transfected into SJL macrophages, what additional classical MHC molecules would be expressed on the transfected macrophages?
- Draw diagrams illustrating the general structure, including the domains, of class I MHC molecules, class II MHC molecules, and membrane-bound antibody on B cells. Label each

chain and the domains within it, the antigen-binding regions, and regions that have the immunoglobulin-fold structure.

8. One of the characteristic features of the MHC is the large number of different alleles at each locus.
 - a. Where are most of the polymorphic amino acid residues located in MHC molecules? What is the significance of this location?
 - b. How is MHC polymorphism thought to be generated?
9. As a student in an immunology laboratory class, you have been given spleen cells from a mouse immunized with the LCM virus. You determine the antigen-specific functional activity of these cells with two different assays. In assay 1, the spleen cells are incubated with macrophages that have been briefly exposed to the LCM virus; the production of interleukin 2 (IL-2) is a positive response. In assay 2, the spleen cells are incubated with LCM-infected target cells; lysis of the target cells represents a positive response in this assay. The results of the assays using macrophages and target cells of different haplotypes are presented in the table below. Note that the experiment has been set up in a way to exclude alloreactive responses (reactions against nonself MHC molecules).
 - a. The activity of which cell population is detected in each of the two assays?
 - b. The functional activity of which MHC molecules is detected in each of the two assays?
 - c. From the results of this experiment, which MHC molecules are required, in addition to the LCM virus, for specific reactivity of the spleen cells in each of the two assays?
 - d. What additional experiments could you perform to unambiguously confirm the MHC molecules required for antigen-specific reactivity of the spleen cells?
- e. Which of the mouse strains listed in the table below could have been the source of the immunized spleen cells tested in the functional assays? Give your reasons.
10. A T_C-cell clone recognizes a particular measles virus peptide when it is presented by H-2D^b. Another MHC molecule has a peptide-binding cleft identical to the one in H-2D^b but differs from H-2D^b at several other amino acids in the $\alpha 1\beta 1$ domain. Predict whether the second MHC molecule could present this measles virus peptide to the T_C-cell clone. Briefly explain your answer.
11. How can you determine if two different inbred mouse strains have identical MHC haplotypes?
12. Human red blood cells are not nucleated and do not express any MHC molecules. Why is this property fortuitous for blood transfusions?
13. The hypothetical allelic combination *HLA-A99* and *HLA-B276* carries a relative risk of 200 for a rare, and yet unnamed, disease that is fatal to pre-adolescent children.
 - a. Will every individual with A99/B276 contract the disease?
 - b. Will everyone with the disease have the A99/B276 combination?
 - c. How frequently will the A99/B276 allelic combination be observed in the general population? Do you think that this combination will be more or less frequent than predicted by the frequency of the two individual alleles? Why?

For use with Question 9.

Mouse strain used as source of macrophages and target cells	MHC haplotype of macrophages and virus-infected target cells				Response of spleen cells	
					IL-2 production in response to LCM-pulsed macrophages (assay 1)	Lysis of LCM-infected cells (assay 2)
C3H	<i>k</i>	<i>k</i>	<i>k</i>	<i>k</i>	+	—
BALB/c	<i>d</i>	<i>d</i>	<i>d</i>	<i>d</i>	—	+
(BALB/c × B10.A)F ₁	<i>d/k</i>	<i>d/k</i>	<i>d/k</i>	<i>d/d</i>	+	+
ATL	<i>s</i>	<i>k</i>	<i>k</i>	<i>d</i>	+	+
B10.A (3R)	<i>b</i>	<i>b</i>	<i>b</i>	<i>d</i>	—	+
B10.A (4R)	<i>k</i>	<i>k</i>	—	<i>b</i>	+	—

chapter 8

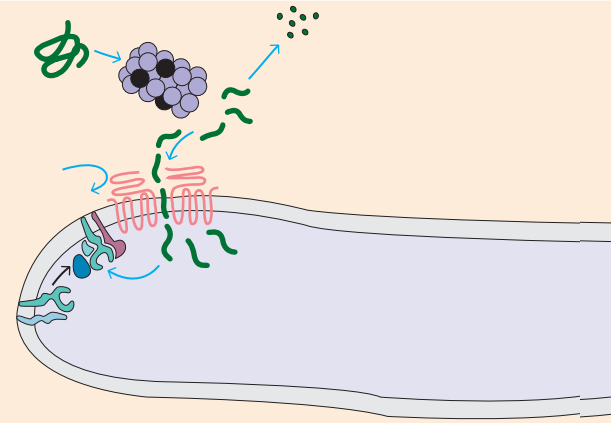
Antigen Processing and Presentation

RECOGNITION OF FOREIGN PROTEIN ANTIGENS BY a T cell requires that peptides derived from the antigen be displayed within the cleft of an MHC molecule on the membrane of a cell. The formation of these peptide-MHC complexes requires that a protein antigen be degraded into peptides by a sequence of events called **antigen processing**. The degraded peptides then associate with MHC molecules within the cell interior, and the peptide-MHC complexes are transported to the membrane, where they are displayed (**antigen presentation**).

Class I and class II MHC molecules associate with peptides that have been processed in different intracellular compartments. Class I MHC molecules bind peptides derived from **endogenous antigens** that have been processed within the cytoplasm of the cell (e.g., normal cellular proteins, tumor proteins, or viral and bacterial proteins produced within infected cells). Class II MHC molecules bind peptides derived from **exogenous antigens** that are internalized by phagocytosis or endocytosis and processed within the endocytic pathway. This chapter examines in more detail the mechanism of antigen processing and the means by which processed antigen and MHC molecules are combined. In addition, a third pathway for the presentation of nonpeptide antigens derived from bacterial pathogens is described.

Self-MHC Restriction of T Cells

Both $CD4^+$ and $CD8^+$ T cells can recognize antigen only when it is presented by a self-MHC molecule, an attribute called *self-MHC restriction*. Beginning in the mid-1970s, experiments conducted by a number of researchers demonstrated self-MHC restriction in T-cell recognition. A. Rosenthal and E. Shevach, for example, showed that antigen-specific proliferation of T_H cells occurred only in response to antigen presented by macrophages of the same MHC haplotype as the T cells. In their experimental system, guinea pig macrophages from strain 2 were initially incubated with an antigen. After the “antigen-pulsed” macrophages had processed the antigen and presented it on their surface, they were mixed with T cells from the same strain (strain 2), a different strain (strain 13), or (2×13) F_1 animals, and the magnitude of T-cell proliferation in response to the antigen-pulsed macrophages was measured.

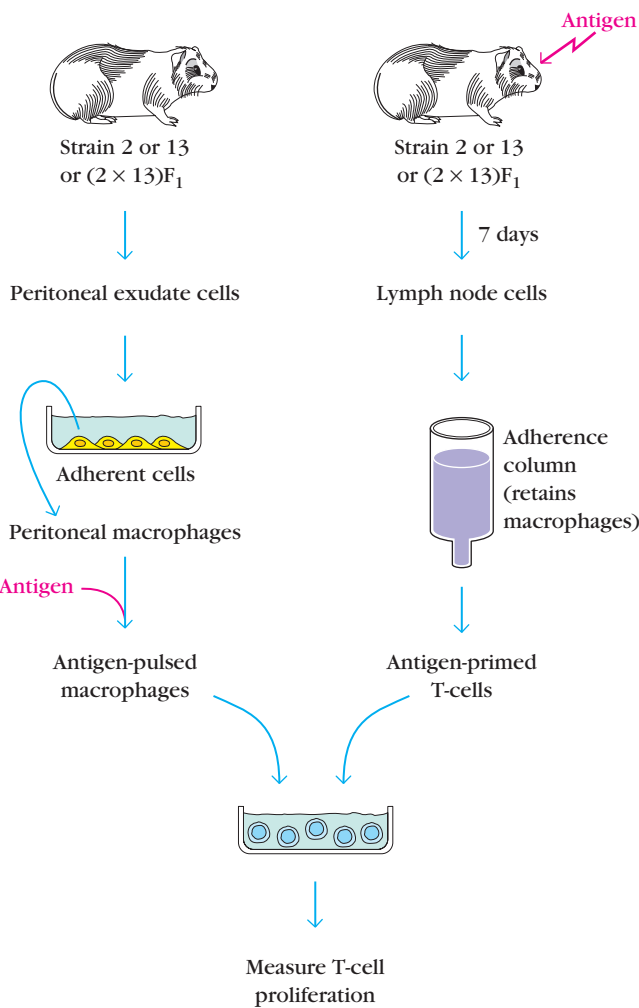


Antigen Processing for Presentation by Class I MHC Molecules

- Self-MHC Restriction of T Cells
- Role of Antigen-Presenting Cells
- Evidence for Two Processing and Presentation Pathways
- Endogenous Antigens: The Cytosolic Pathway
- Exogenous Antigens: The Endocytic Pathway
- Presentation of Nonpeptide Antigens

The results of these experiments, outlined in Figure 8-1, showed that strain-2 antigen-pulsed macrophages activated strain-2 and F_1 T cells but not strain-13 T cells. Similarly, strain-13 antigen-pulsed macrophages activated strain-13 and F_1 T cells but not strain-2 T cells. Subsequently, congenic and recombinant congenic strains of mice, which differed from each other only in selected regions of the H-2 complex, were used as the source of macrophages and T cells. These experiments confirmed that the $CD4^+$ T_H cell is activated and proliferates only in the presence of antigen-pulsed macrophages that share class II MHC alleles. Thus, antigen recognition by the $CD4^+$ T_H cell is *class II MHC restricted*.

In 1974 R. Zinkernagel and P. Doherty demonstrated the self-MHC restriction of $CD8^+$ T cells. In their experiments, mice were immunized with lymphocytic choriomeningitis (LCM) virus; several days later, the animals' spleen cells, which included T_C cells specific for the virus, were isolated and incubated with LCM-infected target cells of the same or different haplotype (Figure 8-2). They found that the T_C cells killed only syngeneic virus-infected target cells. Later studies with congenic and recombinant congenic strains showed



Antigen-primed T cell	Antigen-pulsed macrophages		
	Strain 2	Strain 13	(2 × 13)F ₁
Strain 2	+	–	+
Strain 13	–	+	+
(2 × 13)F ₁	+	+	+

FIGURE 8-1 Experimental demonstration of self-MHC restriction of T_H cells. Peritoneal exudate cells from strain 2, strain 13, or (2 × 13) F₁ guinea pigs were incubated in plastic Petri dishes, allowing enrichment of macrophages, which are adherent cells. The peritoneal macrophages were then incubated with antigen. These “antigen-pulsed” macrophages were incubated in vitro with T cells from strain 2, strain 13, or (2 × 13) F₁ guinea pigs, and the degree of T-cell proliferation was assessed. The results indicated that T_H cells could proliferate only in response to antigen presented by macrophages that shared MHC alleles. [Adapted from A. Rosenthal and E. Shevach, 1974, J. Exp. Med. 138:1194, by copyright permission of the Rockefeller University Press.]

that the T_C cell and the virus-infected target cell must share class I molecules encoded by the K or D regions of the MHC. Thus, antigen recognition by CD8⁺ T_C cells is *class I MHC*

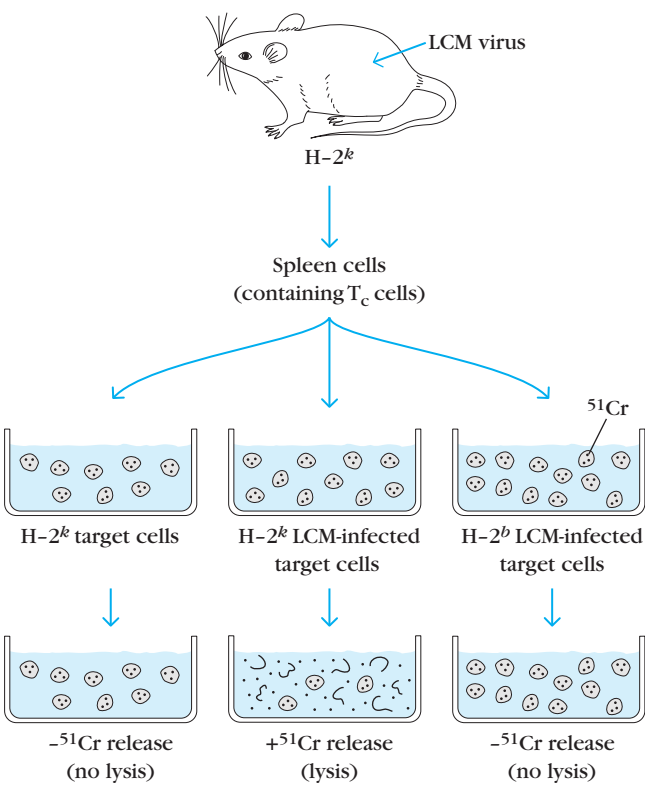


FIGURE 8-2 Classic experiment of Zinkernagel and Doherty demonstrating that antigen recognition by T_C cells exhibits MHC restriction. H-2^k mice were primed with the lymphocytic choriomeningitis (LCM) virus to induce cytotoxic T lymphocytes (CTLs) specific for the virus. Spleen cells from this LCM-primed mouse were then added to target cells of different H-2 haplotypes that were intracellularly labeled with ⁵¹Cr (black dots) and either infected or not with the LCM virus. CTL-mediated killing of the target cells, as measured by the release of ⁵¹Cr into the culture supernatant, occurred only if the target cells were infected with LCM and had the same MHC haplotype as the CTLs. [Adapted from P. C. Doherty and R. M. Zinkernagel, 1975, J. Exp. Med. 141:502.]

restricted. In 1996, Doherty and Zinkernagel were awarded the Nobel prize for their major contribution to the understanding of cell-mediated immunity.

Role of Antigen-Presenting Cells

As early as 1959, immunologists were confronted with data suggesting that T cells and B cells recognized antigen by different mechanisms. The dogma of the time, which persisted until the 1980s, was that cells of the immune system recognize the entire protein in its native conformation. However, experiments by P. G. H. Gell and B. Benacerraf demonstrated that, while a primary antibody response and cell-mediated response were induced by a protein in its native conformation, a secondary antibody response (mediated by B cells) could be induced only by native antigen, whereas a secondary

cell-mediated response could be induced by either the native or the denatured antigen (see Table 3-5). These findings were viewed as an interesting enigma, but implications for antigen presentation were completely overlooked until the early 1980s.

Processing of Antigen Is Required for Recognition by T Cells

The results obtained by K. Ziegler and E. R. Unanue were among those that contradicted the prevailing dogma that antigen recognition by B and T cells was basically similar. These researchers observed that T_H -cell activation by bacterial protein antigens was prevented by treating the antigen-presenting cells with paraformaldehyde prior to antigen exposure. However, if the antigen-presenting cells were first allowed to ingest the antigen and were fixed with paraformaldehyde 1–3 h later, T_H -cell activation still occurred (Figure

8-3a,b). During that interval of 1–3 h, the antigen-presenting cells had processed the antigen and had displayed it on the membrane in a form able to activate T cells.

Subsequent experiments by R. P. Shimonkevitz showed that internalization and processing could be bypassed if antigen-presenting cells were exposed to peptide digests of an antigen instead of the native antigen (Figure 8-3c). In these experiments, antigen-presenting cells were treated with glutaraldehyde (this chemical, like paraformaldehyde, fixes the cell, making the membrane impermeable) and then incubated with native ovalbumin or with ovalbumin that had been subjected to partial enzymatic digestion. The digested ovalbumin was able to interact with the glutaraldehyde-fixed antigen-presenting cells, thereby activating ovalbumin-specific T_H cells, whereas the native ovalbumin failed to do so. These results suggest that antigen processing involves the digestion of the protein into peptides that are recognized by the ovalbumin-specific T_H cells.

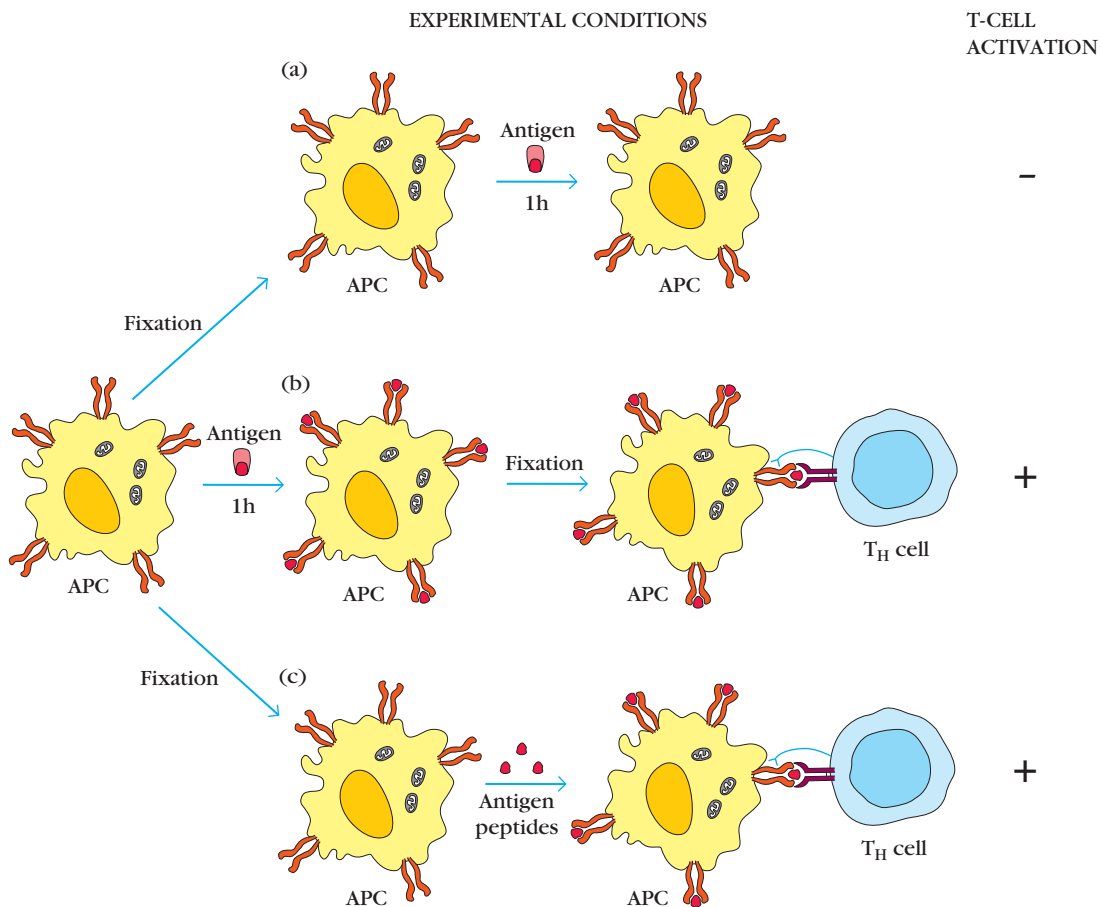


FIGURE 8-3 Experimental demonstration that antigen processing is necessary for T_H -cell activation. (a) When antigen-presenting cells (APCs) are fixed before exposure to antigen, they are unable to activate T_H cells. (b) In contrast, APCs fixed at least 1 h after antigen exposure can activate T_H cells. (c) When APCs are fixed

before antigen exposure and incubated with peptide digests of the antigen (rather than native antigen), they also can activate T_H cells. T_H -cell activation is determined by measuring a specific T_H -cell response (e.g., cytokine secretion).

TABLE 8-1 Antigen-presenting cells

Professional antigen-presenting cells	Nonprofessional antigen-presenting cells	
Dendritic cells (several types)	Fibroblasts (skin)	Thymic epithelial cells
Macrophages	Glial cells (brain)	Thyroid epithelial cells
B cells	Pancreatic beta cells	Vascular endothelial cells

At about the same time, A. Townsend and his colleagues began to identify the proteins of influenza virus that were recognized by T_C cells. Contrary to their expectations, they found that internal proteins of the virus, such as matrix and nucleocapsid proteins, were often recognized by T_C cells better than the more exposed envelope proteins. Moreover, Townsend’s work revealed that T_C cells recognized short linear peptide sequences of the influenza protein. In fact, when noninfected target cells were incubated in vitro with synthetic peptides corresponding to sequences of internal influenza proteins, these cells could be recognized by T_C cells and subsequently lysed just as well as target cells that had been infected with live influenza virus. These findings along with those presented in Figure 8-3 suggest that antigen processing is a metabolic process that digests proteins into peptides, which can then be displayed on the cell membrane together with a class I or class II MHC molecule.

Most Cells Can Present Antigen with Class I MHC; Presentation with Class II MHC Is Restricted to APCs

Since all cells expressing either class I or class II MHC molecules can present peptides to T cells, strictly speaking they all could be designated as antigen-presenting cells. However, by convention, cells that display peptides associated with class I MHC molecules to CD8⁺ T_C cells are referred to as *target cells*; cells that display peptides associated with class II MHC molecules to CD4⁺ T_H cells are called **antigen-presenting cells (APCs)**. This convention is followed throughout this text.

A variety of cells can function as antigen-presenting cells. Their distinguishing feature is their ability to express class II MHC molecules and to deliver a co-stimulatory signal. Three cell types are classified as *professional* antigen-presenting cells: dendritic cells, macrophages, and B lymphocytes. These cells differ from each other in their mechanisms of antigen uptake, in whether they constitutively express class II MHC molecules, and in their co-stimulatory activity:

- Dendritic cells are the most effective of the antigen-presenting cells. Because these cells constitutively express a high level of class II MHC molecules and co-stimulatory activity, they can activate naive T_H cells.

- Macrophages must be activated by phagocytosis of particulate antigens before they express class II MHC molecules or the co-stimulatory B7 membrane molecule.
- B cells constitutively express class II MHC molecules but must be activated before they express the co-stimulatory B7 molecule.

Several other cell types, classified as *nonprofessional* antigen-presenting cells, can be induced to express class II MHC molecules or a co-stimulatory signal (Table 8-1). Many of these cells function in antigen presentation only for short periods of time during a sustained inflammatory response.

Because nearly all nucleated cells express class I MHC molecules, virtually any nucleated cell is able to function as a target cell presenting endogenous antigens to T_C cells. Most often, target cells are cells that have been infected by a virus or some other intracellular microorganism. However, altered self-cells such as cancer cells, aging body cells, or allogeneic cells from a graft can also serve as targets.

Evidence for Two Processing and Presentation Pathways

The immune system uses two different pathways to eliminate intracellular and extracellular antigens. Endogenous antigens (those generated within the cell) are processed in the *cytosolic pathway* and presented on the membrane with class I MHC molecules; exogenous antigens (those taken up by endocytosis) are processed in the *endocytic pathway* and presented on the membrane with class II MHC molecules (Figure 8-4).

Experiments carried out by L. A. Morrison and T. J. Braciale provided early evidence that the antigenic peptides presented by class I and class II MHC molecules are derived from different processing pathways. These researchers based their experimental protocol on the properties of two clones of T_C cells, one that recognized influenza hemagglutinin (HA) associated with a class I MHC molecule, and an atypical T_C line that recognized the same antigen associated with a class II MHC molecule. (In this case, and in some

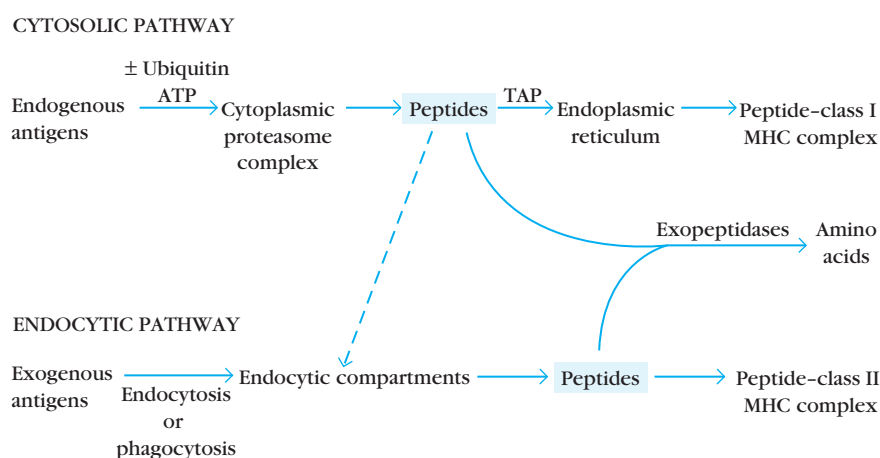


FIGURE 8-4 Overview of cytosolic and endocytic pathways for processing antigen. The proteasome complex contains enzymes that cleave peptide bonds, converting proteins into peptides. The antigenic peptides from proteasome cleavage and those from endocytic compartments associate with class I or class II MHC molecules, and the peptide-MHC complexes are then transported

to the cell membrane. TAP (transporter of antigenic peptides) transports the peptides to the endoplasmic reticulum. It should be noted that the ultimate fate of most peptides in the cell is neither of these pathways, but rather to be degraded completely into amino acids.

others as well, the association of T-cell function with MHC restriction is not absolute). In one set of experiments, target cells that expressed both class I and class II MHC molecules were incubated with infectious influenza virus or with UV-inactivated influenza virus. (The inactivated virus retained its antigenic properties but was no longer capable of replicating within the target cells.) The target cells were then incubated with the class I-restricted or the atypical class II-restricted T_C cells and subsequent lysis of the target cells was determined. The results of their experiments, presented in Table 8-2, show that the class II-restricted T_C cells responded to target cells treated with either infectious or noninfectious influenza virions. The class I-restricted T_C cells responded

only to target cells treated with infectious virions. Similarly, target cells that had been treated with infectious influenza virions in the presence of emetine, which inhibits viral protein synthesis, stimulated the class II-restricted T_C cells but not the class I-restricted T_C cells. Conversely, target cells that had been treated with infectious virions in the presence of chloroquine, a drug that blocks the endocytic processing pathway, stimulated class I- but not class II-restricted T_C cells.

These results support the distinction between the processing of exogenous and endogenous antigens, including the preferential association of exogenous antigens with class II MHC molecules and of endogenous antigens with class I

TABLE 8-2 Effect of antigen presentation on activation of class I and class II MHC-restricted T_C cells

Treatment of target cells*	CTL ACTIVITY†	
	Class I restricted	Class II restricted
Infectious virus	+	+
UV-inactivated virus (noninfectious)	–	+
Infectious virus + emetine	–	+
Infectious virus + chloroquine	+	–

*Target cells, which expressed both class I and class II MHC molecules, were treated with the indicated preparations of influenza virus and other agents. Emetine inhibits viral protein synthesis, and chloroquine inhibits the endocytic processing pathway.

†Determined by lysis (+) and no lysis (–) of the target cells.

SOURCE: Adapted from T. J. Braciale et al., 1987, *Immunol. Rev.* 98:95.

MHC molecules. Association of viral antigen with class I MHC molecules required replication of the influenza virus and viral protein synthesis within the target cells; association with class II did not. These findings suggested that the peptides presented by class I and class II MHC molecules are trafficked through separate intracellular compartments; class I MHC molecules interact with peptides derived from cytosolic degradation of endogenously synthesized proteins, class II molecules with peptides derived from endocytic degradation of exogenous antigens. The next two sections examine these two pathways in detail.

Endogenous Antigens: The Cytosolic Pathway

In eukaryotic cells, protein levels are carefully regulated. Every protein is subject to continuous turnover and is degraded at a rate that is generally expressed in terms of its half-life. Some proteins (e.g., transcription factors, cyclins, and key metabolic enzymes) have very short half-lives; denatured, misfolded, or otherwise abnormal proteins also are degraded rapidly. The pathway by which endogenous antigens are degraded for presentation with class I MHC molecules utilizes the same pathways involved in the normal turnover of intracellular proteins.

Peptides for Presentation Are Generated by Protease Complexes Called Proteasomes

Intracellular proteins are degraded into short peptides by a cytosolic proteolytic system present in all cells. Those proteins targeted for proteolysis often have a small protein, called *ubiquitin*, attached to them (Figure 8-5a). Ubiquitin-protein conjugates can be degraded by a multifunctional protease complex called a **proteasome**. Each proteasome is a large (26S), cylindrical particle consisting of four rings of protein subunits with a central channel of diameter 10–50 Å. A proteasome can cleave peptide bonds between 2 or 3 different amino acid combinations in an ATP-dependent process (Figure 8-5b). Degradation of ubiquitin-protein complexes is thought to occur within the central hollow of the proteasome.

Experimental evidence indicates that the immune system utilizes this general pathway of protein degradation to produce small peptides for presentation with class I MHC molecules. The proteasomes involved in antigen processing include two subunits encoded within the MHC gene cluster, LMP2 and LMP7, and a third non-MHC protein, LMP10 (also called MECL-1). All three are induced by increased levels of the T-cell cytokine IFN- γ . The peptidase activities of proteasomes containing LMP2, LMP7, and LMP10 preferentially generate peptides that bind to MHC class I molecules. Such proteasomes, for example, show increased hydrolysis of peptide bonds that follow basic and/or hydrophobic

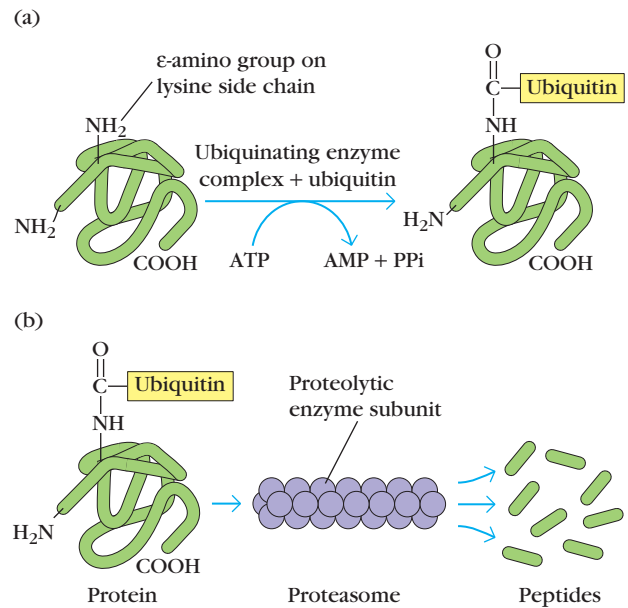


FIGURE 8-5 Cytosolic proteolytic system for degradation of intracellular proteins. (a) Proteins to be degraded are often covalently linked to a small protein called ubiquitin. In this reaction, which requires ATP, a ubiquitinating enzyme complex links several ubiquitin molecules to a lysine-amino group near the amino terminus of the protein. (b) Degradation of ubiquitin-protein complexes occurs within the central channel of proteasomes, generating a variety of peptides. Proteasomes are large cylindrical particles whose subunits catalyze cleavage of peptide bonds.

residues. As described in Chapter 7, peptides that bind to class I MHC molecules terminate almost exclusively with hydrophobic or basic residues.

Peptides Are Transported from the Cytosol to the Rough Endoplasmic Reticulum

Insight into the role that peptide transport, the delivery of peptides to the MHC molecule, plays in the cytosolic processing pathway came from studies of cell lines with defects in peptide presentation by class I MHC molecules. One such mutant cell line, called RMA-S, expresses about 5% of the normal levels of class I MHC molecules on its membrane. Although RMA-S cells synthesize normal levels of class I α chains and β_2 -microglobulin, neither molecule appears on the membrane. A clue to the mutation in the RMA-S cell line was the discovery by A. Townsend and his colleagues that “feeding” these cells peptides restored their level of membrane-associated class I MHC molecules to normal. These investigators suggested that peptides might be required to stabilize the interaction between the class I α chain and β_2 -microglobulin. The ability to restore expression of class I MHC molecules on the membrane by feeding the cells predigested peptides suggested that the RMA-S cell line might have a defect in peptide transport.

Subsequent experiments showed that the defect in the RMA-S cell line occurs in the protein that transports peptides from the cytoplasm to the RER, where class I molecules are synthesized. When RMA-S cells were transfected with a functional gene encoding the transporter protein, the cells began to express class I molecules on the membrane. The transporter protein, designated **TAP** (for **transporter associated with antigen processing**) is a membrane-spanning heterodimer consisting of two proteins: TAP1 and TAP2 (Figure 8-6a). In addition to their multiple transmembrane segments, the TAP1 and TAP2 proteins each have a domain projecting into the lumen of the RER, and an ATP-binding domain that projects into the cytosol. Both TAP1 and TAP2 belong to the family of ATP-binding cassette proteins found in the membranes of many cells, including bacteria; these proteins mediate ATP-dependent transport of amino acids, sugars, ions, and peptides.

Peptides generated in the cytosol by the proteasome are translocated by TAP into the RER by a process that requires the hydrolysis of ATP (Figure 8-6b). TAP has the highest affinity for peptides containing 8–10 amino acids, which is the optimal peptide length for class I MHC binding. In addition, TAP appears to favor peptides with hydrophobic or basic carboxyl-terminal amino acids, the preferred anchor residues for class I MHC molecules. Thus, TAP is optimized to transport peptides that will interact with class I MHC molecules.

The *TAP1* and *TAP2* genes map within the class II MHC region, adjacent to the *LMP2* and *LMP7* genes (see Figure 7-15). Both the transporter genes and these *LMP* genes are polymorphic; that is, different allelic forms of these genes exist within the population. Allelic differences in LMP-mediated proteolytic cleavage of protein antigens or in the transport of different peptides from the cytosol into the RER may contribute to the observed variation among individuals in their response to different endogenous antigens. TAP deficiencies can lead to a disease syndrome that has aspects of both immunodeficiency and autoimmunity (see Clinical Focus).

Peptides Assemble with Class I MHC Aided by Chaperone Molecules

Like other proteins, the α chain and β_2 -microglobulin components of the class I MHC molecule are synthesized on polysomes along the rough endoplasmic reticulum. Assembly of these components into a stable class I MHC molecular complex that can exit the RER requires the presence of a peptide in the binding groove of the class I molecule. The assembly process involves several steps and includes the participation of *molecular chaperones*, which facilitate the folding of polypeptides. The first molecular chaperone involved in class I MHC assembly is *calnexin*, a resident membrane protein of the endoplasmic reticulum. Calnexin associates with the free class I α chain and promotes its folding. When β_2 -microglobulin binds to the α chain, calnexin is released and the class I molecule associ-

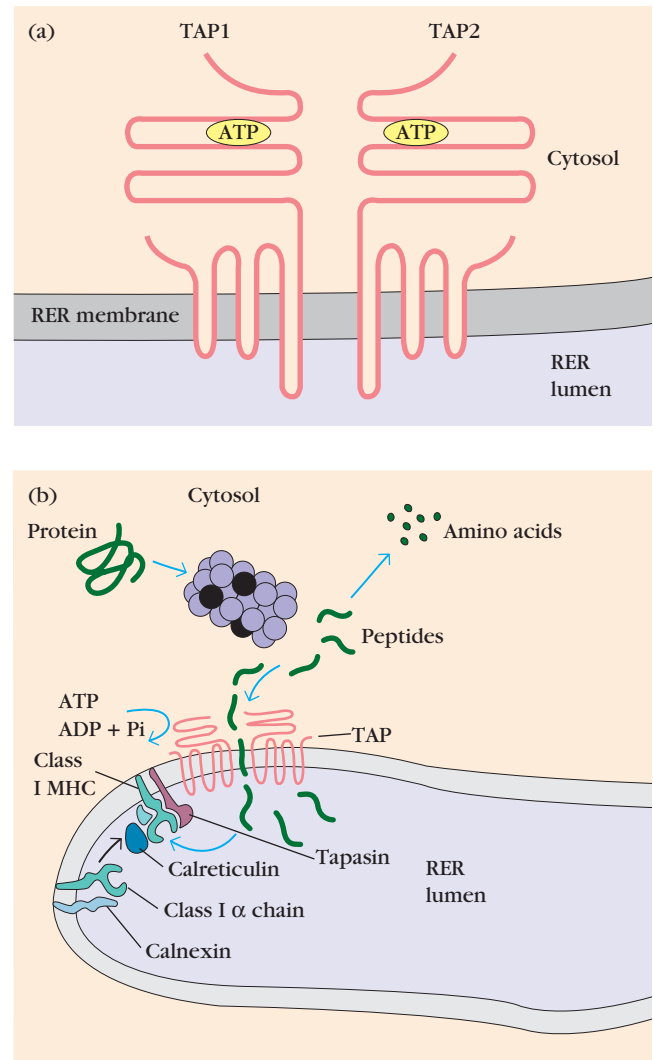


FIGURE 8-6 Generation of antigenic peptide–class I MHC complexes in the cytosolic pathway. (a) Schematic diagram of TAP, a heterodimer anchored in the membrane of the rough endoplasmic reticulum (RER). The two chains are encoded by *TAP1* and *TAP2*. The cytosolic domain in each TAP subunit contains an ATP-binding site, and peptide transport depends on the hydrolysis of ATP. (b) In the cytosol, association of LMP2, LMP7, and LMP10 (black spheres) with a proteasome changes its catalytic specificity to favor production of peptides that bind to class I MHC molecules. Within the RER membrane, a newly synthesized class I α chain associates with calnexin until β_2 -microglobulin binds to the α chain. The class I α chain/ β_2 -microglobulin heterodimer then binds to calreticulin and the TAP-associated protein tapasin. When a peptide delivered by TAP is bound to the class I molecule, folding of MHC class I is complete and it is released from the RER and transported through the Golgi to the surface of the cell.

ates with the chaperone *calreticulin* and with *tapasin*. Tapasin (TAP-associated protein) brings the TAP transporter into proximity with the class I molecule and allows it to acquire an antigenic peptide (Figure 8-7). The physical association of the α chain– β_2 -microglobulin



CLINICAL FOCUS

Deficiency in Transporters Associated with Antigen Presentation (TAP) Leads to a Diverse Disease Spectrum

A relatively rare condition known as bare lymphocyte syndrome, or BLS, has been recognized for more than 22 years. The lymphocytes in BLS patients express MHC molecules at below-normal levels and, in some cases, not at all. In type 1 BLS, a deficiency in MHC class I molecules exists; in type 2 BLS, expression of class II molecules is impaired. The pathogenesis of one type of BLS underscores the importance of the class I family of MHC molecules in their dual roles of preventing autoimmunity as well as defending against pathogens.

Defects in promoter sequences that preclude MHC gene transcription were found for some type 2 BLS cases, but in many instances the nature of the underlying defect is not known. A recent study has identified a group of patients with type 1 BLS due to defects in *TAP1* or *TAP2* genes. Manifestations of the TAP deficiency were consistent in this patient group and define a unique disease. As described earlier in this chapter, TAP proteins are necessary for the loading of peptides onto class I molecules, a step that is essential for expression of class I MHC molecules on the cell surface. Lymphocytes in individuals with TAP deficiency express levels of class I molecules significantly lower than normal controls. Other cellular abnormalities include increased numbers of NK and $\gamma\delta$ T cells, and decreased levels of $CD8^+ \alpha\beta$ T cells. As we shall see, the disease manifestations are reasonably well explained by these deviations in the levels of certain cells involved in immune function.

In early life the TAP-deficient individual suffers frequent bacterial infections

of the upper respiratory tract, and in the second decade begins to have chronic infection of the lungs. It is thought that a post-nasal-drip syndrome common in younger patients promotes the bacterial lung infections in later life. Noteworthy is the absence of any severe viral infection, which is common in immunodeficiencies with T-cell involvement (see Chapter 19). Bronchiectasis (dilation of the bronchial tubes) often occurs and recurring infections can lead to lung damage that may be fatal. The most characteristic mark of the deficiency is the occurrence of necrotizing skin lesions on the extremities and the midface. These lesions ulcerate and may cause disfigurement (see figure). The skin lesions are probably due to activated NK cells and $\gamma\delta$ T cells; NK

cells were isolated from biopsied skin from several patients, supporting this possibility. Normally, the activity of NK cells is limited through the action of killer-cell-inhibitory receptors (KIRs), which deliver a negative signal to the NK cell following interaction with class I molecules (see Chapter 14). The deficiency of class I molecules in TAP-related BLS patients explains the excessive activity of the NK cells. Activation of NK cells further explains the absence of severe virus infections, which are limited by NK and $\gamma\delta$ cells.

The best treatment for the characteristic lung infections appears to be antibiotics and intravenous immunoglobulin. Attempts to limit the skin disease by immunosuppressive regimens, such as steroid treatment or cytotoxic agents, can lead to exacerbation of lesions and is therefore contraindicated. Mutations in the promoter region of *TAP* that preclude expression of the gene were found for several patients, suggesting the possibility of gene therapy, but the cellular distribution of class I is so widespread that it is not clear what cells would need to be corrected to alleviate all symptoms.



Necrotizing granulomatous lesions in the midface of patient with TAP-deficiency syndrome. TAP deficiency leads to a condition with symptoms characteristic of autoimmunity, such as the skin lesions that appear on the extremities and the midface, as well as immunodeficiency that causes chronic sinusitis, leading to recurrent lung infection. [From S. D. Gadola et al., 1999, *Lancet* **354**:1598, and 2000, *Clinical and Experimental Immunology* **121**:173.]

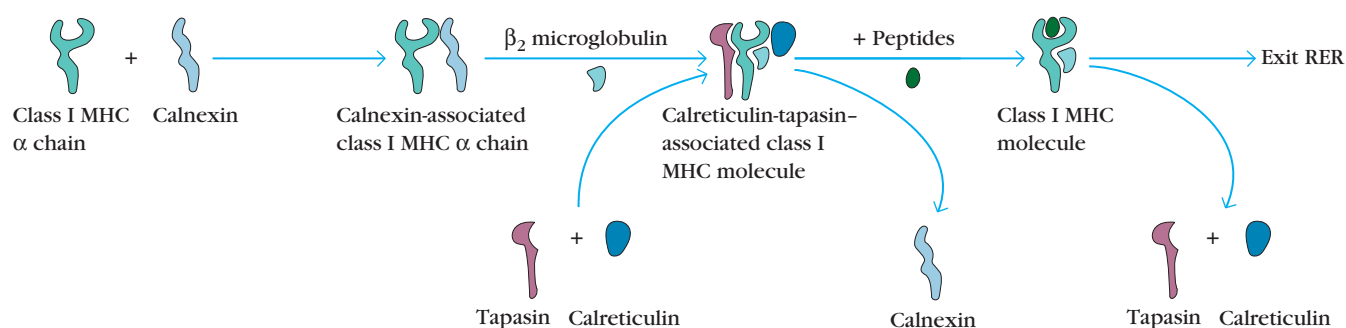


FIGURE 8-7 Assembly and stabilization of class I MHC molecules. Newly formed class I α chains associate with calnexin, a molecular chaperone, in the RER membrane. Subsequent binding to β₂-microglobulin releases calnexin and allows binding to the

chaperonin calreticulin and to tapasin, which is associated with the peptide transporter TAP. This association promotes binding of an antigenic peptide, which stabilizes the class I molecule–peptide complex, allowing its release from the RER.

heterodimer with the TAP protein (see Figure 8-6b) promotes peptide capture by the class I molecule before the peptides are exposed to the luminal environment of the RER. Peptides not bound by class I molecules are rapidly degraded. As a consequence of peptide binding, the class I molecule displays increased stability and can dissociate from calreticulin and tapasin, exit from the RER, and proceed to the cell surface via the Golgi. An additional chaperone protein, ERp57, has been observed in association with calnexin and calreticulin complexes. The precise role of this resident endoplasmic reticulum protein in the class I peptide assembly and loading process has not yet been defined, but it is thought to contribute to the formation of disulfide bonds during the maturation of class I chains. Because its role is not clearly defined, ERp57 is not shown in Figures 8-6 and 8-7.

Exogenous Antigens: The Endocytic Pathway

Figure 8-8 recapitulates the endogenous pathway discussed previously (left side), and compares it with the separate exogenous pathway (right), which we shall now consider. Whether an antigenic peptide associates with class I or with class II molecules is dictated by the mode of entry into the cell, either exogenous or endogenous, and by the site of processing.

Antigen-presenting cells can internalize antigen by phagocytosis, endocytosis, or both. Macrophages internalize antigen by both processes, whereas most other APCs are not phagocytic or are poorly phagocytic and therefore internalize exogenous antigen only by endocytosis (either receptor-mediated endocytosis or pinocytosis). B cells, for example, internalize antigen very effectively by receptor-mediated endocytosis using antigen-specific membrane antibody as the receptor.

Peptides Are Generated from Internalized Molecules in Endocytic Vesicles

Once an antigen is internalized, it is degraded into peptides within compartments of the endocytic processing pathway. As

the experiment shown in Figure 8-3 demonstrated, internalized antigen takes 1–3 h to transverse the endocytic pathway and appear at the cell surface in the form of peptide–class II MHC complexes. The endocytic pathway appears to involve three increasingly acidic compartments: early endosomes (pH 6.0–6.5); late endosomes, or endolysosomes (pH 5.0–6.0); and lysosomes (pH 4.5–5.0). Internalized antigen moves from early to late endosomes and finally to lysosomes, encountering hydrolytic enzymes and a lower pH in each compartment (Figure 8-9). Lysosomes, for example, contain a unique collection of more than 40 acid-dependent hydrolases, including proteases, nucleases, glycosidases, lipases, phospholipases, and phosphatases. Within the compartments of the endocytic pathway, antigen is degraded into oligopeptides of about 13–18 residues, which bind to class II MHC molecules. Because the hydrolytic enzymes are optimally active under acidic conditions (low pH), antigen processing can be inhibited by chemical agents that increase the pH of the compartments (e.g., chloroquine) as well as by protease inhibitors (e.g., leupeptin).

The mechanism by which internalized antigen moves from one endocytic compartment to the next has not been conclusively demonstrated. It has been suggested that early endosomes from the periphery move inward to become late endosomes and finally lysosomes. Alternatively, small transport vesicles may carry antigens from one compartment to the next. Eventually the endocytic compartments, or portions of them, return to the cell periphery, where they fuse with the plasma membrane. In this way, the surface receptors are recycled.

The Invariant Chain Guides Transport of Class II MHC Molecules to Endocytic Vesicles

Since antigen-presenting cells express both class I and class II MHC molecules, some mechanism must exist to prevent class II MHC molecules from binding to the same set of antigenic peptides as the class I molecules. When class II MHC molecule are synthesized within the RER, three pairs of class II αβ chains associate with a preassembled trimer of a



VISUALIZING CONCEPTS

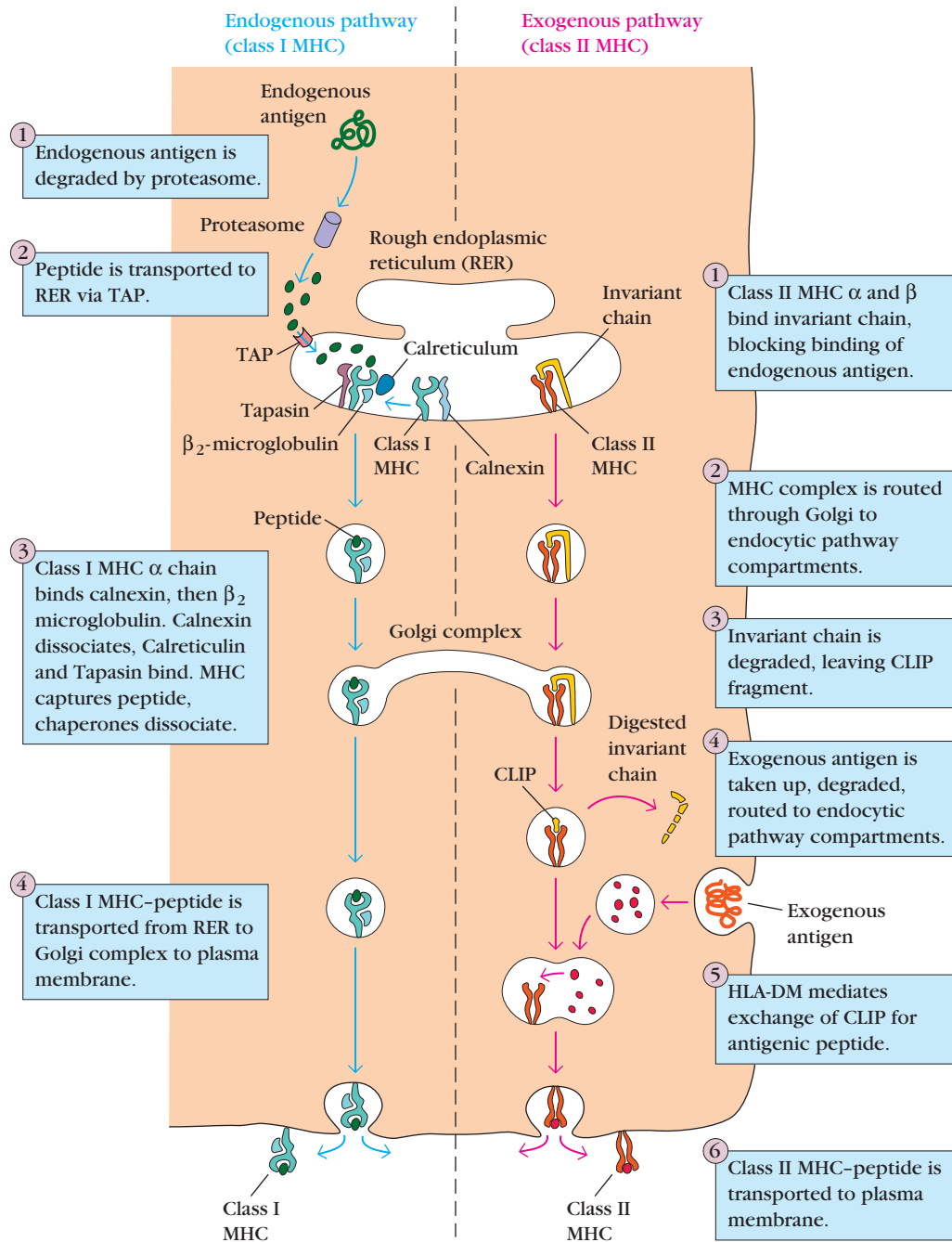


FIGURE 8-8 Separate antigen-presenting pathways are utilized for endogenous (green) and exogenous (red) antigens. The mode of antigen entry into cells and the site of antigen processing de-

termine whether antigenic peptides associate with class I MHC molecules in the rough endoplasmic reticulum or with class II molecules in endocytic compartments.

protein called **invariant chain (Ii, CD74)**. This trimeric protein interacts with the peptide-binding cleft of the class II molecules, preventing any endogenously derived peptides

from binding to the cleft while the class II molecule is within the RER (see right side of Figure 8-8). The invariant chain also appears to be involved in the folding of the class II α and

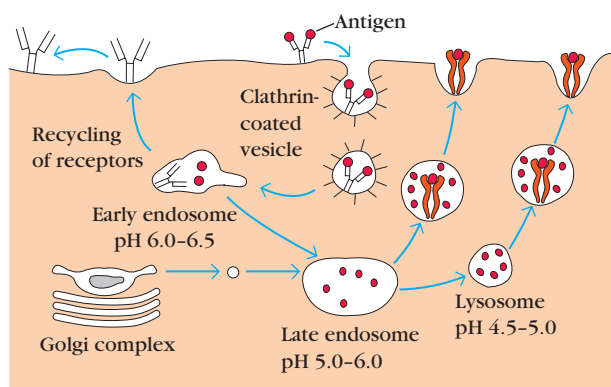


FIGURE 8-9 Generation of antigenic peptides in the endocytic processing pathway. Internalized exogenous antigen moves through several acidic compartments, in which it is degraded into peptides that ultimately associate with class II MHC molecules transported in vesicles from the Golgi complex. The cell shown here is a B cell, which internalizes antigen by receptor-mediated endocytosis, with the membrane-bound antibody functioning as an antigen-specific receptor.

β chains, their exit from the RER, and the subsequent routing of class II molecules to the endocytic processing pathway from the trans-Golgi network.

The role of the invariant chain in the routing of class II molecules has been demonstrated in transfection experiments with cells that lack the genes encoding class II MHC molecules and the invariant chain. Immunofluorescent labeling of such cells transfected only with class II MHC genes revealed class II molecules localized within the Golgi complex. However, in cells transfected with both the class II MHC genes and invariant-chain gene, the class II molecules were localized in the cytoplasmic vesicular structures of the endocytic pathway. The invariant chain contains sorting signals in its cytoplasmic tail that directs the transport of the class II MHC complex from the trans-Golgi network to the endocytic compartments.

Peptides Assemble with Class II MHC Molecules by Displacing CLIP

Recent experiments indicate that most class II MHC–invariant chain complexes are transported from the RER, where they are formed, through the Golgi complex and trans-Golgi network, and then through the endocytic pathway, moving from early endosomes to late endosomes, and finally to lysosomes. As the proteolytic activity increases in each successive compartment, the invariant chain is gradually degraded. However, a short fragment of the invariant chain termed *CLIP* (for *class II–associated invariant chain peptide*) remains bound to the class II molecule after the invariant chain has been cleaved within the endosomal compartment. CLIP physically occupies the peptide-binding groove of the class II MHC molecule, presumably preventing any premature binding of antigenic peptide (see Figure 8-8).

A nonclassical class II MHC molecule called *HLA-DM* is required to catalyze the exchange of CLIP with antigenic peptides (Figure 8-10a). MHC class II genes encoding *HLA-DM* have been identified in the mouse and rabbit, indicating

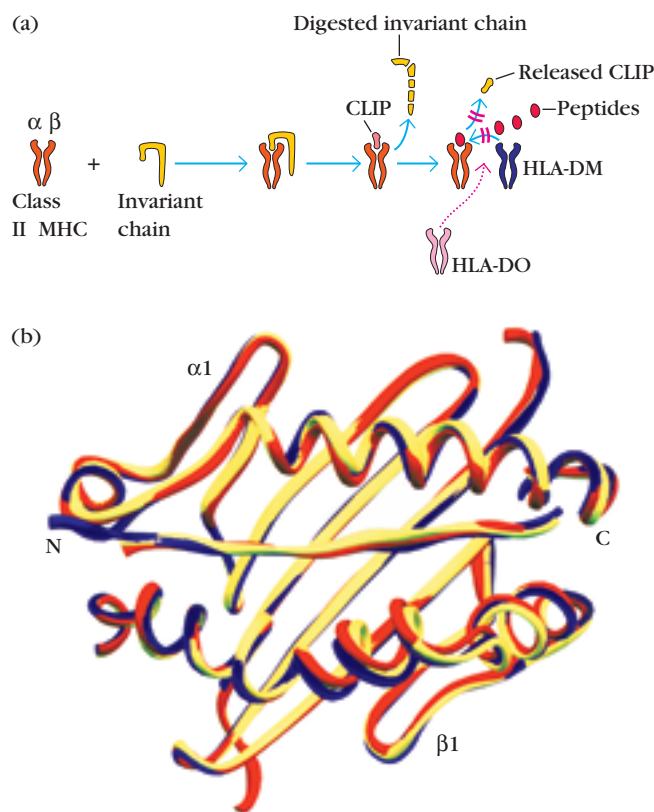


FIGURE 8-10 (a) Assembly of class II MHC molecules. Within the rough endoplasmic reticulum, a newly synthesized class II MHC molecule binds an invariant chain. The bound invariant chain prevents premature binding of peptides to the class II molecule and helps to direct the complex to endocytic compartments containing peptides derived from exogenous antigens. Digestion of the invariant chain leaves CLIP, a small fragment remaining in the binding groove of the class II MHC molecule. *HLA-DM*, a nonclassical class II molecule expressed within endosomal compartments, mediates exchange of antigenic peptides for CLIP. The nonclassical class II molecule *HLA-DO* may act as a negative regulator of class II antigen processing by binding to *HLA-DM* and inhibiting its role in the dissociation of CLIP from class II molecules. (b) Comparison of three-dimensional structures showing the binding groove of HLA class II molecules ($\alpha 1$ and $\beta 1$) containing different antigenic peptides or CLIP. The red lines show DR4 complexed with collagen II peptide, yellow lines are DR1 with influenza hemagglutinin peptide, and blue lines are DR3 associated with CLIP. (N indicates the amino terminus and C the carboxyl terminus of the peptides.) No major differences in the structures of the class II molecules or in the conformation of the bound peptides are seen. This comparison shows that CLIP binds the class II molecule in a manner identical to that of antigenic peptides. [Part (b) from Dessen et al., 1997, *Immunity* 7:473–481; courtesy of Don Wiley, Harvard University.]

that HLA-DM is widely conserved among mammalian species. Like other class II MHC molecules, HLA-DM is a heterodimer of α and β chains. However, unlike other class II molecules, HLA-DM is not polymorphic and is not expressed at the cell membrane but is found predominantly within the endosomal compartment. The *DM α* and *DM β* genes are located near the *TAP* and *LMP* genes in the MHC complex of humans and DM is expressed in cells that express classical class II molecules.

The reaction between HLA-DM and the class II CLIP complex facilitating exchange of CLIP for another peptide is impaired in the presence of HLA-DO, which binds to HLA-DM and lessens the efficiency of the exchange reaction. HLA-DO, like HLA-DM, is a nonclassical and nonpolymorphic class II molecule that is also found in the MHC of other species. HLA-DO differs from HLA-DM in that it is expressed only by B cells and the thymus, and unlike other class II molecules, its expression is not induced by IFN- γ . An additional difference is that the genes encoding the α and the β chains of HLA-DO are not adjacent in the MHC as are all other class II α and β pairs (see Fig 7-15).

An HLA-DR3 molecule associated with CLIP was isolated from a cell line that did not express HLA-DM and was therefore defective in antigen processing. Superimposing the structure of HLA-DR3–CLIP on another DR molecule bound to antigenic peptide reveals that CLIP binds to class II in the same stable manner as the antigenic peptide (Figure 8-10b). The discovery of this stable complex in a cell with defective HLA-DM supports the argument that HLA-DM is required for the replacement of CLIP.

Although it certainly modulates the activity of HLA-DM, the precise role of HLA-DO remains obscure. One possibility is that it acts in the selection of peptides bound to class II MHC molecules in B cells. DO occurs in complex with DM in these cells and this association continues in the endosomal compartments. Conditions of higher acidity weaken the association of DM/DO and increase the possibility of antigenic peptide binding despite the presence of DO. Such a pH-dependent interaction could lead to preferential selection of class II-bound peptides from lysosomal compartments in B cells as compared with other APCs.

As with class I MHC molecules, peptide binding is required to maintain the structure and stability of class II MHC molecules. Once a peptide has bound, the peptide–class II complex is transported to the plasma membrane, where the neutral pH appears to enable the complex to assume a compact, stable form. Peptide is bound so strongly in this compact form that it is difficult to replace a class II–bound peptide on the membrane with another peptide at physiologic conditions.

Presentation of Nonpeptide Antigens

To this point the discussion has been limited to peptide antigens and their presentation by classical class I and II MHC molecules. It is well known that nonprotein antigens also are

recognized by the immune system, and there are reports dating back to the 1980s of T cell proliferation in the presence of nonprotein antigens derived from infectious agents. More recent reports indicate that T cells that express the $\gamma\delta$ TCR (T-cell receptors are dimers of either $\alpha\beta$ or $\gamma\delta$ chains) that react with glycolipid antigens derived from bacteria such as *Mycobacterium tuberculosis*. These nonprotein antigens are presented by members of the CD1 family of nonclassical class I molecules.

The CD1 family of molecules associates with β_2 -microglobulin and has general structural similarity to class I MHC molecules. There are five genes encoding human CD1 molecules (*CD1A-E*, encoding the gene products CD1a-d, with no product yet identified for *E*). These genes are located not within the MHC but on chromosome 1 (Figure 8-11a). The genes are classified into two groups based on sequence homology. Group 1 includes *CD1A*, *B*, *C*, and *E*; *CD1D* is in group 2. All mammalian species studied have CD1 genes, although the number varies. Rodents have only group 2 *CD1* genes, the counterpart of human *CD1D*, whereas rabbits, like humans, have five genes, including both group 1 and 2 types. Sequence identity of CD1 with classical class I molecules is considerably lower than the identity of the class I molecules with each other. Comparison of the three-dimensional structure of the mouse CD1d1 with the class I MHC molecule H-2k^b shows that the antigen-binding groove of the CD1d1 molecules is deeper and more voluminous than that of the classical class I molecule (Fig 8-11b).

Expression of CD1 molecules varies according to subset; *CD1D1* genes are expressed mainly in nonprofessional APCs and on certain B-cell subsets. The mouse CD1d1 is more widely distributed and found on T cells, B cells, dendritic cells, hepatocytes, and some epithelial cells. The *CD1A*, *B*, and *C* genes are expressed on immature thymocytes and professional APCs, mainly those of the dendritic type. *CD1C* gene expression is seen on B cells, whereas the *CD1A* and *B* products are not. *CD1* genes can be induced by exposure to certain cytokines such as GM-CSF or IL-3. The intracellular trafficking patterns of the CD1 molecules differ; for example, CD1a is found mostly in early endosomes or on the cell surface; CD1b and CD1d localize to late endosomes; and CD1c is found throughout the endocytic system.

Certain CD1 molecules are recognized by T cells in the absence of foreign antigens, and self restriction can be demonstrated in these reactions. Examination of antigens presented by CD1 molecules revealed them to be lipid components (mycolic acid) of the *M. tuberculosis* cell wall. Further studies of CD1 presentation indicated that a glycolipid (lipoarabinomannan) from *Mycobacterium leprae* could also be presented by these molecules. The data concerning CD1 antigen presentation point out the existence of a third pathway for the processing of antigens, a pathway with distinct intracellular steps that do not involve the molecules found to facilitate class I antigen processing. For example, CD1 molecules are able to process antigen in TAP-deficient cells. Recent data indicate that the CD1a and 1b molecules traffic differently,

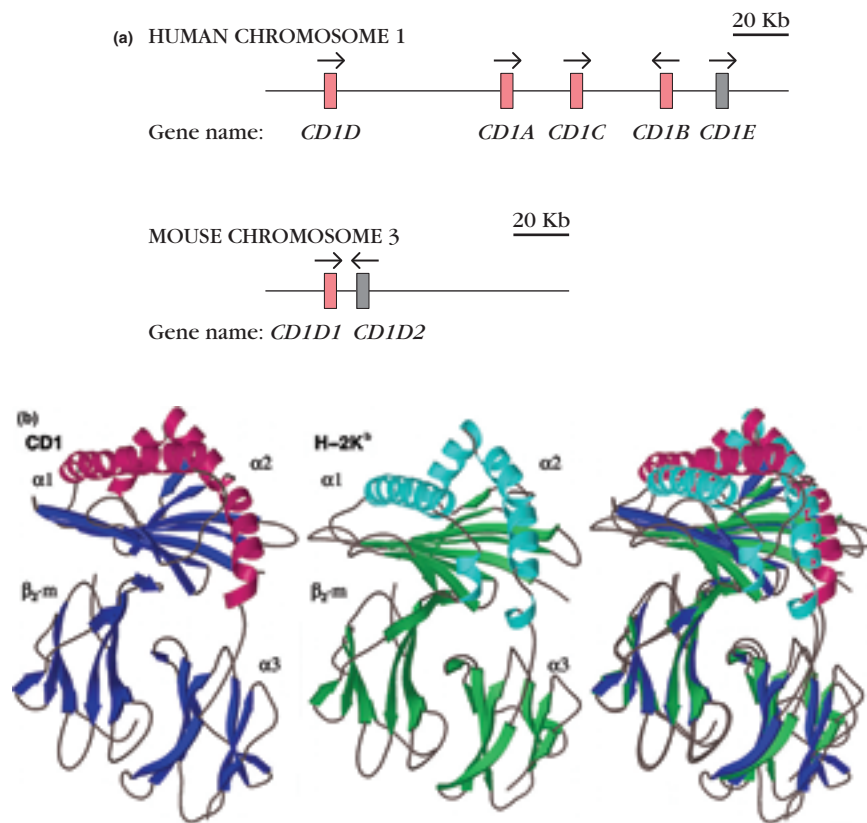


FIGURE 8-11 The CD1 family of genes and structure of a CD1d molecule. (a) The genes encoding the CD1 family of molecules in human (top) and mouse (bottom). The genes are separated into two groups based on sequence identity; *CD1A*, *B*, *C*, and *E* are group 1, *CD1D* genes are group 2. The products of the pink genes have been identified; products of grey genes have not yet been

detected. (b) Comparison of the crystal structures of mouse non-classical CD1 and classical class I molecule H-2K^b. Note the differences in the antigen binding grooves. [Part (b) reprinted from *Trends in Immunology* (formerly *Immunology Today*), Vol. 19, S. A. Porcelli and R. L. Modlin, *The CD1 family of lipid antigen presenting molecules*, pp. 362–368, 1998, with permission from Elsevier Science.]

with CD1a at the surface or in the recycling endocytic compartments and CD1b and CD1d in the lysosomal compartments. Exactly how the CD1 pathway complements or intersects the better understood class I and class II pathways remains an open question. The T-cell types reactive to CD1 were first thought to be limited to T cells expressing the $\gamma\delta$ TCR and lacking both CD4 and CD8, or T cells with a single TCR α chain, but recent reports indicate that a wider range of T-cell types will recognize CD1-presenting cells. Recent evidence indicates that natural killer T cells recognize CD1d molecules presenting autologous antigen. This may represent a mechanism for eliminating cells that are altered by stress, senescence, or neoplasia.

SUMMARY

- T-cells recognize antigen displayed within the cleft of a self-MHC molecule on the membrane of a cell.
- In general, CD4⁺ T_H cells recognize antigen with class II MHC molecules on antigen-processing cells.

- CD8⁺ T_C cells recognize antigen with class I MHC molecules on target cells.
- Complexes between antigenic peptides and MHC molecules are formed by degradation of a protein antigen in one of two different antigen-processing pathways.
- Endogenous antigens are degraded into peptides within the cytosol by proteasomes and assemble with class I molecules in the RER.
- Exogenous antigens are internalized and degraded within the acidic endocytic compartments and subsequently pair with class II molecules.
- Peptide binding to class II molecules involves replacing a fragment of invariant chain in the binding cleft by a process catalyzed by nonclassic MHC molecule HLA-DM.
- Presentation of nonpeptide (lipid and glycolipid) antigens derived from bacteria involves the class I-like CD1 molecules.

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3. L. A. Morrison and T. J. Braciale conducted an experiment to determine whether antigens presented by class I or II MHC molecules are processed in different pathways. Their results are summarized in Table 8-2.
- Explain why the class I–restricted T_C cells did not respond to target cells infected with UV-inactivated influenza virus.
 - Explain why chloroquine inhibited the response of the class II–restricted T_C cells to live virus.
 - Explain why emetine inhibited the response of class I–restricted but not class II–restricted T_C cells to live virus.
4. For each of the following cell components or processes, indicate whether it is involved in the processing and presentation of exogenous antigens (EX), endogenous antigens (EN), or both (B). Briefly explain the function of each item.
- _____ Class I MHC molecules
 - _____ Class II MHC molecules
 - _____ Invariant (Ii) chains
 - _____ Lysosomal hydrolases
 - _____ TAP1 and TAP2 proteins
 - _____ Transport of vesicles from the RER to the Golgi complex
 - _____ Proteasomes
 - _____ Phagocytosis or endocytosis
 - _____ Calnexin
 - _____ CLIP
 - _____ Tapasin
5. Antigen-presenting cells have been shown to present lysozyme peptide 46–61 together with the class II I-A^k molecule. When CD4⁺T_H cells are incubated with APCs and native lysozyme or the synthetic lysozyme peptide 46–61, T_H-cell activation occurs.
- If chloroquine is added to the incubation mixture, presentation of the native protein is inhibited, but the peptide continues to induce T_H-cell activation. Explain why this occurs.
 - If chloroquine addition is delayed for 3 h, presentation of the native protein is not inhibited. Explain why this occurs.
6. Cells that can present antigen to T_H cells have been classified into two groups—professional and nonprofessional APCs.
- Name the three types of professional APCs. For each type indicate whether it expresses class II MHC molecules and a co-stimulatory signal constitutively or must be activated before doing so.
 - Give three examples of nonprofessional APCs. When are these cells most likely to function in antigen presentation?
7. Predict whether T_H-cell proliferation or CTL-mediated cytotoxicity of target cells will occur with the following mixtures of cells. The CD4⁺ T_H cells are from lysozyme-primed mice, and the CD8⁺ CTLs are from influenza-infected mice. Use R to indicate a response and NR to indicate no response.
- _____ H-2^k T_H cells + lysozyme-pulsed H-2^k macrophages
 - _____ H-2^k T_H cells + lysozyme-pulsed H-2^{b/k} macrophages

Study Questions

CLINICAL FOCUS QUESTION Patients with TAP deficiency have partial immunodeficiency as well as autoimmune manifestations. How do the profiles for patients' immune cells explain the partial immunodeficiency? Why is it difficult to design a gene therapy treatment for this disease, despite the fact that a single gene defect is implicated?

- Explain the difference between the terms *antigen-presenting cell* and *target cell*, as they are commonly used in immunology.
- Define the following terms:
 - Self-MHC restriction
 - Antigen processing
 - Endogenous antigen
 - Exogenous antigen

- _____ H-2^k T_H cells + lysozyme-pulsed H-2^k macrophages
- _____ H-2^k T_H cells + lysozyme-pulsed H-2^{b/k} macrophages

- c. _____H-2^k T_H cells + lysozyme-primed H-2^d macrophages
 - d. _____H-2^k CTLs + influenza-infected H-2^k macrophages
 - e. _____H-2^k CTLs + influenza-infected H-2^d macrophages
 - f. _____H-2^d CTLs + influenza-infected H-2^{d/k} macrophages
8. HLA-DM and HLA-DO are termed nonclassical MHC class II molecules. How do they differ from the classical MHC class II? How do they differ from each other?
9. Molecules of the CD1 family were recently shown to present nonpeptide antigens.
- a. What is a major source of nonpeptide antigens?
 - b. Why are CD1 molecules not classified as members of the MHC family even though they associate with β_2 -microglobulin?
 - c. What evidence suggests that the CD1 pathway is different from that utilized by classical class I MHC molecules?

T-Cell Receptor

THE ANTIGEN-SPECIFIC NATURE OF T-CELL RESPONSES clearly implies that T cells possess an antigen-specific and clonally restricted receptor. However, the identity of this receptor remained unknown long after the B-cell receptor (immunoglobulin molecule) had been identified. Relevant experimental results were contradictory and difficult to conceptualize within a single model because the T-cell receptor (TCR) differs from the B-cell antigen-binding receptor in important ways. First, the T-cell receptor is membrane bound and does not appear in a soluble form as the B-cell receptor does; therefore, assessment of its structure by classic biochemical methods was complicated, and complex cellular assays were necessary to determine its specificity. Second, most T-cell receptors are specific not for antigen alone but for antigen combined with a molecule encoded by the major histocompatibility complex (MHC). This property precludes purification of the T-cell receptor by simple antigen-binding techniques and adds complexity to any experimental system designed to investigate the receptor.

A combination of immunologic, biochemical, and molecular-biological manipulations has overcome these problems. The molecule responsible for T-cell specificity was found to be a heterodimer composed of either α and β or γ and δ chains. Cells that express TCRs have approximately 10^5 TCR molecules on their surface. The genomic organization of the T-cell receptor gene families and the means by which the diversity of the component chains is generated were found to resemble those of the B-cell receptor chains. Further, the T-cell receptor is associated on the membrane with a signal-transducing complex, CD3, whose function is similar to that of the Ig- α /Ig- β complex of the B-cell receptor.

Important new insights concerning T-cell receptors have been gained by recent structure determinations using x-ray crystallography, including new awareness of differences in how TCRs bind to class I or class II MHC molecules. This chapter will explore the nature of the T-cell receptor molecules that specifically recognize MHC-antigen complexes, as well as some that recognize native antigens.

Early Studies of the T-Cell Receptor

By the early 1980s, investigators had learned much about T-cell function but were thwarted in their attempts to

ART TO COME

Interaction of $\alpha\beta$ TCR with Class II MHC–Peptide

- Early Studies of the T-Cell Receptor
- $\alpha\beta$ and $\gamma\delta$ T-Cell Receptors: Structure and Roles
- Organization and Rearrangement of TCR Genes
- T-Cell Receptor Complex: TCR-CD3
- T-Cell Accessory Membrane Molecules
- Three-Dimensional Structures of TCR–Peptide–MHC Complexes
- Alloreactivity of T Cells

identify and isolate its antigen-binding receptor. The obvious parallels between the recognition functions of T cells and B cells stimulated a great deal of experimental effort to take advantage of the anticipated structural similarities between immunoglobulins and T-cell receptors. Reports published in the 1970s claimed discovery of immunoglobulin isotypes associated exclusively with T cells (IgT) and of antisera that recognize variable-region markers (idiotypes) common to antibodies and T-cell receptors with similar specificity. These experiments could not be reproduced and were proven to be incorrect when it was demonstrated that the T-cell receptor and immunoglobulins do not have common recognition elements and are encoded by entirely separate gene families. As the following sections will show, a sequence of well-designed experiments using cutting-edge technology was required to correctly answer questions about the structure of the T-cell receptor, the genes that encode it, and the manner in which it recognizes antigen.

Classic Experiments Demonstrated the Self-MHC Restriction of the T-Cell Receptor

By the early 1970s, immunologists had learned to generate cytotoxic T lymphocytes (CTLs) specific for virus-infected target cells. For example, when mice were infected with lymphocytic choriomeningitis (LCM) virus, they would produce CTLs that could lyse LCM-infected target cells *in vitro*. Yet these same CTLs failed to bind free LCM virus or viral antigens. Why didn't the CTLs bind the virus or viral antigens directly as immunoglobulins did? The answer began to emerge in the classic experiments of R. M. Zinkernagel and P. C. Doherty in 1974 (see Figure 8-2). These studies demonstrated that antigen recognition by T cells is specific not for viral antigen alone but for antigen associated with an MHC molecule (Figure 9-1). T cells were shown to recognize antigen only when presented on the membrane of a cell by a self-MHC molecule. This attribute, called *self-MHC restriction*, distinguishes recognition of antigen by T cells and B cells. In 1996, Doherty and Zinkernagel were awarded the Nobel Prize for this work.

Two models were proposed to explain the MHC restriction of the T-cell receptor. The *dual-receptor model* envisioned a T cell with two separate receptors, one for antigen and one for class I or class II MHC molecules. The *altered-self model* proposed that a single receptor recognizes an alteration in self-MHC molecules induced by their association with foreign antigens. The debate between proponents of these two models was waged for a number of years, until an elegant experiment by J. Kappler and P. Marrack demonstrated that specificity for both MHC and antigen resides in a single receptor. An overwhelming amount of structural and functional data has since been added in support of the altered-self model.

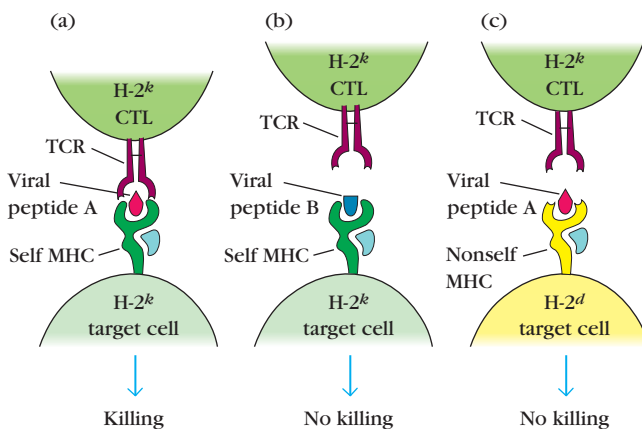


FIGURE 9-1 Self-MHC restriction of the T-cell receptor (TCR). A particular TCR is specific for both an antigenic peptide and a self-MHC molecule. In this example, the H-2^k CTL is specific for viral peptide A presented on an H-2^k target cell (a). Antigen recognition does not occur when peptide B is displayed on an H-2^k target cell (b) nor when peptide A is displayed on an H-2^d target cell (c).

T-Cell Receptors Were Isolated by Using Clonotypic Antibodies

Identification and isolation of the T-cell receptor was accomplished by producing large numbers of monoclonal antibodies to various T-cell clones and then screening the antibodies to find one that was clone specific, or *clonotypic*. This approach assumes that, since the T-cell receptor is specific for both an antigen and an MHC molecule, there should be significant structural differences in the receptor from clone to clone; each T-cell clone should have an antigenic marker similar to the idiotype markers that characterize monoclonal antibodies. Using this approach, researchers in the early 1980s isolated the receptor and found that it was a heterodimer consisting of α and β chains.

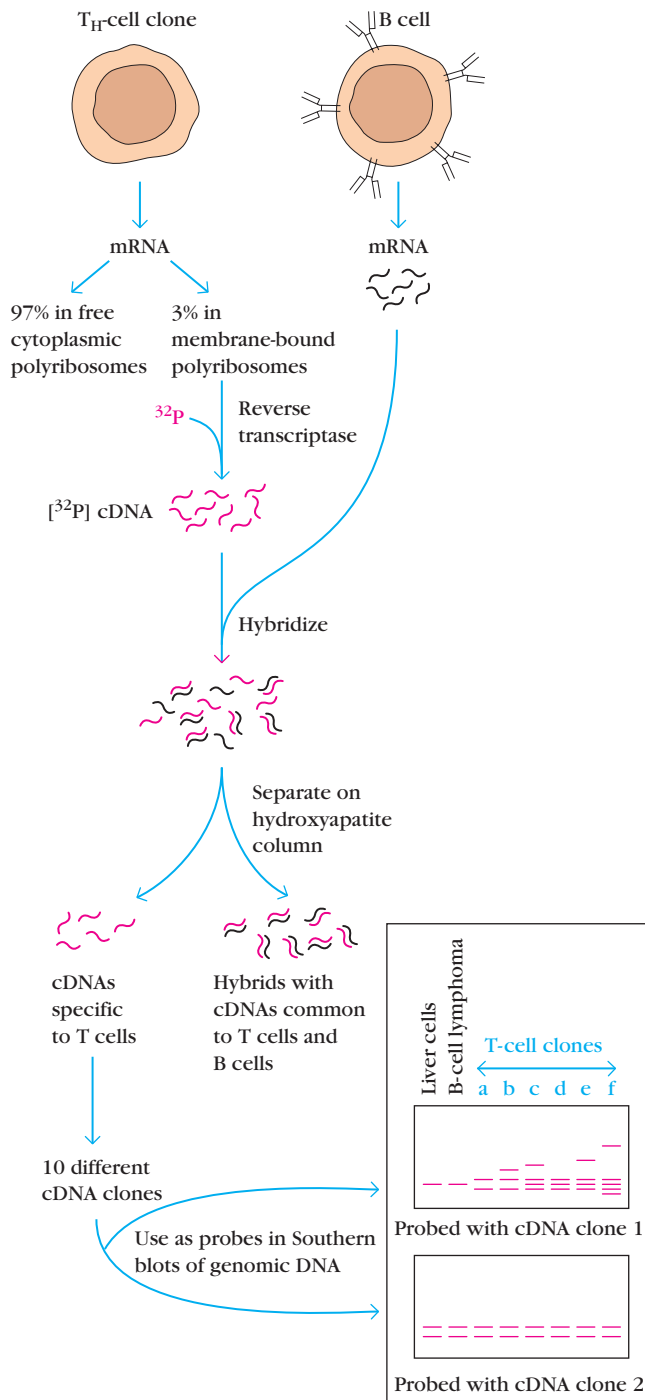
When antisera were prepared using $\alpha\beta$ heterodimers isolated from membranes of various T-cell clones, some antisera bound to $\alpha\beta$ heterodimers from all the clones, whereas other antisera were clone specific. This finding suggested that the amino acid sequences of the TCR α and β chains, like those of the immunoglobulin heavy and light chains, have constant and variable regions. Later, a second type of TCR heterodimer consisting of δ and γ chains was identified. In human and mouse, the majority of T cells express the $\alpha\beta$ heterodimer; the remaining T cells express the $\gamma\delta$ heterodimer. As described below, the exact proportion of T cells expressing $\alpha\beta$ or $\gamma\delta$ TCRs differs by organ and species, but $\alpha\beta$ T cells normally predominate.

The TCR β -Chain Gene Was Cloned by Use of Subtractive Hybridization

In order to identify and isolate the TCR genes, S. M. Hedrick and M. M. Davis sought to isolate mRNA that encodes the α and β chains from a T_H-cell clone. This was no easy task because the receptor mRNA represents only a minor fraction of the total cell mRNA. By contrast, in the plasma cell, immunoglobulin is a major secreted cell product, and mRNAs encoding the heavy and light chains are abundant and easy to purify.

The successful scheme of Hedrick and Davis assumed that the TCR mRNA—like the mRNAs that encode other integral membrane proteins—would be associated with membrane-bound polyribosomes rather than with free cytoplasmic ribosomes. They therefore isolated the membrane-bound polyribosomal mRNA from a T_H-cell clone and used reverse transcriptase to synthesize ³²P-labeled cDNA probes (Figure 9-2). Because only 3% of lymphocyte mRNA is in the membrane-bound polyribosomal fraction, this step eliminated 97% of the cell mRNA.

Hedrick and Davis next used a technique called *DNA subtractive hybridization* to remove from their preparation the [³²P]cDNA that was not unique to T cells. Their rationale for this step was based on earlier measurements by Davis showing that 98% of the genes expressed in lymphocytes are common to B cells and T cells. The 2% of the expressed genes that



is unique to T cells should include the genes encoding the T-cell receptor. Therefore, by hybridizing B-cell mRNA with their T_H-cell [³²P]cDNA, they were able to remove, or subtract, all the cDNA that was common to B cells and T cells. The unhybridized [³²P]cDNA remaining after this step presumably represented the expressed polyribosomal mRNA that was unique to the T_H-cell clone, including the mRNA encoding its T-cell receptor.

Cloning of the unhybridized [³²P]cDNA generated a library from which 10 different cDNA clones were identified. To determine which of these T-cell-specific cDNA clones

FIGURE 9-2 Production and identification of a cDNA clone encoding the T-cell receptor. The flow chart outlines the procedure used by S. Hedrick and M. Davis to obtain [³²P]cDNA clones corresponding to T-cell-specific mRNAs. The technique of DNA subtractive hybridization enabled them to isolate [³²P]cDNA unique to the T cell. The labeled T_H-cell cDNA clones were used as probes (*inset*) in Southern-blot analyses of genomic DNA from liver cells, B-lymphoma cells, and six different T_H-cell clones (a–f). Probing with cDNA clone 1 produced a distinct blot pattern for each T-cell clone, whereas probing with cDNA clone 2 did not. Assuming that liver cells and B cells contained unrearranged germ-line TCR DNA, and that each of the T-cell clones contained different rearranged TCR genes, the results using cDNA clone 1 as the probe identified clone 1 as the T-cell-receptor gene. The cDNA of clone 2 identified the gene for another T-cell membrane molecule encoded by DNA that does not undergo rearrangement. [Based on S. Hedrick et al., 1984, *Nature* **308**:149.]

represented the T-cell receptor, all were used as probes to look for genes that rearranged in mature T cells. This approach was based on the assumption that, since the αβ T-cell receptor appeared to have constant and variable regions, its genes should undergo DNA rearrangements like those observed in the Ig genes of B cells. The two investigators tested DNA from T cells, B cells, liver cells, and macrophages by Southern-blot analysis using the 10 [³²P]cDNA probes to identify unique T-cell genomic DNA sequences. One clone showed bands indicating DNA rearrangement in T cells but not in the other cell types. This cDNA probe identified six different patterns for the DNA from six different mature T-cell lines (see Figure 9-2 inset, upper panel). These different patterns presumably represented rearranged TCR genes. Such results would be expected if rearranged TCR genes occur only in mature T cells. The observation that each of the six T-cell lines showed different Southern-blot patterns was consistent with the predicted differences in TCR specificity in each T-cell line.

The cDNA clone 1 identified by the Southern-blot analyses shown in Figure 9-2 has all the hallmarks of a putative TCR gene: it represents a gene sequence that rearranges, is expressed as a membrane-bound protein, and is expressed only in T cells. This cDNA clone was found to encode the β chain of the T-cell receptor. Later, cDNA clones were identified encoding the α chain, the γ chain, and finally the δ chain. These findings opened the way to understanding the T-cell receptor and made possible subsequent structural and functional studies.

αβ and γδ T-Cell Receptors: Structure and Roles

The domain structures of αβ and γδ TCR heterodimers are strikingly similar to that of the immunoglobulins;

thus, they are classified as members of the immunoglobulin superfamily (see Figure 4-19). Each chain in a TCR has two domains containing an intrachain disulfide bond that spans 60–75 amino acids. The amino-terminal domain in both chains exhibits marked sequence variation, but the sequences of the remainder of each chain are conserved. Thus the TCR domains—one variable (V) and one constant (C)—are structurally homologous to the V and C domains of immunoglobulins, and the TCR molecule resembles an Fab fragment (Figure 9-3). The TCR variable domains have three hypervariable regions, which appear to be equivalent to the complementarity determining regions (CDRs) in immunoglobulin light and heavy chains. There is an additional area of hypervariability (HV4) in the β chain that does not normally contact antigen and therefore is not considered a CDR.

In addition to the constant domain, each TCR chain contains a short connecting sequence, in which a cysteine residue forms a disulfide link with the other chain of the heterodimer. Following the connecting region is a transmembrane region of 21 or 22 amino acids, which anchors each chain in the plasma membrane. The transmembrane domains of both chains are unusual in that they contain positively charged amino acid residues. These residues enable the chains of the TCR heterodimer to interact with chains of the signal-transducing CD3 complex. Finally, each TCR chain

contains a short cytoplasmic tail of 5–12 amino acids at the carboxyl-terminal end.

$\alpha\beta$ and $\gamma\delta$ T-cell receptors were initially difficult to investigate because, like all transmembrane proteins, they are insoluble. This problem was circumvented by expressing modified forms of the protein *in vitro* that had been engineered to contain premature in-frame stop codons that preclude translation of the membrane-binding sequence that makes the molecule insoluble.

The majority of T cells in the human and the mouse express T-cell receptors encoded by the $\alpha\beta$ genes. These receptors interact with peptide antigens processed and presented on the surface of antigen-presenting cells. Early indications that certain T cells reacted with *nonpeptide* antigens were puzzling until some light was shed on the problem when products of the CD1 family of genes were found to present carbohydrates and lipids. More recently, it has been found that certain $\gamma\delta$ cells react with antigen that is neither processed nor presented in the context of a MHC molecule.

Differences in the antigen-binding regions of $\alpha\beta$ and $\gamma\delta$ were expected because of the different antigens they recognize, but no extreme dissimilarities were expected. However, the recently completed three-dimensional structure for a $\gamma\delta$ receptor that reacts with a phosphoantigen, reported by Allison, Garboczi, and their coworkers, reveals significant

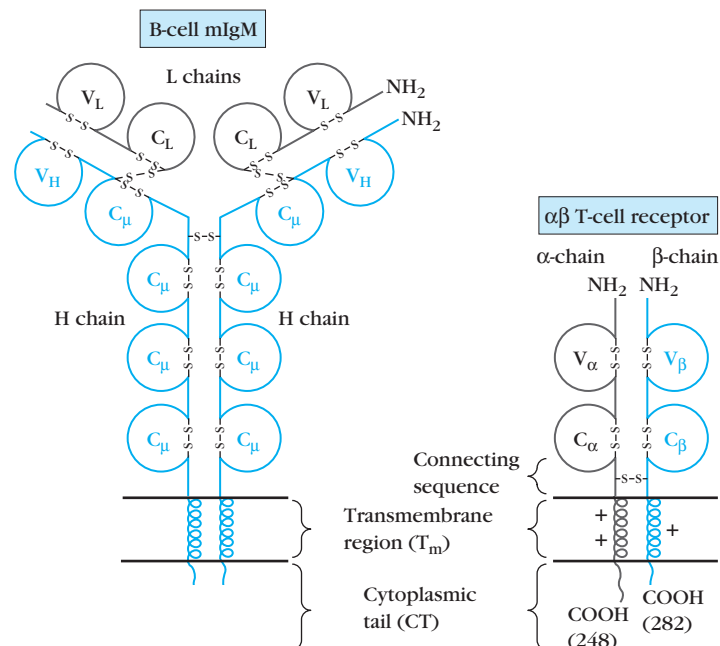


FIGURE 9-3 Schematic diagram illustrating the structural similarity between the $\alpha\beta$ T-cell receptor and membrane-bound IgM on B cells. The TCR α and β chain each contains two domains with the immunoglobulin-fold structure. The amino-terminal domains (V_α and V_β) exhibit sequence variation and contain three hypervariable regions equivalent to the CDRs in antibodies. The sequence of the constant domains (C_α and C_β) does not vary. The two TCR chains are connected by a disulfide bond between their constant sequences; the

IgM H chains are connected to one another by a disulfide bond in the hinge region of the H chain, and the L chains are connected to the H chains by disulfide links between the C termini of the L chains and the C_μ region. TCR molecules interact with CD3 via positively charged amino acid residues (indicated by +) in their transmembrane regions. Numbers indicate the length of the chains in the TCR molecule. Unlike the antibody molecule, which is bivalent, the TCR is monovalent.

differences in the overall structures of the two receptor types, pointing to possible functional variation. The receptor they studied was composed of the $\gamma 9$ and $\delta 2$ chains, which are those most frequently expressed in human peripheral blood. A deep cleft on the surface of the molecule accommodates the microbial phospholipid for which the $\gamma\delta$ receptor is specific. This antigen is recognized without MHC presentation.

The most striking feature of the structure is how it differs from the $\alpha\beta$ receptor in the orientation of its V and C regions. The so-called elbow angle between the long axes of the V and C regions of $\gamma\delta$ TCR is 111° ; in the $\alpha\beta$ TCR, the elbow angle is 149° , giving the molecules distinct shapes (Figure 9-4). The full significance of this difference is not known, but it could contribute to differences in signaling mechanisms and in how the molecules interact with coreceptor molecules.

The number of $\gamma\delta$ cells in circulation is small compared with cells that have $\alpha\beta$ receptors, and the V gene segments of $\gamma\delta$ receptors exhibit limited diversity. As seen from the data in Table 9-1, the majority of $\gamma\delta$ cells are negative for both CD4 and CD8, and most express a single $\gamma\delta$ -chain subtype. In humans the predominant receptor expressed on circulating $\gamma\delta$ cells recognizes a microbial phospholipid antigen, 3-formyl-1-butyl pyrophosphate, found on *M. tuberculosis* and other bacteria and parasites. This specificity for frequently encountered pathogens led to speculation that $\gamma\delta$ cells may function as an arm of the innate immune response, allowing rapid reactivity to certain antigens without the need for a processing step. Interestingly, the specificity of circulating $\gamma\delta$ cells in the mouse and of other species studied does not parallel that of humans, suggesting that the $\gamma\delta$ response may be directed against pathogens commonly encountered by a given species. Furthermore, data indicating that $\gamma\delta$ cells can secrete a spectrum of cytokines suggest that they may play a regulatory role in recruiting $\alpha\beta$ T cells to the site of invasion by pathogens. The recruited $\alpha\beta$ T cells would presumably display a broad spectrum of receptors; those with the highest

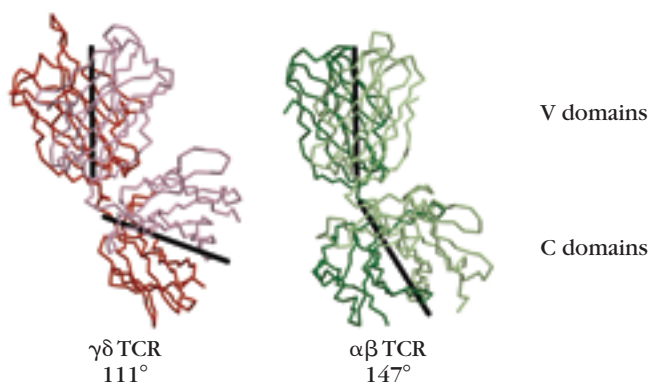


FIGURE 9-4 Comparison of the $\gamma\delta$ TCR and $\alpha\beta$ TCR. The difference in the elbow angle is highlighted with black lines. [From T. Allison et al., 2001, *Nature* **411**: 820.]

TABLE 9-1 Comparison of $\alpha\beta$ and $\gamma\delta$ T cells

Feature	$\alpha\beta$ T cells	$\gamma\delta$ T cells
Proportion of CD3 ⁺ cells	90–99%	1–10%
TCR V gene germ-line repertoire	Large	Small
CD4/CD8 phenotype		
CD4 ⁺	~60%	<1%
CD8 ⁺	~30%	~30%
CD4 ⁺ CD8 ⁺	<1%	<1%
CD4 ⁺ CD8 [−]	<1%	~60%
MHC restriction	CD4 ⁺ : MHC class II CD8 ⁺ : MHC class I	No MHC restriction
Ligands	Peptide + MHC	Phospholipid antigen

SOURCE: D. Kabelitz et al., 1999, *Springer Seminars in Immunopathology* 21:55, p. 36.

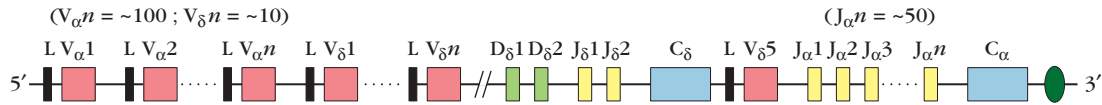
affinity would be selectively activated and amplified to deal with the pathogen.

Organization and Rearrangement of TCR Genes

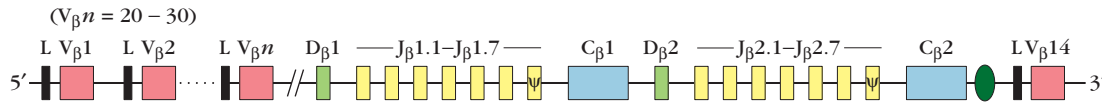
The genes that encode the $\alpha\beta$ and $\gamma\delta$ T-cell receptors are expressed only in cells of the T-cell lineage. The four TCR loci (α , β , γ , and δ) are organized in the germ line in a manner that is remarkably similar to the multigene organization of the immunoglobulin (Ig) genes (Figure 9-5). As in the case of Ig genes, functional TCR genes are produced by rearrangements of V and J segments in the α -chain and γ -chain families and V, D, and J segments in the β -chain and δ -chain families. In the mouse, the α -, β -, and γ -chain gene segments are located on chromosomes 14, 6, and 13, respectively. The δ -gene segments are located on chromosome 14 between the V_α and J_α segments. The location of the δ -chain gene family is significant: a productive rearrangement of the α -chain gene segments deletes C_δ , so that, in a given T cell, the $\alpha\beta$ TCR receptor cannot be coexpressed with the $\gamma\delta$ receptor.

Mouse germ-line DNA contains about 100 V_α and 50 J_α gene segments and a single C_α segment. The δ -chain gene family contains about 10 V gene segments, which are largely distinct from the V_α gene segments, although some sharing

Mouse TCR α -chain and δ -chain DNA (chromosome 14)



Mouse TCR β -chain DNA (chromosome 6)



Mouse TCR γ -chain DNA (chromosome 13)

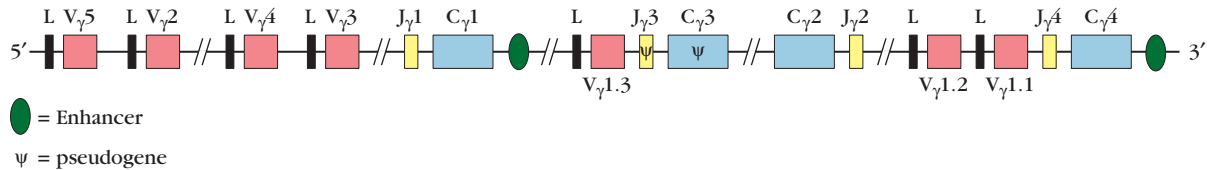


FIGURE 9-5 Germ-line organization of the mouse TCR α -, β -, γ -, and δ -chain gene segments. Each C gene segment is composed of a series of exons and introns, which are not shown. The organization of TCR gene segments in humans is similar, although the number of

the various gene segments differs in some cases (see Table 9-2). [Adapted from D. Raulet, 1989, *Annu. Rev. Immunol.* **7**:175, and M. Davis, 1990, *Annu. Rev. Biochem.* **59**:475.]

of V segments has been observed in rearranged α - and δ -chain genes. Two D_δ and two J_δ gene segments and one C_δ segment have also been identified. The β -chain gene family has 20–30 V gene segments and two almost identical repeats of D, J, and C segments, each repeat consisting of one D_β , six J_β , and one C_β . The γ -chain gene family consists of seven V_γ segments and three different functional J_γ - C_γ repeats. The organization of the TCR multigene families in humans is generally similar to that in mice, although the number of segments differs (Table 9-2).

TABLE 9-2 TCR Multigene families in humans

Gene	Chromosome location	NO. OF GENE SEGMENTS			
		V	D	J	C
α Chain	14	50		70	1
δ Chain*	14	3	3	3	1
β Chain†	7	57	2	13	2
γ Chain‡	7	14		5	2

*The δ -chain gene segments are located between the V_α and J_α segments.

†There are two repeats, each containing 1 D_β , 6 or 7 J_β , and 1 C_β .

‡There are two repeats, each containing 2 or 3 J_γ and 1 C_γ .

SOURCE: Data from P. A. H. Moss et al., 1992, *Annu. Rev. Immunol.* **10**:71.

TCR Variable-Region Genes Rearrange in a Manner Similar to Antibody Genes

The α chain, like the immunoglobulin L chain, is encoded by V, J, and C gene segments. The β chain, like the immunoglobulin H chain, is encoded by V, D, J, and C gene segments. Rearrangement of the TCR α - and β -chain gene segments results in VJ joining for the α chain and VDJ joining for the β chain (Figure 9-6).

After transcription of the rearranged TCR genes, RNA processing, and translation, the α and β chains are expressed as a disulfide-linked heterodimer on the membrane of the T cell. Unlike immunoglobulins, which can be membrane bound or secreted, the $\alpha\beta$ heterodimer is expressed only in a membrane-bound form; thus, no differential RNA processing is required to produce membrane and secreted forms. Each TCR constant region includes a connecting sequence, a transmembrane sequence, and a cytoplasmic sequence.

The germ-line DNA encoding the TCR α and β chain constant regions is much simpler than the immunoglobulin heavy-chain germ-line DNA, which has multiple C gene segments encoding distinct isotypes with different effector functions. TCR α -chain DNA has only a single C gene segment; the β -chain DNA has two C gene segments, but their protein products differ by only a few amino acids and have no known functional differences.

MECHANISM OF TCR DNA REARRANGEMENTS

The mechanisms by which TCR germ-line DNA is rearranged to form functional receptor genes appear to be



VISUALIZING CONCEPTS

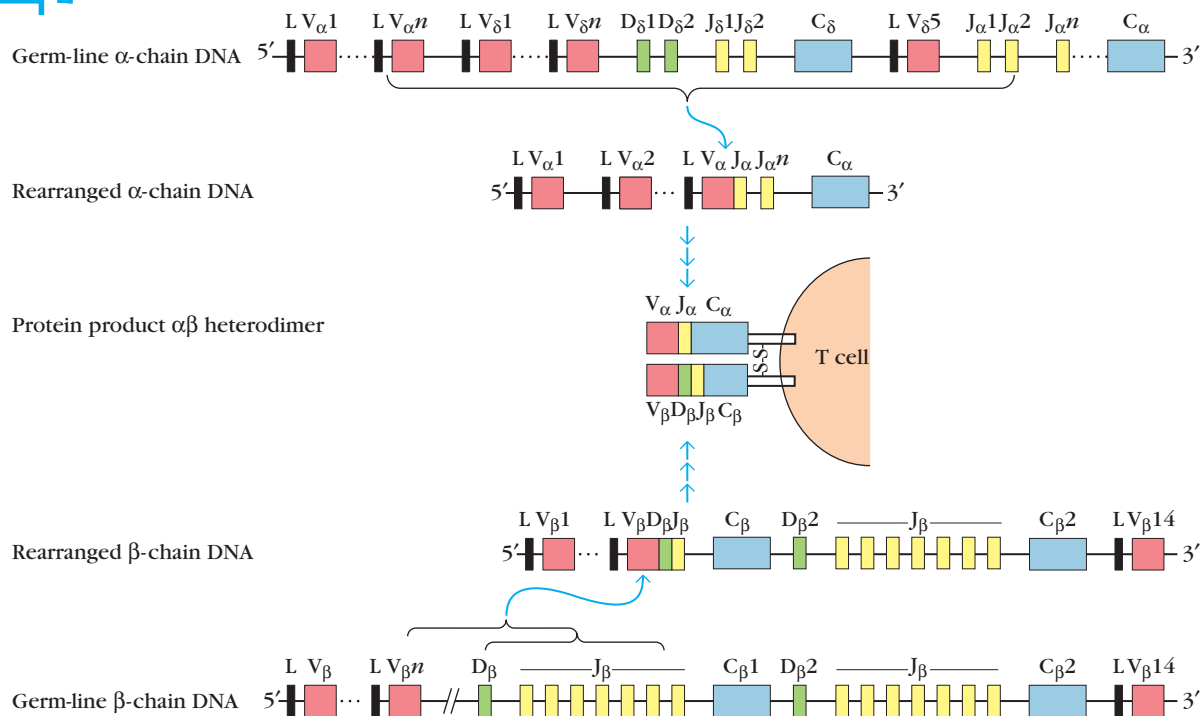


FIGURE 9-6 Example of gene rearrangements that yield a functional gene encoding the $\alpha\beta$ T-cell receptor. The α -chain DNA, analogous to immunoglobulin light-chain DNA, undergoes a variable-region V α -J α joining. The β -chain DNA, analogous to immunoglobulin heavy-chain DNA, undergoes two variable-region joinings: first D β to J β and then V β to D β J β . Transcription of the rearranged genes yields primary transcripts, which are processed to give mRNAs encoding the α and β chains of the membrane-

bound TCR. The leader sequence is cleaved from the nascent polypeptide chain and is not present in the finished protein. As no secreted TCR is produced, differential processing of the primary transcripts does not occur. Although the β -chain DNA contains two C genes, the gene products of these two C genes exhibit no known functional differences. The C genes are composed of several exons and introns, which are not individually shown here (see Figure 9-7).

similar to the mechanisms of Ig-gene rearrangements. For example, conserved heptamer and nonamer recombination signal sequences (RSSs), containing either 12-bp (one-turn) or 23-bp (two-turn) spacer sequences, have been identified flanking each V, D, and J gene segment in TCR germ-line DNA (see Figure 5-6). All of the TCR-gene rearrangements follow the one-turn/two-turn joining rule observed for the Ig genes, so recombination can occur only between the two different types of RSSs.

Like the pre-B cell, the pre-T cell expresses the recombination-activating genes (*RAG-1* and *RAG-2*). The RAG-1/2 recombinase enzyme recognizes the heptamer and nonamer recognition signals and catalyzes V-J and V-D-J joining during TCR-gene rearrangement by the same deletional or inversional mechanisms that occur in the Ig genes (see Figure 5-7). As described in Chapter 5 for the

immunoglobulin genes, RAG-1/2 introduces a nick on one DNA strand between the coding and signal sequences. The recombinase then catalyzes a transesterification reaction that results in the formation of a hairpin at the coding sequence and a flush 5' phosphorylated double-strand break at the signal sequence. Circular excision products thought to be generated by looping-out and deletion during TCR-gene rearrangement have been identified in thymocytes (see Figure 5-8).

Studies with SCID mice, which lack functional T and B cells, provide evidence for the similarity in the mechanisms of Ig-gene and TCR-gene rearrangements. As explained in Chapter 19, SCID mice have a defect in a gene required for the repair of double-stranded DNA breaks. As a result of this defect, D and J gene segments are not joined during rearrangement of either Ig or TCR DNA (see Figure 5-10). This

finding suggests that the same double-stranded break-repair enzymes are involved in V-D-J rearrangements in B cells and in T cells.

Although B cells and T cells use very similar mechanisms for variable-region gene rearrangements, the Ig genes are not normally rearranged in T cells and the TCR genes are not rearranged in B cells. Presumably, the recombinase enzyme system is regulated in each cell lineage, so that only rearrangement of the correct receptor DNA occurs. Rearrangement of the gene segments in both T and B cell creates a DNA sequence unique to that cell and its progeny. The large number of possible configurations of the rearranged genes makes this new sequence a marker that is specific for the cell clone. These unique DNA sequences have been used to aid in diagnoses and in treatment of lymphoid leukemias and lymphomas, cancers that involve clonal proliferation of T or B cells (see Clinical Focus on page 208).

ALLELIC EXCLUSION OF TCR GENES

As mentioned above, the δ genes are located within the α -gene complex and are deleted by α -chain rearrangements. This event provides an irrevocable mode of exclusion for the δ genes located on the same chromosome as the rearranging α genes. Allelic exclusion of genes for the TCR α and β chains occurs as well, but exceptions have been observed.

The organization of the β -chain gene segments into two clusters means that, if a nonproductive rearrangement occurs, the thymocyte can attempt a second rearrangement. This increases the likelihood of a productive rearrangement for the β chain. Once a productive rearrangement occurs for one β -chain allele, the rearrangement of the other β allele is inhibited.

Exceptions to allelic exclusion are most often seen for the TCR α -chain genes. For example, analyses of T-cell clones that express a functional $\alpha\beta$ T-cell receptor revealed a number of clones with productive rearrangements of both α -chain alleles. Furthermore, when an immature T-cell lymphoma that expressed a particular $\alpha\beta$ T-cell receptor was subcloned, several subclones were obtained that expressed the same β -chain allele but an α -chain allele different from the one expressed by the original parent clone. Studies with transgenic mice also indicate that allelic exclusion is less stringent for TCR α -chain genes than for β -chain genes. Mice that carry a productively rearranged $\alpha\beta$ -TCR transgene do not rearrange and express the endogenous β -chain genes. However, the endogenous α -chain genes sometimes are expressed at various levels in place of the already rearranged α -chain transgene.

Since allelic exclusion is not complete for the TCR α chain, there are rare occasions when more than one α chain is expressed on the membrane of a given T cell. The obvious question is how do the rare T cells that express two $\alpha\beta$ T-cell receptors maintain a single antigen-binding specificity? One proposal suggests that when a T cell expresses two different $\alpha\beta$ T-cell receptors, only one is likely to be self-MHC restricted and therefore functional.

Rearranged TCR Genes Are Assembled from V, J, and D Gene Segments

The general structure of rearranged TCR genes is shown in Figure 9-7. The variable regions of T-cell receptors are, of course, encoded by rearranged VDJ and VJ sequences. In TCR genes, combinatorial joining of V gene segments appears to generate CDR1 and CDR2, whereas junctional flexibility and N-region nucleotide addition generate CDR3. Rearranged TCR genes also contain a short leader (L) exon upstream of the joined VJ or VDJ sequences. The amino acids encoded by the leader exon are cleaved as the nascent polypeptide enters the endoplasmic reticulum.

The constant region of each TCR chain is encoded by a C gene segment that has multiple exons (see Figure 9-7) corresponding to the structural domains in the protein (see Figure 9-3). The first exon in the C gene segment encodes most of the C domain of the corresponding chain. Next is a short exon that encodes the connecting sequence, followed by exons that encode the transmembrane region and the cytoplasmic tail.

TCR Diversity Is Generated Like Antibody Diversity but Without Somatic Mutation

Although TCR germ-line DNA contains far fewer V gene segments than Ig germ-line DNA, several mechanisms that operate during TCR gene rearrangements contribute to a high degree of diversity among T-cell receptors. Table 9-3 (page 210) and Figure 9-8 (page 211) compare the generation of diversity among antibody molecules and TCR molecules.

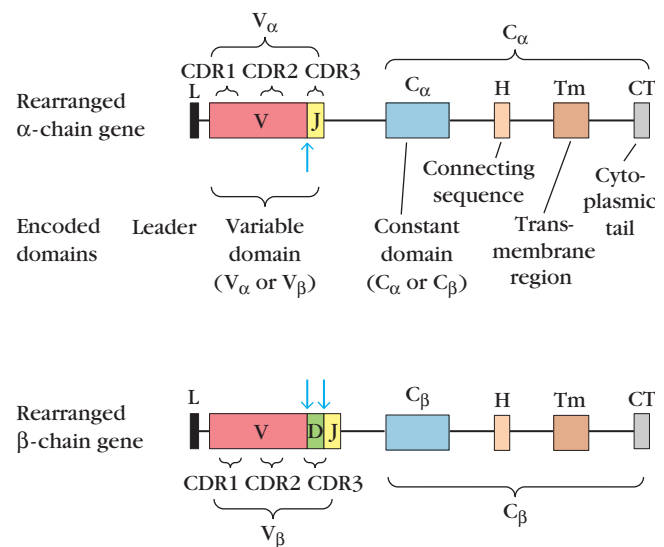


FIGURE 9-7 Schematic diagram of rearranged $\alpha\beta$ -TCR genes showing the exons that encode the various domains of the $\alpha\beta$ T-cell receptor and approximate position of the CDRs. Junctional diversity (vertical arrows) generates CDR3 (see Figure 9-8). The structures of the rearranged γ - and δ -chain genes are similar, although additional junctional diversity can occur in δ -chain genes.



CLINICAL FOCUS

T-Cell Rearrangements as Markers for Cancerous Cells

T-cell cancers, which include leukemia and lymphoma, involve the uncontrolled proliferation of a clonal population of T cells. Successful treatment requires quick and certain diagnosis in order to apply the most effective treatment. Once treatment is initiated, reliable tests are needed to determine whether the treatment regimen was successful. In principle, because T-cell cancers are clonal in nature, the cell population that is cancerous could be identified and monitored by the expression of its unique T-cell receptor molecules. However, this approach is rarely practical because detection of a specific TCR molecule requires the tedious and lengthy preparation of a specific antibody directed against its variable region (an anti-idiotypic antibody). Also, surface expression of the TCR molecule occurs somewhat late in the development of the T cell, so cancers stemming from T cells that have not progressed beyond an early stage of development will not display a TCR molecule and will not be detected by the antibody. An alternative means of identifying a clonal population of T cells is to look at their DNA rather than protein products. The pattern resulting from rearrangement of the TCR genes can provide a unique marker for the cancerous T cell. Because rearrangement of

the TCR genes in the T cells occurs before the product molecule is expressed, T cells in early stages of development can be detected. The unique gene fragments that result from TCR gene rearrangement can be detected by simple molecular-biological techniques and provide a true fingerprint for a clonal cell population.

DNA patterns that result from rearrangement of the genes in the TCR β region are used most frequently as markers. There are approximately 50 V_β gene segments that can rearrange to one of two D-region gene segments and subsequently to one of 12 J gene segments (see Figure 9-8). Because each of the 50 or so V-region genes is flanked by unique sequences, this process creates new DNA sequences that are unique to each cell that undergoes the rearrangement; these new sequences may be detected by Southern-blot techniques or by PCR (polymerase chain reaction). Since the entire sequence of the D, J, and C region of the TCR gene β complex is known, the appropriate probes and restriction enzymes are easily chosen for Southern blotting (see diagram).

Detection of rearranged TCR DNA may be used as a diagnostic tool when abnormally enlarged lymph nodes persist; this condition could result either from inflammation due to chronic infec-

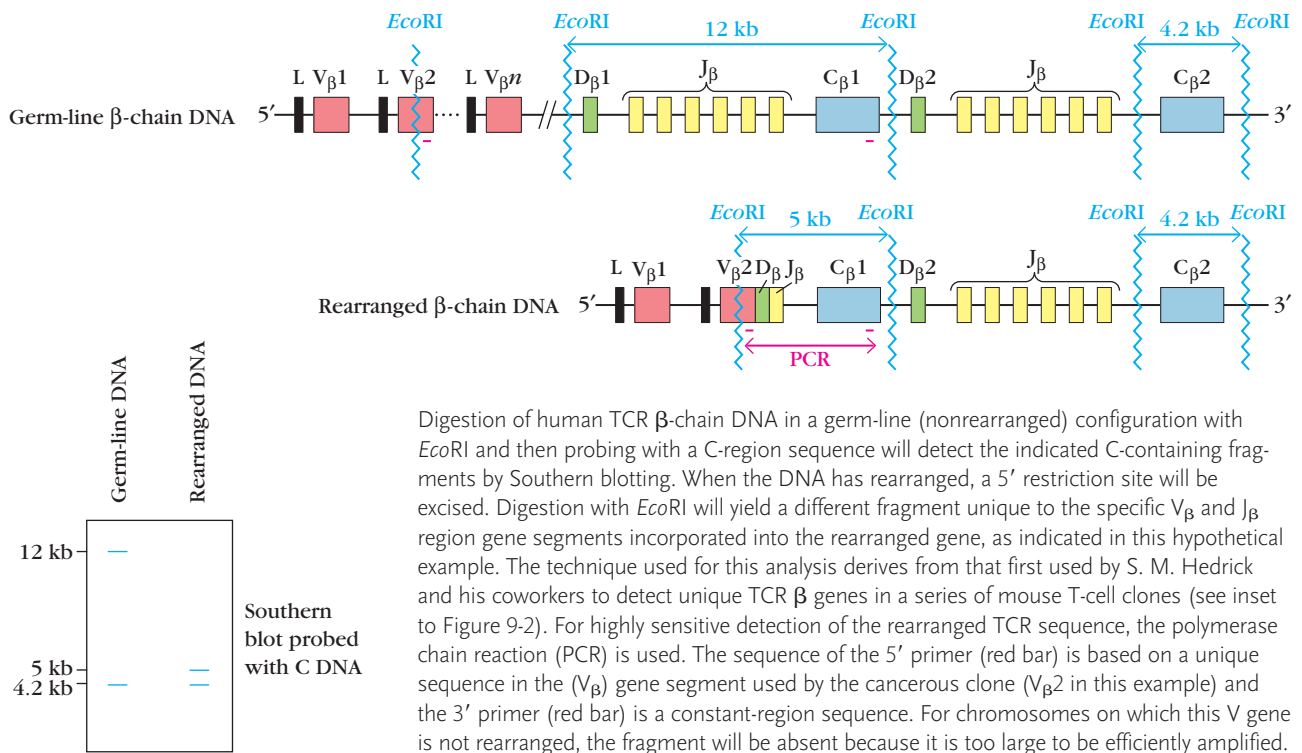
tion or from proliferation of a cancerous lymphoid cell. If inflammation is the cause, the cells would come from a variety of clones, and the DNA isolated from them would be a mixture of many different TCR sequences resulting from multiple rearrangements; no unique fragments would be detected. If the persistent enlargement of the nodes represents a clonal proliferation, there would be a detectable DNA fragment, because the cancerous cells would all contain the same TCR DNA sequence produced by DNA rearrangement in the parent cell. Thus the question whether the observed enlargement was due to the cancerous growth of T cells could be answered by the presence of a single new gene fragment in the DNA from the cell population. Because Ig genes rearrange in the same fashion as the TCR genes, similar techniques use Ig probes to detect clonal B-cell populations by their unique DNA patterns. The technique, therefore, has value for a wide range of lymphoid-cell cancers.

Although the detection of a unique DNA fragment resulting from rearranged TCR or Ig genes indicates clonal proliferation and possible malignancy of T or B cells, the absence of such a fragment does not rule out cancer of a population of lymphoid cells. The cell involved may not contain rearranged TCR or Ig genes that can be detected by the method used, either because of its developmental stage or because it is of another lineage ($\gamma\delta$ T cells, for example).

If the DNA fragment test and other diagnostic criteria indicate that the patient has a lymphoid cell cancer, treatment by

Combinatorial joining of variable-region gene segments generates a large number of random gene combinations for all the TCR chains, as it does for the Ig heavy- and light-chain genes. For example, 100 V_α and 50 J_α gene segments can generate 5×10^3 possible VJ combinations for the TCR α chain. Similarly, 25 V_β , 2 D_β , and 12 J_β gene segments can give 6×10^2 possible combinations. Although there are

fewer TCR V_α and V_β gene segments than immunoglobulin V_H and V_α segments, this difference is offset by the greater number of J segments in TCR germ-line DNA. Assuming that the antigen-binding specificity of a given T-cell receptor depends upon the variable region in both chains, random association of 5×10^3 V_α combinations with 6×10^2 V_β combinations can generate 3×10^6 possible combinations



Digestion of human TCR β -chain DNA in a germ-line (nonrearranged) configuration with *Eco*RI and then probing with a C-region sequence will detect the indicated C-containing fragments by Southern blotting. When the DNA has rearranged, a 5' restriction site will be excised. Digestion with *Eco*RI will yield a different fragment unique to the specific V_{β} and J_{β} region gene segments incorporated into the rearranged gene, as indicated in this hypothetical example. The technique used for this analysis derives from that first used by S. M. Hedrick and his coworkers to detect unique TCR β genes in a series of mouse T-cell clones (see inset to Figure 9-2). For highly sensitive detection of the rearranged TCR sequence, the polymerase chain reaction (PCR) is used. The sequence of the 5' primer (red bar) is based on a unique sequence in the (V_{β}) gene segment used by the cancerous clone ($V_{\beta 2}$ in this example) and the 3' primer (red bar) is a constant-region sequence. For chromosomes on which this V gene is not rearranged, the fragment will be absent because it is too large to be efficiently amplified.

radiation therapy or chemotherapy would follow. The success of this treatment can be monitored by probing DNA from the patient for the unique sequence found in the cancerous cell. If the treatment regimen is successful, the number of cancerous cells will decline greatly. If the number of cancerous cells falls below 1% or 2% of the total T-cell population, analysis by Southern blot may no longer detect the unique fragment. In this case, a more sensitive technique, PCR, may be used. (With PCR it is possible to am-

plify, or synthesize multiple copies of, a specific DNA sequence in a sample; primers can hybridize to the two ends of that specific sequence and thus direct a DNA polymerase to copy it; see Figure 23-13 for details.) To detect a portion of the rearranged TCR DNA, amplification using a sequence from the rearranged V region as one primer and a sequence from the β -chain C region as the other primer will yield a rearranged TCR DNA fragment of predicted size in sufficient quantity to be detected by electrophore-

sis (see red arrow in the diagram). Recently, quantitative PCR methods have been used to follow patients who are in remission in order to make decisions about resuming treatment if the number of cancerous cells, as estimated by these techniques, has risen above a certain level. Therefore, the presence of the rearranged DNA in the clonal population of T cells gives the clinician a valuable tool for diagnosing lymphoid-cell cancer and for monitoring the progress of treatment.

for the $\alpha\beta$ T-cell receptor. Additional means to generate diversity in the TCR V genes are described below, so 3×10^6 combinations represents a minimum estimate.

As illustrated in Figure 9-8b, the location of one-turn (12-bp) and two-turn (23-bp) recombination signal sequences (RSSs) in TCR β - and δ -chain DNA differs from that in Ig heavy-chain DNA. Because of the arrangement of

the RSSs in TCR germ-line DNA, alternative joining of D gene segments can occur while the one-turn/two-turn joining rule is observed. Thus, it is possible for a V_{β} gene segment to join directly with a J_{β} or a D_{β} gene segment, generating a (VJ) $_{\beta}$ or (VDJ) $_{\beta}$ unit.

Alternative joining of δ -chain gene segments generates similar units; in addition, one D_{δ} can join with another,

TABLE 9-3 Sources of possible diversity in mouse immunoglobulin and TCR genes

Mechanism of diversity	IMMUNOGLOBULINS		$\alpha\beta$ T-CELL RECEPTOR		$\gamma\delta$ T-CELL RECEPTOR	
	H Chain	κ Chain	α Chain	β Chain	γ Chain	δ Chain
ESTIMATED NUMBER OF SEGMENTS						
Multiple germ-line gene segments						
V	134	85	100	25	7	10
D	13	0	0	2	0	2
J	4	4	50	12	3	2
POSSIBLE NUMBER OF COMBINATIONS*						
Combinatorial V-J	$134 \times 13 \times 4$	85×4	100×50	$25 \times 2 \times 12$	7×3	$10 \times 2 \times 2$
and V-D-J joining	$= 7 \times 10^3$	$= 3.4 \times 10^2$	$= 5 \times 10^3$	$= 6 \times 10^2$	$= 21$	$= 40$
Alternative joining	—	—	—	+	—	+
of D gene segments				(some)		(often)
Junctional flexibility	+	+	+	+	+	+
N-region nucleotide addition [†]	+	—	+	+	+	+
P-region nucleotide addition	+	+	+	+	+	+
Somatic mutation	+	+	—	—	—	—
Combinatorial association of chains		+		+		+

*A plus sign (+) indicates mechanism makes a significant contribution to diversity but to an unknown extent.

A minus sign (—) indicates mechanism does not operate.

[†]See Figure 9-8d for theoretical number of combinations generated by N-region addition.

yielding (VDDJ)₈ and, in humans, (VDDDJ)₈. This mechanism, which cannot operate in Ig heavy-chain DNA, generates considerable additional diversity in TCR genes.

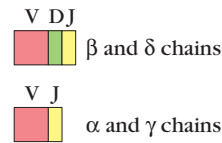
The joining of gene segments during TCR-gene rearrangement exhibits **junctional flexibility**. As with the Ig genes, this flexibility can generate many nonproductive rearrangements, but it also increases diversity by encoding several alternative amino acids at each junction (see Figure 9-8c). In both Ig and TCR genes, nucleotides may be added at the junctions between some gene segments during rearrangement (see Figure 5-15). Variation in endonuclease cleavage leads to the addition of further nucleotides that are palindromic. Such **P-region nucleotide addition** can occur in the genes encoding all the TCR and Ig chains. Addition of **N-region nucleotides**, catalyzed by a terminal deoxynucleotidyl transferase, generates additional junctional diversity. Whereas the addition of N-region nucleotides in immunoglobulins occurs only in the Ig heavy-chain genes, it occurs in the genes encoding all the TCR chains. As many as six nucleotides can be added by this mechanism at each junction, generating up to 5461 possible combinations, assuming

random selection of nucleotides (see Figure 9-8d). Some of these combinations, however, lead to nonproductive rearrangements by inserting in-frame stop codons that prematurely terminate the TCR chain, or by substituting amino acids that render the product nonfunctional. Although each junctional region in a TCR gene encodes only 10–20 amino acids, enormous diversity can be generated in these regions. Estimates suggest that the combined effects of P- and N-region nucleotide addition and joining flexibility can generate as many as 10^{13} possible amino acid sequences in the TCR junctional regions alone.

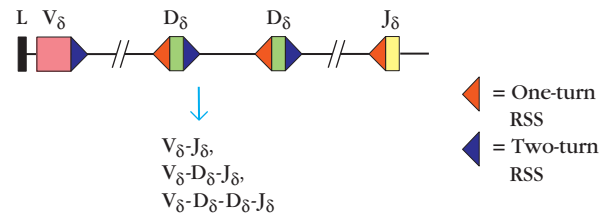
The mechanism by which diversity is generated for the TCR must allow the receptor to recognize a very large number of different processed antigens while restricting its MHC-recognition repertoire to a much smaller number of self-MHC molecules. TCR DNA has far fewer V gene segments than Ig DNA (see Table 9-3). It has been postulated that the smaller number of V gene segments in TCR DNA have been selected to encode a limited number of CDR1 and CDR2 regions with affinity for regions of the α helices of MHC molecules. Although this is an attractive idea, it is

T-CELL RECEPTOR

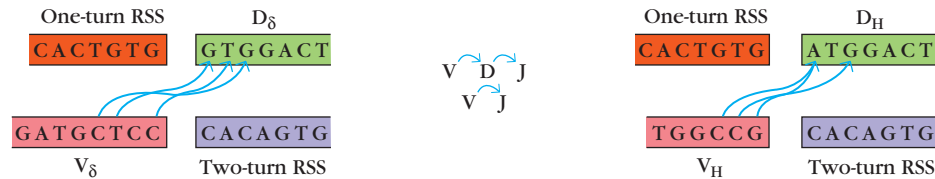
(a) Combinatorial V-J and V-D-J joining



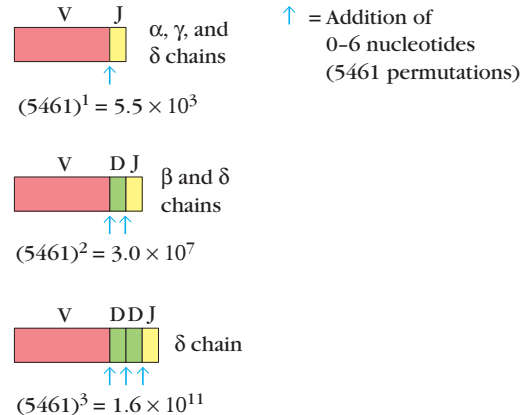
(b) Alternative joining of D gene segments



(c) Junctional flexibility



(d) N-region nucleotide addition



IMMUNOGLOBULIN

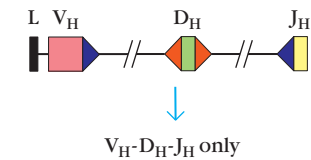
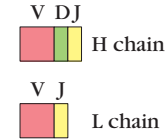


FIGURE 9-8 Comparison of mechanisms for generating diversity in TCR genes and immunoglobulin genes. In addition to the mechanisms shown, P-region nucleotide addition occurs in both TCR and

Ig genes, and somatic mutation occurs in Ig genes. Combinatorial association of the expressed chains generates additional diversity among both TCR and Ig molecules.

made unlikely by recent data on the structure of the TCR-peptide-MHC complex showing contact between peptide and CDR1 as well as CDR3. Therefore the TCR residues that bind to peptide versus those that bind MHC are not confined solely to the highly variable CDR3 region.

In contrast to the limited diversity of CDR1 and CDR2, the CDR3 of the TCR has even greater diversity than that seen in immunoglobulins. Diversity in CDR3 is generated by junctional diversity in the joining of V, D, and J segments,

joining of multiple D gene segments, and the introduction of P and N nucleotides at the V-D-J and V-J junctions (see Figure 9-7).

Unlike the Ig genes, the TCR genes do not appear to undergo extensive somatic mutation. That is, the functional TCR genes generated by gene rearrangements during T-cell maturation in the thymus have the same sequences as those found in the mature peripheral T-cell population. The absence of somatic mutation in T cells ensures that T-cell

specificity does not change after thymic selection and therefore reduces the possibility that random mutation might generate a self-reactive T cell. Although a few experiments have provided evidence for somatic mutation of receptor genes in T cells in the germinal center, this appears to be the exception and not the rule.

T-Cell Receptor Complex: TCR-CD3

As explained in Chapter 4, membrane-bound immunoglobulin on B cells associates with another membrane protein, the Ig- α /Ig- β heterodimer, to form the B-cell antigen receptor (see Figure 4-18). Similarly, the T-cell receptor associates with **CD3**, forming the TCR-CD3 membrane complex. In both cases, the accessory molecule participates in signal transduction *after* interaction of a B or T cell with antigen; it does not influence interaction with antigen.

The first evidence suggesting that the T-cell receptor is associated with another membrane molecule came from experiments in which fluorescent antibody to the receptor was shown to cause aggregation of another membrane protein designated CD3. Later experiments by J. P. Allison and

L. Lanier using cross-linking reagents demonstrated that the two chains must be within 12 Å. Subsequent experiments demonstrated not only that CD3 is closely associated with the $\alpha\beta$ heterodimer but also that its expression is required for membrane expression of $\alpha\beta$ and $\gamma\delta$ T-cell receptors—each heterodimer forms a complex with CD3 on the T-cell membrane. Loss of the genes encoding either CD3 or the TCR chains results in loss of the entire molecular complex from the membrane.

CD3 is a complex of five invariant polypeptide chains that associate to form three dimers: a heterodimer of gamma and epsilon chains ($\gamma\epsilon$), a heterodimer of delta and epsilon chains ($\delta\epsilon$), and a homodimer of two zeta chains ($\zeta\zeta$) or a heterodimer of zeta and eta chains ($\zeta\eta$) (Figure 9-9). The ζ and η chains are encoded by the same gene, but differ in their carboxyl-terminal ends because of differences in RNA splicing of the primary transcript. About 90% of the CD3 complexes examined to date incorporate the ($\zeta\zeta$) homodimer; the remainder have the ($\zeta\eta$) heterodimer. The T-cell receptor complex can thus be envisioned as four dimers: the $\alpha\beta$ or $\gamma\delta$ TCR heterodimer determines the ligand-binding specificity, whereas the CD3 dimers ($\gamma\epsilon$, $\delta\epsilon$, and $\zeta\zeta$ or $\zeta\eta$) are required for membrane expression of the T-cell receptor and for signal transduction.

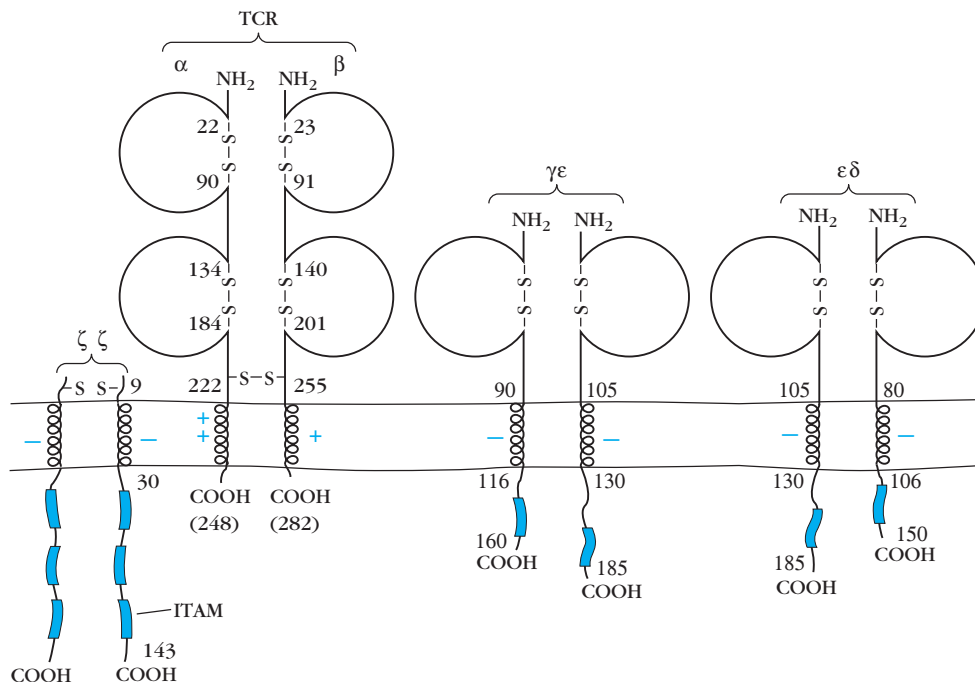


FIGURE 9-9 Schematic diagram of the TCR-CD3 complex, which constitutes the T-cell antigen-binding receptor. The CD3 complex consists of the $\zeta\zeta$ homodimer (alternately, a $\zeta\eta$ heterodimer) plus $\gamma\epsilon$ and $\delta\epsilon$ heterodimers. The external domains of the γ , δ , and ϵ chains of CD3 are similar to the immunoglobulin fold, which facilitates their interaction with the T-cell receptor and each other. Ionic interactions

also may occur between the oppositely charged transmembrane regions in the TCR and CD3 chains. The long cytoplasmic tails of the CD3 chains contain a common sequence, the immunoreceptor tyrosine-based activation motif (ITAM), which functions in signal transduction.

The γ , δ , and ϵ chains of CD3 are members of the immunoglobulin superfamily, each containing an immunoglobulin-like extracellular domain followed by a transmembrane region and a cytoplasmic domain of more than 40 amino acids. The ζ chain has a distinctly different structure, with a very short external region of only 9 amino acids, a transmembrane region, and a long cytoplasmic tail containing 113 amino acids. The transmembrane region of all the CD3 polypeptide chains contains a negatively charged aspartic acid residue that interacts with one or two positively charged amino acids in the transmembrane region of each TCR chain.

The cytoplasmic tails of the CD3 chains contain a motif called the **immunoreceptor tyrosine-based activation motif (ITAM)**. ITAMs are found in a number of other receptors, including the Ig- α /Ig- β heterodimer of the B-cell receptor complex and the Fc receptors for IgE and IgG. The ITAM sites have been shown to interact with tyrosine kinases and to play an important role in signal transduction. In CD3, the γ , δ , and ϵ chains each contain a single copy of ITAM, whereas the ζ and η chains contain three copies (see Figure 9-9). The function of CD3 in signal transduction is described more fully in Chapter 10.

T-Cell Accessory Membrane Molecules

Although recognition of antigen-MHC complexes is mediated solely by the TCR-CD3 complex, various other membrane molecules play important accessory roles in antigen recognition and T-cell activation (Table 9-4). Some of these molecules strengthen the interaction between T cells and

antigen-presenting cells or target cells, some act in signal transduction, and some do both.

CD4 and CD8 Coreceptors Bind to Conserved Regions of MHC Class II or I Molecules.

T cells can be subdivided into two populations according to their expression of CD4 or CD8 membrane molecules. As described in preceding chapters, CD4⁺ T cells recognize antigen that is combined with class II MHC molecules and function largely as helper cells, whereas CD8⁺ T cells recognize antigen that is combined with class I MHC molecules and function largely as cytotoxic cells. CD4 is a 55-kDa monomeric membrane glycoprotein that contains four extracellular immunoglobulin-like domains (D₁–D₄), a hydrophobic transmembrane region, and a long cytoplasmic tail (Figure 9-10) containing three serine residues that can be phosphorylated. CD8 generally takes the form of a disulfide-linked $\alpha\beta$ heterodimer or of an $\alpha\alpha$ homodimer. Both the α and β chains of CD8 are small glycoproteins of approximately 30–38 kDa. Each chain consists of a single extracellular immunoglobulin-like domain, a hydrophobic transmembrane region, and a cytoplasmic tail (Figure 9-10) containing 25–27 residues, several of which can be phosphorylated.

CD4 and CD8 are classified as *coreceptors* based on their abilities to recognize the peptide-MHC complex and their roles in signal transduction. The extracellular domains of CD4 and CD8 bind to the conserved regions of MHC molecules on antigen-presenting cells (APCs) or target cells. Crystallographic studies of a complex composed of the class I MHC molecule HLA-A2, an antigenic peptide, and a CD8 $\alpha\alpha$ homodimer indicate that CD8 binds to class I molecules

TABLE 9-4 Selected T-cell accessory molecules

Name	Ligand	FUNCTION		
		Adhesion	Signal transduction	Member of Ig superfamily
CD4	Class II MHC	+	+	+
CD8	Class I MHC	+	+	+
CD2 (LFA-2)	CD58 (LFA-3)	+	+	+
LFA-1 (CD11a/CD18)	ICAM-1 (CD54)	+	?	+/(–)
CD28	B7	?	+	+
CTLA-4	B7	?	+	–
CD45R	CD22	+	+	+
CD5	CD72	?	+	–

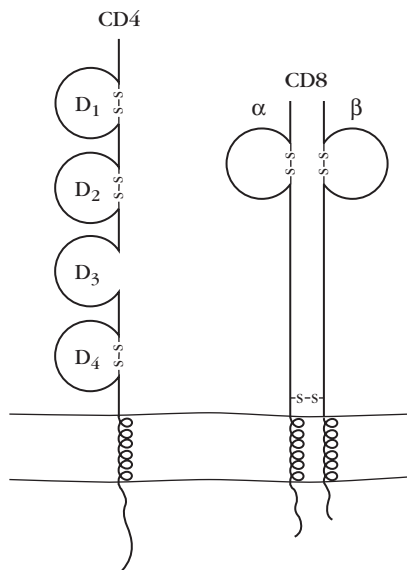


FIGURE 9-10 General structure of the CD4 and CD8 coreceptors. CD8 may take the form of an $\alpha\beta$ heterodimer, or an $\alpha\alpha$ homodimer. The monomeric CD4 molecule contains four Ig-fold domains; each chain in the CD8 molecule contains one.

by contacting the MHC class I $\alpha 2$ and $\alpha 3$ domains as well as having some contact with β_2 -microglobulin (Figure 9-11a). The orientation of the class I $\alpha 3$ domain changes slightly upon binding to CD8. This structure is consistent with a single MHC molecule binding to CD8; no evidence for the possibility of multimeric class I–CD8 complexes was observed. Similar structural data document the mode by which CD4 binds to the class II molecule. The contact between CD4 and MHC II involves contact of the membrane-distal domain of CD4 with a hydrophobic pocket formed by residues from the $\alpha 2$ and $\beta 2$ domains of MHC II. CD4 facilitates signal transduction and T-cell activation of cells recognizing class II–peptide complexes (Figure 9-11b).

Whether there are differences between the roles played by the CD4 and CD8 coreceptors remains open to speculation. Despite the similarities in structure, recall that the nature of the binding of peptide to class I and class II molecules differs in that class I has a closed groove that binds a short peptide with a higher degree of specificity. Recent data shown below indicate that the angle at which the TCR approaches the peptide MHC complex differs between class I and II. The differences in roles played by the CD4 and CD8 coreceptors may be due to these differences in binding requirements. As will be explained in Chapter 10, binding of the CD4 and CD8 molecules serves to transmit stimulatory signals to the T cells; the signal-transduction properties of both CD4 and

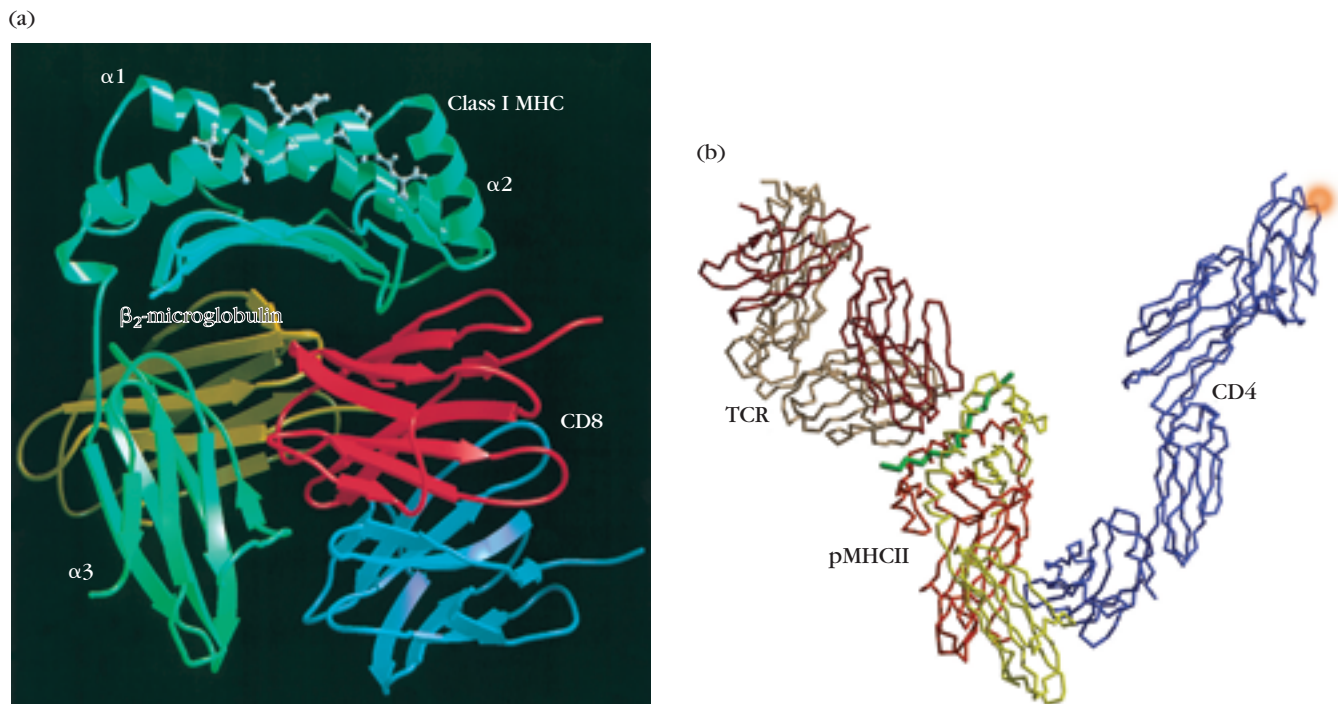


FIGURE 9-11 Interactions of coreceptors with TCR and MHC molecules. (a) Ribbon diagram showing three-dimensional structure of an HLA-A2 MHC class I molecule bound to a CD8 $\alpha\alpha$ homodimer. The HLA-A2 heavy chain is shown in green, β_2 -microglobulin in gold, the CD8 $\alpha 1$ in red, the CD8 $\alpha 2$ in blue, and the bound peptide in white. A flexible loop of the $\alpha 3$ domain (residues 223–229) is in con-

tact with the two CD8 subunits. In this model, the right side of CD8 would be anchored in the T-cell membrane, and the lower left end of the class I MHC molecule (the $\alpha 3$ domain) is attached to the surface of the target cell. (b) Interaction of CD4 with the class II MHC peptide complex (pMHCII). [Part (a) from Gao et al., 1997, *Nature*, **387**:630; part (b) from Wang et al., 2001, *PNAS*, **98**(19): 10799.]

CD8 are mediated through their cytoplasmic domains. Recent data on the interaction between CD4 and the peptide-class II complex indicates that there is very weak affinity between them, suggesting that recruitment of molecules involved in signal transduction may be the major role for CD4.

Affinity of TCR for Peptide-MHC Complexes Is Weak Compared with Antibody Binding

The affinity of T-cell receptors for peptide-MHC complexes is low to moderate, with K_d values ranging from 10^{-4} to 10^{-7} M. This level of affinity is weak compared with

antigen-antibody interactions, which generally have K_d values ranging from 10^{-6} to 10^{-10} M (Figure 9-12a). However, T-cell interactions do not depend solely on binding by the TCR; *cell-adhesion molecules* strengthen the bond between a T cell and an antigen-presenting cell or a target cell. Several accessory membrane molecules, including CD2, LFA-1, CD28, and CD45R bind independently to other ligands on antigen-presenting cells or target cells (see Table 9-4 and Figure 9-12b). Once cell-to-cell contact has been made by the adhesion molecules, the T-cell receptor may scan the membrane for peptide-MHC complexes. During activation of a T cell by a particular peptide-MHC complex, there is a transient increase in the membrane expression of

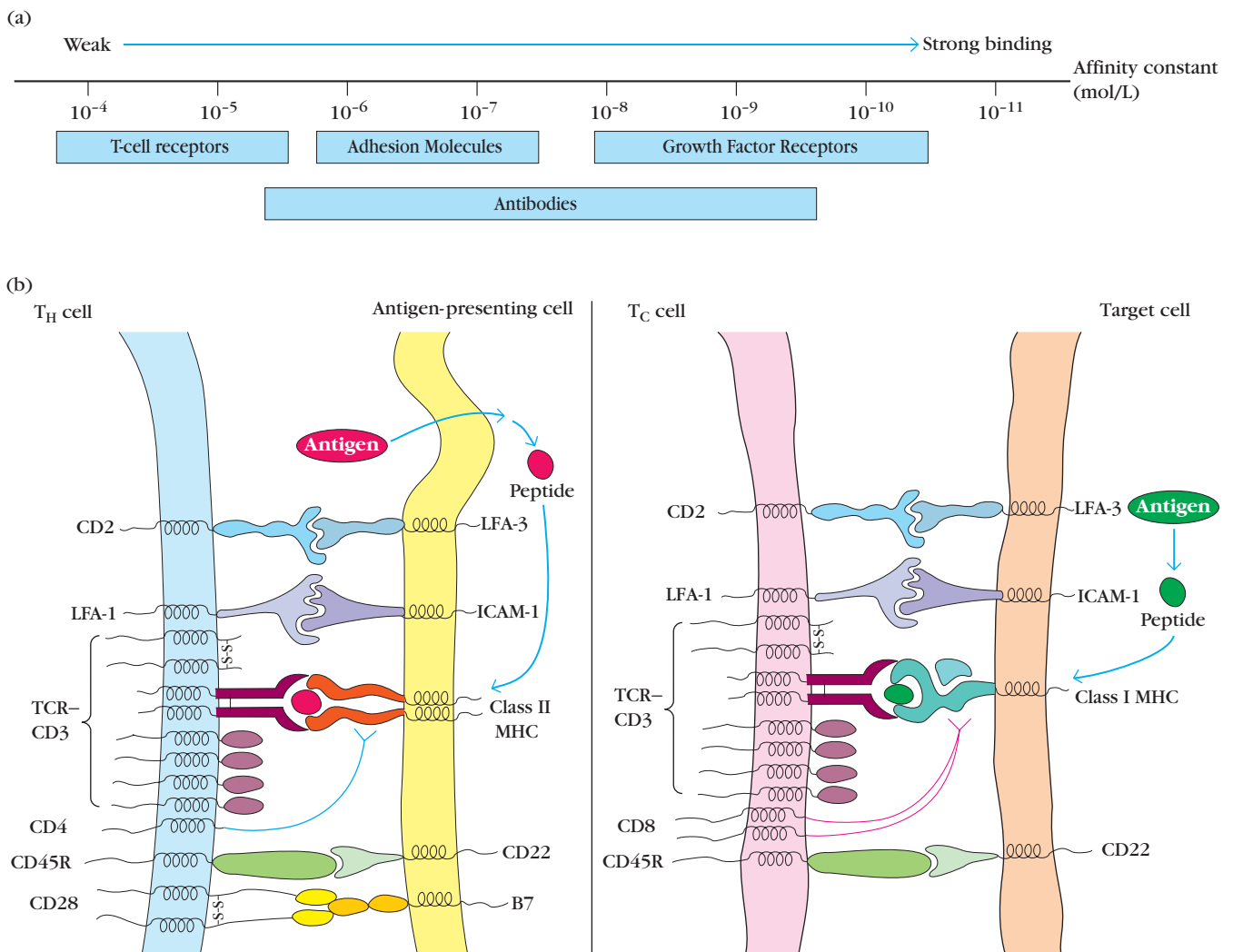


FIGURE 9-12 Role of coreceptors in TCR binding affinity. (a) Affinity constants for various biologic systems. (b) Schematic diagram of the interactions between the T-cell receptor and the peptide-MHC complex and of various accessory molecules with their ligands on an antigen-presenting cell (left) or target cell (right).

Binding of the coreceptors CD4 and CD8 and the other accessory molecules to their ligands strengthens the bond between the interacting cells and/or facilitates the signal transduction that leads to activation of the T cell.

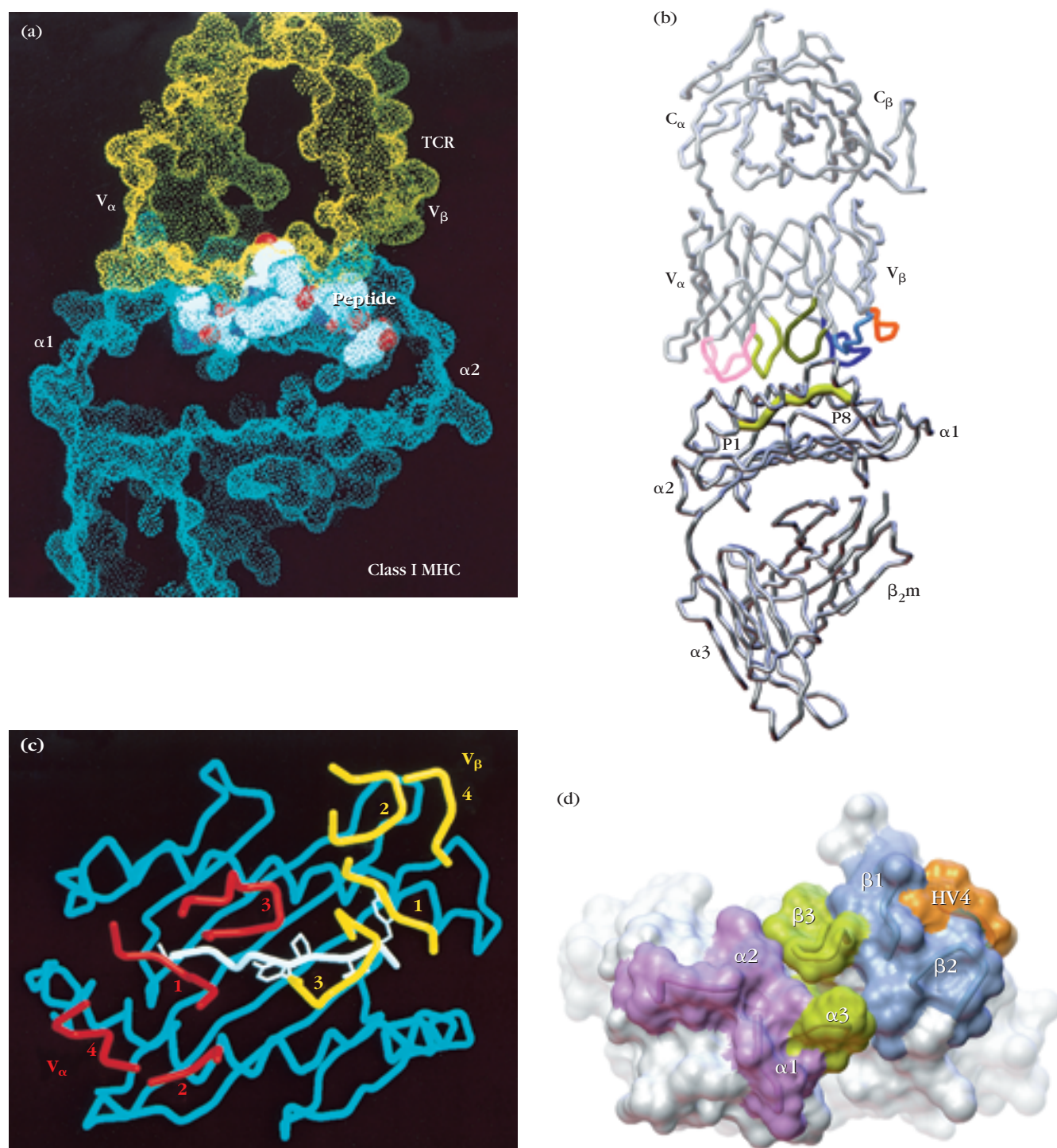


FIGURE 9-13 Three-dimensional structures for the TCR-MHC-peptide complex. (a) Model showing the interaction between the human TCR (top, yellow) and the HLA-A2 class I MHC molecule (bottom, blue) with bound HTLV-I Tax peptide (white and red). (b) Backbone tube diagram of the ternary complex of mouse TCR bound to the class I MHC H-2K^b molecule and peptide (green tube numbered P1–P8). CDR1 and 2 of the TCR α -chain variable domain (V_α) are colored pink; CDR 1 and 2 of the β -chain variable domain (V_β) are blue, and the CDR3s of both chains are green. The HV4 of the β chain is orange. (c) MHC molecule viewed from above (i.e., from top of part (a)), with the hypervariable loops (1–4)

of the human TCR α (red) and β (yellow) variable chains superimposed on the Tax peptide (white) and the $\alpha 1$ and $\alpha 2$ domains of the HLA-A2 MHC class I molecule (blue). (d) CDR regions of mouse TCR α and β chains viewed from above, showing the surface that is involved in binding the MHC-peptide complex. The CDRs are labeled according to their origin (for example, $\alpha 1$ is CDR1 from the α chain). HV4 is the fourth hypervariable region of the β chain. [Parts (a) and (c) from D. N. Garboczi et al., 1996, *Nature* **384**:134–141, courtesy of D. C. Wiley, Harvard University; parts (b) and (d) from C. Garcia et al., 1996, *Science* **274**:209, courtesy of C. Garcia, Scripps Research Institute.]

cell-adhesion molecules, causing closer contact between the interacting cells, which allows cytokines or cytotoxic substances to be transferred more effectively. Soon after activation, the degree of adhesion declines and the T cell detaches from the antigen-presenting cell or target cell. Like CD4 and CD8, some of these other molecules also function as signal-transducers. Their important role is demonstrated by the ability of monoclonal antibodies specific for the binding sites of the cell-adhesion molecules to block T-cell activation.

Three-Dimensional Structures of TCR-Peptide-MHC Complexes

The interaction between the T-cell receptor and an antigen bound to an MHC molecule is central to both humoral and cell-mediated responses. The molecular elements of this interaction have now been described in detail by x-ray crystallography for TCR molecules binding to peptide-MHC class I and class II complexes. A three-dimensional structure has been determined for the trimolecular complex, including TCR α and β chains and an HLA-A2 molecule to which an antigenic peptide is bound. Separate studies describe a mouse TCR molecule bound to peptides complexed with the mouse class I molecule H-2K^b and with the mouse class II IA^k molecule. The comparisons of the TCR complexed with either class I or class II suggest that there are differences in how the TCR contacts the MHC-peptide complex. Newly added to our library of TCR structures is that of a $\gamma\delta$ receptor bound to an antigen that does not require processing.

From x-ray analysis, the TCR-peptide-MHC complex consists of a single TCR molecule bound to a single MHC molecule and its peptide. The TCR contacts the MHC molecule through the TCR variable domains (Figure 9-13 a,b). Although the structures of the constant region of the TCR α chain and the MHC $\alpha 3$ domain were not clearly established by studies of the crystallized human complexes (see Figure 9-13a), the overall area of contact and the structure of the complete TCR variable regions were clear. The constant regions were established by studies of the mouse complex, which showed the orientation proposed for the human models (see Figure 9-13b). Viewing the MHC molecule with its bound peptide from above, we can see that the TCR is situated across it diagonally, relative to the long dimension of the peptide (Figure 9-13c). The CDR3 loops of the TCR α and β chains meet in the center of the peptide; and the CDR1 loop of the TCR α chain is at the N terminus of the peptide, while CDR1 of the β chain is at the C terminus of the peptide. The CDR2 loops are in contact with the MHC molecule; CDR2 α is over the $\alpha 2$ domain alpha helix and CDR2 β over the $\alpha 1$ domain alpha helix (Figure 9-13c). A space-filling model of the binding site viewed from above (looking down into the MHC cleft) indicates that the peptide is buried beneath the TCR and therefore is not seen

from this angle (Figure 9-13d). The data also show that the fourth hypervariable regions of the α and β chains are not in contact with the antigenic peptide.

As predicted from data for immunoglobulins, the recognition of the peptide-MHC complex occurs through the variable loops in the TCR structure. CDR1 and CDR3 from both the TCR α and the TCR β chain contact the peptide and a large area of the MHC molecule. The peptide is buried (see Figure 9-13d) more deeply in the MHC molecule than it is in the TCR, and the TCR molecule fits across the MHC molecule, contacting it through a flat surface of the TCR at the “high points” on the MHC molecule. The fact that the CDR1 region contacts both peptide and MHC suggests that regions other than CDR3 are involved in peptide binding.

TCRs Interact Differently with Class I and Class II Molecules

Can the conclusions drawn from the three-dimensional structure of TCR-peptide-class I complexes be extrapolated to interactions of TCR with class II complexes? Ellis Reinherz and his colleagues resolved this question by analysis of a TCR molecule in complex with a mouse class II molecule and its specific antigen. While the structures of the peptide-binding regions in class I and class II molecules are similar, Chapter 7 showed that there are differences in how they accommodate bound peptide (see Figures 7-10a and b). A comparison of the interactions of a TCR with class I MHC-peptide and class II-peptide reveals a significant difference in the angle at which the TCR molecule sits on the MHC complexes (Figure 9-14). Also notable is a greater number of contact residues between TCR and class II MHC, which is consistent with the known higher affinity of interaction. However, it remains to be seen whether the evident difference in the number of contact points will be true for all class I and II structures.

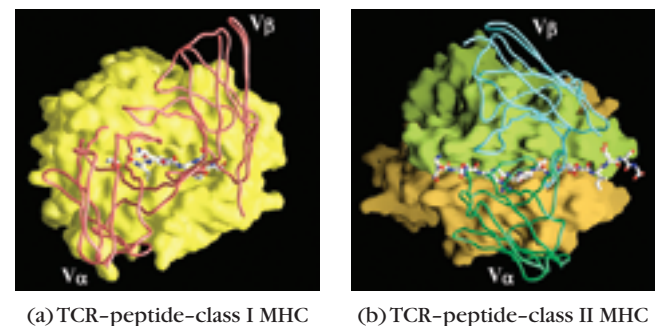


FIGURE 9-14 Comparison of the interactions between $\alpha\beta$ TCR and (a) class I MHC-peptide, and (b) class II MHC-peptide. The TCR (wire diagram) is red in (a), blue-green in (b); the MHC molecules are shown as surface models; peptide is shown as ball and stick. [From Reinherz et al., 1999, *Science* **286**:1913.]

Alloreactivity of T Cells

The preceding sections have focused on the role of MHC molecules in the presentation of antigen to T cells and the interactions of TCRs with peptide-MHC complexes. However, as noted in Chapter 7, MHC molecules were first identified because of their role in rejection of foreign tissue. Graft-rejection reactions result from the direct response of T cells to MHC molecules, which function as **histocompatibility antigens**. Because of the extreme polymorphism of the MHC, most individuals of the same species have unique sets of MHC molecules, or histocompatibility antigens, and are considered to be **allogeneic**, a term used to describe genetically different individuals of the same species (see Chapter 21). Therefore, T cells respond even to **allografts** (grafts from members of the same species), and MHC molecules are considered **alloantigens**. Generally, CD4⁺ T cells are alloreactive to class II alloantigens, and CD8⁺ T cells respond to class I alloantigens.

The alloreactivity of T cells is puzzling for two reasons. First, the ability of T cells to respond to allogeneic histocompatibility antigens alone appears to contradict all the evidence indicating that T cells can respond only to foreign antigen plus *self*-MHC molecules. In responding to allogeneic grafts, however, T cells recognize a *foreign* MHC molecule directly. A second problem posed by the T-cell response to allogeneic MHC molecules is that the frequency of alloreactive T cells is quite high; it has been estimated that 1%–5% of all T cells are reactive to a given alloantigen, which is higher than the normal frequency of T cells reactive with any particular foreign antigenic peptide plus self-MHC molecule. This high frequency of alloreactive T cells appears to contradict the basic tenet of clonal selection. If 1 T cell in 20 reacts with a given alloantigen and if one assumes there are on the order of 100 distinct H-2 haplotypes in mice, then there are not enough distinct T-cell specificities to cover all the unique H-2 alloantigens, let alone foreign antigens displayed by self-MHC molecules.

One possible and biologically satisfying explanation for the high frequency of alloreactive T cells is that a particular T-cell receptor specific for a foreign antigenic peptide plus a self-MHC molecule can also cross-react with certain allogeneic MHC molecules. In other words, if an allogeneic MHC molecule plus allogeneic peptide structurally resembles a processed foreign peptide plus self-MHC molecule, the same T-cell receptor may recognize both peptide-MHC complexes. Since allogeneic cells express on the order of 10⁵ class I MHC molecules per cell, T cells bearing low-affinity cross-reactive receptors might be able to bind by virtue of the high density of membrane alloantigen. Foreign antigen, on the other hand, would be sparsely displayed on the membrane of an antigen-presenting cell or altered self-cell associated with class I or class II MHC molecules, limiting responsiveness to only those T cells bearing high-affinity receptors.

Information relevant to mechanisms for alloreactivity was gained by Reiser and colleagues, who determined the structure of a mouse TCR complexed with an allogeneic class I molecule containing a bound octapeptide. This analysis revealed a structure similar to those reported for TCR bound to class I self-MHC complexes, leading the authors to conclude that allogeneic recognition is not unlike recognition of self-MHC antigens. The absence of negative selection for the peptides contained in the foreign MHC molecules can contribute to the high frequency of alloreactive T cells. This condition, coupled with the differences in the structure of the exposed portions of the allogeneic MHC molecule, may account for the phenomenon of alloreactivity. An explanation for the large number of alloreactive cells can be found in the large number of potential antigens provided by the foreign molecule plus the possible peptide antigens bound by them.

SUMMARY

- Most T-cell receptors, unlike antibodies, do not react with soluble antigen but rather with processed antigen bound to a self-MHC molecule; certain $\gamma\delta$ receptors recognize antigens not processed and presented with MHC.
- T-cell receptors, first isolated by means of clonotypic monoclonal antibodies, are heterodimers consisting of an α and β chain or a γ and δ chain.
- The membrane-bound T-cell receptor chains are organized into variable and constant domains. TCR domains are similar to those of immunoglobulins and the V region has hypervariable regions.
- TCR germ-line DNA is organized into multigene families corresponding to the α , β , γ , and δ chains. Each family contains multiple gene segments.
- The mechanisms that generate TCR diversity are generally similar to those that generate antibody diversity, although somatic mutation does not occur in TCR genes, as it does in immunoglobulin genes.
- The T-cell receptor is closely associated with the CD3, a complex of polypeptide chains involved in signal transduction.
- T cells express membrane molecules, including CD4, CD8, CD2, LFA-1, CD28, and CD45R, that play accessory roles in T-cell function or signal transduction.
- Formation of the ternary complex TCR-antigen-MHC requires binding of a peptide to the MHC molecule and binding of the complex by the T-cell receptor.
- Interactions between TCR and MHC class I/peptide differ from those with MHC class II/peptide in the contact points between the TCR and MHC molecules.
- The $\gamma\delta$ T-cell receptor is distinguished by ability to bind native antigens and by differences in the orientation of the variable and constant regions.

- In addition to reaction with self MHC plus foreign antigens, T cells also respond to foreign MHC molecules, a reaction that leads to rejection of allogeneic grafts.

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USEFUL WEB SITES

<http://imgt.cines.fr>

A comprehensive database of genetic information on TCRs, MHC molecules, and immunoglobulins, from the International ImmunoGenetics Database, University of Montpellier, France.

<http://www.bioscience.org/knockout/tcrab.htm>

This location presents a brief summary of the effects of TCR knockouts.

Study Questions

CLINICAL FOCUS QUESTION A patient presents with an enlarged lymph node, and a T-cell lymphoma is suspected. However, DNA sampled from biopsied tissue shows no evidence of a predominant gene rearrangement when probed with α and β TCR genes. What should be done next to rule out lymphocyte malignancy?

1. Indicate whether each of the following statements is true or false. If you think a statement is false, explain why.
 - a. Monoclonal antibody specific for CD4 will coprecipitate the T-cell receptor along with CD4.
 - b. Subtractive hybridization can be used to enrich for mRNA that is present in one cell type but absent in another cell type within the same species.
 - c. Clonotypic monoclonal antibody was used to isolate the T-cell receptor.
 - d. The T cell uses the same set of V, D, and J gene segments as the B cell but uses different C gene segments.
 - e. The $\alpha\beta$ TCR is bivalent and has two antigen-binding sites.
 - f. Each $\alpha\beta$ T cell expresses only one β -chain and one α -chain allele.
 - g. Mechanisms for generation of diversity of T-cell receptors are identical to those used by immunoglobulins.
 - h. The Ig- α /Ig- β heterodimer and CD3 serve analogous functions in the B-cell receptor and T-cell receptor, respectively.
2. What led Zinkernagel and Doherty to conclude that T-cell receptor recognition requires both antigen and MHC molecules?
3. Draw the basic structure of the $\alpha\beta$ T-cell receptor and compare it with the basic structure of membrane-bound immunoglobulin.
4. Several membrane molecules, in addition to the T-cell receptor, are involved in antigen recognition and T-cell activation. Describe the properties and distinct functions of the following T-cell membrane molecules: (a) CD3, (b) CD4 and CD8, and (c) CD2.
5. Indicate whether each of the properties listed below applies to the T-cell receptor (TCR), B-cell immunoglobulin (Ig), or both (TCR/Ig).
 - a. _____ Is associated with CD3
 - b. _____ Is monovalent
 - c. _____ Exists in membrane-bound and secreted forms
 - d. _____ Contains domains with the immunoglobulin-fold structure
 - e. _____ Is MHC restricted
 - f. _____ Exhibits diversity generated by imprecise joining of gene segments
 - g. _____ Exhibits diversity generated by somatic mutation
6. A major obstacle to identifying and cloning TCR genes is the low level of TCR mRNA in T cells.
 - a. To overcome this obstacle, Hedrick and Davis made three important assumptions that proved to be correct. Describe each assumption and how it facilitated identification of the genes that encode the T-cell receptor.



- b. Suppose, instead, that Hedrick and Davis wanted to identify the genes that encode IL-4. What changes in the three assumptions should they make?
7. Hedrick and Davis used the technique of subtractive hybridization to isolate cDNA clones that encode the T-cell receptor. You wish to use this technique to isolate cDNA clones that encode several gene products and have available clones of various cell types to use as the source of cDNA or mRNA for hybridization. For each gene product listed in the left column of the table below, select the most appropriate.

Gene product	cDNA source	mRNA source
IL-2		
CD8		
J chain		
IL-1		
CD3		

cDNA and mRNA source clones are from the following cell types: T_H1 cell line (A); T_H2 cell line (B); T_C cell line (C); macrophage (D); IgA-secreting myeloma cell (E); IgG-secreting myeloma cell (F); myeloid progenitor cell (G); and B-cell line (H). More than one cell type may be correct in some cases.

8. Mice from different inbred strains listed in the left column of the accompanying table were infected with LCM virus. Spleen cells derived from these LCM-infected mice were then tested for their ability to lyse LCM-infected ⁵¹Cr-labeled target cells from the strains listed across the top of the table. Indicate with (+) or (−) whether you would expect to see ⁵¹Cr released from the labeled target cells.

Source of spleen cells from LCM-infected mice	Release of ⁵¹ Cr from LCM-infected target cells			
	B10.D2 (H-2 ^d)	B10 (H-2 ^b)	B10.BR (H-2 ^k)	(BALB/c × B10) F ₁ (H-2 ^{b/d})
B10.D2 (H-2 ^d)				
B10 (H-2 ^b)				
BALB/c (H-2 ^d)				
BALB/b (H-2 ^b)				

9. The γδ T-cell receptor differs from the αβ in both structural and functional parameters. Describe how they are *similar* to one another and different from the B-cell antigen receptors.

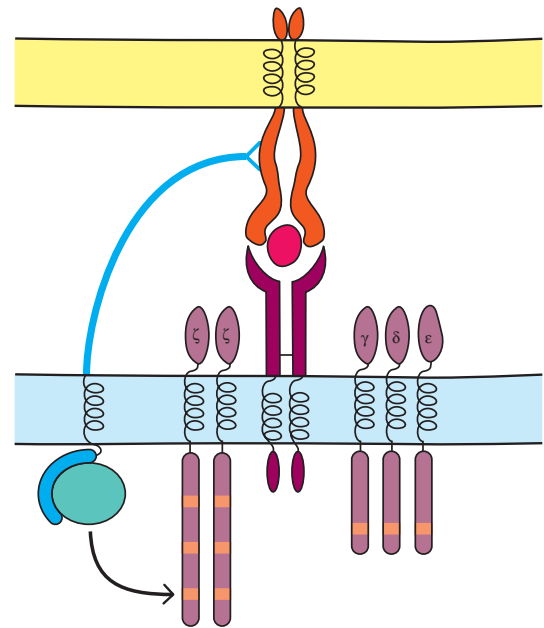
T-Cell Maturation, Activation, and Differentiation

THE ATTRIBUTE THAT DISTINGUISHES ANTIGEN recognition by most T cells from recognition by B cells is MHC restriction. In most cases, both the maturation of progenitor T cells in the thymus and the activation of mature T cells in the periphery are influenced by the involvement of MHC molecules. The potential antigenic diversity of the T-cell population is reduced during maturation by a selection process that allows only MHC-restricted and nonself-reactive T cells to mature. The final stages in the maturation of most T cells proceed along two different developmental pathways, which generate functionally distinct $CD4^+$ and $CD8^+$ subpopulations that exhibit class II and class I MHC restriction, respectively.

Activation of mature peripheral T cells begins with the interaction of the T-cell receptor (TCR) with an antigenic peptide displayed in the groove of an MHC molecule. Although the specificity of this interaction is governed by the TCR, its low avidity necessitates the involvement of coreceptors and other accessory membrane molecules that strengthen the TCR-antigen-MHC interaction and transduce the activating signal. Activation leads to the proliferation and differentiation of T cells into various types of effector cells and memory T cells. Because the vast majority of thymocytes and peripheral T cells express the $\alpha\beta$ T-cell receptor rather than the $\gamma\delta$ T-cell receptor, all references to the T-cell receptor in this chapter denote the $\alpha\beta$ receptor unless otherwise indicated. Similarly, unless otherwise indicated, all references to T cells denote those $\alpha\beta$ receptor-bearing T cells that undergo MHC restriction.

T-Cell Maturation and the Thymus

Progenitor T cells from the early sites of hematopoiesis begin to migrate to the thymus at about day 11 of gestation in mice and in the eighth or ninth week of gestation in humans. In a manner similar to B-cell maturation in the bone marrow, T-cell maturation involves rearrangements of the germ-line TCR genes and the expression of various membrane markers. In the thymus, developing T cells, known as **thymocytes**, proliferate and differentiate along developmental pathways that generate functionally distinct subpopulations of mature T cells.



Engagement of TcR by Peptide: MHC Initiates Signal Transduction

- T-Cell Maturation and the Thymus
- Thymic Selection of the T-Cell Repertoire
- T_H -Cell Activation
- T-Cell Differentiation
- Cell Death and T-Cell Populations
- Peripheral $\gamma\delta$ T-Cells

As indicated in Chapter 2, the thymus occupies a central role in T-cell biology. Aside from being the main source of all T cells, it is where T cells diversify and then are shaped into an effective primary T-cell repertoire by an extraordinary pair of selection processes. One of these, **positive selection**, permits the survival of only those T cells whose TCRs are capable of recognizing self-MHC molecules. It is thus responsible for the creation of a self-MHC-restricted repertoire of T cells. The other, **negative selection**, eliminates T cells that react too strongly with self-MHC or with self-MHC plus self-peptides. It is an extremely important factor in generating a primary T-cell repertoire that is self-tolerant.

As shown in Figure 10-1, when T-cell precursors arrive at the thymus, they do not express such signature surface markers of T cells as the T-cell receptor, the CD3 complex, or the coreceptors CD4 and CD8. In fact, these progenitor cells have



VISUALIZING CONCEPTS

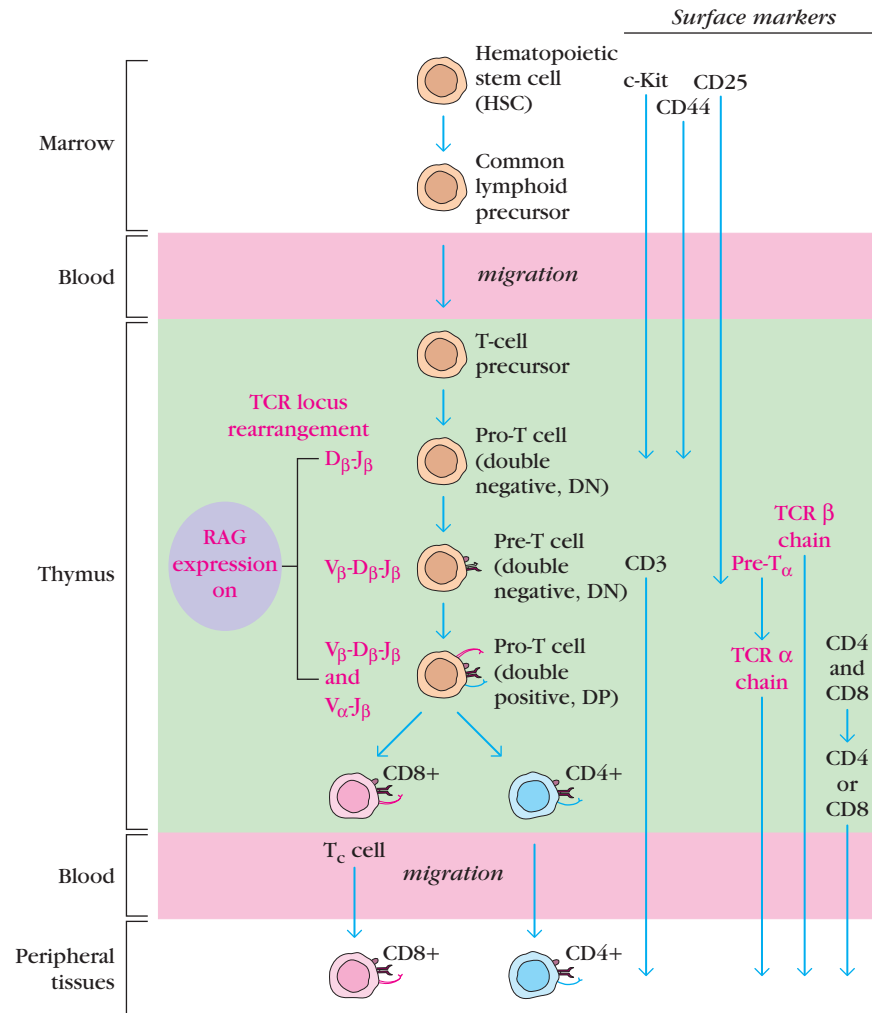


FIGURE 10-1 Development of $\alpha\beta$ T cells in the mouse. T-cell precursors arrive at the thymus from bone marrow via the bloodstream, undergo development to mature T cells, and are exported to the periphery where they can undergo antigen-induced activation and differentiation into effector cells and memory cells. Each stage of development is characterized by stage-specific intracellular events and the display of distinctive cell-surface markers.

not yet rearranged their TCR genes and do not express proteins, such as RAG-1 and RAG-2, that are required for rearrangement. After arriving at the thymus, these T-cell precursors enter the outer cortex and slowly proliferate. During approximately three weeks of development in the thymus, the differentiating T cells progress through a series of stages that are marked by characteristic changes in their cell-surface phenotype. For example, as mentioned previously, thymocytes early in development lack detectable CD4 and CD8. Because these cells are CD4⁻CD8⁻, they are referred to as **double-negative (DN)** cells.

Even though these coreceptors are not expressed during the DN early stages, the differentiation program is progressing and is marked by changes in the expression of such cell surface molecules as c-Kit, CD44, and CD25. The initial thymocyte population displays c-Kit, the receptor for stem-cell growth factor, and CD44, an adhesion molecule involved in homing; CD25, the β -chain of the IL-2 receptor, also appears

on early-stage DN cells. During this period, the cells are proliferating but the TCR genes remain unrearranged. Then the cells stop expressing c-Kit, markedly reduce CD44 expression, turn on expression of the recombinase genes RAG-1 and RAG-2 and begin to rearrange their TCR genes. Although it is not shown in Figure 10-1, a small percentage (<5%) of thymocytes productively rearrange the γ - and δ -chain genes and develop into double-negative CD3⁺ $\gamma\delta$ T cells. In mice, this thymocyte subpopulation can be detected by day 14 of gestation, reaches maximal numbers between days 17 and 18, and then declines until birth (Figure 10-2).

Most double-negative thymocytes progress down the $\alpha\beta$ developmental pathway. They stop proliferating and begin to rearrange the TCR β -chain genes, then express the β chain. Those cells of the $\alpha\beta$ lineage that fail to productively rearrange and express β chains die. Newly synthesized β chains combine with a 33-kDa glycoprotein known as the pre-T α chain and associate with the CD3 group to form a novel com-

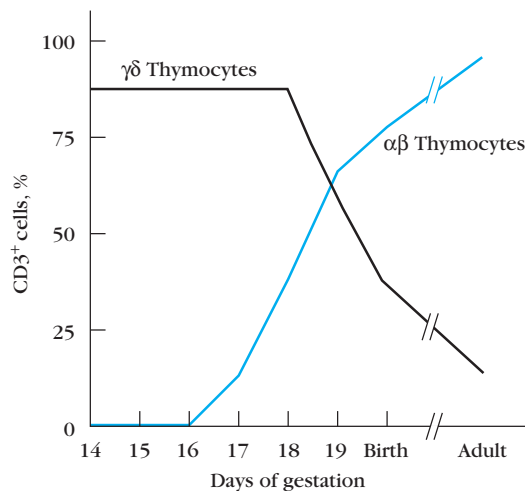


FIGURE 10-2 Time course of appearance of $\gamma\delta$ thymocytes and $\alpha\beta$ thymocytes during mouse fetal development. The graph shows the percentage of $CD3^+$ cells in the thymus that are double-negative ($CD4^-8^-$) and bear the $\gamma\delta$ T-cell receptor (black) or are double-positive ($CD4^+8^+$) and bear the $\alpha\beta$ T-cell receptor (blue).

plex called the **pre-T-cell receptor** or **pre-TCR** (Figure 10-3). Some researchers have suggested that the pre-TCR recognizes some intra-thymic ligand and transmits a signal through the CD3 complex that activates signal-transduction pathways that have several effects:

- Indicates that a cell has made a productive TCR β -chain rearrangement and signals its further proliferation and maturation.

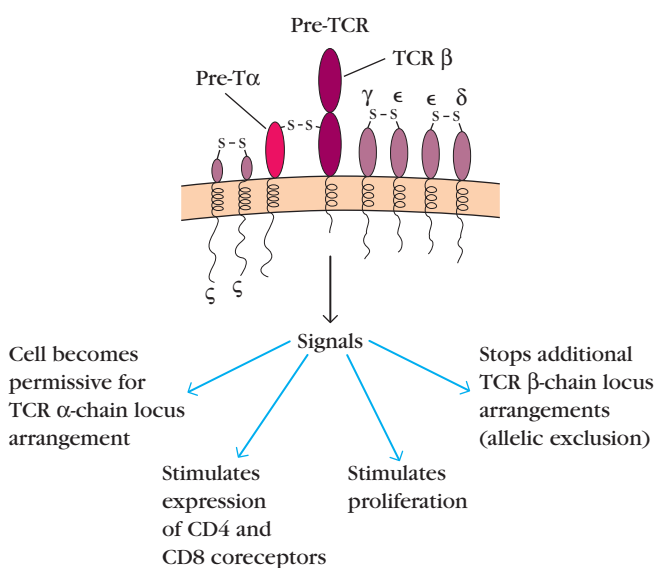


FIGURE 10-3 Structure and activity of the pre-T-cell receptor (pre-TCR). Binding of ligands yet to be identified to the pre-TCR generates intracellular signals that induce a variety of processes.

- Suppresses further rearrangement of TCR β -chain genes, resulting in allelic exclusion.
- Renders the cell permissive for rearrangement of the TCR α chain.
- Induces developmental progression to the $CD4^+8^+$ *double-positive* state.

After advancing to the **double-positive (DP)** stage, where both CD4 and CD8 coreceptors are expressed, the thymocytes begin to proliferate. However, during this proliferative phase, TCR α -chain gene rearrangement does not occur; both the *RAG-1* and *RAG-2* genes are transcriptionally active, but the RAG-2 protein is rapidly degraded in proliferating cells, so rearrangement of the α -chain genes cannot take place. The rearrangement of α -chain genes does not begin until the double-positive thymocytes stop proliferating and RAG-2 protein levels increase. The proliferative phase prior to the rearrangement of the α -chain increases the diversity of the T-cell repertoire by generating a clone of cells with a single TCR β -chain rearrangement. Each of the cells within this clone can then rearrange a different α -chain gene, thereby generating a much more diverse population than if the original cell had first undergone rearrangement at both the β - and α -chain loci before it proliferated. In mice, the TCR α -chain genes are not expressed until day 16 or 17 of gestation; double-positive cells expressing both CD3 and the $\alpha\beta$ T-cell receptor begin to appear at day 17 and reach maximal levels about the time of birth (see Figure 10-2). The possession of a complete TCR enables DP thymocytes to undergo the rigors of positive and negative selection.

T-cell development is an expensive process for the host. An estimated 98% of all thymocytes do not mature—they die by apoptosis within the thymus either because they fail to make a productive TCR-gene rearrangement or because they fail to survive thymic selection. Double-positive thymocytes that express the $\alpha\beta$ TCR-CD3 complex and survive thymic selection develop into immature **single-positive $CD4^+$** thymocytes or **single-positive $CD8^+$** thymocytes. These single-positive cells undergo additional negative selection and migrate from the cortex to the medulla, where they pass from the thymus into the circulatory system.

Thymic Selection of the T-Cell Repertoire

Random gene rearrangement within TCR germ-line DNA combined with junctional diversity can generate an enormous TCR repertoire, with an estimated potential diversity exceeding 10^{15} for the $\alpha\beta$ receptor and 10^{18} for the $\gamma\delta$ receptor. Gene products encoded by the rearranged TCR genes have no inherent affinity for foreign antigen plus a self-MHC molecule; they theoretically should be capable of recognizing soluble antigen (either foreign or self), self-MHC molecules, or

antigen plus a nonself-MHC molecule. Nonetheless, the most distinctive property of mature T cells is that they recognize only foreign antigen combined with self-MHC molecules.

As noted, thymocytes undergo two selection processes in the thymus:

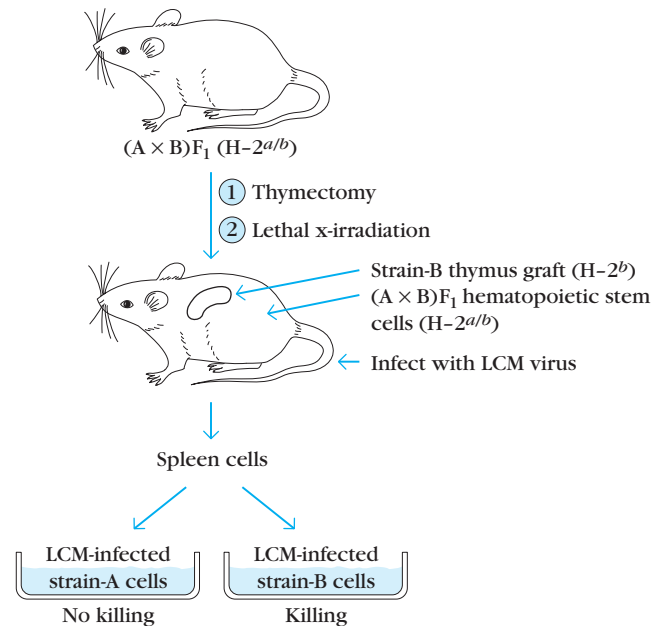
- Positive selection for thymocytes bearing receptors capable of binding self-MHC molecules, which results in **MHC restriction**. Cells that fail positive selection are eliminated within the thymus by apoptosis.
- Negative selection that eliminates thymocytes bearing high-affinity receptors for self-MHC molecules alone or self-antigen presented by self-MHC, which results in **self-tolerance**.

Both processes are necessary to generate mature T cells that are self-MHC restricted and self-tolerant. As noted already, some 98% or more of all thymocytes die by apoptosis within the thymus. The bulk of this high death rate appears to reflect a weeding out of thymocytes that fail positive selection because their receptors do not specifically recognize foreign antigen plus self-MHC molecules.

Early evidence for the role of the thymus in selection of the T-cell repertoire came from chimeric mouse experiments by R. M. Zinkernagel and his colleagues (Figure 10-4). These researchers implanted thymectomized and irradiated $(A \times B) F_1$ mice with a B-type thymus and then reconstituted the animal's immune system with an intravenous infusion of F_1 bone-marrow cells. To be certain that the thymus graft did not contain any mature T cells, it was irradiated before being transplanted. In such an experimental system, T-cell progenitors from the $(A \times B) F_1$ bone-marrow transplant mature within a thymus that expresses only B-haplotype MHC molecules on its stromal cells. Would these $(A \times B) F_1$ T cells now be MHC-restricted for the haplotype of the thymus? To answer this question, the chimeric mice were infected with LCM virus and the immature T cells were then tested for their ability to kill LCM-infected target cells from the strain A or strain B mice. As shown in Figure 10-4, when T_C cells from the chimeric mice were tested on LCM virus infected target cells from strain A or strain B mice, they could only lyse LCM-infected target cells from strain B mice. These mice have the same MHC haplotype, B, as the implanted thymus. Thus, the MHC haplotype of the thymus in which T cells develop determines their MHC restriction.

Thymic stromal cells, including epithelial cells, macrophages, and dendritic cells, play essential roles in positive and negative selection. These cells express class I MHC molecules and can display high levels of class II MHC also. The interaction of immature thymocytes that express the TCR-CD3 complex with populations of thymic stromal cells results in positive and negative selection by mechanisms that are under intense investigation. First, we'll examine the details of each selection process and then study some experiments that provide insights into the operation of these processes.

EXPERIMENT



CONTROL

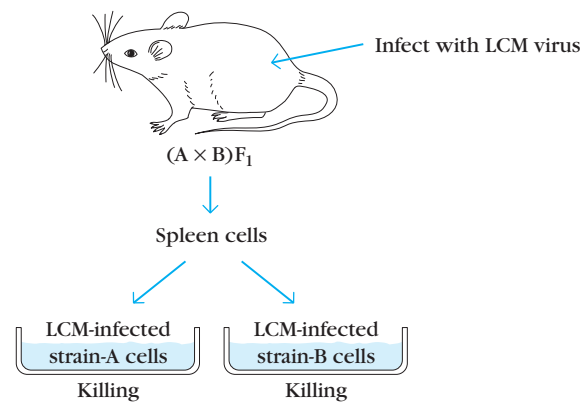


FIGURE 10-4 Experimental demonstration that the thymus selects for maturation only those T cells whose T-cell receptors recognize antigen presented on target cells with the haplotype of the thymus. Thymectomized and lethally irradiated $(A \times B) F_1$ mice were grafted with a strain-B thymus and reconstituted with $(A \times B) F_1$ bone-marrow cells. After infection with the LCM virus, the CTL cells were assayed for their ability to kill ^{51}Cr -labeled strain-A or strain-B target cells infected with the LCM virus. Only strain-B target cells were lysed, suggesting that the $H-2^b$ grafted thymus had selected for maturation only those T cells that could recognize antigen combined with $H-2^b$ MHC molecules.

Positive Selection Ensures MHC Restriction

Positive selection takes place in the cortical region of the thymus and involves the interaction of immature thymocytes with cortical epithelial cells (Figure 10-5). There is evidence that the T-cell receptors on thymocytes tend to cluster with

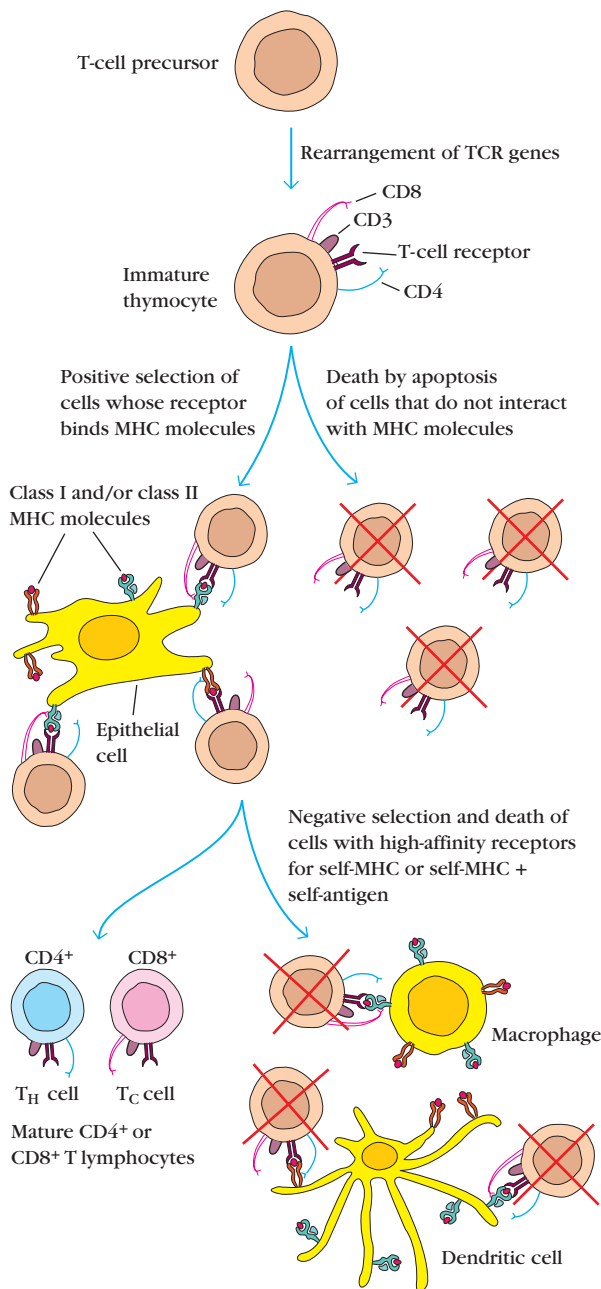


FIGURE 10-5 Positive and negative selection of thymocytes in the thymus. Thymic selection involves thymic stromal cells (epithelial cells, dendritic cells, and macrophages), and results in mature T cells that are both self-MHC restricted and self-tolerant.

MHC molecules on the cortical cells at sites of cell-cell contact. Some researchers have suggested that these interactions allow the immature thymocytes to receive a protective signal that prevents them from undergoing cell death; cells whose receptors are not able to bind MHC molecules would not interact with the thymic epithelial cells and consequently would not receive the protective signal, leading to their death by apoptosis.

During positive selection, the RAG-1, RAG-2, and TdT proteins required for gene rearrangement and modification continue to be expressed. Thus each of the immature thymocytes in a clone expressing a given β chain have an opportunity to rearrange different TCR α -chain genes, and the resulting TCRs are then selected for self-MHC recognition. Only those cells whose $\alpha\beta$ TCR heterodimer recognizes a self-MHC molecule are selected for survival. Consequently, the presence of more than one combination of $\alpha\beta$ TCR chains among members of the clone is important because it increases the possibility that some members will “pass” the test for positive selection. Any cell that manages to rearrange an α chain that allows the resulting $\alpha\beta$ TCR to recognize self-MHC will be spared; all members of the clone that fail to do so will die by apoptosis within 3 to 4 days.

Negative Selection Ensures Self-Tolerance

The population of MHC-restricted thymocytes that survive positive selection comprises some cells with low-affinity receptors for self-antigen presented by self-MHC molecules and other cells with high-affinity receptors. The latter thymocytes undergo negative selection by an interaction with thymic stromal cells. During negative selection, dendritic cells and macrophages bearing class I and class II MHC molecules interact with thymocytes bearing high-affinity receptors for self-antigen plus self-MHC molecules or for self-MHC molecules alone (see Figure 10-5). However, the precise details of the process are not yet known. Cells that experience negative selection are observed to undergo death by apoptosis. Tolerance to self-antigens encountered in the thymus is thereby achieved by eliminating T cells that are reactive to these antigens.

Experiments Revealed the Essential Elements of Positive and Negative Selection

Direct evidence that binding of thymocytes to class I or class II MHC molecules is required for positive selection in the thymus came from experimental studies with knockout mice incapable of producing functional class I or class II MHC molecules (Table 10-1). Class I-deficient mice were found to have a normal distribution of double-negative, double-positive, and $CD4^+$ thymocytes, but failed to produce $CD8^+$ thymocytes. Class II-deficient mice had double-negative, double-positive, and $CD8^+$ thymocytes but lacked $CD4^+$ thymocytes. Not surprisingly, the lymph nodes of these class II-deficient mice lacked $CD4^+$ T cells. Thus, the absence of class I or II MHC molecules prevents positive selection of $CD8^+$ or $CD4^+$ T cells, respectively.

Further experiments with transgenic mice provided additional evidence that interaction with MHC molecules plays a role in positive selection. In these experiments, rearranged $\alpha\beta$ -TCR genes derived from a $CD8^+$ T-cell clone specific for influenza antigen plus H-2^k class I MHC molecules were injected into fertilized eggs from two different mouse strains,

TABLE 10-1 Effect of class I or II MHC deficiency on thymocyte populations*

Cell type	KNOCKOUT MICE		
	Control mice	Class I deficient	Class II deficient
CD4 [−] CD8 [−]	+	+	+
CD4 ⁺ CD8 ⁺	+	+	+
CD4 ⁺	+	+	−
CD8 ⁺	+	−	−

*Plus sign indicates normal distribution of indicated cell types in thymus.
Minus sign indicates absence of cell type.

one with the H-2^k haplotype and one with the H-2^d haplotype (Figure 10-6). Since the receptor transgenes were already rearranged, other TCR-gene rearrangements were suppressed in the transgenic mice; therefore, a high percentage of the thymocytes in the transgenic mice expressed the T-cell receptor encoded by the transgene. Thymocytes expressing the TCR transgene were found to mature into CD8⁺ T cells only in the transgenic mice with the H-2^k class I MHC haplotype (i.e., the haplotype for which the transgene receptor was restricted). In transgenic mice with a different MHC haplotype (H-2^d), immature, double-positive thymocytes expressing the transgene were present, but these thymocytes failed to mature into CD8⁺ T cells. These findings confirmed that interaction between T-cell receptors on immature thymocytes and self-MHC molecules is required for positive selection. In the absence of self-MHC molecules, as in the H-2^d transgenic mice, positive selection and subsequent maturation do not occur.

Evidence for deletion of thymocytes reactive with self-antigen plus MHC molecules comes from a number of experimental systems. In one system, thymocyte maturation was analyzed in transgenic mice bearing an αβ TCR transgene specific for the class I D^b MHC molecule plus H-Y antigen, a small protein that is encoded on the Y chromosome and is therefore a self-molecule only in male mice. In this experiment, the MHC haplotype of the transgenic mice was H-2^b, the same as the MHC restriction of the transgene-encoded receptor. Therefore any differences in the selection of thymocytes in male and female transgenics would be related to the presence or absence of H-Y antigen.

Analysis of thymocytes in the transgenic mice revealed that female mice contained thymocytes expressing the H-Y-specific TCR transgene, but male mice did not (Figure 10-7). In other words, H-Y-reactive thymocytes were self-reactive in the male mice and were eliminated. However, in the female transgenics, which did not express the H-Y antigen, these cells were not self-reactive and thus were not eliminated. When thymocytes from these male transgenic mice were cul-

tured in vitro with antigen-presenting cells expressing the H-Y antigen, the thymocytes were observed to undergo apoptosis, providing a striking example of negative selection.

Some Central Issues in Thymic Selection Remain Unresolved

Although a great deal has been learned about the developmental processes that generate mature CD4⁺ and CD8⁺ T cells, some mysteries persist. Prominent among them is a seeming paradox: If positive selection allows only thymocytes reactive with self-MHC molecules to survive, and negative selection eliminates the self-MHC-reactive thymocytes, then no T cells would be allowed to mature. Since this is not the outcome of T-cell development, clearly, other factors operate to prevent these two MHC-dependent processes from eliminating the entire repertoire of MHC-restricted T cells.

Experimental evidence from fetal thymic organ culture (FTOC) has been helpful in resolving this puzzle. In this system, mouse thymic lobes are excised at a gestational age of day 16 and placed in culture. At this time, the lobes consist predominantly of CD4[−]8[−] thymocytes. Because these immature, double-negative thymocytes continue to develop in the organ culture, thymic selection can be studied under conditions that permit a range of informative experiments. Particular use has

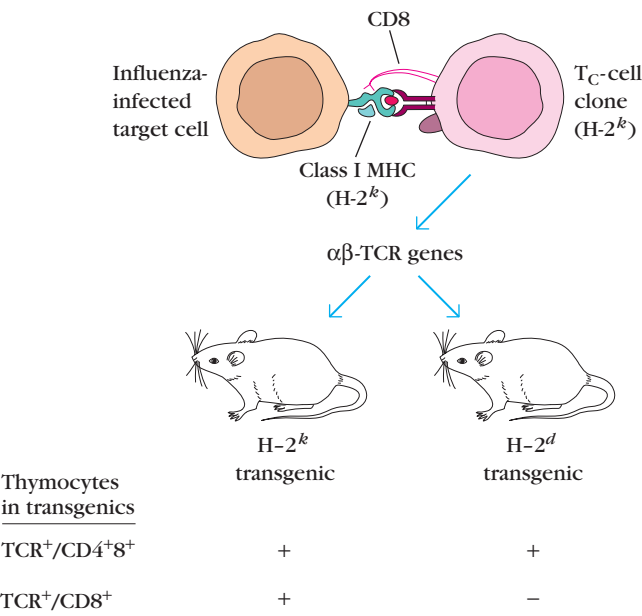


FIGURE 10-6 Effect of host haplotype on T-cell maturation in mice carrying transgenes encoding an H-2^b class I-restricted T-cell receptor specific for influenza virus. The presence of the rearranged TCR transgenes suppressed other gene rearrangements in the transgenics; therefore, most of the thymocytes in the transgenics expressed the αβ T-cell receptor encoded by the transgene. Immature double-positive thymocytes matured into CD8⁺ T cells only in transgenics with the haplotype (H-2^k) corresponding to the MHC restriction of the TCR transgene.

while others can deliver a complete signal. In this model, positive selection takes place when the TCRs of developing thymocytes encounter MHC-peptide complexes that deliver weak or partial signals to their receptors, and negative selection results when the signal is complete. At this point it is not possible to decide between the avidity model and the differential-signaling model; both have experimental support. It may be that in some cases, one of these mechanisms operates to the complete exclusion of the other. It is also possible that no single mechanism accounts for all the outcomes in the cellular interactions that take place in the thymus and more than one mechanism may play a significant role. Further work is required to complete our understanding of this matter.

The differential expression of the coreceptor CD8 also can affect thymic selection. In an experiment in which CD8 ex-

pression was artificially raised to twice its normal level, the concentration of mature CD8⁺ cells in the thymus was one-thirteenth of the concentration in control mice bearing normal levels of CD8 on their surface. Since the interaction of T cells with class I MHC molecules is strengthened by participation of CD8, perhaps the increased expression of CD8 would increase the avidity of thymocytes for class I molecules, possibly making their negative selection more likely.

Another important open question in thymic selection is how double-positive thymocytes are directed to become either CD4⁺8⁻ or CD4⁻8⁺ T cells. Selection of CD4⁺8⁺ thymocytes gives rise to class I MHC-restricted CD8⁺ T cells and class II-restricted CD4⁺ T cells. Two models have been proposed to explain the transformation of a double-positive precursor into one of two different single-positive lineages

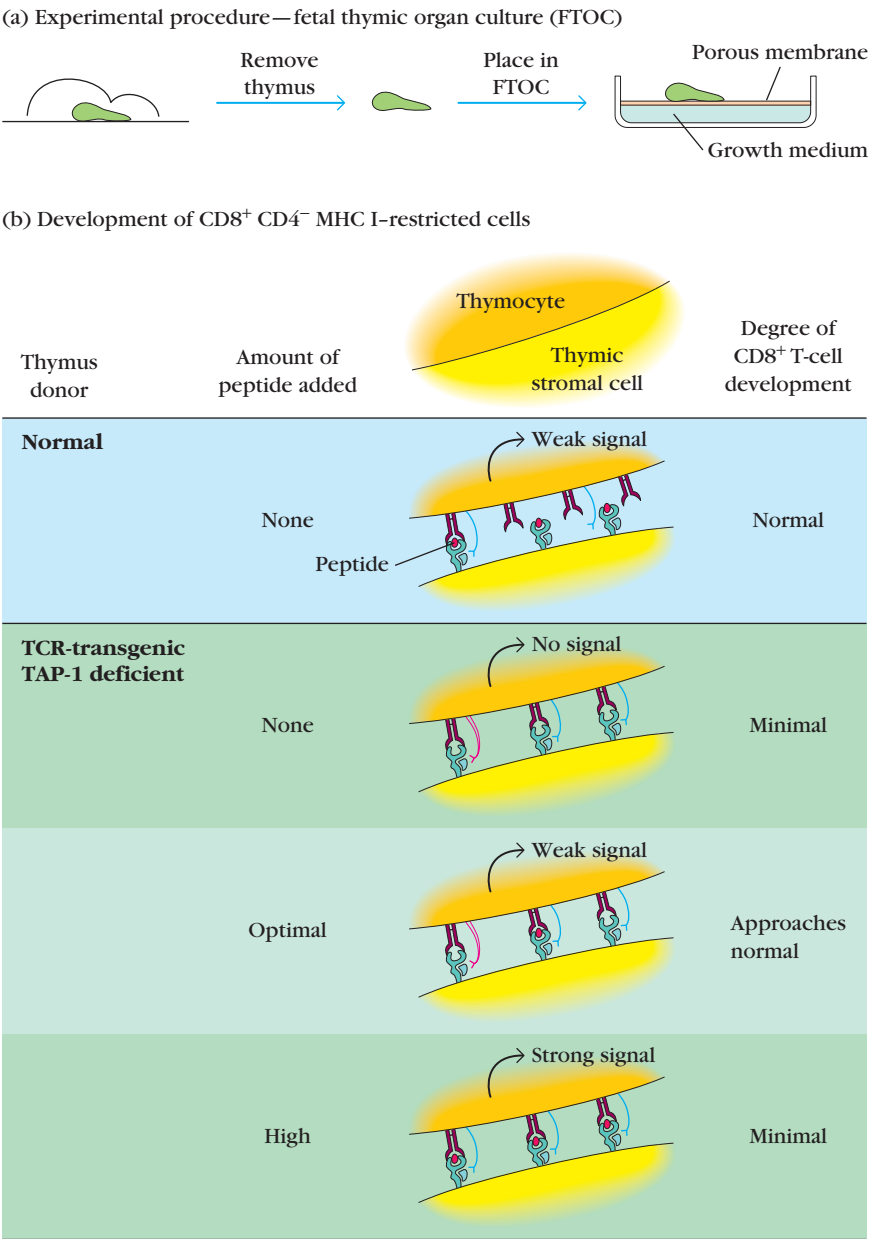


FIGURE 10-8 Role of peptides in selection. Thymuses harvested before their thymocyte populations have undergone positive and negative selection allow study of the development and selection of single positive (CD4⁺CD8⁻ and CD4⁻CD8⁺) T cells. (a) Outline of the experimental procedure for in vitro fetal thymic organ culture (FTOC). (b) The development and selection of CD8⁺CD4⁻ class I-restricted T cells depends on TCR peptide-MHC I interactions. TAP-1 knockout mice are unable to form peptide-MHC complexes unless peptide is added. The mice used in this study were transgenic for the α and β chains of a TCR that recognizes the added peptide bound to MHC I molecules of the TAP-1 knockout/TCR transgenic mice. Varying the amount of added peptide revealed that low concentrations of peptide, producing low avidity of binding, resulted in positive selection and nearly normal levels of CD4⁻CD8⁺ cells. High concentrations of peptide, producing high avidity of binding to the TCR, caused negative selection, and few CD4⁻CD8⁺ T cells appeared. [Adapted from Ashton Rickardt et al. (1994) *Cell* 25:651.]

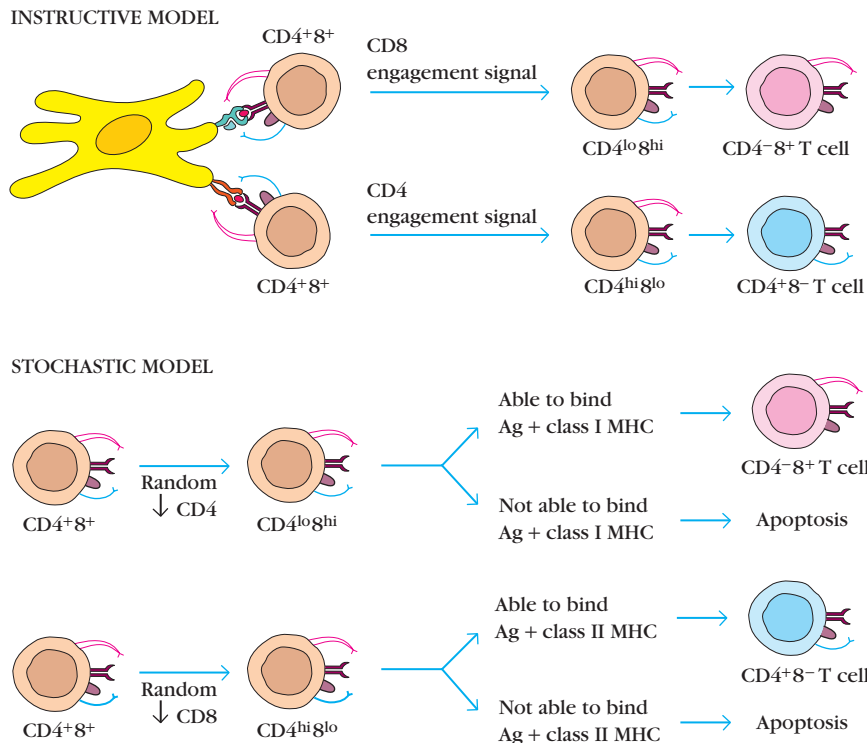


FIGURE 10-9 Proposed models for the role of the CD4 and CD8 coreceptors in thymic selection of double positive thymocytes leading to single positive T cells. According to the instructive model, interaction of one coreceptor with MHC molecules on stromal cells results in down-regulation of the other coreceptor. According to the stochastic model, down-regulation of CD4 or CD8 is a random process.

(Figure 10-9). The *instructional model* postulates that the multiple interactions between the TCR, CD8⁺ or CD4⁺ coreceptors, and class I or class II MHC molecules instruct the cells to differentiate into either CD8⁺ or CD4⁺ single-positive cells, respectively. This model would predict that a class I MHC–specific TCR together with the CD8 coreceptor would generate a signal that is different from the signal induced by a class II MHC–specific TCR together with the CD4 coreceptor. The *stochastic model* suggests that CD4 or CD8 expression is switched off randomly with no relation to the specificity of the TCR. Only those thymocytes whose TCR and remaining coreceptor recognize the same class of MHC molecule will mature. At present, it is not possible to choose one model over the other.

T_H-Cell Activation

The central event in the generation of both humoral and cell-mediated immune responses is the activation and clonal expansion of T_H cells. Activation of T_C cells, which is generally similar to T_H-cell activation, is described in Chapter 14. T_H-cell activation is initiated by interaction of the TCR-CD3 complex with a processed antigenic peptide bound to a class II MHC molecule on the surface of an antigen-presenting cell. This interaction and the resulting activating signals also involve various accessory membrane molecules on the T_H cell and the antigen-presenting cell. Interaction of a T_H cell with antigen initiates a cascade of biochemical events that induces the resting T_H cell to enter the cell cycle, proliferating

and differentiating into memory cells or effector cells. Many of the gene products that appear upon interaction with antigen can be grouped into one of three categories depending on how early they can be detected after antigen recognition (Table 10-2):

- *Immediate genes*, expressed within half an hour of antigen recognition, encode a number of transcription factors, including c-Fos, c-Myc, c-Jun, NFAT, and NF-κB
- *Early genes*, expressed within 1–2 h of antigen recognition, encode IL-2, IL-2R (IL-2 receptor), IL-3, IL-6, IFN-γ, and numerous other proteins
- *Late genes*, expressed more than 2 days after antigen recognition, encode various adhesion molecules

These profound changes are the result of signal-transduction pathways that are activated by the encounter between the TCR and MHC-peptide complexes. An overview of some of the basic strategies of cellular signaling will be useful background for appreciating the specific signaling pathways used by T cells.

Signal-Transduction Pathways Have Several Features in Common

The detection and interpretation of signals from the environment is an indispensable feature of all cells, including those of the immune system. Although there are an enormous number of different signal-transduction pathways, some common themes are typical of these crucial integrative processes:

TABLE 10-2 Time course of gene expression by T_H cells following interaction with antigen

Gene product	Function	Time mRNA expression begins	Location	Ratio of activated to nonactivated cells
IMMEDIATE				
c-Fos	Protooncogene; nuclear-binding protein	15 min	Nucleus	> 100
c-Jun	Cellular oncogene; transcription factor	15–20 min	Nucleus	?
NFAT	Transcription factor	20 min	Nucleus	50
c-Myc	Cellular oncogene	30 min	Nucleus	20
NF-κB	Transcription factor	30 min	Nucleus	> 10
EARLY				
IFN-γ	Cytokine	30 min	Secreted	> 100
IL-2	Cytokine	45 min	Secreted	> 1000
Insulin receptor	Hormone receptor	1 h	Cell membrane	3
IL-3	Cytokine	1–2 h	Secreted	> 100
TGF-β	Cytokine	< 2 h	Secreted	> 10
IL-2 receptor (p55)	Cytokine receptor	2 h	Cell membrane	> 50
TNF-β	Cytokine	1–3 h	Secreted	> 100
Cyclin	Cell-cycle protein	4–6 h	Cytoplasmic	> 10
IL-4	Cytokine	< 6 h	Secreted	> 100
IL-5	Cytokine	< 6 h	Secreted	> 100
IL-6	Cytokine	< 6 h	Secreted	> 100
c-Myb	Protooncogene	16 h	Nucleus	100
GM-CSF	Cytokine	20 h	Secreted	?
LATE				
HLA-DR	Class II MHC molecule	3–5 days	Cell membrane	10
VLA-4	Adhesion molecule	4 days	Cell membrane	> 100
VLA-1, VLA-2, VLA-3, VLA-5	Adhesion molecules	7–14 days	Cell membrane	> 100, ?, ?, ?

SOURCE: Adapted from G. Crabtree, *Science* 243:357.

- *Signal transduction begins with the interaction between a signal and its receptor.* Signals that cannot penetrate the cell membrane bind to receptors on the surface of the cell membrane. This group includes water-soluble signaling molecules and membrane-bound ligands (MHC-peptide complexes, for example). Hydrophobic signals, such as steroids, that can diffuse through the cell membrane are bound by intracellular receptors.
- *Signals are often transduced through G proteins,* membrane-linked macromolecules whose activities are controlled by binding of the guanosine nucleotides GTP and GDP, which act as molecular switches. Bound GTP turns on the signaling capacities of the G protein;

hydrolysis of GTP or exchange for GDP turns off the signal by returning the G protein to an inactive form. There are two major categories of G proteins. *Small G proteins* consist of a single polypeptide chain of about 21 kDa. An important small G protein, known as Ras, is a key participant in the activation of an important proliferation-inducing signal-transduction cascade triggered by binding of ligands to their receptor tyrosine kinases. *Large G proteins* are composed of α, β, and γ subunits and are critically involved in many processes, including vision, olfaction, glucose metabolism, and phenomena of immunological interest such as leukocyte chemotaxis.

- *Signal reception often leads to the generation within the cell of a “second messenger,”* a molecule or ion that can diffuse to other sites in the cell and evoke changes. Examples are cyclic nucleotides (cAMP, cGMP), calcium ion (Ca^{2+}), and membrane phospholipid derivatives such as diacylglycerol (DAG) and inositol triphosphate (IP_3).
- *Protein kinases and protein phosphatases are activated or inhibited.* Kinases catalyze the phosphorylation of target residues (tyrosine, serine, or threonine) of key elements in many signal-transduction pathways. Phosphatases catalyze dephosphorylation, reversing the effect of kinases. These enzymes play essential roles in many signal-transduction pathways of immunological interest.
- *Many signal transduction pathways involve the signal-induced assembly of some components of the pathway.* Molecules known as *adaptor proteins* bind specifically and simultaneously to two or more different molecules with signaling roles, bringing them together and promoting their combined activity.
- *Signals are amplified by enzyme cascades.* Each enzyme in the cascade catalyzes the activation of many copies of the next enzyme in the sequence, greatly amplifying the signal at each step and offering many opportunities to modulate the intensity of a signal along the way.
- *The default setting for signal-transduction pathways is OFF.* In the absence of an appropriately presented signal, transmission through the pathway does not take place.

Multiple Signaling Pathways Are Initiated by TCR Engagement

The events that link antigen recognition by the T-cell receptor to gene activation echo many of the themes just reviewed. The key element in the initiation of T-cell activation is the recognition by the TCR of MHC-peptide complexes on antigen-presenting cells.

As described in Chapter 9, the TCR consists of a mostly extracellular ligand-binding unit, a predominantly intracellular signaling unit, the CD3 complex, and the homodimer of ζ (zeta) chains. Experiments with knockout mice have shown that all of these components are essential for signal transduction. Two phases can be recognized in the antigen-mediated induction of T-cell responses:

- *Initiation.* The engagement of MHC-peptide by the TCR leads to clustering with CD4 or CD8 coreceptors as these coreceptors bind to invariant regions of the MHC molecule (Figure 10-10). Lck, a protein tyrosine kinase associated with the cytoplasmic tails of the coreceptors, is brought close to the cytoplasmic tails of the TCR complex and phosphorylates the immunoreceptor tyrosine-based activation motifs (ITAMs, described in Chapter 9). The phosphorylated tyrosines in the ITAMs

of the zeta chain provide docking sites to which a protein tyrosine kinase called ZAP-70 attaches (step 2 in Figure 10-10) and becomes active. ZAP-70 then catalyzes the phosphorylation of a number of membrane-associated adaptor molecules (step 3), which act as anchor points for the recruitment of several intracellular signal transduction pathways. One set of pathways involves a form of the enzyme phospholipase C (PLC), which anchors to an adaptor molecule, is activated by phosphorylation and cleaves a membrane phospholipid to generate second messengers. Another set activates small G proteins.

- *Generation of multiple intracellular signals.* Many signaling pathways are activated as a consequence of the steps that occur in the initiation phase, as shown to the right in Figure 10-10, and described below.

We shall consider several of the signaling pathways recruited by T-cell activation, but the overall process is quite complex and many of the details will not be presented here. The review articles suggested at the end of this chapter provide extensive coverage of this very active research area.

Phospholipase C γ (PLC γ): PLC γ is activated by phosphorylation and gains access to its substrate by binding to a membrane-associated adaptor protein (Figure 10-11a). PLC γ hydrolyzes a phospholipid component of the membrane to generate inositol 1,4,5-triphosphate (IP_3) and diacylglycerol (DAG). IP_3 causes a rapid release of Ca^{2+} from the endoplasmic reticulum and opens Ca^{2+} channels in the cell membrane (Figure 10-11b). DAG activates protein kinase C, a multifunctional kinase that phosphorylates many different targets (Figure 10-11c).

Ca^{2+} : Calcium ion is involved in an unusually broad range of processes, including vision, muscle contraction, and many others. It is an essential element in many T-cell responses, including a pathway that leads to the movement of a major transcription factor, NFAT, from the cytoplasm into the nucleus (Figure 10-11b). In the nucleus, NFAT supports the transcription of genes required for the expression of the T-cell growth-promoting cytokines IL-2, IL-4, and others.

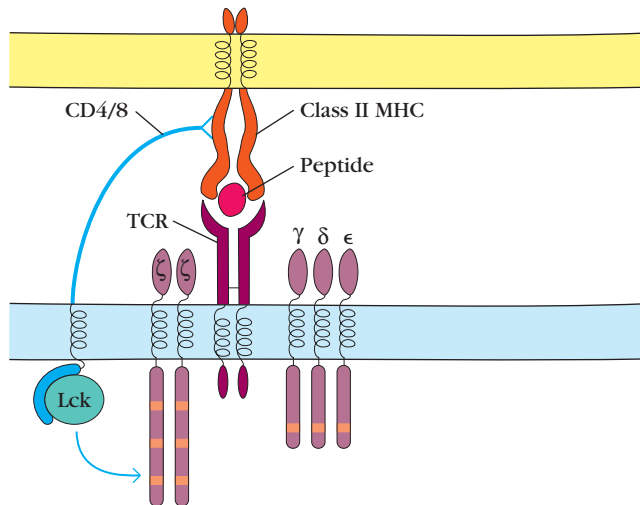
Protein kinase C (PKC): This enzyme, which affects many pathways, causes the release of an inhibitory molecule from the transcription factor NF- κ B, allowing NF- κ B to enter the nucleus, where it promotes the expression of genes required for T-cell activation (Figure 10-11c). NF- κ B is essential for a variety of T-cell responses and provides survival signals that protect T cells from apoptotic death.

The Ras/MAP kinase pathway: Ras is a pivotal component of a signal-transduction pathway that is found in many cell types and is evolutionarily conserved across a spectrum of eukaryotes from yeasts to humans. Ras is a small G protein whose activation by GTP initiates a cascade of protein kinases known as the mitogen-activated protein kinase (MAP kinase) pathway. As shown in Figure 10-12, phosphorylation of the end product of this cascade, MAP kinase (also called

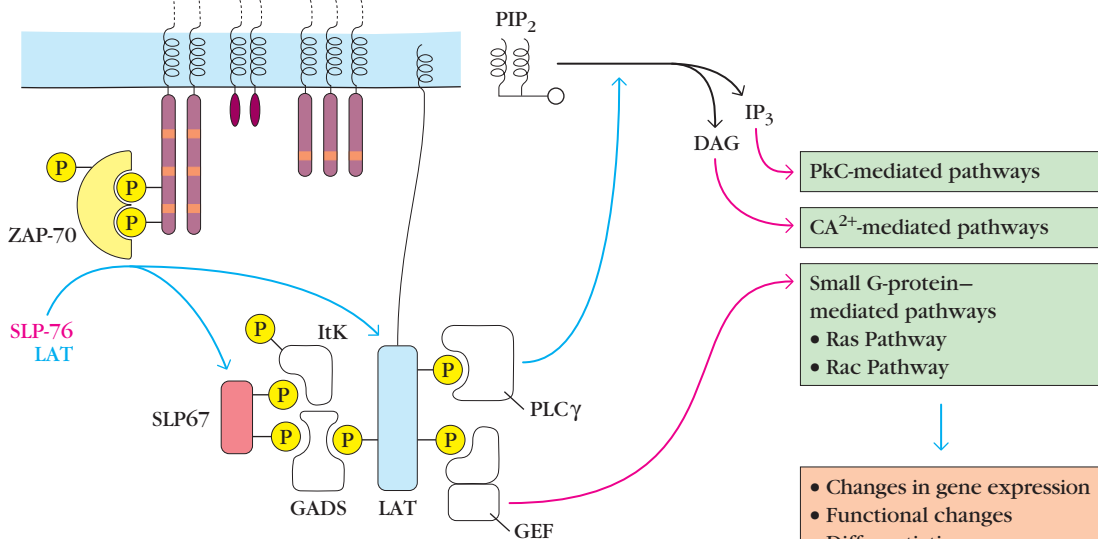


VISUALIZING CONCEPTS

- Engagement of MHC-peptide initiates processes that lead to assembly of signaling complex.



- CD4/8-associated Lck phosphorylates ITAMs of coreceptors, creates docking site for ZAP-70



- ZAP-70 phosphorylates adaptor molecules that recruit components of several signaling pathways

FIGURE 10-10 Overview of TCR-mediated signaling. TCR engagement by peptide-MHC complexes initiates the assembly of a signaling complex. An early step is the Lck-mediated phosphorylation of ITAMs on the zeta (ζ) chains of the TCR complex, creating docking sites to which the protein kinase ZAP-70 attaches and becomes activated by phosphorylation. A series of ZAP-70-catalyzed protein phosphorylations enable the generation of a variety of signals. (Abbreviations: DAG = diacylglycerol; GADS =

Grb2-like adaptor downstream of Shc; GEF = guanine nucleotide exchange factor; ITAM = immunoreceptor tyrosine-based activation motif; Itk = inducible T cell kinase; IP₃ = inositol 1,4,5 triphosphate; LAT = linker of activated T cells; PIP₂ = phosphoinositol biphosphate; PLC γ = phospholipase C gamma; Lck = lymphocyte kinase; SLP-76 = SH2-containing leukocyte-specific protein of 76 kDa; ZAP-70 = zeta associated protein of 70 kDa.)

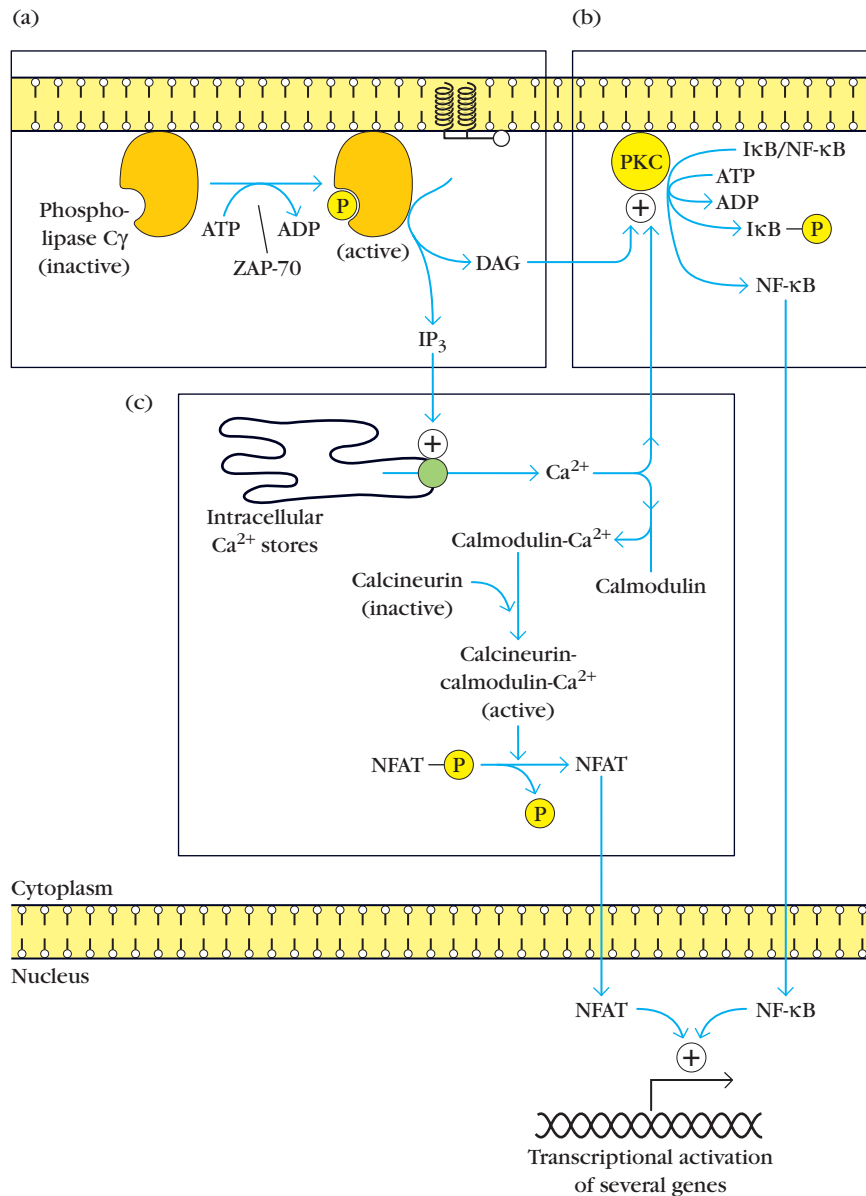


FIGURE 10-11 Signal-transduction pathways associated with T-cell activation. (a) Phospholipase Cγ (PLC) is activated by phosphorylation. Active PLC hydrolyzes a phospholipid component of the plasma membrane to generate the second messengers, DAG and IP₃. (b) Protein kinase C (PKC) is activated by DAG and Ca²⁺. Among the numerous effects of PKC is phosphorylation of IκB, a cytoplasmic protein that binds the transcription factor NF-κB and prevents it from entering the nucleus. Phosphorylation of IκB releases NF-κB, which then translocates into the nucleus. (c) Ca²⁺-dependent activation of calcineurin. Calcineurin is a Ca²⁺/calmodulin-dependent phosphatase. IP₃ mediates the release of Ca²⁺ from the endoplasmic reticulum. Ca²⁺ binds the protein calmodulin, which then associates with and activates the Ca²⁺/calmodulin-dependent phosphatase calcineurin. Active calcineurin removes a phosphate group from NFAT, which allows this transcription factor to translocate into the nucleus.

ERK), allows it to activate Elk, a transcription factor necessary for the expression of Fos. Phosphorylation of Fos by MAP kinase allows it to associate with Jun to form AP-1, which is an essential transcription factor for T-cell activation.

Co-Stimulatory Signals Are Required for Full T-Cell Activation

T-cell activation requires the dynamic interaction of multiple membrane molecules described above, but this interaction, by itself, is not sufficient to fully activate naive T cells. Naive T cells require more than one signal for activation and subsequent proliferation into effector cells:

- *Signal 1*, the initial signal, is generated by interaction of an antigenic peptide with the TCR-CD3 complex.

- A subsequent antigen-nonspecific co-stimulatory signal, *signal 2*, is provided primarily by interactions between CD28 on the T cell and members of the B7 family on the APC.

There are two related forms of B7, B7-1 and B7-2 (Figure 10-13). These molecules are members of the immunoglobulin superfamily and have a similar organization of extracellular domains but markedly different cytosolic domains. Both B7 molecules are constitutively expressed on dendritic cells and induced on activated macrophages and activated B cells. The ligands for B7 are CD28 and CTLA-4 (also known as CD152), both of which are expressed on the T-cell membrane as disulfide-linked homodimers; like B7, they are members of the immunoglobulin superfamily (Figure 10-13). Although CD28 and CTLA-4 are structurally similar glycoproteins, they act antagonistically. Signaling through

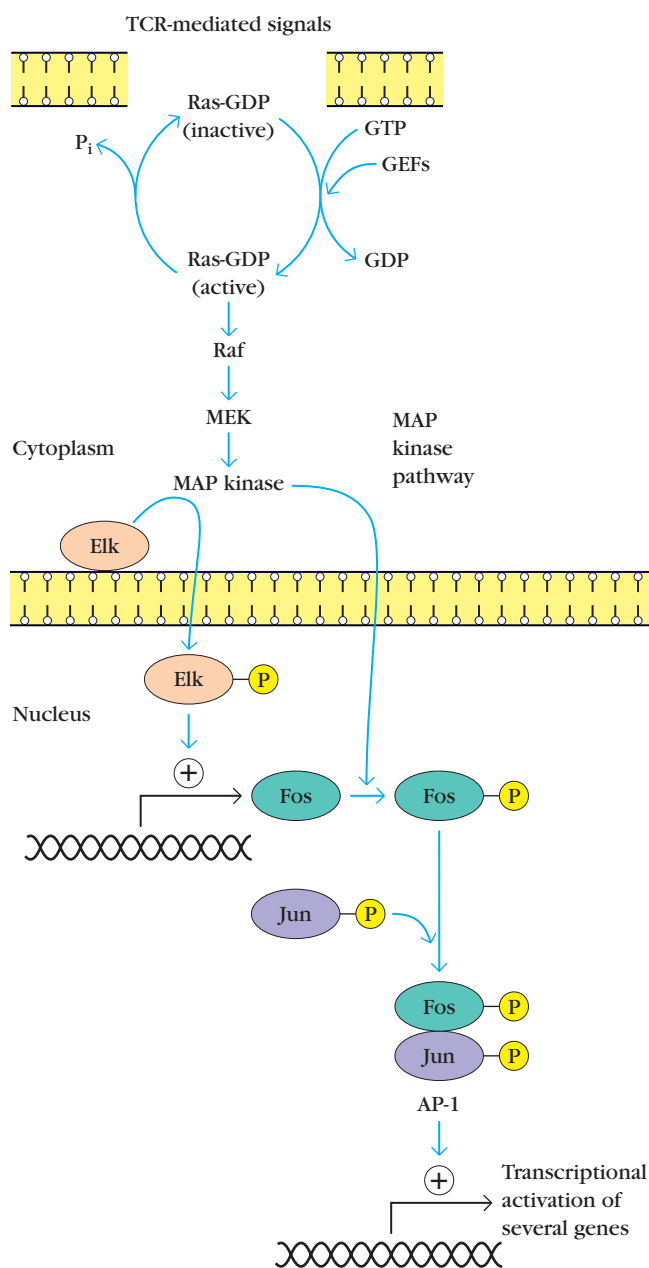


FIGURE 10-12 Activation of the small G protein, Ras. Signals from the T-cell receptor result in activation of Ras via the action of specific guanine nucleotide exchange factors (GEFs) that catalyze the exchange of GDP for GTP. Active Ras causes a cascade of reactions that result in the increased production of the transcription factor Fos. Following their phosphorylation, Fos and Jun dimerize to yield the transcription factor AP-1. Note that all these pathways have important effects other than the specific examples shown in the figure.

CD28 delivers a positive co-stimulatory signal to the T cell; signaling through CTLA-4 is inhibitory and down-regulates the activation of the T cell. CD28 is expressed by both resting and activated T cells, but CTLA-4 is virtually undetectable on resting cells. Typically, engagement of the TCR causes the induction of CTLA-4 expression, and CTLA-4 is readily de-

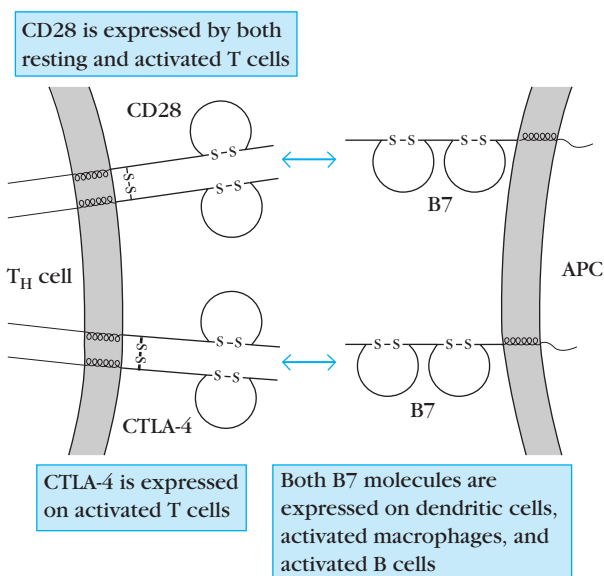


FIGURE 10-13 T_H -cell activation requires a co-stimulatory signal provided by antigen-presenting cells (APCs). Interaction of B7 family members on APCs with CD28 delivers the co-stimulatory signal. Engagement of the closely related CTLA-4 molecule with B7 produces an inhibitory signal. All of these molecules contain at least one immunoglobulin-like domain and thus belong to the immunoglobulin superfamily. [Adapted from P. S. Linsley and J. A. Ledbetter, 1993, *Annu. Rev. Immunol.* **11**:191.]

tectable within 24 hours of stimulation, with maximal expression within 2 or 3 days post-stimulation. Even though the peak surface levels of CTLA-4 are lower than those of CD28, it still competes favorably for these molecules because it has a significantly higher avidity for these molecules than CD28 does. Interestingly, the level of CTLA-4 expression is increased by CD28-generated co-stimulatory signals. This provides regulatory braking via CTLA-4 in proportion to the acceleration received from CD28. Some of the importance of CTLA-4 in the regulation of lymphocyte activation and proliferation is revealed by experiments with CTLA-4 knockout mice. T cells in these mice proliferate massively, which leads to lymphadenopathy (greatly enlarged lymph nodes), splenomegaly (enlarged spleen), and death at 3 to 4 weeks after birth. Clearly, the production of inhibitory signals by engagement of CTLA-4 is important in lymphocyte homeostasis.

CTLA-4 can effectively block CD28 co-stimulation by competitive inhibition at the B7 binding site, an ability that holds promise for clinical use in autoimmune diseases and transplantation. As shown in Figure 10-14, an ingeniously engineered chimeric molecule has been designed to explore the therapeutic potential of CTLA-4. The Fc portion of human IgG has been fused to the B7-binding domain of CTLA-4 to produce a chimeric molecule called CTLA-4Ig. The human Fc region endows the molecule with a longer half-life in the body and the presence of B7 binding domains

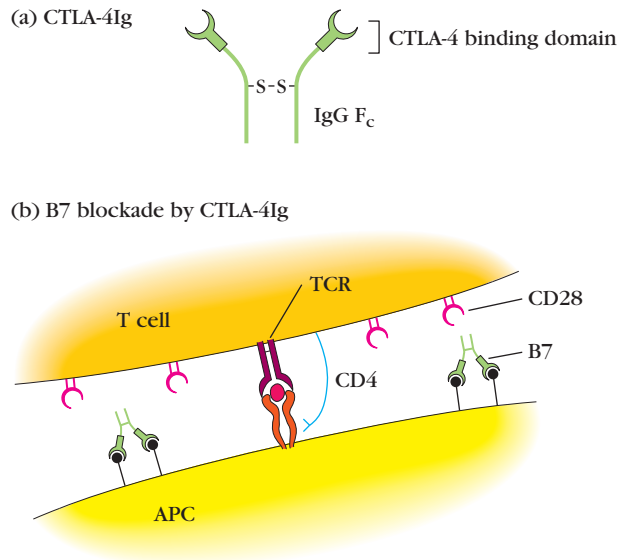


FIGURE 10-14 CTLA-4Ig, a chimeric suppressor of co-stimulation. (a) CTLA-4Ig, a genetically engineered molecule in which the Fc portion of human IgG is joined to the B7-binding domain of CTLA-4. (b) CTLA-4Ig blocks costimulation by binding to B7 on antigen presenting cells and preventing the binding of CD28, a major co-stimulatory molecule of T cells.

allows it to bind to B7. A promising clinical trial of CTLA-4 has been conducted in patients with psoriasis vulgaris, a T-cell-mediated autoimmune inflammatory skin disease. During the course of this trial, forty-three patients received four doses of CTLA-4Ig, and 46% of this group experienced a 50% or greater sustained improvement in their skin condition. Further studies of the utility of CTLA-4Ig are also being pursued in other areas.

Clonal Anergy Ensues If a Co-Stimulatory Signal Is Absent

T_H -cell recognition of an antigenic peptide–MHC complex sometimes results in a state of nonresponsiveness called **clonal anergy**, marked by the inability of cells to proliferate in response to a peptide–MHC complex. Whether clonal expansion or clonal anergy ensues is determined by the presence or absence of a co-stimulatory signal (signal 2), such as that produced by interaction of CD28 on T_H cells with B7 on antigen-presenting cells. Experiments with cultured cells show that, if a resting T_H cell receives the TCR-mediated signal (signal 1) in the absence of a suitable co-stimulatory signal, then the T_H cell will become anergic. Specifically, if resting T_H cells are incubated with glutaraldehyde-fixed APCs, which do not express B7 (Figure 10-15a), the fixed APCs are able to present peptides together with class II MHC molecules, thereby providing signal 1, but they are unable to provide the necessary co-stimulatory signal 2. In the absence of a co-stimulatory signal, there is minimal production of cy-

tokines, especially of IL-2. Anergy can also be induced by incubating T_H cells with normal APCs in the presence of the Fab portion of anti-CD28, which blocks the interaction of CD28 with B7 (Figure 10-15b).

Two different control experiments demonstrate that fixed APCs bearing appropriate peptide–MHC complexes can deliver an effective signal mediated by T-cell receptors. In one experiment, T cells are incubated both with fixed APCs bearing peptide–MHC complexes recognized by the TCR of the T cells and with normal APCs, which express B7 (Figure 10-15d). The fixed APCs engage the TCRs of the T cells, and the B7 molecules on the surface of the normal APCs cross-link the CD28 of the T cell. These T cells thus receive both signals and undergo activation. The addition of bivalent anti-CD28 to mixtures of fixed APCs and T cells also provides effective co-stimulation by crosslinking CD28 (Figure 10-15e). Well-controlled systems for studying anergy in vitro have stimulated considerable interest in this phenomenon. However, more work is needed to develop good animal systems for establishing anergy and studying its role in vivo.

Superantigens Induce T-Cell Activation by Binding the TCR and MHC II Simultaneously

Superantigens are viral or bacterial proteins that bind simultaneously to the V_β domain of a T-cell receptor and to the α chain of a class II MHC molecule. Both exogenous and endogenous superantigens have been identified. Crosslinkage of a T-cell receptor and class II MHC molecule by either type of superantigen produces an activating signal that induces T-cell activation and proliferation (Figure 10-16).

Exogenous superantigens are soluble proteins secreted by bacteria. Among them are a variety of **exotoxins** secreted by gram-positive bacteria, such as staphylococcal enterotoxins, toxic-shock-syndrome toxin, and exfoliative-dermatitis toxin. Each of these exogenous superantigens binds particular V_β sequences in T-cell receptors (Table 10-3) and crosslinks the TCR to a class II MHC molecule.

Endogenous superantigens are cell-membrane proteins encoded by certain viruses that infect mammalian cells. One group, encoded by mouse mammary tumor virus (MTV), can integrate into the DNA of certain inbred mouse strains; after integration, retroviral proteins are expressed on the membrane of the infected cells. These viral proteins, called **minor lymphocyte stimulating (Mls)** determinants, bind particular V_β sequences in T-cell receptors and crosslink the TCR to a class II MHC molecule. Four Mls superantigens, originating in different MTV strains, have been identified.

Because superantigens bind outside of the TCR antigen-binding cleft, any T cell expressing a particular V_β sequence will be activated by a corresponding superantigen. Hence, the activation is polyclonal and can affect a significant percentage (5% is not unusual) of the total T_H population. The massive activations that follow crosslinkage by a superantigen results in overproduction of T_H -cell cytokines, leading to systemic toxicity. The food poisoning induced by staphy-

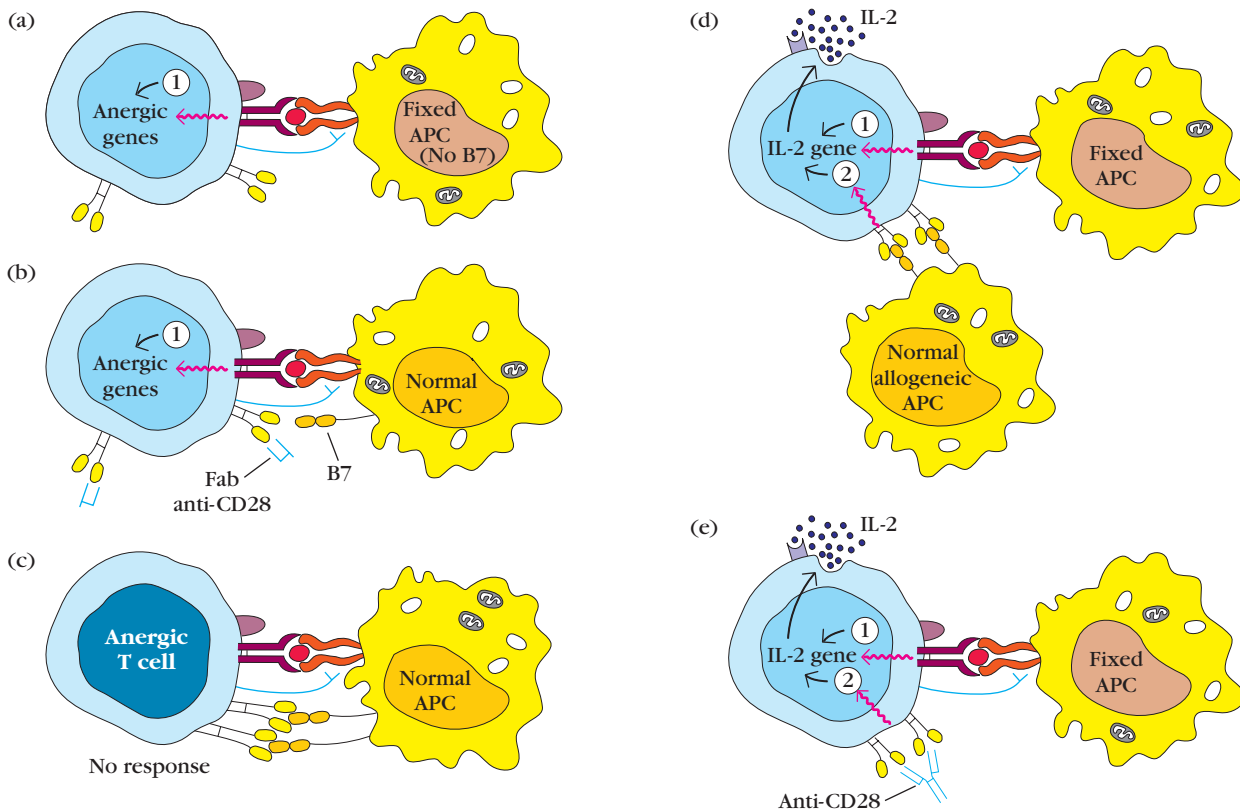


FIGURE 10-15 Experimental demonstration of clonal anergy versus clonal expansion. (a,b) Only signal 1 is generated when resting T_H cells are incubated with glutaraldehyde-fixed antigen-presenting cells (APCs) or with normal APCs in the presence of the Fab portion

of anti-CD28. (c) The resulting anergic T cells cannot respond to normal APCs. (d,e) In the presence of normal allogeneic APCs or anti-CD28, both of which produce the co-stimulatory signal 2, T cells are activated by fixed APCs.

lococcal enterotoxins and the toxic shock induced by toxic-shock-syndrome toxin are two examples of the consequences of cytokine overproduction induced by superantigens.

Superantigens can also influence T-cell maturation in the thymus. A superantigen present in the thymus during thymic processing will induce the negative selection of all thymocytes bearing a TCR V_β domain corresponding to the superantigen specificity. Such massive deletion can be caused by exogenous or endogenous superantigens and is characterized by the absence of all T cells whose receptors possess V_β domains targeted by the superantigen.

T-Cell Differentiation

$CD4^+$ and $CD8^+$ T cells leave the thymus and enter the circulation as resting cells in the G_0 stage of the cell cycle. There are about twice as many $CD4^+$ T cells as $CD8^+$ T cells in the periphery. T cells that have not yet encountered antigen (naïve T cells) are characterized by condensed chromatin, very little cytoplasm, and little transcriptional activity. Naïve T cells continually recirculate between the blood and lymph

systems. During recirculation, naïve T cells reside in secondary lymphoid tissues such as lymph nodes. If a naïve cell does not encounter antigen in a lymph node, it exits through the efferent lymphatics, ultimately draining into the thoracic duct and rejoining the blood. It is estimated that each naïve T cell recirculates from the blood to the lymph nodes and back again every 12–24 hours. Because only about 1 in 10^5 naïve T cells is specific for any given antigen, this large-scale recirculation increases the chances that a naïve T cell will encounter appropriate antigen.

Activated T Cells Generate Effector and Memory T Cells

If a naïve T cell recognizes an antigen-MHC complex on an appropriate antigen-presenting cell or target cell, it will be activated, initiating a *primary response*. About 48 hours after activation, the naïve T cell enlarges into a blast cell and begins undergoing repeated rounds of cell division. As described earlier, activation depends on a signal induced by engagement of the TCR complex and a co-stimulatory signal induced by the CD28-B7 interaction (see Figure 10-13). These signals trigger entry of the T cell into the G_1 phase of the cell

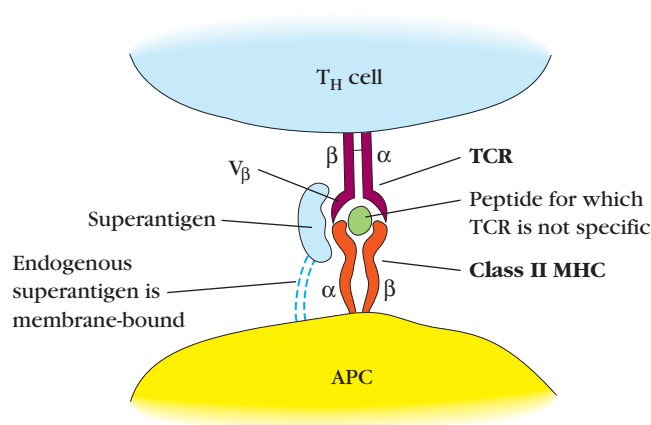


FIGURE 10-16 Superantigen-mediated crosslinkage of T-cell receptor and class II MHC molecules. A superantigen binds to all TCRs bearing a particular V_β sequence regardless of their antigenic specificity. Exogenous superantigens are soluble secreted bacterial proteins, including various exotoxins. Endogenous superantigens are membrane-embedded proteins produced by certain viruses; they include MIs antigens encoded by mouse mammary tumor virus.

cycle and, at the same time, induce transcription of the gene for IL-2 and the α chain of the high-affinity IL-2 receptor. In addition, the co-stimulatory signal increases the half-life of the IL-2 mRNA. The increase in IL-2 transcription, together with stabilization of the IL-2 mRNA, increases IL-2 produc-

tion by 100-fold in the activated T cell. Secretion of IL-2 and its subsequent binding to the high-affinity IL-2 receptor induces the activated naive T cell to proliferate and differentiate (Figure 10-17). T cells activated in this way divide 2–3 times per day for 4–5 days, generating a large clone of progeny cells, which differentiate into memory or effector T-cell populations.

The various *effector T cells* carry out specialized functions such as cytokine secretion and B-cell help (activated $CD4^+$ T_H cells) and cytotoxic killing activity ($CD8^+$ CTLs). The generation and activity of CTL cells are described in detail in Chapter 14. Effector cells are derived from both naive and memory cells after antigen activation. Effector cells are short-lived cells, whose life spans range from a few days to a few weeks. The effector and naive populations express different cell-membrane molecules, which contribute to different recirculation patterns.

As described in more detail in Chapter 12, $CD4^+$ effector T cells form two subpopulations distinguished by the different panels of cytokines they secrete. One population, called the **T_H1 subset**, secretes IL-2, IFN- γ , and TNF- β . The T_H1 subset is responsible for classic cell-mediated functions, such as delayed-type hypersensitivity and the activation of cytotoxic T lymphocytes. The other subset, called the **T_H2 subset**, secretes IL-4, IL-5, IL-6, and IL-10. This subset functions more effectively as a helper for B-cell activation.

The *memory T-cell* population is derived from both naive T cells and from effector cells after they have encountered antigen. Memory T cells are antigen-generated, generally

TABLE 10-3 Exogenous superantigens and their V_β specificity

Superantigen	Disease*	V _β SPECIFICITY	
		Mouse	Human
Staphylococcal enterotoxins			
SEA	Food poisoning	1, 3, 10, 11, 12, 17	nd
SEB	Food poisoning	3, 8.1, 8.2, 8.3	3, 12, 14, 15, 17, 20
SEC1	Food poisoning	7, 8.2, 8.3, 11	12
SEC2	Food poisoning	8.2, 10	12, 13, 14, 15, 17, 20
SEC3	Food poisoning	7, 8.2	5, 12
SED	Food poisoning	3, 7, 8.3, 11, 17	5, 12
SEE	Food poisoning	11, 15, 17	5.1, 6.1–6.3, 8, 18
Toxic-shock-syndrome toxin (TSST1)	Toxic-shock syndrome	15, 16	2
Exfoliative-dermatitis toxin (ExFT)	Scalded-skin syndrome	10, 11, 15	2
Mycoplasma-arthritis supernatant (MAS)	Arthritis, shock	6, 8.1–8.3	nd
Streptococcal pyrogenic exotoxins (SPE-A, B, C, D)	Rheumatic fever, shock	nd	nd

*Disease results from infection by bacteria that produce the indicated superantigens.

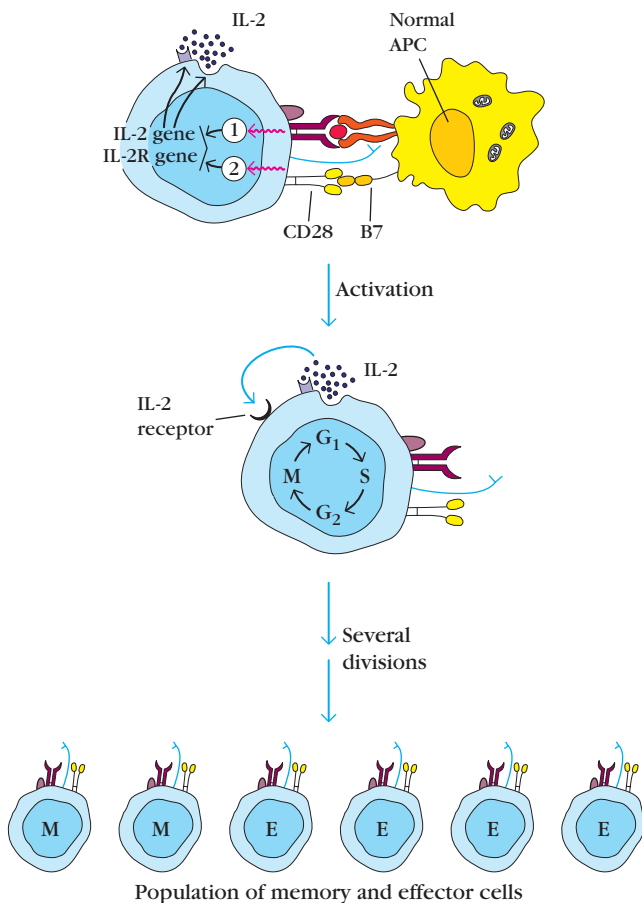


FIGURE 10-17 Activation of a T_H cell by both signal 1 and co-stimulatory signal 2 up-regulates expression of IL-2 and the high-affinity IL-2 receptor, leading to the entry of the T cell into the cell cycle and several rounds of proliferation. Some of the cells differentiate into effector cells, others into memory cells.

long-lived, quiescent cells that respond with heightened reactivity to a subsequent challenge with the same antigen, generating a *secondary response*. An expanded population of memory T cells appears to remain long after the population of effector T cells has declined. In general, memory T cells express many of the same cell-surface markers as effector T cells; no cell-surface markers definitively identify them as memory cells.

Like naive T cells, most memory T cells are resting cells in the G₀ stage of the cell cycle, but they appear to have less stringent requirements for activation than naive T cells do. For example, naive T_H cells are activated only by dendritic cells, whereas memory T_H cells can be activated by macrophages, dendritic cells, and B cells. It is thought that the expression of high levels of numerous adhesion molecules by memory T_H cells enables these cells to adhere to a broad spectrum of antigen-presenting cells. Memory cells also display recirculation patterns that differ from those of naive or effector T cells.

A CD4⁺CD25⁺ Subpopulation of T cells Negatively Regulates Immune Responses

Investigators first described T cell populations that could suppress immune responses during the early 1970s. These cells were called suppressor T cells (T_s) and were believed to be CD8⁺ T cells. However, the cellular and molecular basis of the observed suppression remained obscure, and eventually great doubt was cast on the existence of CD8⁺ suppressor T cells. Recent research has shown that there are indeed T cells that suppress immune responses. Unexpectedly, these cells have turned out to be CD4⁺ rather than CD8⁺ T cells. Within the population of CD4⁺CD25⁺ T cells, there are regulatory T cells that can inhibit the proliferation of other T cell populations *in vitro*. Animal studies show that members of the CD4⁺CD25⁺ population inhibit the development of autoimmune diseases such as experimentally induced inflammatory bowel disease, experimental allergic encephalitis, and autoimmune diabetes. The suppression by these regulatory cells is antigen specific because it depends upon activation through the T cell receptor. Cell contact between the suppressing cells and their targets is required. If the regulatory cells are activated by antigen but separated from their targets by a permeable barrier, no suppression occurs. The existence of regulatory T cells that specifically suppress immune responses has clinical implications. The depletion or inhibition of regulatory T cells followed by immunization may enhance the immune responses to conventional vaccines. In this regard, some have suggested that elimination of T cells that suppress responses to tumor antigens may facilitate the development of anti-tumor immunity. Conversely, increasing the suppressive activity of regulatory T cell populations could be useful in the treatment of allergic or autoimmune diseases. The ability to increase the activity of regulatory T cell populations might also be useful in suppressing organ and tissue rejection. Future work on this regulatory cell population will seek deeper insights into the mechanisms by which members of CD4⁺CD25⁺ T cell populations regulate immune responses. There will also be determined efforts to discover ways in which the activities of these populations can be increased to diminish unwanted immune responses and decreased to promote desirable ones.

Antigen-Presenting Cells Have Characteristic Co-Stimulatory Properties

Only professional antigen-presenting cells (dendritic cells, macrophages, and B cells) are able to present antigen together with class II MHC molecules and deliver the co-stimulatory signal necessary for complete T-cell activation that leads to proliferation and differentiation. The principal co-stimulatory molecules expressed on antigen-presenting cells are the glycoproteins B7-1 and B7-2 (see Figure 10-13). The professional antigen-presenting cells differ in their ability to display antigen and also differ in their ability to deliver the co-stimulatory signal (Figure 10-18).

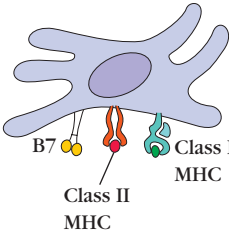
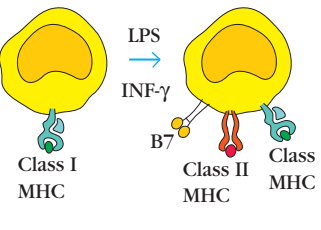
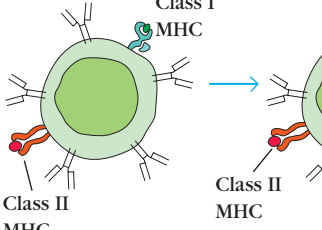
	Dendritic cell	Macrophage		B Lymphocyte	
		Resting 	Activated	Resting 	Activated
Antigen uptake	Endocytosis phagocytosis (by Langerhans cells)	Phagocytosis	Phagocytosis	Receptor-mediated endocytosis	Receptor-mediated endocytosis
Class II MHC expression	Constitutive (+++)	Inducible (-)	Inducible (++)	Constitutive (++)	Constitutive (+++)
Co-stimulatory activity	Constitutive B7 (+++)	Inducible B7 (-)	Inducible B7 (++)	Inducible B7 (-)	Inducible B7 (++)
T-cell activation	Naive T cells Effector T cells Memory T cells	(-)	Effector T cells Memory T cells	Effector T cells Memory T cells	Naive T cells Effector T cells Memory T cells

FIGURE 10-18 Differences in the properties of professional antigen-presenting cells affect their ability to present antigen and

induce T-cell activation. Note that activation of effector and memory T cells does not require the co-stimulatory B7 molecule.

Dendritic cells constitutively express high levels of class I and class II MHC molecules as well as high levels of B7-1 and B7-2. For this reason, dendritic cells are very potent activators of naive, memory, and effector T cells. In contrast, all other professional APCs require activation for expression of co-stimulatory B7 molecules on their membranes; consequently, resting macrophages are not able to activate naive T cells and are poor activators of memory and effector T cells. Macrophages can be activated by phagocytosis of bacteria or by bacterial products such as LPS or by $\text{INF-}\gamma$, a $\text{T}_\text{H}1$ -derived cytokine. Activated macrophages up-regulate their expression of class II MHC molecules and co-stimulatory B7 molecules. Thus, activated macrophages are common activators of memory and effector T cells, but their effectiveness in activating naive T cells is considered minimal.

B cells also serve as antigen-presenting cells in T-cell activation. Resting B cells express class II MHC molecules but fail to express co-stimulatory B7 molecules. Consequently, resting B cells cannot activate naive T cells, although they can activate the effector and memory T-cell populations. Upon activation, B cells up-regulate their expression of class II MHC molecules and begin expressing B7. These activated B cells can now activate naive T cells as well as the memory and effector populations.

Cell Death and T-Cell Populations

Cell death is an important feature of development in all multicellular organisms. During fetal life it is used to mold and sculpt, removing unnecessary cells to provide shape and form. It also is an important feature of lymphocyte homeostasis, returning T- and B-cell populations to their appropriate levels after bursts of antigen-induced proliferation. Apoptosis also plays a crucial role in the deletion of potentially autoreactive thymocytes during negative selection and in the removal of developing T cells unable to recognize self (failure to undergo positive selection).

Although the induction of apoptosis involves different signals depending on the cell types involved, the actual death of the cell is a highly conserved process amongst vertebrates and invertebrates. For example, T cells may be induced to die by many different signals, including the withdrawal of growth factor, treatment with glucocorticoids, or TCR signaling. Each of these signals engages unique signaling pathways, but in all cases, the actual execution of the cell involves the activation of a specialized set of proteases known as *caspases*. The role of these proteases was first revealed by studies of developmentally programmed cell deaths in the nematode

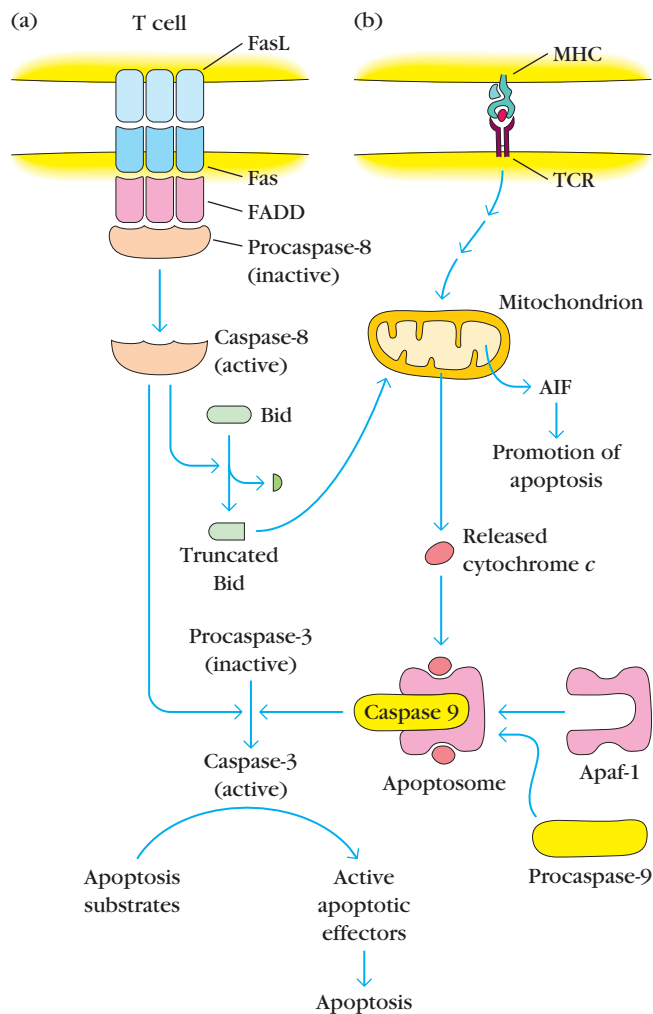


FIGURE 10-19 Two pathways to apoptosis in T cells. (a) Activated peripheral T cells are induced to express high levels of Fas and FasL. FasL induces the trimerization of Fas on a neighboring cell. FasL can also engage Fas on the same cell, resulting in a self-induced death signal. Trimerization of Fas leads to the recruitment of FADD, which leads in turn to the cleavage of associated molecules of procaspase 8 to form active caspase 8. Caspase 8 cleaves procaspase 3, producing active caspase 3, which results in the death of the cell. Caspase 8 can also cleave Bid to a truncated form that can activate the mitochondrial death pathway. (b) Other signals, such as the engagement of the TCR by peptide-MHC complexes on an APC, result in the activation of the mitochondrial death pathway. A key feature of this pathway is the release of AIF (apoptosis inducing factor) and cytochrome *c* from the inner mitochondrial membrane into the cytosol. Cytochrome *c* interacts with Apaf-1 and subsequently with procaspase 9 to form the active apoptosome. The apoptosome initiates the cleavage of procaspase 3, producing active caspase 3, which initiates the execution phase of apoptosis by proteolysis of substances whose cleavage commits the cell to apoptosis. [Adapted in part from S. H. Kaufmann and M. O. Hengartner, 2001. Trends Cell Biol. 11:526.]

C. elegans, where the death of cells was shown to be totally dependent upon the activity of a gene that encoded a cysteine protease with specificity for aspartic acid residues. We now know that in mammals there are at least 14 cysteine proteases or caspases, and all cell deaths require the activity of at least a subset of these molecules. We also know that essentially every cell in the body produces caspase proteins, suggesting that every cell has the potential to initiate its own death.

Cells protect themselves from apoptotic death under normal circumstances by keeping caspases in an inactive form within a cell. Upon reception of the appropriate death signal, certain caspases are activated by proteolytic cleavage and then activate other caspases in turn, leading to the activation of *effector caspases*. This catalytic cascade culminates in cell death. Although it is not well understood how caspase activation directly results in apoptotic death of the cell, presumably it is through the cleavage of critical targets necessary for cell survival.

T cells use two different pathways to activate caspases (Figure 10-19). In peripheral T cells, antigen stimulation results in proliferation of the stimulated T cell and production of several cytokines including IL-2. Upon activation, T cells increase the expression of two key cell-surface proteins involved in T-cell death, Fas and Fas ligand (FasL). When Fas binds its ligand, FasL, FADD (Fas-associated protein with death domain) is recruited and binds to Fas, followed by the recruitment of procaspase 8, an inactive form of caspase 8. The association of FADD with procaspase 8 results in the proteolytic cleavage of procaspase 8 to its active form; caspase 8 then initiates a proteolytic cascade that leads to the death of the cell.

Outside of the thymus, most of the TCR-mediated apoptosis of mature T cells is mediated by the Fas pathway. Repeated or persistent stimulation of peripheral T cells results in the coexpression of both Fas and Fas ligand, followed by the apoptotic death of the cell. The Fas/FasL mediated death of T cells as a consequence of activation is called *activation-induced cell death* (AICD) and is a major homeostatic mechanism, regulating the size of the pool of T cells and removing T cells that repeatedly encounter self antigens.

The importance of Fas and FasL in the removal of activated T cells is underscored by *lpr/lpr* mice, a naturally occurring mutation that results in non-functional Fas. When T cells become activated in these mice, the Fas/FasL pathway is not operative; the T cells continue to proliferate, producing IL-2 and maintaining an activated state. These mice spontaneously develop autoimmune disease, have excessive numbers of T cells, and clearly demonstrate the consequences of a failure to delete activated T cells. An additional mutation, *gld/gld*, is also informative. These mice lack functional FasL and display much the same abnormalities found in the *lpr/lpr* mice. Recently, humans with defects in Fas have been reported. As expected, these individuals display characteristics of autoimmune disease. (See the Clinical Focus box.)

Fas and FasL are members of a family of related receptor/ligands including tumor necrosis factor (TNF) and its

ligand, TNFR (tumor necrosis factor receptor). Like Fas and FasL, membrane-bound TNFR interacts with TNF to induce apoptosis. Also similar to Fas/FasL-induced apoptosis, TNF/TNFR-induced death is the result of the activation of caspase 8 followed by the activation of effector caspases such as caspase 3.

In addition to the activation of apoptosis through death receptor proteins like Fas and TNFR, T cells can die through other pathways that do not activate procaspase 8. For example, negative selection in the thymus induces the apoptotic death of developing T cells via a signaling pathway that originates at the TCR. We still do not completely understand why some signals through the TCR induce positive selection and others induce negative selection, but we know that the strength of the signal plays a critical role. A strong, negatively selecting signal induces a route to apoptosis in which mitochondria play a central role. In mitochondrially dependent apoptotic pathways, cytochrome *c*, which normally resides in the inner mitochondrial membrane, leaks into the cytosol. Cytochrome *c* binds to a protein known as Apaf-1 (apoptotic protease-activating factor-1) and undergoes an ATP-dependent conformational change and oligomerization. Binding of the oligomeric form of Apaf-1 to procaspase 9 results in its transformation to active caspase 9. The complex of cytochrome *c*/Apaf-1/caspase 9, called the *apoptosome*, proteolytically cleaves procaspase 3 generating active caspase 3, which initiates a cascade of reactions that kills the cell (Figure 10-19). Finally, mitochondria also release another molecule, AIF (apoptosis inducing factor), which plays a role in the induction of cell death.

Cell death induced by Fas/FasL is swift, with rapid activation of the caspase cascade leading to cell death in 2–4 hours. On the other hand, TCR-induced negative selection appears to be a more circuitous process, requiring the activation of several processes including mitochondrial membrane failure, the release of cytochrome *c*, and the formation of the apoptosome before caspases become involved. Consequently, TCR-mediated negative selection can take as long as 8–10 hours.

An important feature in the mitochondrially induced cell death pathway is the regulatory role played by Bcl-2 family members. Bcl-2 and Bcl-XL both reside in the mitochondrial membrane. These proteins are strong inhibitors of apoptosis, and while it is not clear how they inhibit cell death, one hypothesis is that they somehow regulate the release of cytochrome *c* from the mitochondria. There are at least three groups of Bcl-2 family members. Group I members are anti-apoptotic and include Bcl-2 and Bcl-xL. Group II and Group III members are pro-apoptotic and include Bax and Bak in Group II and Bid and Bim in Group III. There is clear evidence that levels of anti-apoptotic Bcl-2 family members play an important role in regulating apoptosis in lymphocytes. Bcl-2 family members dimerize, and the anti-apoptotic group members may control apoptosis by dimerizing with pro-apoptotic members, blocking their activity. As indicated in Figure 10-19, cleavage of Bid, catalyzed by caspase 8 gen-

erated by the Fas pathway, can turn on the mitochondrial pathway. Thus signals initiated through Fas can also involve the mitochondrial death pathway.

While it is apparent there are several ways a lymphocyte can be signaled to die, all of these pathways to cell death converge upon the activation of caspases. This part of the cell-death pathway, the execution phase, is common to almost all death pathways known in both vertebrates and invertebrates, demonstrating that apoptosis is an ancient process that has been conserved throughout evolution.

Peripheral $\gamma\delta$ T Cells

In 1986, a small population of peripheral-blood T cells was discovered that expressed CD3 but failed to stain with monoclonal antibody specific for the $\alpha\beta$ T-cell receptor, indicating an absence of the $\alpha\beta$ heterodimer. Many of these cells eventually were found to express the $\gamma\delta$ receptor.

$\gamma\delta$ T Cells Are Far Less Pervasive Than $\alpha\beta$ T Cells

In humans, less than 5% of T cells bear the $\gamma\delta$ heterodimer, and the percentage of $\gamma\delta$ T cells in the lymphoid organs of mice has been reported to range from 1% to 3%. In addition to their presence in blood and lymphoid tissues, they also appear in the skin, intestinal epithelium, and pulmonary epithelium. Up to 1% of the epidermal cells in the skin of mice are $\gamma\delta$ T cells. In general, $\gamma\delta$ T cells are not MHC-restricted, and most do not express the coreceptors CD4 and CD8 present on populations of $\alpha\beta$ T cells. Although the potential of the γ and δ TCR loci to generate diversity is great, very little diversity is found in this type of T cell. In fact, as pointed out in Chapter 9, most of the $\gamma\delta$ T cells in humans have an identical combination of $\gamma\delta$ chains ($\gamma 9$ and $\delta 2$).

$\gamma\delta$ T Cells Recognize Nonpeptide Ligands

Not all T cells are self-MHC restricted and recognize only peptide antigens displayed in the cleft of the self-MHC molecule. Indeed, Chapters 2 and 8 describe $\alpha\beta$ TCR-bearing T cells (NK1-T cells and CD1-restricted T cells) that are not restricted by conventional MHC molecules. In one study, a $\gamma\delta$ T-cell clone was found to bind directly to a herpes-virus protein without requiring antigen processing and presentation together with MHC. Human $\gamma\delta$ T cells have been reported that display MHC-independent binding of a phospholipid derived from *M. tuberculosis*, the organism responsible for tuberculosis (see Chapter 9). This finding suggests that in many cases the TCR receptors of $\gamma\delta$ T cells bind to epitopes in much the same way that the immunoglobulin receptors of B cells do. The fact that most human $\gamma\delta$ T cells all have the same specificity suggests that like other components of the innate immune system, they recognize and respond to



CLINICAL FOCUS

Failure of Apoptosis Causes Defective Lymphocyte Homeostasis

The maintenance of appropriate numbers of various types of lymphocytes is extremely important to an effective immune system. One of the most important elements in this regulation is apoptosis mediated by the Fas/FasL ligand system. The following excerpts from medical histories show what can happen when this key regulatory mechanism fails.

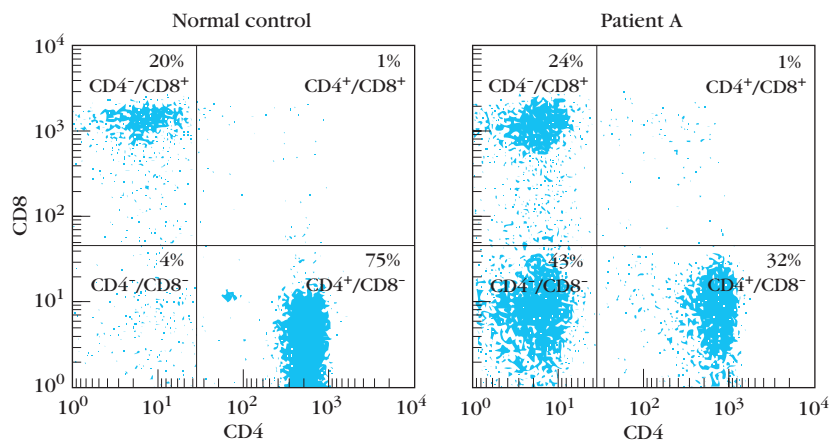
Patient A: A woman, now 43, has had a long history of immunologic imbalances and other medical problems. By age 2, she was diagnosed with the Canale-Smith syndrome (CSS), a severe enlargement of such lymphoid tissues as lymph nodes (lymphadenopathy) and spleen (splenomegaly). Biopsy of lymph nodes showed that, in common with many other CSS patients, she had greatly increased numbers of lymphocytes. She had reduced numbers of platelets (thrombocytopenia) and, because her red blood cells were being lysed, she was anemic (hemolytic anemia). The reduction in numbers of platelets and the lysis of red blood cells could be traced to the action of circulating antibodies that reacted with these host components. At age 21, she was diagnosed with grossly enlarged pelvic lymph nodes that had to be removed. Ten years later, she was again found to have an enlarged abdominal mass, which on surgical removal turned out to be a half-pound lymph-node aggregate. She has continued to have mild lymphadenopathy and, typical of these patients, the lymphocyte populations of enlarged nodes had elevated numbers of T cells (87% as opposed to a normal range of 48%–67% T cells). Ex-

amination of these cells by flow cytometry and fluorescent antibody staining revealed an excess of double-negative T cells (see illustration below). Also, like many patients with Canale-Smith syndrome, she has had cancer, breast cancer at age 22 and skin cancer at ages 22 and 41.

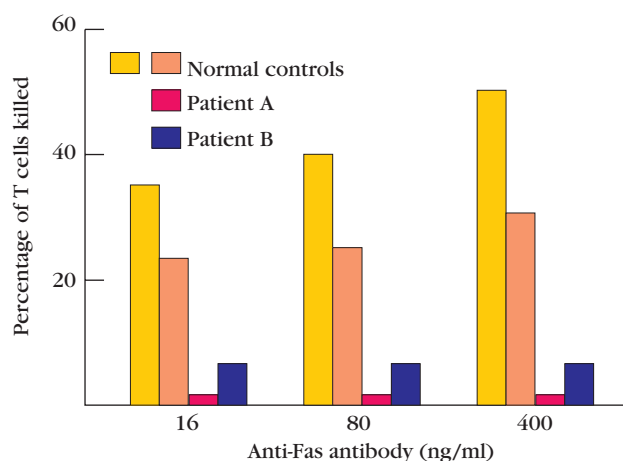
Patient B: A man who was eventually diagnosed with Canale-Smith syndrome had severe lymphadenopathy and splenomegaly as an infant and child. He was treated from age 4 to age 12 with corticosteroids and the immunosuppressive drug mercaptopurine. These appeared to help, and the swelling of lymphoid tissues became milder during adolescence and adulthood. At age 42, he died of liver cancer.

Patient C: An 8-year-old boy, the son of patient B, was also afflicted with Canale-Smith syndrome and showed elevated T-cell counts and severe lymphadenopathy at the age of seven months. At age 2 his spleen became so enlarged that it had to be removed. He also developed hemolytic anemia and thrombocytopenia. However, although he continued to have elevated T-cell counts, the severity of his hemolytic anemia and thrombocytopenia have so far been controlled by treatment with methotrexate, a DNA-synthesis-inhibiting drug used for immunosuppression and cancer chemotherapy.

Recognition of the serious consequences of a failure to regulate the number of lymphocytes, as exemplified by these case histories, emerged from detailed study of several children whose enlarged lymphoid tissues attracted medical attention. In each of these cases of Canale-Smith syndrome, examination revealed grossly enlarged lymph nodes that were 1–2 cm in girth and sometimes large enough to distort the local anatomy. In four of a group of five children who were studied intensively, the



Flow-cytometric analysis of T cells in the blood of Patient A and a control subject. The relative staining by an anti-CD8 antibody is shown on the y axis and the relative staining by an anti-CD4 antibody appears on the x axis. Mature T cells are either CD4⁺ or CD8⁺. While almost all of the T cells in the control subject are CD4⁺ or CD8⁺, the CSS patient shows high numbers of double-negative T cells (43%), which express neither CD4 nor CD8. The percentage of each category of T cells is indicated in the quadrants. [Adapted from Drappa et al., 1996, New England Journal of Medicine **335**:1643.]



Fas-mediated killing takes place when Fas is crosslinked by FasL, its normal ligand, or by treatment with anti-Fas antibody, which artificially crosslinks Fas molecules. This experiment shows the reduction in numbers of T cells after induction of apoptosis in T cells from patients and controls by crosslinking Fas with increasing amounts of an anti-Fas monoclonal antibody. T cells from the Canale-Smith patients (A and B) are resistant to Fas-mediated death. [Adapted from Drappa *et al.*, 1996, *New England Journal of Medicine* **335**:1643.]

spleens were so massive that they had to be removed.

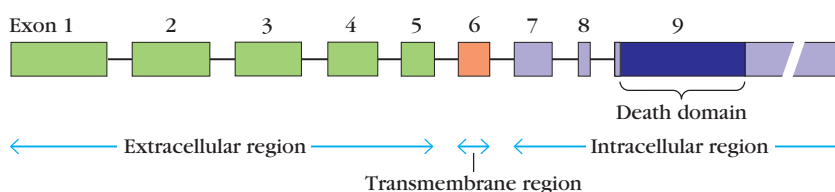
Even though the clinical picture in Canale-Smith syndrome can vary from person to person, with some individuals suffering severe chronic affliction and others only sporadic episodes of illness, there is a common feature, a failure of activated lymphocytes to undergo Fas-mediated apoptosis. Isolation and sequencing of Fas genes from a number of patients and more than 100 controls reveals that CSS patients are heterozygous ($fas^{+/-}$) at the *fas* locus and thus carry one copy of a defective *fas* gene. A comparison of Fas-mediated cell death in T cells from normal controls who do not carry mutant Fas genes with death induced in T cells from CSS patients, shows a marked defect in Fas-induced death (see illustration above). Characterization of the Fas genes so far seen in CSS patients reveals that they have mutations in or around the region encoding the death-inducing domain (the “death domain”) of this protein (see illustration

below). Such mutations result in the production of Fas protein that lacks biological activity but still competes with normal Fas molecules for interactions with essential components of the Fas-mediated death pathway. Other mutations have been found in the extracellular domain of Fas, often associated with milder forms of CSS or no disease at all.

A number of research groups have conducted detailed clinical studies of CSS patients, and the following general observations have been made:

- The cell populations of the blood and lymphoid tissues of CSS patients show dramatic elevations (5-fold to as much as 20-fold) in the numbers of lymphocytes of all sorts, including T cells, B cells, and NK cells.
- Most of the patients have elevated levels of one or more classes of immunoglobulin (hyper-gammaglobulinemia).
- Immune hyperactivity is responsible for such autoimmune phenomena as the production of autoantibodies against red blood cells, resulting in hemolytic anemia, and a depression in platelet counts due to the activity of anti-platelet auto-antibodies.

These observations establish the importance of the death-mediated regulation of lymphocyte populations in lymphocyte homeostasis. Such death is necessary because the immune response to antigen results in a sudden and dramatic increase in the populations of responding clones of lymphocytes and temporarily distorts the representation of these clones in the repertoire. In the absence of cell death, the periodic stimulation of lymphocytes that occurs in the normal course of life would result in progressively increasing, and ultimately unsustainable, lymphocyte levels. As the Canale-Smith syndrome demonstrates, without the essential culling of lymphocytes by apoptosis, severe and life-threatening disease can result.



Map of *fas* locus. The *fas* gene is composed of 9 exons separated by 8 introns. Exons 1–5 encode the extracellular part of the protein, exon 6 encodes the transmembrane region, and exons 7–9 encode the intracellular region of the molecule. Much of exon 9 is responsible for encoding the critical death domain. [Adapted from G. H. Fisher *et al.*, 1995, *Cell* **81**:935.]

molecular patterns that are found in certain pathogens but not in humans. Thus they may play a role as first lines of defense against certain pathogens, expressing effector functions that help control infection and secreting cytokines that promote an adaptive immune response by $\alpha\beta$ T cells and B cells.

SUMMARY

- Progenitor T cells from the bone marrow enter the thymus and rearrange their TCR genes. In most cases these thymocytes rearrange $\alpha\beta$ TCR genes and become $\alpha\beta$ T cells. A small minority rearrange $\gamma\delta$ TCR genes and become $\gamma\delta$ T cells.
- The earliest thymocytes lack detectable CD4 and CD8 and are referred to as double-negative cells. During development, the majority of double-negative thymocytes develop into $CD4^+CD8^-$ $\alpha\beta$ T cells or $CD4^-CD8^+$ $\alpha\beta$ T cells.
- Positive selection in the thymus eliminates T cells unable to recognize self-MHC and is the basis of MHC restriction. Negative selection eliminates thymocytes bearing high-affinity receptors for self-MHC molecules alone or self-antigen plus self-MHC and produces self-tolerance.
- T_H -cell activation is initiated by interaction of the TCR-CD3 complex with a peptide-MHC complex on an antigen-presenting cell. Activation also requires the activity of accessory molecules, including the coreceptors CD4 and CD8. Many different intracellular signal-transduction pathways are activated by the engagement of the TCR.
- T-cells that express CD4 recognize antigen combined with a class II MHC molecule and generally function as T_H cells; T cells that express CD8 recognize antigen combined with a class I MHC molecule and generally function as T_C cells.
- In addition to the signals mediated by the T-cell receptor and its associated accessory molecules (signal 1), activation of the T_H cell requires a co-stimulatory signal (signal 2) provided by the antigen-presenting cell. The co-stimulatory signal is commonly induced by interaction between molecules of the B7 family on the membrane of the APC with CD28 on the T_H cell. Engagement of CTLA-4, a close relative of CD28, by B7 inhibits T-cell activation.
- TCR engagement with antigenic peptide-MHC may induce activation or clonal anergy. The presence or absence of the co-stimulatory signal (signal 2) determines whether activation results in clonal expansion or clonal anergy.
- Naive T cells are resting cells (G_0) that have not encountered antigen. Activation of naive cells leads to the generation of effector and memory T cells. Memory T cells, which are more easily activated than naive cells, are responsible for secondary responses. Effector cells are short lived and perform helper, cytotoxic, or delayed-type hypersensitivity functions.
- The T-cell repertoire is shaped by apoptosis in the thymus and periphery.
- $\gamma\delta$ T cells are not MHC restricted. Most in humans bind free antigen, and most have the same specificity. They may function as part of the innate immune system.

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USEFUL WEB SITES

<http://www.ncbi.nlm.nih.gov/Omim/>
<http://www.ncbi.nlm.nih.gov/htbinpost/Omim/getmim>

The Online Mendelian Inheritance in Man Web site contains a subsite that features ten different inherited diseases associated with defects in the TCR complex or associated proteins.

<http://www.ultranet.com/~jkimball/BiologyPages/A/Apoptosis.html>

http://www.ultranet.com/~jkimball/BiologyPages/B/B_and_Tcells.html

These subsites of John Kimball's Biology Pages Web site provide a clear introduction to T-cell biology and a good basic discussion of apoptosis.

<http://www.bioscience.org/knockout/knochohome.htm>

Within the Frontiers in Bioscience Database of Gene Knockouts, one can find information on the effects of knockouts of many genes involved in the development and function of cells of the T cells.

Study Questions

CLINICAL FOCUS QUESTION Over a period of several years, a group of children and adolescents are regularly dosed with compound X, a life-saving drug. However, in addition to its beneficial effects, this drug interferes with Fas-mediated signaling.

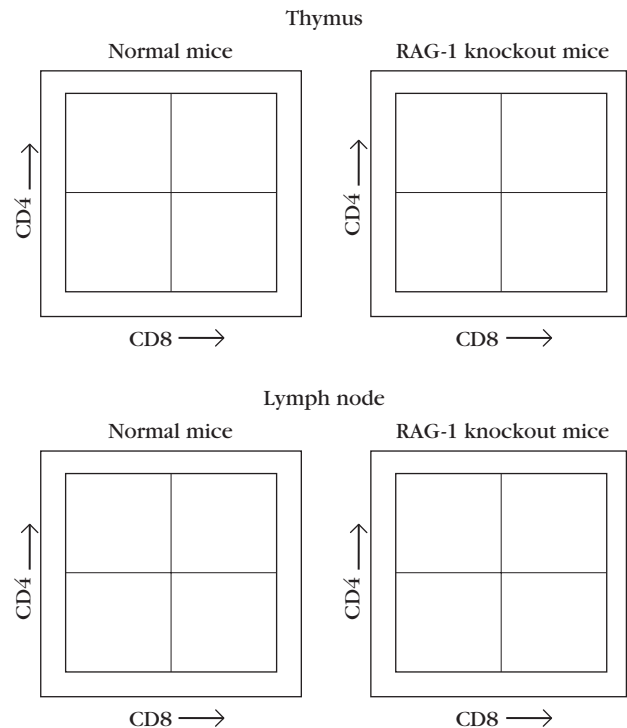
- What clinical manifestations of this side effect of compound X might be seen in these patients?
- If white blood cells from an affected patient are stained with a fluorescein-labeled anti-CD4 and a phycoerythrin-labeled anti-CD8 antibody, what might be seen in the flow-cytometric analysis of these cells? What pattern would be expected if the same procedure were performed on white blood cells from a healthy control?

- You have a $CD8^+$ CTL clone (from an $H-2^k$ mouse) that has a T-cell receptor specific for the H-Y antigen. You clone the $\alpha\beta$ TCR genes from this cloned cell line and use them to prepare transgenic mice with the $H-2^k$ or $H-2^d$ haplotype.
 - How can you distinguish the immature thymocytes from the mature $CD8^+$ thymocytes in the transgenic mice?
 - For each transgenic mouse listed in the table below, indicate with (+) or (−) whether the mouse would have immature double-positive and mature $CD8^+$ thymocytes bearing the transgenic T-cell receptor.

Transgenic mouse	Immature thymocytes	Mature $CD8^+$ thymocytes
$H-2^k$ female		
$H-2^k$ male		
$H-2^d$ female		
$H-2^d$ male		

- Explain your answers for the $H-2^k$ transgenics.
 - Explain your answers for the $H-2^d$ transgenics.
- Cyclosporin and FK506 are powerful immunosuppressive drugs given to transplant recipients. Both drugs prevent the formation of a complex between calcineurin and Ca^{2+} /calmodulin. Explain why these compounds suppress T-cell-mediated aspects of transplant rejection. Hint: see Figure 10-11.

- Antigenic activation of T_H cells leads to the release or induction of various nuclear factors that activate gene transcription.
 - What transcription factors that support proliferation of activated T_H cells are present in the cytoplasm of resting T_H cells in inactive forms?
 - Once in the nucleus, what might these transcription factors do?
- You have fluorescein-labeled anti-CD4 and rhodamine-labeled anti-CD8. You use these antibodies to stain thymocytes and lymph-node cells from normal mice and from RAG-1 knockout mice. In the diagrams below, draw the FACS plots that you would expect.



- In order to demonstrate positive thymic selection experimentally, researchers analyzed the thymocytes from normal $H-2^b$ mice, which have a deletion of the class II *IE* gene, and from $H-2^b$ mice in which the class II *IA* gene had been knocked out.
 - What MHC molecules would you find on antigen-presenting cells from the normal $H-2^b$ mice?
 - What MHC molecules would you find on antigen-presenting cells from the *IA* knockout $H-2^b$ mice?
 - Would you expect to find $CD4^+$ T cells, $CD8^+$ T cells, or both in each type of mouse? Why?
- In his classic chimeric-mouse experiments, Zinkernagel took bone marrow from mouse 1 and a thymus from mouse 2 and transplanted them into mouse 3, which was thymectomized and lethally irradiated. He then challenged the reconstituted mouse with LCM virus and removed its spleen cells. These spleen cells were then incubated with LCM-infected target cells with different MHC haplotypes, and the lysis of the target cells was monitored. The results of two

Experiment	Bone-marrow donor	Thymectomized, x-irradiated recipient	Lysis of LCM-infected target cells		
			H-2 ^d	H-2 ^k	H-2 ^b
A	C57BL/6 × BALB/c	C57BL/6 × BALB/c	+	—	—
B	C57BL/6 × BALB/c	C57BL/6 × BALB/c	—	—	+

such experiments using H-2^b strain C57BL/6 mice and H-2^d strain BALB/c mice are shown in the table on the above.

- What was the haplotype of the thymus-donor strain in experiment A and experiment B?
 - Why were the H-2^b target cells not lysed in experiment A but were lysed in experiment B?
 - Why were the H-2^k target cells not lysed in either experiment?
7. Fill in the blank(s) in each statement below (a–k) with the most appropriate term(s) from the following list. Terms may be used once, more than once, or not at all.

protein phosphatase(s)	CD8	Class I MHC	CD45
protein kinase(s)	CD4	Class II MHC	B7
CD28	IL-2	IL-6	CTLA-4

- Lck and ZAP-70 are _____.
- _____ is a T-cell membrane protein that has cytosolic domains with phosphatase activity.
- Dendritic cells express _____ constitutively, whereas B cells must be activated before they express this membrane molecule.
- Calcineurin, a _____, is involved in generating the active form of the transcription factor NFAT.
- Activation of T_H cells results in secretion of _____ and expression of its receptor, leading to proliferation and differentiation.
- The co-stimulatory signal needed for complete T_H-cell activation is triggered by interaction of _____ on the T cell and _____ on the APC.
- Knockout mice lacking class I MHC molecules fail to produce thymocytes bearing _____.
- Macrophages must be activated before they express _____ molecules and _____ molecules.
- T cells bearing _____ are absent from the lymph nodes of knockout mice lacking class II MHC molecules.
- PIP₂ is split by a _____ to yield DAG and IP₃.
- In activated T_H cells, DAG activates a _____, which acts to generate the transcription factor NF-κB.
- _____ stimulates and _____ inhibits T-cell activation when engaged by _____ or on antigen-presenting cells.

8. You wish to determine the percentage of various types of thymocytes in a sample of cells from mouse thymus using the indirect immunofluorescence method.

- You first stain the sample with goat anti-CD3 (primary antibody) and then with rabbit FITC-labeled anti-goat Ig (secondary antibody), which emits a green color. Analysis of the stained sample by flow cytometry indicates that 70% of the cells are stained. Based on this result, how many of the thymus cells in your sample are expressing antigen-binding receptors on their surface? Would all be expressing the same type of receptor? Explain your answer. What are the remaining unstained cells likely to be?
 - You then separate the CD3⁺ cells with the fluorescence-activated cell sorter (FACS) and restain them. In this case, the primary antibody is hamster anti-CD4 and the secondary antibody is rabbit PE-labeled anti-hamster-Ig, which emits a red color. Analysis of the stained CD3⁺ cells shows that 80% of them are stained. From this result, can you determine how many T_C cells are present in this sample? If yes, then how many T_C cells are there? If no, what additional experiment would you perform in order to determine the number of T_C cells that are present?
9. Many of the effects of engaging the TCR with MHC-peptide can be duplicated by the administration of ionomycin plus a phorbol ester. Ionomycin is a Ca²⁺ ionophore, a compound that allows calcium ions in the medium to cross the plasma membrane and enter the cell. Phorbol esters are analogues of diacylglycerol (DAG). Why does the administration of phorbol and calcium ionophores mimic many effects of TCR engagement?
10. What effects on cell death would you expect to observe in mice carrying the following genetic modifications? Justify your answers.
- Mice that are transgenic for BCL-2 and over-express this protein.
 - Mice in which caspase 8 has been knocked out.
 - Mice in which caspase 3 has been knocked out.

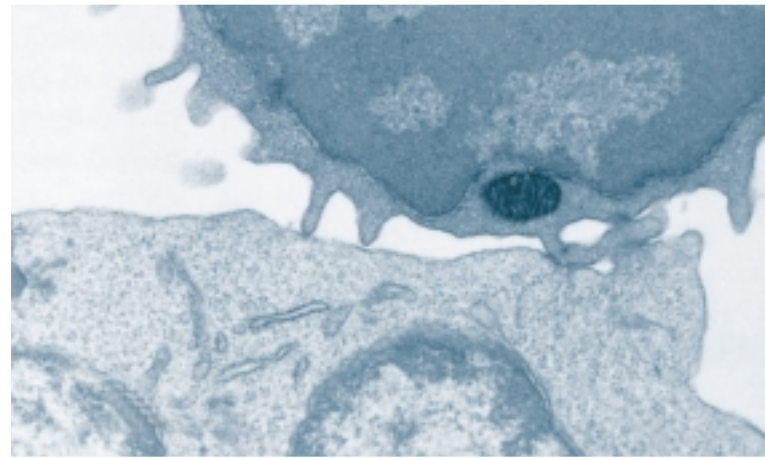
11. Several basic themes of signal transduction were identified and discussed in this chapter. What are these themes? Consider the signal-transduction processes of T-cell activation and provide an example for each of six of the seven themes discussed.

B-Cell Generation, Activation, and Differentiation

THE DEVELOPMENTAL PROCESS THAT RESULTS IN production of plasma cells and memory B cells can be divided into three broad stages: generation of mature, immunocompetent B cells (maturation), activation of mature B cells when they interact with antigen, and differentiation of activated B cells into plasma cells and memory B cells. In many vertebrates, including humans and mice, the bone marrow generates B cells. This process is an orderly sequence of Ig-gene rearrangements, which progresses in the absence of antigen. This is the antigen-independent phase of B-cell development.

A mature B cell leaves the bone marrow expressing membrane-bound immunoglobulin (mIgM and mIgD) with a single antigenic specificity. These **naïve B cells**, which have not encountered antigen, circulate in the blood and lymph and are carried to the secondary lymphoid organs, most notably the spleen and lymph nodes (see Chapter 2). If a B cell is activated by the antigen specific to its membrane-bound antibody, the cell proliferates (clonal expansion) and differentiates to generate a population of antibody-secreting plasma cells and memory B cells. In this activation stage, **affinity maturation** is the progressive increase in the average affinity of the antibodies produced and **class switching** is the change in the isotype of the antibody produced by the B cell from μ to γ , α , or ϵ . Since B cell activation and differentiation in the periphery require antigen, this stage comprises the antigen-dependent phase of B-cell development.

Many B cells are produced in the bone marrow throughout life, but very few of these cells mature. In mice, the size of the recirculating pool of B cells is about 2×10^8 cells. Most of these cells circulate as naïve B cells, which have short life spans (half lives of less than 3 days to about 8 weeks) if they fail to encounter antigen or lose in the competition with other B cells for residence in a supportive lymphoid environment. Given that the immune system is able to generate a total antibody diversity that exceeds 10^9 , clearly only a small fraction of this potential repertoire is displayed at any time by membrane immunoglobulin on recirculating B cells. Indeed, throughout the life span of an animal, only a small fraction of the possible antibody diversity is ever generated.



Initial Contact Between B and T Cells

- B-Cell Maturation
- B-Cell Activation and Proliferation
- The Humoral Response
- In Vivo Sites for Induction of Humoral Responses
- Germinal Centers and Antigen-Induced B-Cell Differentiation
- Regulation of B-Cell Development
- Regulation of the Immune Effector Response

Some aspects of B-cell developmental processes have been described in previous chapters. The overall pathway, beginning with the earliest distinctive B-lineage cell, is described in sequence in this chapter. Figure 11-1 presents an overview of the major events in humans and mice. Most of this chapter applies to humans and mice, but important departures from these developmental pathways have been shown to occur in some other vertebrates. Finally, this chapter will consider the regulation of B-cell development at various stages.

B-Cell Maturation

The generation of mature B cells first occurs in the embryo and continues throughout life. Before birth, the yolk sac, fetal liver, and fetal bone marrow are the major sites of B-cell maturation; after birth, generation of mature B cells occurs in the bone marrow.



VISUALIZING CONCEPTS

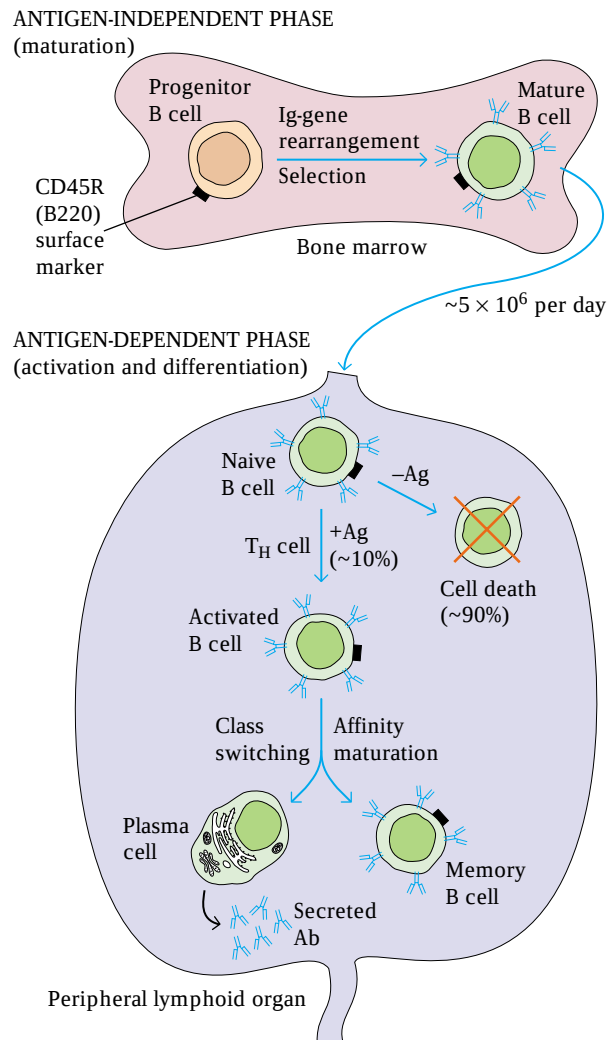


FIGURE 11-1 Overview of B-cell development. During the antigen-independent maturation phase, immunocompetent B cells expressing membrane IgM and IgD are generated in the bone marrow. Only about 10% of the potential B cells reach maturity and exit the bone marrow. Naive B cells in the periphery die within a few days unless they encounter soluble protein antigen and ac-

tivated T_H cells. Once activated, B cells proliferate within secondary lymphoid organs. Those bearing high-affinity mIg differentiate into plasma cells and memory B cells, which may express different isotypes because of class switching. The numbers cited refer to B-cell development in the mouse, but the overall principles apply to humans as well.

Progenitor B Cells Proliferate in Bone Marrow

B-cell development begins as lymphoid stem cells differentiate into the earliest distinctive B-lineage cell—the **progenitor B cell (pro-B cell)**—which expresses a transmembrane tyrosine phosphatase called CD45R (sometimes called B220 in mice). Pro-B cells proliferate within the bone marrow, filling the extravascular spaces between large sinusoids in the

shaft of a bone. Proliferation and differentiation of pro-B cells into **precursor B cells (pre-B cells)** requires the microenvironment provided by the bone-marrow stromal cells. If pro-B cells are removed from the bone marrow and cultured in vitro, they will not progress to more mature B-cell stages unless stromal cells are present. The stromal cells play two important roles: they interact directly with pro-B and pre-B cells, and they secrete various cytokines, notably IL-7, that support the developmental process.

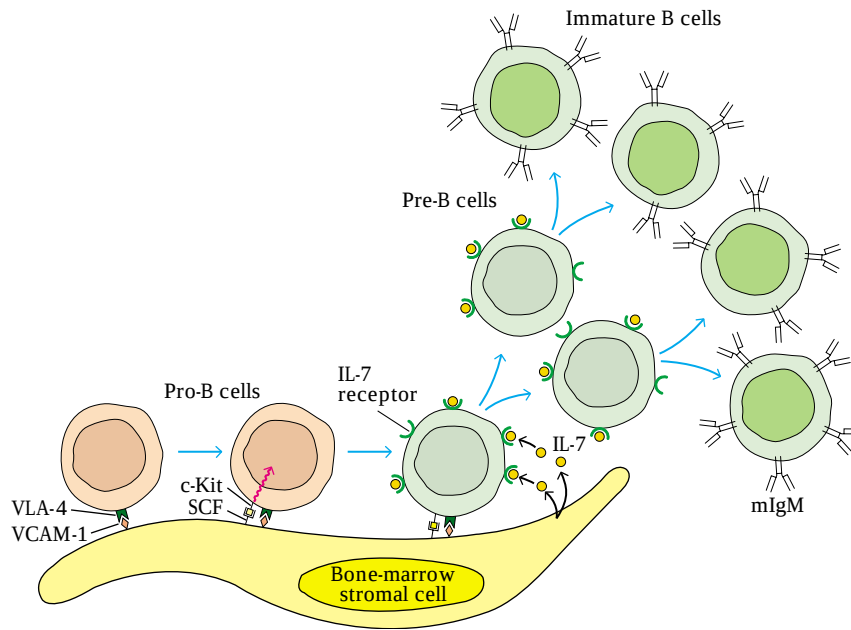


FIGURE 11-2 Bone-marrow stromal cells are required for maturation of progenitor B cells into precursor B cells. Pro-B cells bind to stromal cells by means of an interaction between VCAM-1 on the stromal cell and VLA-4 on the pro-B cell. This interaction promotes the binding of c-Kit on the pro-B cell to stem cell factor (SCF) on the

stromal cell, which triggers a signal, mediated by the tyrosine kinase activity of c-Kit, that stimulates the pro-B cell to express receptors for IL-7. IL-7 released from the stromal cell then binds to the IL-7 receptors, inducing the pro-B cell to mature into a pre-B cell. Proliferation and differentiation eventually produces immature B cells.

At the earliest developmental stage, pro-B cells require direct contact with stromal cells in the bone marrow. This interaction is mediated by several cell-adhesion molecules, including VLA-4 on the pro-B cell and its ligand, VCAM-1, on the stromal cell (Figure 11-2). After initial contact is made, a receptor on the pro-B cell called c-Kit interacts with a stromal-cell surface molecule known as stem-cell factor (SCF). This interaction activates c-Kit, which is a tyrosine kinase, and the pro-B cell begins to divide and differentiate into a pre-B cell and begins expressing a receptor for IL-7. The IL-7 secreted by the stromal cells drives the maturation process, eventually inducing down-regulation of the adhesion molecules on the pre-B cells, so that the proliferating cells can detach from the stromal cells. At this stage, pre-B cells no longer require direct contact with stromal cells but continue to require IL-7 for growth and maturation.

Ig-Gene Rearrangement Produces Immature B Cells

B-cell maturation depends on rearrangement of the immunoglobulin DNA in the lymphoid stem cells. The mechanisms of Ig-gene rearrangement were described in Chapter 5. First to occur in the pro-B cell stage is a heavy-chain D_H -to- J_H gene rearrangement; this is followed by a V_H -to- D_HJ_H rearrangement (Figure 11-3). If the first heavy-chain rearrangement is not productive, then V_H - D_H - J_H rearrange-

ment continues on the other chromosome. Upon completion of heavy-chain rearrangement, the cell is classified as a pre-B cell. Continued development of a pre-B cell into an immature B cell requires a productive light-chain gene rearrangement. Because of allelic exclusion, only one light-chain isotype is expressed on the membrane of a B cell. Completion of a productive light-chain rearrangement commits the now immature B cell to a particular antigenic specificity determined by the cell's heavy-chain VDJ sequence and light-chain VJ sequence. Immature B cells express mIgM (membrane IgM) on the cell surface.

As would be expected, the recombinase enzymes RAG-1 and RAG-2, which are required for both heavy-chain and light-chain gene rearrangements, are expressed during the pro-B and pre-B cell stages (see Figure 11-3). The enzyme terminal deoxynucleotidyl transferase (TdT), which catalyzes insertion of N-nucleotides at the D_H - J_H and V_H - D_HJ_H coding joints, is active during the pro-B cell stage and ceases to be active early in the pre-B-cell stage. Because TdT expression is turned off during the part of the pre-B-cell stage when light-chain rearrangement occurs, N-nucleotides are not usually found in the V_L - J_L coding joints.

The bone-marrow phase of B-cell development culminates in the production of an IgM-bearing immature B cell. At this stage of development the B cell is not fully functional, and antigen induces death or unresponsiveness (anergy) rather than division and differentiation. Full maturation is signaled

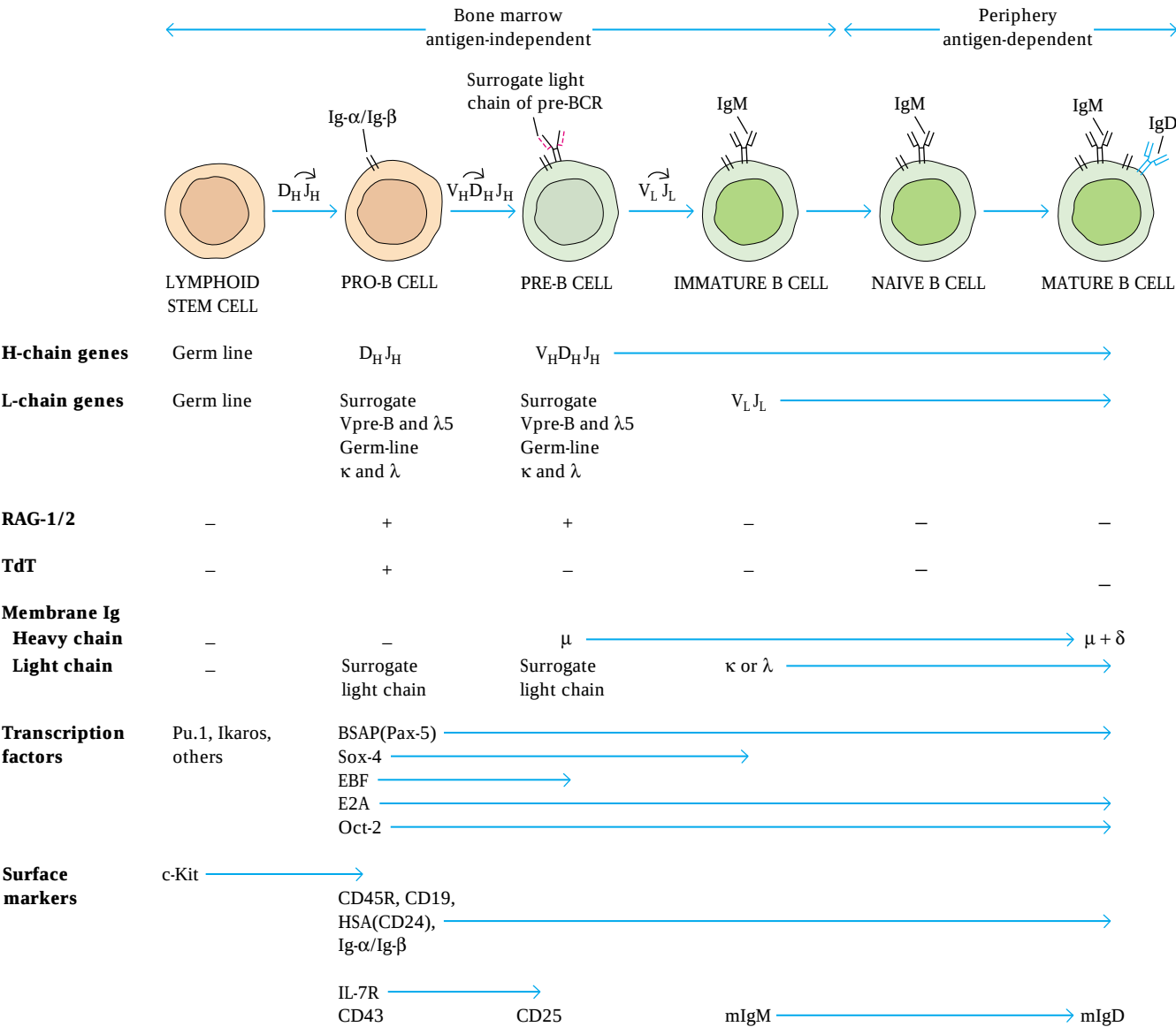


FIGURE 11-3 Sequence of events and characteristics of the stages in B-cell maturation in the bone marrow. The pre-B cell expresses a membrane immunoglobulin consisting of a heavy (H) chain and surrogate light chains, Vpre-B and $\lambda 5$. Changes in the RNA processing of heavy-chain transcripts following the pre-B cell stage lead to syn-

thesis of both membrane-bound IgM and IgD by mature B cells. RAG-1/2 = two enzymes encoded by recombination-activating genes; TdT = terminal deoxyribonucleotidyl transferase. A number of B-cell-associated transcription factors are important at various stages of B-cell development; some are indicated here.

by the co-expression of IgD and IgM on the membrane. This progression involves a change in RNA processing of the heavy-chain primary transcript to permit production of two mRNAs, one encoding the membrane form of the μ chain and the other encoding the membrane form of the δ chain (see Figure 5-19). Although IgD is a characteristic cell-surface marker of mature naive B cells, its function is not clear. However, since immunoglobulin δ knockout mice have essentially normal numbers of fully functional B cells, IgD is not essential to either B-cell development or antigen responsiveness.

The Pre-B-Cell Receptor Is Essential for B-Cell Development

As we saw in Chapter 10, during one stage in T-cell development, the β chain of the T-cell receptor associates with pre-T α to form the pre-T-cell receptor (see Figure 10-1). A parallel situation occurs during B-cell development. In the pre-B cell, the membrane μ chain is associated with the **surrogate light chain**, a complex consisting of two proteins: a V-like sequence called **Vpre-B** and a C-like sequence called

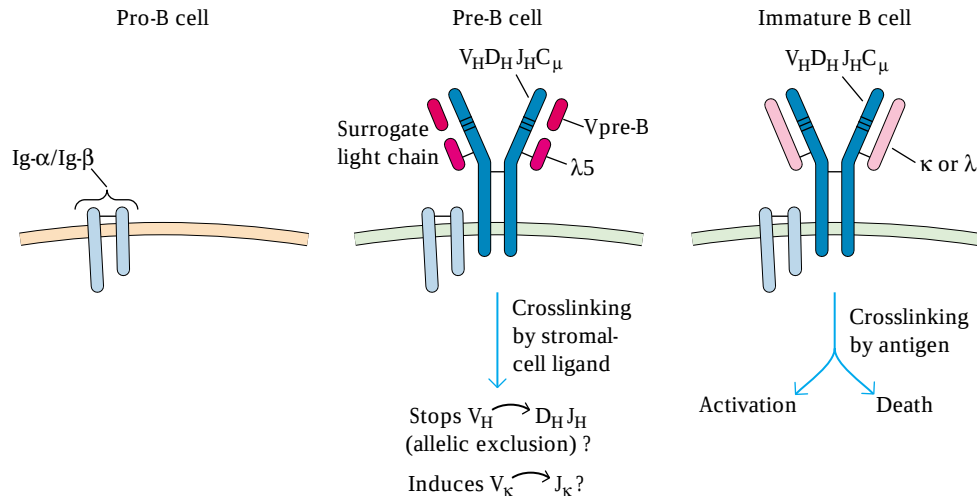


FIGURE 11-4 Schematic diagram of sequential expression of membrane immunoglobulin and surrogate light chain at different stages of B-cell differentiation in the bone marrow. The pre-B-cell receptor contains a surrogate light chain consisting of a V_{pre-B} polypeptide

and a $\lambda 5$ polypeptide, which are noncovalently associated. The immature B cell no longer expresses the surrogate light chain and instead expresses the κ or λ light chain together with the μ heavy chain.

$\lambda 5$, which associate noncovalently to form a light-chain-like structure.

The membrane-bound complex of μ heavy chain and surrogate light chain appears on the pre-B cell associated with the Ig- α /Ig- β heterodimer to form the pre-B-cell receptor (Figure 11-4). Only pre-B cells that are able to express membrane-bound μ heavy chains in association with surrogate light chains are able to proceed along the maturation pathway.

There is speculation that the pre-B-cell receptor recognizes a not-yet-identified ligand on the stromal-cell membrane, thereby transmitting a signal to the pre-B cell that prevents V_H to $D_H J_H$ rearrangement of the other heavy-chain allele, thus leading to allelic exclusion. Following the establishment of an effective pre-B-cell receptor, each pre-B cell undergoes multiple cell divisions, producing 32 to 64 descendants. Each of these progeny pre-B cells may then rearrange different light-chain gene segments, thereby increasing the overall diversity of the antibody repertoire.

The critical role of the pre-B-cell receptor was demonstrated with knockout mice in which the gene encoding the $\lambda 5$ protein of the receptor was disrupted. B-cell development in these mice was shown to be blocked at the pre-B stage, which suggests that a signal generated through the receptor is necessary for pre-B cells to proceed to the immature B-cell stage.

Knockout Experiments Identified Essential Transcription Factors

As described in Chapter 2, many different transcription factors act in the development of hematopoietic cells. Nearly a dozen of them have so far been shown to play roles in B-cell development. Experiments in which particular transcription

factors are knocked out by gene disruption have shown that four such factors, **E2A**, **early B-cell factor (EBF)**, **B-cell-specific activator protein (BSAP)**, and **Sox-4** are particularly important for B-cell development (see Figure 11-3). Mice that lack E2A do not express RAG-1, are unable to make $D_H J_H$ rearrangements, and fail to express $\lambda 5$, a critical component of the surrogate light chain. A similar pattern is seen in EBF-deficient mice. These findings point to important roles for both of these transcription factors early in B-cell development, and they may play essential roles in the early stages of commitment to the B-cell lineage. Knocking out the **Pax-5** gene, whose product is the transcription factor BSAP, also results in the arrest of B-cell development at an early stage. Binding sites for BSAP are found in the promoter regions of a number of B-cell-specific genes, including V_{pre-B} and $\lambda 5$, in a number of Ig switch regions, and in the Ig heavy-chain enhancer. This indicates that BSAP plays a role beyond the early stages of B-cell development. This factor is also expressed in the central nervous system, and its absence results in severe defects in mid-brain development. Although the exact site of action of Sox-4 is not known, it affects early stages of B-cell activation. While Figure 11-3 shows that all of these transcription factors affect development at an early stage, some of them are active at later stages also.

Cell-Surface Markers Identify Development Stages

The developmental progression from progenitor to mature B cell is typified by a changing pattern of surface markers (see Figure 11-3). At the pro-B stage, the cells do not display the heavy or light chains of antibody but they do express CD45R,

which is a form of the protein tyrosine phosphatase found on leukocytes, and the signal-transducing molecules Ig- α /Ig- β , which are found in association with the membrane forms of antibody in later stages of B-cell development. Pro-B cells also express CD19 (part of the B-cell coreceptor), CD43 (leukosialin), and CD24, a molecule also known as heat-stable antigen (HSA) on the surface. At this stage, c-Kit, a receptor for a growth-promoting ligand present on stromal cells, is also found on the surface of pro-B cells. As cells progress from the pro-B to the pre-B stage, they express many of the same markers that were present during the pro-B stage; however, they cease to express c-Kit and CD43 and begin to express CD25, the α chain of the IL-2 receptor. The display of the pre-B-cell receptor (pre-BCR) is a salient feature of the pre-B cell stage. After rearrangement of the light chain, surface immunoglobulin containing both heavy and light chains appears, and the cells, now classified as immature B cells, lose the pre-BCR and no longer express CD25. Monoclonal antibodies are available that can recognize all of these antigenic markers, making it possible to recognize and isolate the various stages of B-cell development by the techniques of immunohistology and flow cytometry described in Chapter 6.

B-1 B Cells Are a Self-Renewing B-Cell Subset

There is a subset of B cells, called B-1 B cells, that arise before B-2 B cells, the major group of B cells in humans and mice. In humans and mice, B-1 B cells compose about 5% of the total B-cell population. They appear during fetal life, express surface IgM but little or no IgD, and are marked by the display of CD5. However, CD5 is not an indispensable component of the B-1 lineage, it does not appear on the B-1 cells of rats, and mice that lack a functional CD5 gene still produce B-1 cells. In animals whose B-2 B cells are the major B-cell population, B-1 cells are minor populations in such secondary tissues as lymph nodes and spleen. Despite their scarcity in many lymphoid sites, they are the major B-cell type found in the peritoneum.

Although there is not a great deal of definitive information on the function of B-1 cells, several features set them apart from the B-2 B cells of humans and mice. During fetal life, B-1 cells arise from stem cells in bone marrow. However, in postnatal life this population renews itself by the prolifer-

ation of some B-1 cells in sites outside the bone marrow to form additional naive B-1 cells. The B-1 population responds poorly to protein antigens but much better to carbohydrate ones. Most of its members are IgM-bearing cells, and this population undergoes much less somatic hypermutation and class switching than the B-2 set of B cells does. Consequently, the antibodies produced by a high proportion of B-1 cells are of rather low affinity.

Self-Reactive B Cells Are Selected Against in Bone Marrow

It is estimated that in the mouse the bone marrow produces about 5×10^7 B cells/day but only 5×10^6 (or about 10%) are actually recruited into the recirculating B-cell pool. This means that 90% of the B cells produced each day die without ever leaving the bone marrow. Some of this loss is attributable to **negative selection** and subsequent elimination (**clonal deletion**) of immature B cells that express auto-antibodies against self-antigens in the bone marrow.

It has long been established that the crosslinkage of mIgM on immature B cells, demonstrated experimentally by treating immature B cells with antibody against the μ constant region, can cause the cells to die by apoptosis within the bone marrow. A similar process is thought to occur in vivo when immature B cells that express self-reactive mIgM bind to self-antigens in the bone marrow. For example, D. A. Nemazee and K. Burki introduced a transgene encoding the heavy and light chains of an IgM antibody specific for K^k , an H-2^k class I MHC molecule, into H-2^d and H-2^{d/k} mice (Figure 11-5a,b). Since class I MHC molecules are expressed on the membrane of all nucleated cells, the endogenous H-2^k and H-2^d class I MHC molecules would be present on bone-marrow stromal cells in the transgenic mice. In the H-2^d mice, which do not express K^k , 25%–50% of the mature, peripheral B cells expressed the transgene-encoded anti- K^k both as a membrane antibody and as secreted antibody. In contrast, in the H-2^{d/k} mice, which express K^k , no mature, peripheral B cells expressed the transgene-encoded antibody to H-2^k (Table 11-1). These results suggest that there is negative selection against any immature B cells expressing auto-antibodies on their membranes because these antibodies react with self-antigen

TABLE 11-1 Expression of transgene encoding IgM antibody to H-2^k class I MHC molecules

Experimental animal	Number of animals tested	EXPRESSION OF TRANSGENE	
		As membrane Ab	As secreted Ab (μg/ml)
Nontransgenics	13	(–)	<0.3
H-2 ^d transgenics	7	(+)	93.0
H-2 ^{d/k} transgenics	6	(–)	<0.3

SOURCE: Adapted from D. A. Nemazee and K. Burki, 1989, *Nature* 337:562.

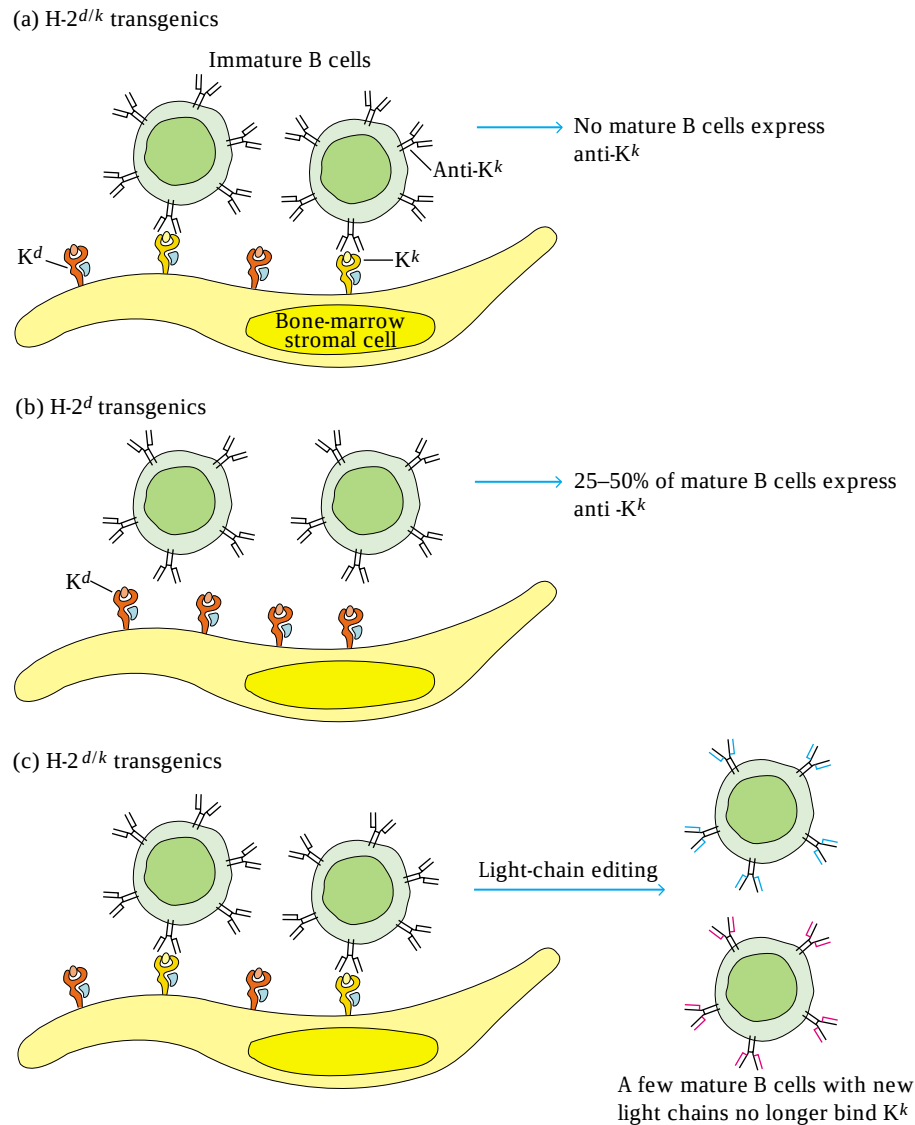


FIGURE 11-5 Experimental evidence for negative selection (clonal deletion) of self-reactive B cells during maturation in the bone marrow. The presence or absence of mature peripheral B cells expressing a transgene-encoded IgM against the H-2 class I molecule K^k was determined in H-2^{d/k} mice (a) and H-2^d mice (b). In the H-2^{d/k} transgenics, the immature B cells recognized the self-antigen K^k and were deleted by negative selection. In the H-2^d transgenics, the immature B cells did not bind to a self-antigen and consequently went on to

mature, so that 25%–50% of the splenic B cells expressed the transgene-encoded anti-K^k as membrane Ig. More detailed analysis of the H-2^{d/k} transgenics revealed a few peripheral B cells that expressed the transgene-encoded μ chain but a different light chain (c). Apparently, a few immature B cells underwent light-chain editing, so they no longer bound the K^k molecule and consequently escaped negative selection. [Adapted from D. A. Nemazee and K. Burki, 1989, *Nature* **337**: 562; S. L. Tiegs et al., 1993, *JEM* **177**:1009.]

(e.g., the K^k molecule in H-2^{d/k} transgenics) present on stromal cells, leading to crosslinking of the antibodies and subsequent death of the immature B cells.

Self-Reactive B Cells May Be Rescued by Editing of Light-Chain Genes

Later work using the transgenic system described by Nemazee and Burki showed that negative selection of immature B cells

does not always result in their immediate deletion (Figure 11-5c). Instead, maturation of the self-reactive cell is arrested while the B cell “edits” the light-chain gene of its receptor. In this case, the H-2^{d/k} transgenics produced a few mature B cells that expressed mIgM containing the μ chain encoded in the transgene, but different, endogenous light chains. One explanation for these results is that when some immature B cells bind a self-antigen, maturation is arrested; the cells up-regulate RAG-1 and RAG-2 expression and begin further

rearrangement of their endogenous light-chain DNA. Some of these cells succeed in replacing the κ light chain of the self-antigen reactive antibody with a λ chain encoded by endogenous λ -chain gene segments. As a result, these cells will begin to express an “edited” mIgM with a different light chain and a specificity that is not self-reactive. These cells escape negative selection and leave the bone marrow.

B-Cell Activation and Proliferation

After export of B cells from the bone marrow, activation, proliferation, and differentiation occur in the periphery and require antigen. Antigen-driven activation and clonal selection of naive B cells leads to generation of plasma cells and memory B cells. In the absence of antigen-induced activation, naive B cells in the periphery have a short life span, dying within a few weeks by apoptosis (see Figure 11-1).

Thymus-Dependent and Thymus-Independent Antigen Have Different Requirements for Response

Depending on the nature of the antigen, B-cell activation proceeds by two different routes, one dependent upon T_H cells, the other not. The B-cell response to **thymus-dependent (TD) antigens** requires direct contact with T_H cells, not simply exposure to T_H -derived cytokines. Antigens that can activate B cells in the absence of this kind of direct participation by T_H cells are known as **thymus-independent (TI) antigens**. TI antigens are divided into types 1 and 2, and they activate B cells by different mechanisms. Some bacterial cell-wall components, including lipopolysaccharide (LPS), function as *type 1 thymus-independent (TI-1) antigens*. *Type 2 thymus-independent (TI-2) antigens* are highly repetitious molecules such as polymeric proteins (e.g., bacterial flagellin) or bacterial cell-wall polysaccharides with repeating polysaccharide units.

Most TI-1 antigens are polyclonal B-cell activators (**mitogens**); that is, they are able to activate B cells regardless of their antigenic specificity. At high concentrations, some TI-1 antigens will stimulate proliferation and antibody secretion by as many as one third of all B cells. The mechanism by which TI-1 antigens activate B cells is not well understood. When B cells are exposed to lower concentrations of TI-1 antigens, only those B cells specific for epitopes of the antigen will be activated. These antigens can stimulate antibody production in nude mice (which lack a thymus and thus are greatly deficient in T cells), and the response is not greatly augmented by transferring T cells into these athymic mice, indicating that TI-1 antigens are truly T-cell independent. The prototypic TI-1 antigen is **lipopolysaccharide (LPS)**, a major component of the cell walls of gram-negative bacteria. At low concentrations, LPS stimulates the production of antibodies specific for LPS. At high concentrations, it is a polyclonal B-cell activator.

TI-2 antigens activate B cells by extensively crosslinking the mIg receptor. However, TI-2 antigens differ from TI-1 antigens in three important respects. First, they are not B-cell mitogens and so do not act as polyclonal activators. Second, TI-1 antigens will activate both mature and immature B cells, but TI-2 antigens activate mature B cells and inactivate immature B cells. Third, although the B-cell response to TI-2 antigens does not require direct involvement of T_H cells, cytokines derived from T_H cells are required for efficient B-cell proliferation and for class switching to isotypes other than IgM. This can be shown by comparing the effect of TI-2 antigens in mice made T-cell-deficient in various ways. In nude mice, which lack thymus-derived T cells but do contain a few T cells that arise from other sites that probably lie in the intestine, TI-2 antigens do elicit B-cell responses. TI-2 antigens do not induce antibody production in mice that cannot express either $\alpha\beta$ or $\gamma\delta$ TCRs because the genes encoding the TCR β and δ chains have been knocked out. Administration of a few T cells to these TCR-knockout mice restores their ability to elicit B-cell responses to TI-2 antigens.

The humoral response to thymus-independent antigens is different from the response to thymus-dependent antigens (Table 11-2). The response to TI antigens is generally weaker, no memory cells are formed, and IgM is the predominant antibody secreted, reflecting a low level of class switching. These differences highlight the important role played by T_H cells in generating memory B cells, affinity maturation, and class switching to other isotypes.

Two Types of Signals Drive B Cells into and Through the Cell Cycle

Naive, or resting, B cells are nondividing cells in the G_0 stage of the cell cycle. Activation drives the resting cell into the cell cycle, progressing through G_1 into the S phase, in which DNA is replicated. The transition from G_1 to S is a critical restriction point in the cell cycle. Once a cell has reached S, it completes the cell cycle, moving through G_2 and into mitosis (M).

Analysis of the progression of lymphocytes from G_0 to the S phase revealed similarities with the parallel sequence in fibroblast cells. These events could be grouped into two categories, competence signals and progression signals. Competence signals drive the B cell from G_0 into early G_1 , rendering the cell competent to receive the next level of signals. Progression signals then drive the cell from G_1 into S and ultimately to cell division and differentiation. Competence is achieved by not one but two distinct signaling events, which are designated *signal 1* and *signal 2*. These signaling events are generated by different pathways with thymus-independent and thymus-dependent antigens, but both pathways include signals generated when multivalent antigen binds and cross-links mIg (Figure 11-6). Once the B cell has acquired an effective competence signal in early activation, the interaction of cytokines and possibly other ligands with the B-cell membrane receptors provides progression signals.

TABLE 11-2 Properties of thymus-dependent and thymus-independent antigens

Property	TD antigens	TI ANTIGENS	
		Type 1	Type 2
Chemical nature	Soluble protein	Bacterial cell-wall components (e.g., LPS)	Polymeric protein antigens; capsular polysaccharides
Humoral response			
Isotype switching	Yes	No	Limited
Affinity maturation	Yes	No	No
Immunologic memory	Yes	No	No
Polyclonal activation	No	Yes (high doses)	No

Transduction of Activating Signals Involves Ig- α /Ig- β Heterodimers

For many years, immunologists questioned how engagement of the Ig receptor by antigen could activate intracellular signaling pathways. All isotypes of mIg have very short cytoplasmic tails. Both mIgM and mIgD on B cells extend into the cytoplasm by only three amino acids; the mIgA tail consists of 14 amino acids; and the mIgG and mIgE tails contain 28 amino acids. In each case, the cytoplasmic tail is too short to be able to generate a signal by associating with intracellular signaling molecules, such as tyrosine kinases and G proteins. The discovery that membrane Ig is associated with the disulfide-linked heterodimer Ig- α /Ig- β , forming the **B-cell receptor (BCR)**, solved this longstanding puzzle. Though it was originally thought that two Ig- α /Ig- β heterodimers associated with one mIg to form the B-cell receptor, careful biochemical analysis has shown that only one Ig- α /Ig- β het-

erodimer associates with a single mIg molecule to form the receptor complex. (Figure 11-7). Thus the BCR is functionally divided into the ligand-binding immunoglobulin molecule and the signal-transducing Ig- α /Ig- β heterodimer. A similar functional division marks the pre-BCR, which transduces signals via a complex consisting of an Ig- α /Ig- β heterodimer and μ heavy chains combined with the surrogate light chain (see Figure 11-4). The Ig- α chain has a long cytoplasmic tail containing 61 amino acids; the tail of the Ig- β chain contains 48 amino acids. The cytoplasmic tails of both Ig- α and Ig- β contain the 18-residue motif termed the **immunoreceptor tyrosine-based activation motif (ITAM)**; see Figure 11-7) which has already been described in several molecules of the T-cell-receptor complex (see Figure 10-11). Interactions with the cytoplasmic tails of Ig- α /Ig- β transduce the stimulus produced by crosslinking of mIg molecules into an effective intracellular signal.

In the BCR and the TCR, as well as in a number of receptors for the Fc regions of particular Ig classes (Fc ϵ RI for IgE; Fc γ RIIA, Fc γ RIIC, Fc γ RIIIA for IgG), ligand binding and signal transduction are mediated by a multimeric complex of proteins that is functionally compartmentalized. The ligand-binding portions of these complexes (mIg in the case of the BCR) is on the surface of the cell, and the signal-transducing portion is mostly or wholly within the cell. As is true of the TCR, signaling from the BCR is mediated by protein tyrosine kinases (PTKs). Furthermore, like the TCR, the BCR itself has no PTK activity; this activity is acquired by recruitment of a number of different kinases, from nearby locations within the cell, to the cytoplasmic tails of the signal. Phosphorylation of tyrosines within the ITAMs of the BCR by receptor-associated PTKs is among the earliest events in B-cell activation and plays a key role in bringing other critical PTKs to the BCR and in their activation. The antigen-mediated crosslinking of BCRs initiates a number of signaling cascades that ultimately result in the cell's responses to the crosslinking of its surface immunoglobulin by antigen. The crosslinking of BCRs results in the induction of many signal-transduction pathways

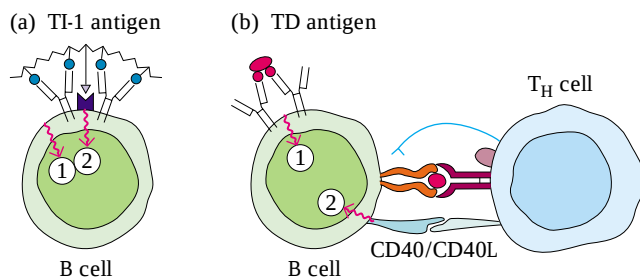


FIGURE 11-6 An effective signal for B-cell activation involves two distinct signals induced by membrane events. Binding of a type 1 thymus-independent (TI-1) antigen to a B cell provides both signals. A thymus-dependent (TD) antigen provides signal 1 by crosslinking mIg, but a separate interaction between CD40 on the B cell and CD40L on an activated T_H cell is required to generate signal 2.

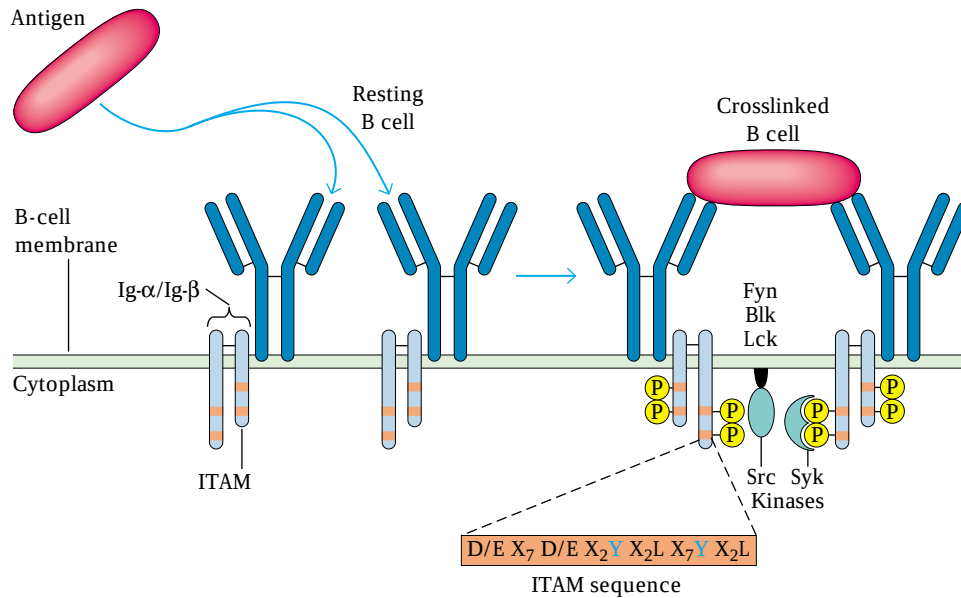


FIGURE 11-7 The initial stages of signal transduction by an activated B-cell receptor (BCR). The BCR comprises an antigen-binding mlg and one signal-transducing Ig-α/Ig-β heterodimer. Following antigen crosslinkage of the BCR, the immunoreceptor tyrosine-based activation motifs (ITAMs) interact with several members of the Src family of tyrosine kinases (Fyn, Blk, and Lck), activating the kinases. The activated

enzymes phosphorylate tyrosine residues on the cytoplasmic tails of the Ig-α/Ig-β heterodimer, creating docking sites for Syk kinase, which is then also activated. The highly conserved sequence motif of ITAMs is shown with the tyrosines (Y) in blue. D/E indicates that an aspartate or a glutamate can appear at the indicated position, and X indicates that the position can be occupied by any amino acid.

and the activation of the B cell. Figure 11-8 shows many parallels between B-cell and T-cell activation. These include:

- **Compartmentalization of function within receptor subunits:** Both the B-cell and T-cell pathways begin with antigen receptors that are composed of an antigen-binding and a signaling unit. The antigen-binding unit confers specificity, but has cytoplasmic tails too short to transduce signals to the cytoplasm of the cell. The signaling unit has long cytoplasmic tails that are the signal transducers of the receptor complex.
- **Activation by membrane-associated Src protein tyrosine kinases:** The receptor-associated PTKs (Lck in T cells and Lyn, Blk, and Fyn in B cells) catalyze phosphorylations during the early stages of signal transduction that are essential to the formation of a functional receptor signaling complex.
- **Assembly of a large signaling complex with protein-tyrosine-kinase activity:** The phosphorylated tyrosines in the ITAMs of the BCR and TCR provide docking sites for the molecules that endow these receptors with PTK activity; ZAP-70 in T cells and Syk in B cells.
- **Recruitment of other signal-transduction pathways:** Signals from the BCR and TCR result in the production

of the second messengers IP₃ and DAG. IP₃ causes the release of Ca²⁺ from intracellular stores, and DAG activates PKC. A third important set of signaling pathways are those governed by the small G proteins Ras and Rac that are also activated by signals received through the TCR or BCR.

- **Changes in gene expression:** One of the important outcomes of signal-transduction processes set in motion with engagement of the BCR or the TCR is the generation or translocation to the nucleus of active transcription factors that stimulate or inhibit the transcription of specific genes.

Failures in signal transduction can have severe consequences for the immune system. The Clinical Focus on X-linked agammaglobulinemia describes the effect of defective signal transduction on the development of B cells.

The B-Cell–Coreceptor Complex Can Enhance B-Cell Responses

Stimulation through antigen receptors can be modified significantly by signals through coreceptors. Recall that co-stimulation through CD28 is an essential feature of effective positive stimulation of T lymphocytes, while signaling through CTLA-4 inhibits the response of the T cell. In B cells



VISUALIZING CONCEPTS

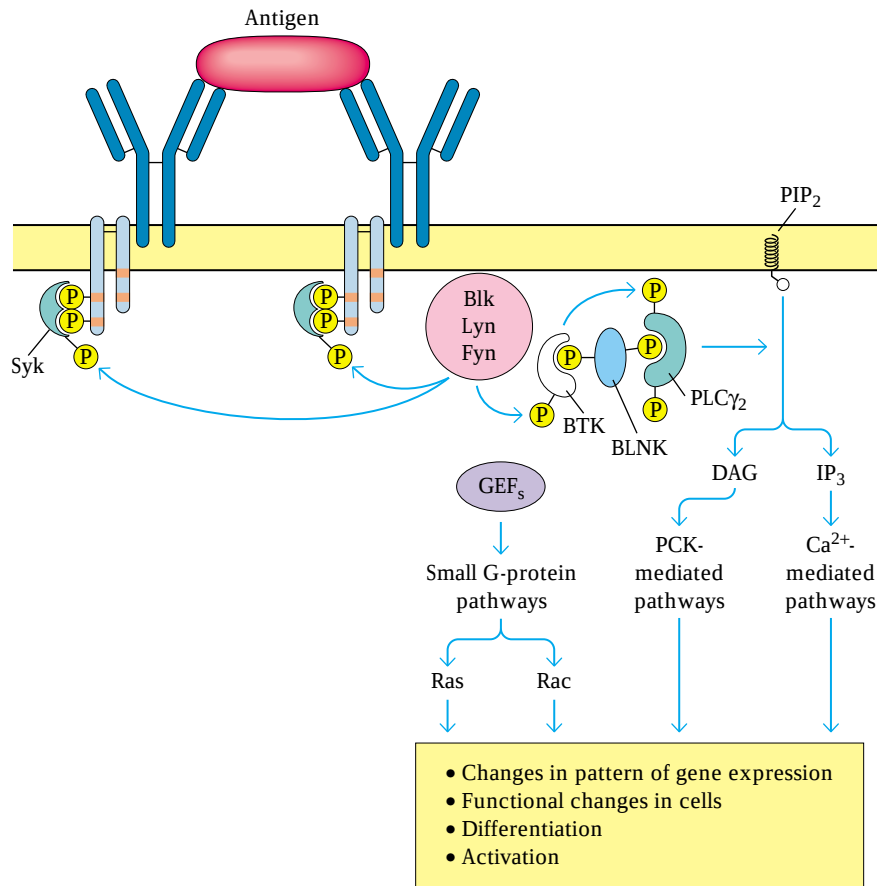


FIGURE 11-8 Some of the many signal-transduction pathways activated by the BCR. In one pathway, Syk activates PLC γ_2 by tyrosine phosphorylation. PLC γ_2 then hydrolyzes PIP $_2$, a membrane phospholipid, to produce the second messengers DAG and IP $_3$. DAG and Ca $^{2+}$ released by the action of IP $_3$ collaboratively activate

the PKC, which induces additional signal-transduction pathways. The activated receptor complex also generates signals that activate the Ras pathway. Activated Ras initiates a cascade of phosphorylations that culminates in the activation of transcription factors that up-regulate the expression of many genes.

a component of the B-cell membrane, called the **B-cell coreceptor**, provides stimulatory modifying signals.

The B-cell coreceptor is a complex of three proteins: CD19, CR2 (CD21), and TAPA-1 (CD81) (Figure 11-9). CD19, a member of the immunoglobulin superfamily, has a long cytoplasmic tail and three extracellular domains. The CR2 component is a receptor of C3d, a breakdown product of the complement system, which is an important effector mechanism for destroying invaders (Chapter 13); note that the involvement of C3d in the pathway for coreceptor activity reveals different arms of the immune system interacting with each other. CR2 also functions as a receptor for a membrane molecule and the transmembrane protein TAPA-1. In addition to the stimulatory coreceptor, another molecule, CD22, which

is constitutively associated with the B-cell receptor in resting B cells, delivers a negative signal that makes B-cells more difficult to activate. As shown in Figure 11-9, the CR2 component of the coreceptor complex binds to complement-coated antigen that has been captured by the mIg on the B cell. This crosslinks the coreceptor to the BCR and allows the CD19 component of the coreceptor to interact with the Ig- α /Ig- β component of the BCR. CD19 contains six tyrosine residues in its long cytoplasmic tail and is a major substrate of the protein tyrosine kinase activity that is mediated by crosslinkage of the BCR. Phosphorylation of CD19 permits it to bind a number of signaling molecules, including the protein tyrosine kinase Lyn.

The delivery of these signaling molecules to the BCR complex contributes to the activation process, and the coreceptor



CLINICAL FOCUS

X-Linked Agammaglobulinemia: A Failure in Signal Transduction and B-Cell Development

X-linked

agammaglobulinemia is a genetically determined immunodeficiency disease characterized by the inability to synthesize all classes of antibody. It was discovered in 1952 by O. C. Bruton in what is still regarded as an outstanding example of research in clinical immunology. Bruton's investigation involved a young boy who had mumps 3 times and experienced 19 different episodes of serious bacterial infections during a period of just over 4 years. Because pneumococcus bacteria were isolated from the child's blood during

10 of the episodes of bacterial infection, attempts were made to induce immunity to pneumococcus by immunization with pneumococcus vaccine. The failure of these efforts to induce antibody responses prompted Bruton to determine whether the patient could mount antibody responses when challenged with other antigens. Surprisingly, immunization with diphtheria and typhoid vaccine preparations did not raise humoral responses in this patient. Electrophoretic analysis of the patient's serum revealed that although normal amounts of albumin and other typical serum proteins were present, gamma

globulin, the major antibody fraction of serum, was absent. Having traced the immunodeficiency to a lack of antibody, Bruton tried a bold new treatment. He administered monthly doses of human immune serum globulin. The patient's experience of a fourteen-month period free of bacterial sepsis established the usefulness of the immunoglobulin replacement for the treatment of immunodeficiency.

Though initially called Bruton's agammaglobulinemia, this hereditary immunodeficiency disease was renamed X-linked agammaglobulinemia, or X-LA, after the discovery that the responsible gene lies on the X chromosome. The disease has the following clinical features:

- Because this defect is X-linked, almost all afflicted individuals are male.
- Signs of immunodeficiency may appear as early as 9 months after birth, when the supply of

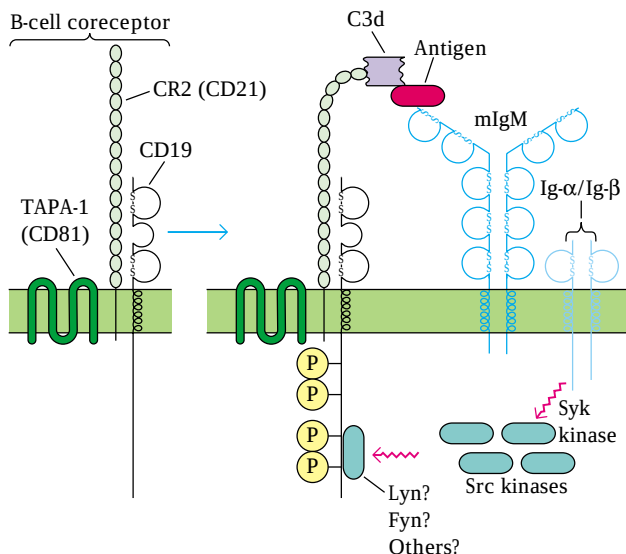


FIGURE 11-9 The B-cell coreceptor is a complex of three cell membrane molecules: TAPA-1 (CD81), CR2 (CD21), and CD19. Binding of the CR2 component to complement-derived C3d that has coated antigen captured by mIg results in the phosphorylation of CD19. The Src-family tyrosine kinase Lyn binds to phosphorylated CD19. The resulting activated Lyn and Fyn can trigger the signal-transduction pathways shown in Figure 11-8 that begin with phospholipase C.

complex serves to amplify the activating signal transmitted through the BCR. In one experimental in vitro system, for example, 10^4 molecules of mIgM had to be engaged by antigen for B-cell activation to occur when the coreceptor was not involved. But when CD19/CD2/TAPA-1 coreceptor was crosslinked to the BCR, only 10^2 molecules of mIgM had to be engaged for B-cell activation. Another striking experiment highlights the role played by the B-cell coreceptor. Mice were immunized with either unmodified lysozyme or a hybrid protein in which genetic engineering was used to join hen's egg lysozyme to C3d. The fusion protein bearing 2 or 3 copies of C3d produced anti-lysozyme responses that were 1000 to 10,000 times greater than those to lysozyme alone. Perhaps coreceptor phenomena such as these explain how naive B cells that often express mIg with low affinity for antigen are able to respond to low concentrations of antigen in a primary response. Such responses, even though initially of low affinity, can play a significant role in the ultimate generation of high-affinity antibody. As described later in this chapter, response to an antigen can lead to affinity maturation, resulting in higher average affinity of the B-cell population. Finally, two experimental observations indicate that the CD19 component of the B-cell coreceptor can play a role independent of CR2, the complement receptor. In normal mice, artificially crosslinking the BCR with anti-BCR antibodies results in the

maternal antibody acquired in utero has decreased below protective levels.

- There is a high frequency of infection by *Streptococcus pneumoniae* and *Haemophilus influenzae*; bacterial pneumonia, sinusitis, meningitis, or septicemia are often seen in these patients.
- Although infection by many viruses is no more severe in these patients than in normal individuals, long-term antiviral immunity is usually not induced.
- Analysis by fluorescence microscopy or flow cytometry shows few or no mature B cells in the blood.

Studies of this disease at the cellular and molecular level provide insights into the workings of the immune system. A scarcity of B cells in the periphery explained the inability of X-LA patients to

make antibody. Studies of the cell populations in bone marrow traced the lack of B cells to failures in B-cell development. The samples displayed a ratio of pro-B cells to pre-B cells 10 times normal, suggesting inhibition of the transition from the pro- to the pre-B-cell stage. The presence of very few mature B cells in the marrow indicated a more profound blockade in the development of B cells from pre-B cells.

In the early 1990s, the gene responsible for X-LA was cloned. The normal counterpart of this gene encodes a protein tyrosine kinase that has been named Bruton's tyrosine kinase (Btk) in honor of the resourceful and insightful physician who discovered X-LA and devised a treatment for it. Parallel studies in mice have shown that the absence of Btk causes a syndrome known as *xid*, an immunodeficiency disease that is essentially identical to its human counterpart, X-LA. Btk has turned out to play important roles in B-cell signaling. For example, crosslinking of the

B-cell receptor results in the phosphorylation of a tyrosine residue in the catalytic domain of Btk. This activates the protein-tyrosine-kinase activity of Btk, which then phosphorylates phospholipase C- γ_2 (PLC- γ_2); in vitro studies of cell cultures in which Btk has been knocked out show compromised PLC- γ_2 activation. Once activated, PLC- γ_2 hydrolyzes membrane phospholipids, liberating the potent second messengers IP₃ and DAG. As mentioned earlier, IP₃ causes a rise in intracellular Ca²⁺, and DAG is an activator of protein kinase C (PKC). Thus, Btk plays a pivotal role in activating a network of intracellular signals vital to the function of mature B cells and earlier members of the B-cell lineage. Research has shown that it belongs to a family of PTKs known as Tec kinases; its counterpart in T cells is Itk. The insights gained from studies of X-LA, *xid*, and Btk are impressive examples of how the study of pathological states can clarify the workings of normal cells.

stimulation of some of the signal-transduction pathways characteristic of B-cell activation. On the other hand, treatment of B cells from mice in which CD19 has been knocked out with anti-BCR antibody fails to induce these pathways. Furthermore, CD19 knockout mice make greatly diminished antibody response to most antigens.

T_H Cells Play Essential Roles in Most B-Cell Responses

As noted already, activation of B cells by soluble protein antigens requires the involvement of T_H cells. Binding of antigen to B-cell mIg does not itself induce an effective competence signal without additional interaction with membrane molecules on the T_H cell. In addition, a cytokine-mediated progression is required for B-cell proliferation. Figure 11-10 outlines the probable sequence of events in B-cell activation by a thymus-dependent (TD) antigen. This process is considerably more complex than activation induced by thymus-independent (TI) antigens.

FORMATION OF T-B CONJUGATE

After binding of antigen by mIg on B cells, the antigen is internalized by receptor-mediated endocytosis and processed within the endocytic pathway into peptides. Antigen binding

also initiates signaling through the BCR that induces the B cell to up-regulate a number of cell-membrane molecules, including class II MHC molecules and the co-stimulatory ligand B7 (see Figure 11-10a). Increased expression of both of these membrane proteins enhances the ability of the B cell to function as an antigen-presenting cell in T_H-cell activation. B-cells could be regarded as helping their helpers because the antigenic peptides produced within the endocytic processing pathway associate with class II MHC molecules and are presented on the B-cell membrane to the T_H cell, inducing its activation. It generally takes 30–60 min after internalization of antigen for processed antigenic peptides to be displayed on the B-cell membrane in association with class II MHC molecules.

Because a B cell recognizes and internalizes antigen specifically, by way of its membrane-bound Ig, a B cell is able to present antigen to T_H cells at antigen concentrations that are 100 to 10,000 times lower than what is required for presentation by macrophages or dendritic cells. When antigen concentrations are high, macrophages and dendritic cells are effective antigen-presenting cells, but, as antigen levels drop, B cells take over as the major presenter of antigen to T_H cells.

Once a T_H cell recognizes a processed antigenic peptide displayed by a class II MHC molecule on the membrane of a

- (a) Antigen crosslinks mIg, generating signal ①, which leads to increased expression of class II MHC and co-stimulatory B7. Antigen–antibody complexes are internalized by receptor-mediated endocytosis and degraded to peptides, some of which are bound by class II MHC and presented on the membrane as peptide–MHC complexes.
- (b) T_H cell recognizes antigen–class II MHC on B-cell membrane. This plus co-stimulatory signal activates T_H cell.
- (c) 1. T_H cell begins to express CD40L.
2. Interaction of CD40 and CD40L provides signal ②.
3. B7-CD28 interactions provide co-stimulation to the T_H cell.
- (d) 1. B cell begins to express receptors for various cytokines.
2. Binding of cytokines released from T_H cell in a directed fashion sends signals that support the progression of the B cell to DNA synthesis and to differentiation.

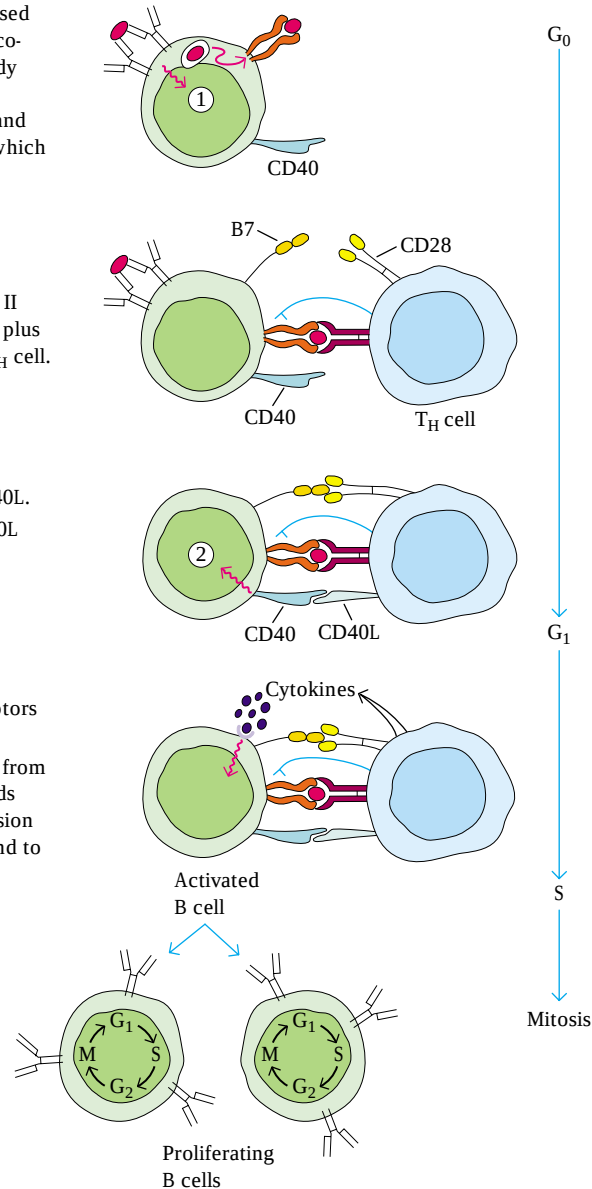


FIGURE 11-10 Sequence of events in B-cell activation by a thymus-dependent antigen. The cell-cycle phase of the interacting B cell is indicated on the right.

B cell, the two cells interact to form a T-B conjugate (Figure 11-11). Micrographs of T-B conjugates reveal that the T_H cells in antigen-specific conjugates have reorganized the Golgi apparatus and the microtubular-organizing center toward the junction with the B cell. This structural adjustment facilitates the release of cytokines toward the antigen-specific B cell.

CONTACT-DEPENDENT HELP MEDIATED BY CD40/CD40L INTERACTION

Formation of a T-B conjugate not only leads to the directional release of T_H -cell cytokines, but also to the up-regulation of

CD40L (CD154), a T_H -cell membrane protein that then interacts with CD40 on B cells to provide an essential signal for T-cell–dependent B-cell activation. CD40 belongs to the tumor necrosis factor (TNF) family of cell-surface proteins and soluble cytokines that regulate cell proliferation and programmed cell death by apoptosis. CD40L belongs to the TNF receptor (TNFR) family. Interaction of CD40L with CD40 on the B cell delivers a signal (signal 2) to the B cell that, in concert with the signal generated by mIg crosslinkage (signal 1), drives the B cell into G_1 (see Figure 11-10c). The signals from CD40 are transduced by a number of intracellular signaling pathways, ultimately resulting in changes in gene expression. Stud-

ies have shown that although CD40 does not have kinase activity, its crosslinking is followed by the activation of protein tyrosine kinases such as Lyn and Syk. Crosslinking of CD40 also results in the activation of phospholipase C and the subsequent generation of the second messengers IP_3 and DAG, and the activation of a number of transcription factors. *Ligation* of CD40 also results in its association with members of the TNFR-associated factor (TRAF) family. A consequence of this interaction is the activation of the transcription factor NF- κ B.

Several lines of evidence have identified the CD40/CD40L interaction as the mediator of contact-dependent help. The role of an inducible T_H -cell membrane protein in B-cell activation was first revealed by experiments in which naive B cells were incubated with antigen and plasma membranes prepared from either activated or resting T_H -cell clones. Only the membranes from the activated T_H cells induced B-cell proliferation, suggesting that one or more molecules expressed on the membrane of an activated T_H cell engage receptors on the B cell to provide contact-dependent help. Furthermore, when antigen-stimulated B cells are treated with anti-CD40 monoclonal antibodies in the absence of T_H cells, they become activated and proliferate. Thus, engagement of CD40, whether by antibodies to CD40 or by CD40L, is critical in providing signal 2 to the B cell. If appropriate cytokines are also added to this experimental system, then the proliferating B cells will differentiate into plasma cells. Conversely, antibodies to CD40L have been shown to block B-cell activation by blocking the CD40/CD40L interaction.

SIGNALS PROVIDED BY T_H -CELL CYTOKINES

Although B cells stimulated with membrane proteins from activated T_H cells are able to proliferate, they fail to dif-

ferentiate unless cytokines are also present; this finding suggests that both a membrane-contact signal and cytokine signals are necessary to induce B-cell proliferation and differentiation. As noted already, electron micrographs of T-B conjugates reveal that the antigen-specific interaction between a T_H and a B cell induces a redistribution of T_H -cell membrane proteins and cytoskeletal elements that results in the polarized release of cytokines toward the interacting B cell.

Once activated, the B cell begins to express membrane receptors for various cytokines, such as IL-2, IL-4, IL-5, and others. These receptors then bind the cytokines produced by the interacting T_H cell. The signals produced by these cytokine-receptor interactions support B-cell proliferation and can induce differentiation into plasma cells and memory B cells, class switching, and affinity maturation. Each of these events is described in a later section.

Mature Self-Reactive B Cells Can Be Negatively Selected in the Periphery

Because some self-antigens do not have access to the bone marrow, B cells expressing mIgM specific for such antigens cannot be eliminated by the negative-selection process in the bone marrow described earlier. To avoid autoimmune responses from such mature self-reactive B cells, some process for deleting them or rendering them inactive must occur in peripheral lymphoid tissue.

A transgenic system developed by C. Goodnow and his coworkers has helped to clarify the process of negative selection of mature B cells in the periphery. Goodnow's experimental system included two groups of transgenic mice

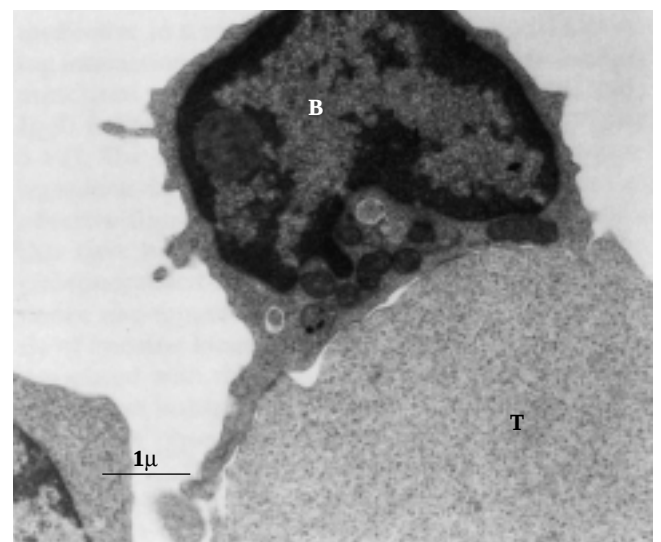
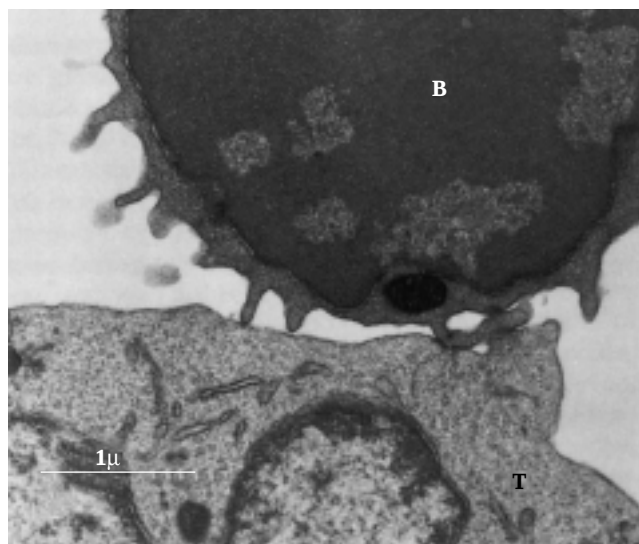


FIGURE 11-11 Transmission electron micrographs of initial contact between a T cell and B cell (left) and of a T-B conjugate (right). Note the

broad area of membrane contact between the cells after formation of the conjugate. [From V. M. Sanders *et al.*, 1986, *J. Immunol.* **137**:2395.]

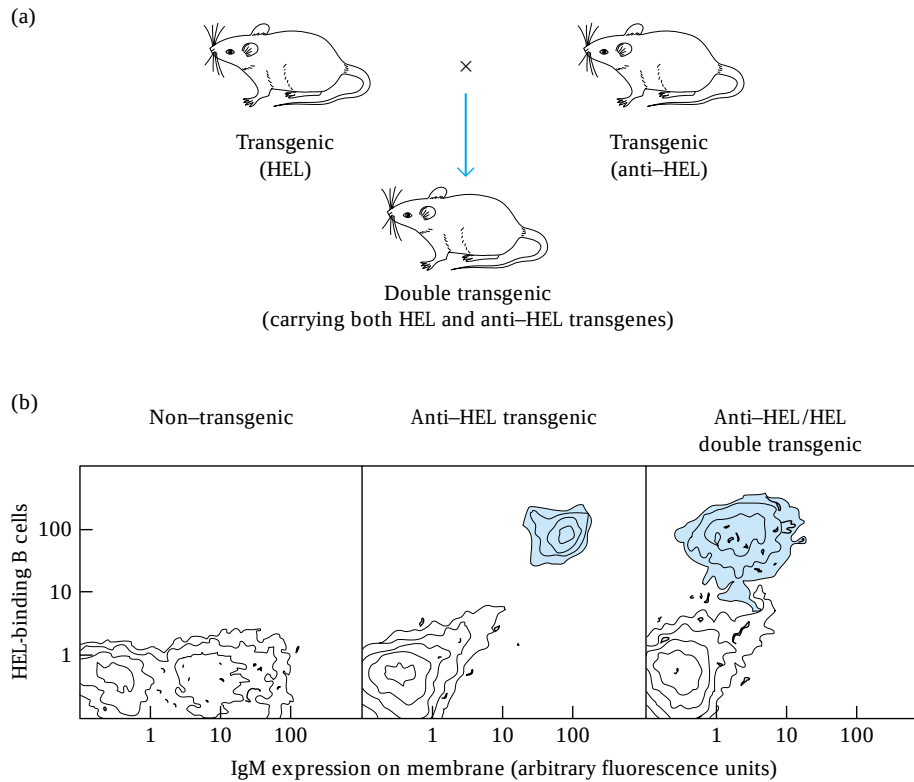


FIGURE 11-12 Goodnow's experimental system for demonstrating clonal anergy in mature peripheral B cells. (a) Production of double-transgenic mice carrying transgenes encoding HEL (hen egg-white lysozyme) and anti-HEL antibody. (b) Flow cytometric analysis of peripheral B cells that bind HEL compared with membrane IgM levels. The number of B cells binding HEL was measured by determining how many cells bound fluorescently labeled HEL. Levels of membrane IgM were determined by incubating the cells with anti-mouse IgM antibody labeled with a fluorescent label different from that used to label HEL. Measurement of the flu-

orescence emitted from this label indicated the level of membrane IgM expressed by the B cells. The nontransgenics (*left*) had many B cells that expressed high levels of surface IgM but almost no B cells that bound HEL above the background level of 1. Both anti-HEL transgenics (*middle*) and anti-HEL/HEL double transgenics (*right*) had large numbers of B cells that bound HEL (blue), although the level of membrane IgM was about twentyfold lower in the double transgenics. The data in Table 11-3 indicate that the B cells expressing anti-HEL in the double transgenics cannot mount a humoral response to HEL.

(Figure 11-12a). One group carried a hen's egg-white lysozyme (HEL) transgene linked to a metallothioneine promoter, which placed transcription of the HEL gene under the control of zinc levels. The other group of transgenic mice carried rearranged immunoglobulin heavy- and light-chain transgenes encoding anti-HEL antibody; in normal mice, the frequency of HEL-reactive B cells is on the order of 1 in 10^3 , but in these transgenic mice the rearranged anti-HEL transgene is expressed by 60%–90% of the mature peripheral B cells. Goodnow mated the two groups of transgenics to produce “double-transgenic” offspring carrying both the HEL and anti-HEL transgenes. Goodnow then asked what effect HEL, which is expressed in the periphery but not in the bone marrow, would have upon the development of B cells expressing the anti-HEL transgene.

The Goodnow double-transgenic system has yielded several interesting findings concerning negative selection of

B cells (Table 11-3). He found that double-transgenic mice expressing high levels of HEL (10^{-9} M) continued to have mature, peripheral B cells bearing anti-HEL membrane antibody, but these B cells were functionally nonresponsive; that is, they were **anergic**. The flow-cytometric analysis of B cells from double-transgenic mice showed that, while large numbers of anergic anti-HEL cells were present, they expressed IgM at levels about 20-fold lower than anti-HEL single transgenics (Figure 11-12b). Further study demonstrated that the double transgenics had both surface IgM and IgD, indicating that the anergy was induced in mature rather than immature B cells. When these mice were given an immunizing dose of HEL, few anti-HEL plasma cells were induced and the serum anti-HEL titer was low.

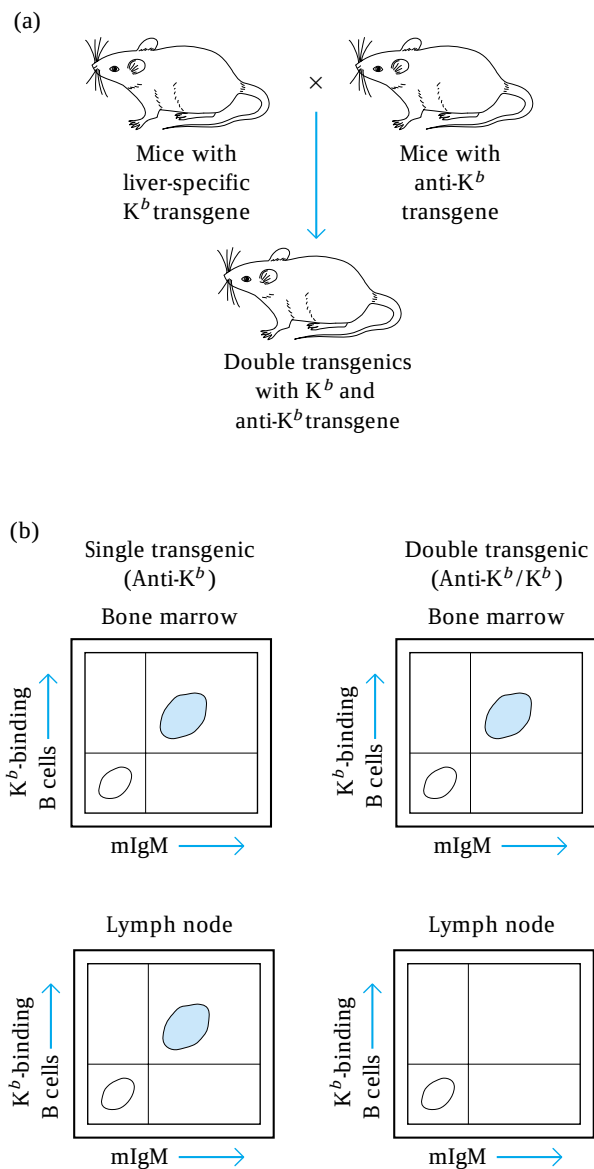
To study what would happen if a class I MHC self-antigen were expressed only in the periphery, Nemazee and Burki modified the transgenic system used in the experiments on

TABLE 11-3 Expression of anti-HEL transgene by mature peripheral B cells in single and double-transgenic mice

Experimental group	HEL level	Membrane anti-HEL	Anti-HEL PFC/spleen*	Anti-HEL serum titer*
Anti-HEL single transgenics	None	+	High	High
Anti-HEL/HEL single transgenics (Group 1)	10^{-9} M	+	Low	Low

* Experimental animals were immunized with hen egg-white lysozyme (HEL). Several days later, hemolytic plaque assays for the number of plasma cells secreting anti-HEL antibody were performed and the serum anti-HEL titers were determined. PFC = plaque-forming cells; see Figure 23-1 for a description of the plaque assay.

SOURCE: Adapted from C. C. Goodnow, 1992, *Annu. Rev. Immunol.* 10:489.



negative selection in the bone marrow described previously (Figure 11-5a). They first produced a transgene consisting of the class I K^b gene linked to a liver-specific promoter, so that the class I K^b molecule could be expressed only in the liver. Transgenic mice expressing an anti- K^b antibody on their B cells also were produced, and the two groups of transgenic mice were then mated (Figure 11-13a). In the resulting double-transgenic mice, the immature B cells expressing anti- K^b mIgM would not encounter class I K^b molecules in the bone marrow. Flow-cytometric analysis of the B cells in the double transgenics showed that immature B cells expressing the transgene-encoded anti- K^b cells were present in the bone marrow but not in the peripheral lymphoid organs (Figure 11-13b). In the previous experiments of Nemazee and Burki, the class I MHC self-antigen ($H-2^k$) was expressed on all nucleated cells, and immature B cells expressing the transgene-encoded antibody to this class I molecule were selected against and deleted in the bone marrow (see Figure 11-5a). In their second system, however, the class I self-antigen (K^b) was expressed only in the liver, so that negative selection and deletion occurred at the mature B-cell stage in the periphery.

FIGURE 11-13 Experimental demonstration of clonal deletion of self-reactive mature peripheral B cells by Nemazee and Burki. (a) Production of double-transgenic mice expressing the class I K^b molecule and anti- K^b antibody. Because the K^b transgene contained a liver-specific promoter, K^b was not expressed in the bone marrow of the transgenics. (b) Flow-cytometric analysis of bone marrow and peripheral (lymph node) B cells for K^b binding versus membrane IgM (mIgM). In the double transgenics, B cells expressing anti- K^b (blue) were present in the bone marrow but were absent in the lymph nodes, indicating that mature self-reactive B cells were deleted in the periphery.

The Humoral Response

This section considers the differences between the primary and secondary humoral response and the use of hapten-carrier conjugates in studying the humoral response.

Primary and Secondary Responses Differ Significantly

The kinetics and other characteristics of the humoral response differ considerably depending on whether the humoral response results from activation of naive lymphocytes (primary response) or memory lymphocytes (secondary response). In both cases, activation leads to production of secreted antibodies of various isotypes, which differ in their ability to mediate specific effector functions (see Table 4-2).

The first contact of an exogenous antigen with an individual generates a primary humoral response, characterized by the production of antibody-secreting plasma cells and memory B cells. As Chapter 3 showed, the kinetics of the primary response, as measured by serum antibody level, depend on the nature of the antigen, the route of antigen administration, the presence or absence of adjuvants, and the species or strain being immunized.

In all cases, however, a primary response to antigen is characterized by a lag phase, during which naive B cells undergo clonal selection, subsequent clonal expansion, and dif-

ferentiation into memory cells or plasma cells (Figure 11-14). The lag phase is followed by a logarithmic increase in serum antibody level, which reaches a peak, plateaus for a variable time, and then declines. The duration of the lag phase varies with the nature of the antigen. Immunization of mice with an antigen such as sheep red blood cells (SRBCs) typically results in a lag phase of 3–4 days. Eight or nine successive cell divisions of activated B cells during days 4 and 5 then generate plasma and memory cells. Peak plasma-cell levels are attained at day 4–5; peak serum antibody levels are attained by around day 7–10. For soluble protein antigens, the lag phase is a little longer, often lasting about a week, peak plasma-cell levels are attained by 9–10 days, and peak serum titers are present by around 14 days. During a primary humoral response, IgM is secreted initially, often followed by a switch to an increasing proportion of IgG. Depending on the persistence of the antigen, a primary response can last for various periods, from only a few days to several weeks.

The memory B cells formed during a primary response stop dividing and enter the G_0 phase of the cell cycle. These cells have variable life spans, with some persisting for the life of the individual. The capacity to develop a secondary humoral response (see Figure 11-14) depends on the existence of this population of memory B cells as well as memory T cells. Activation of memory cells by antigen results in a secondary antibody response that can be distinguished from the primary response in several ways (Table 11-4). The secondary response has a shorter lag period, reaches a greater mag-



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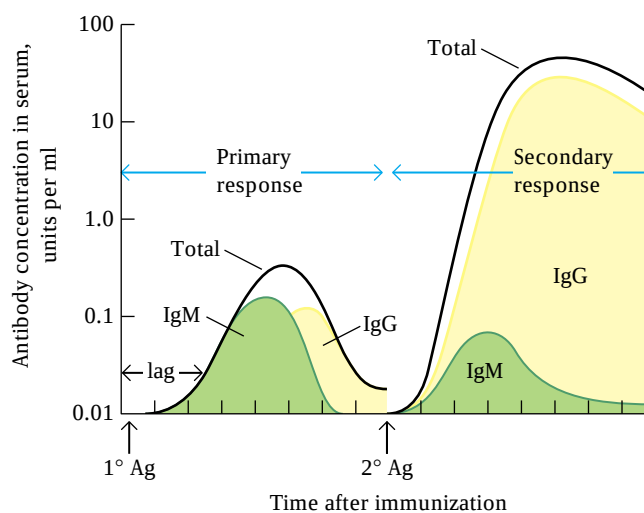


FIGURE 11-14 Concentration and isotype of serum antibody following primary (1°) and secondary (2°) immunization with antigen. The antibody concentrations are plotted on a logarithmic

scale. The time units are not specified because the kinetics differ somewhat with type of antigen, administration route, presence or absence of adjuvant, and the species or strain of animal.

TABLE 11-4 Comparison of primary and secondary antibody responses

Property	Primary response	Secondary response
Responding B cell	Naive (virgin) B cell	Memory B cell
Lag period following antigen administration	Generally 4–7 days	Generally 1–3 days
Time of peak response	7–10 days	3–5 days
Magnitude of peak antibody response	Varies depending on antigen	Generally 100–1000 times higher than primary response
Isotype produced	IgM predominates early in the response	IgG predominates
Antigens	Thymus-dependent and thymus-independent	Thymus-dependent
Antibody affinity	Lower	Higher

nitude, and lasts longer. The secondary response is also characterized by secretion of antibody with a higher affinity for the antigen, and isotypes other than IgM predominate.

A major factor in the more rapid onset and greater magnitude of secondary responses is the fact that the population of memory B cells specific for a given antigen is considerably larger than the population of corresponding naive B cells. Furthermore, memory cells are more easily activated than naive B cells. The processes of affinity maturation and class switching are responsible for the higher affinity and different isotypes exhibited in a secondary response. The higher levels of antibody coupled with the overall higher affinity provide an effective host defense against reinfection. The change in isotype provides antibodies whose effector functions are particularly suited to a given pathogen.

The existence of long-lived memory B cells accounts for a phenomenon called “original antigenic sin,” which was first observed when the antibody response to influenza vaccines was monitored in adults. Monitoring revealed that immunization with an influenza vaccine of one strain elicited an antibody response to that strain but, paradoxically, also elicited an antibody response of greater magnitude to another influenza strain that the individual had been exposed to during childhood. It was as if the memory of the first antigen exposure had left a life-long imprint on the immune system. This phenomenon can be explained by the presence of a memory-cell population, elicited by the influenza strain encountered in childhood, that is activated by cross-reacting epitopes on the vaccine strain encountered later. This process then generates a secondary response, characterized by antibodies with higher affinity for the earlier viral strain.

T Helper Cells Play a Critical Role in the Humoral Response to Hapten-Carrier Conjugates

As Chapter 3 described, when animals are immunized with small organic compounds (haptens) conjugated with large

proteins (carriers), the conjugate induces a humoral immune response consisting of antibodies both to hapten epitopes and to unaltered epitopes on the carrier protein. Hapten-carrier conjugates provided immunologists with an ideal system for studying cellular interactions of the humoral response, and such studies demonstrated that the generation of a humoral antibody response requires recognition of the antigen by both T_H cells and B cells, each recognizing different epitopes on the same antigen. A variety of different hapten-carrier conjugates have been used in immunologic research (Table 11-5).

One of the earliest findings with hapten-carrier conjugates was that a hapten had to be chemically coupled to a larger carrier molecule to induce a humoral response to the hapten. If an animal was immunized with both hapten and carrier separately, very little or no hapten-specific antibody was generated. A second important observation was that, in order to generate a secondary antibody response to a hapten,

TABLE 11-5 Common hapten-carrier conjugates used in immunologic research

Hapten-carrier acronym	Hapten	Carrier protein
DNP-BGG	Dinitrophenol	Bovine gamma globulin
TNP-BSA	Trinitrophenyl	Bovine serum albumin
NIP-KLH	5-Nitrophenyl acetic acid	Keyhole limpet hemocyanin
ARS-OVA	Azophenylarsonate	Ovalbumin
LAC-HGG	Phenyl lactoside	Human gamma globulin

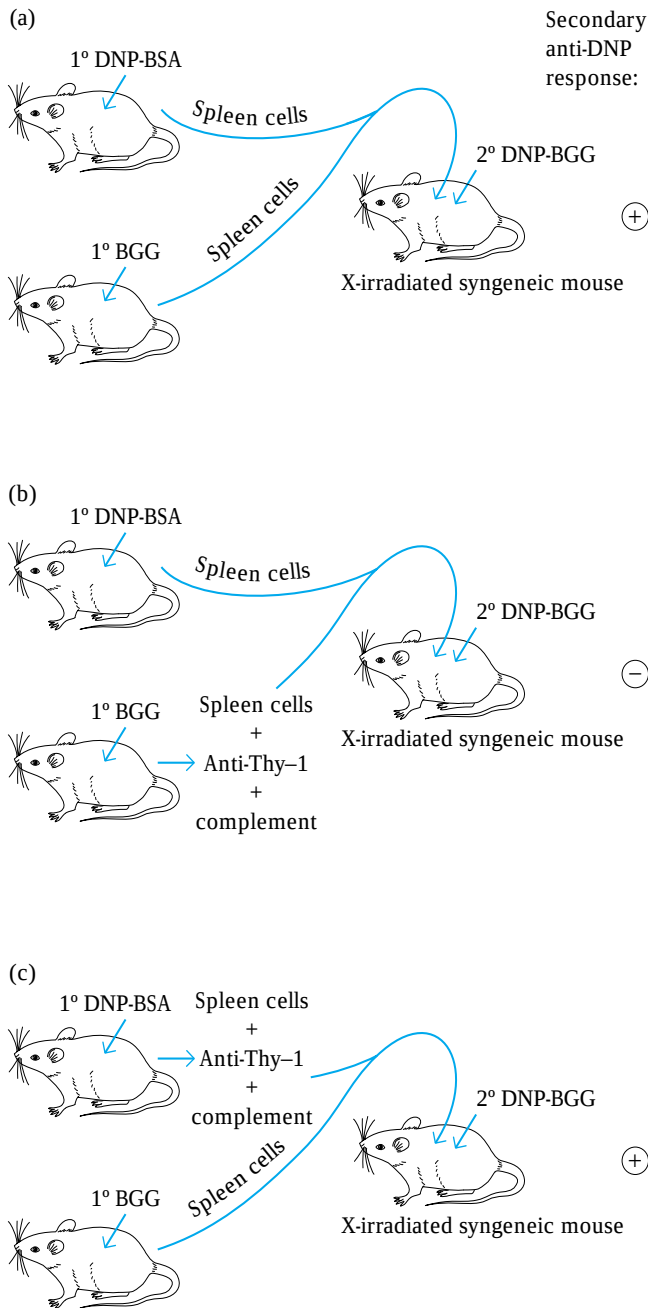


FIGURE 11-15 Cell-transfer experiments demonstrating that hapten-primed and carrier-primed cells are separate populations. (a) X-irradiated syngeneic mice reconstituted with spleen cells from both DNP-BSA-primed mice and BGG-primed mice and challenged with DNP-BGG generated a secondary anti-DNP response. (b) Removal of T cells from the BGG-primed spleen cells, by treatment with anti-Thy-1 antiserum, abolished the secondary anti-DNP response. (c) Removal of T cells from the DNP-BSA-primed spleen cells had no effect on the secondary response to DNP. These experiments show that carrier-primed cells are T cells and hapten-primed cells are B cells.

the animal had to be again immunized with the same hapten-carrier conjugate used for the primary immunization. If the secondary immunization was with the same hapten but conjugated to a different, unrelated carrier, no secondary anti-hapten response occurred. This phenomenon, called the **carrier effect**, could be circumvented by priming the animal separately with the unrelated carrier.

Similar experiments conducted with a cell-transfer system showed that cells immunized against the hapten and cells immunized against the carrier were distinct populations. In these studies, one mouse was primed with the DNP-BSA conjugate and another was primed with the unrelated carrier BGG, which was not conjugated to the hapten. In one experiment, spleen cells from both mice were mixed and injected into a lethally irradiated syngeneic recipient. When this mouse was challenged with DNP conjugated to the unrelated carrier BGG, there was a secondary anti-hapten response to DNP (Figure 11-15a). In a second experiment, spleen cells from the BGG-immunized mice were treated with anti-T-cell antiserum (anti-Thy-1) and complement to lyse the T cells. When this T-cell-depleted sample was mixed with the DNP-BSA-primed spleen cells and injected into an irradiated mouse, no secondary anti-hapten response was observed upon immunizing with DNP-BGG (Figure 11-15b). However, similar treatment of the DNP-BSA-primed spleen cells with anti-Thy-1 and complement did not abolish the secondary anti-hapten response to DNP-BGG (Figure 11-15c). Later experiments, in which antisera were used to specifically deplete $CD4^+$ or $CD8^+$ T cells, showed that the $CD4^+$ T-cell subpopulation was responsible for the carrier effect. These experiments demonstrate that the response of hapten-primed B cells to the hapten-carrier conjugate requires the presence of carrier-primed $CD4^+$ T_H cells specific for carrier epitopes. (It is important to keep in mind that the B-cell response is not limited to the hapten determinant; in fact some B cells do react to epitopes on the carrier; however, the assay can be conducted in such a manner as to detect only anti-hapten responses.)

The experiments with hapten-carrier conjugates revealed that both T_H cells and B cells must recognize antigenic determinants on the same molecule for B-cell activation to occur. This feature of the T- and B-cell interaction in the humoral response is called associative, or linked, recognition. The conclusions drawn from hapten-carrier experiments apply to the humoral response to antigens in general and support the requirement for T-cell help in B-cell activation described earlier in this chapter.

In Vivo Sites for Induction of Humoral Responses

In vivo activation and differentiation of B cells occurs in defined anatomic sites whose structure places certain restrictions on the kinds of cellular interactions that can take place.

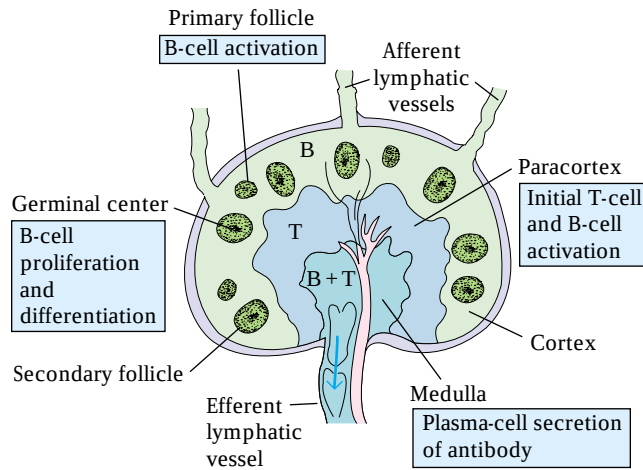


FIGURE 11-16 Schematic diagram of a peripheral lymph node showing anatomic sites at which various steps in B-cell activation, proliferation, and differentiation occur. The cortex is rich in B cells and the paracortex in T cells; both B and T cells are present in large numbers in the medulla. A secondary follicle contains the follicular mantle and a germinal center.

When an antigen is introduced into the body, it becomes concentrated in various peripheral lymphoid organs. Blood-borne antigen is filtered by the spleen, whereas antigen from tissue spaces drained by the lymphatic system is filtered by regional lymph nodes or lymph nodules. The following description focuses on the generation of the humoral response in lymph nodes.

A lymph node is an extremely efficient filter capable of trapping more than 90% of any antigen carried into it by the afferent lymphatics. Antigen or antigen-antibody complexes enter the lymph nodes either alone or associated with antigen-transporting cells (e.g., Langerhans cells or dendritic cells) and macrophages. As antigen percolates through the cellular architecture of a node, it will encounter one of three types of antigen-presenting cells: interdigitating dendritic cells in the paracortex, macrophages scattered throughout the node, or specialized follicular dendritic cells in the follicles and germinal centers. Antigenic challenge leading to a humoral immune response involves a complex series of events, which take place in distinct microenvironments within a lymph node (Figure 11-16). Slightly different pathways may operate during a primary and secondary response because much of the tissue antigen is complexed with circulating antibody in a secondary response.

Once antigen-mediated B-cell activation takes place, small foci of proliferating B cells form at the edges of the T-cell-rich zone. These B cells differentiate into plasma cells secreting IgM and IgG isotypes. Most of the antibody produced during

a primary response comes from plasma cells in these foci. (A similar sequence of events takes place in the spleen, where initial B-cell activation takes place in the T-cell-rich periarterial lymphatic sheath, PALS; see Figure 2-19).

A few days after the formation of foci within lymph nodes, a few activated B cells, together with a few T_H cells, are thought to migrate from the foci to primary follicles. These follicles then develop into secondary follicles, which provide a specialized microenvironment favorable for interactions between B cells, activated T_H cells, and follicular dendritic cells. Note that although they share the highly branched morphology of dendritic cells derived from bone marrow, follicular dendritic cells do not arise in bone marrow, do not express class II MHC molecules, and do not present antigen to $CD4^+$ T cells. Follicular dendritic cells have long extensions, along which are arrayed Fc receptors and complement receptors. These receptors allow follicular dendritic cells to retain and present antigen-antibody complexes for long periods of time, even months, on the surface of the cell. Activated B cells (together with some activated T_H cells) may migrate towards the center of the secondary follicle, forming a germinal center.

Germinal Centers and Antigen-Induced B-Cell Differentiation

Germinal centers arise within 7–10 days after initial exposure to a thymus-dependent antigen. During the first stage of germinal-center formation, activated B cells undergo intense proliferation. These proliferating B cells, known as centroblasts, appear in human germinal centers as a well-defined **dark zone** (Figure 11-17). Centroblasts are distinguished by their large size, expanded cytoplasm, diffuse chromatin, and absence or near absence of surface Ig. Centroblasts eventually give rise to centrocytes, which are small, nondividing B cells that now express membrane Ig. The centrocytes move from the dark zone into a region containing follicular dendritic cells called the **light zone**, where some centrocytes make contact with antigen displayed as antigen-antibody complexes on the surface of follicular dendritic cells. Three important B-cell differentiation events take place in germinal centers: affinity maturation, class switching, and formation of plasma cells and memory B cells. In general, affinity maturation and memory-cell formation require germinal centers. However some class switching and significant plasma-cell formation occur outside germinal centers.

Affinity Maturation Is the Result of Repeated Mutation and Selection

The average affinity of the antibodies produced during the course of the humoral response increases remarkably during the process of affinity maturation, an effect first noticed by H. N. Eisen and G. W. Siskind when they immunized rabbits



VISUALIZING CONCEPTS

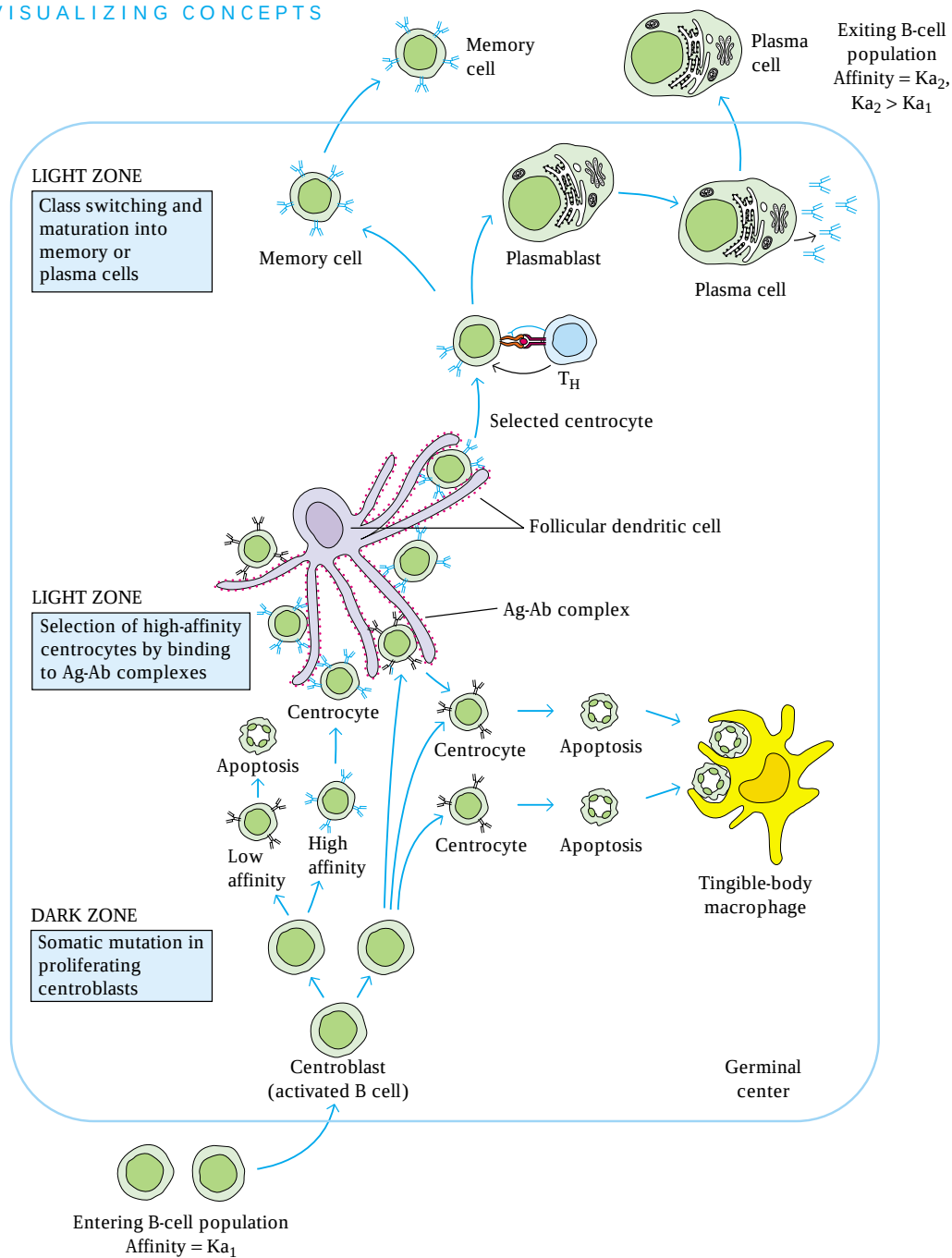


FIGURE 11-17 Overview of cellular events within germinal centers. Antigen-stimulated B cells migrate into germinal centers, where they reduce expression of surface Ig and undergo rapid cell division and mutation of rearranged immunoglobulin V region genes within the dark zone. Subsequently, division stops and the B cells migrate to the light zone and increase their expression of surface Ig. At this stage they are called centrocytes. Within the light zone centrocytes must interact with follicular dendritic cells and T helper cells to survive. Follicular dendritic cells bind antigen-antibody complexes along their long extensions and the centro-

cytes must compete with each other to bind antigen. B cells bearing high-affinity membrane immunoglobulin (antibodies shown in blue) are most likely to compete successfully. Those that fail this antigen-mediated selection (antibodies shown in black) die by apoptosis. B cells that pass antigen selection and receive a second survival signal from T_H cells differentiate into either memory B cells or antibody-secreting plasma cells. The encounter with T_H cells may also induce class switching. A major outcome of the germinal center is to generate higher affinity B cells (K_{A2}) from B cells of lower affinity (K_{A1}).

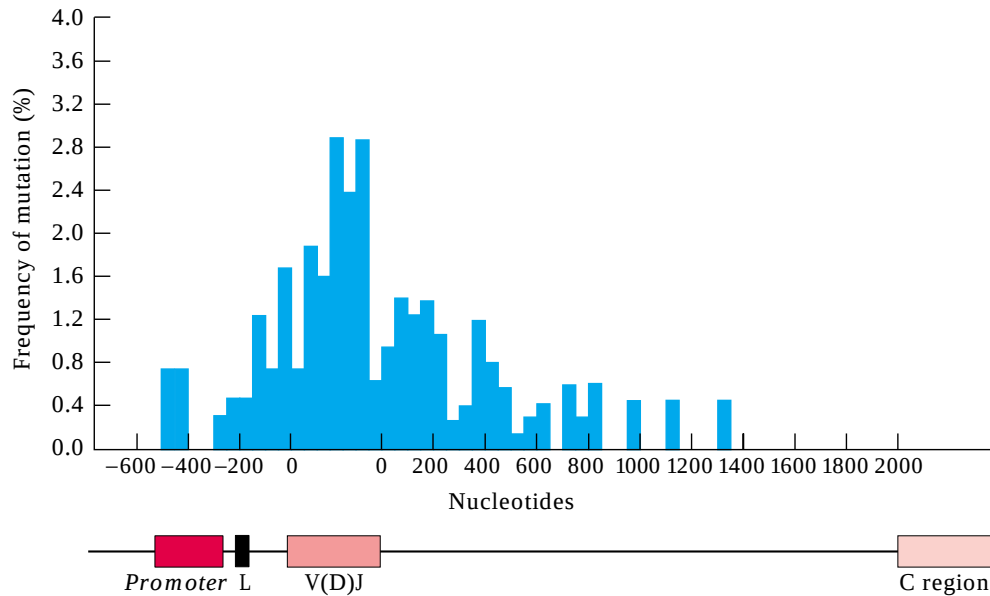


FIGURE 11-18 The frequency of somatic hypermutation decreases with the distance from the rearranged V(D)J gene. Experimental measurement of the mutation frequency shows that few if any mutations are seen upstream of the promoter of the rearranged

gene. Mutations do not extend into the portion of the gene encoding the constant region because there are no mutations at positions more than about 1.5 kb 3' of the rearranged gene. [Adapted from P. Gearhart, in *Fundamental Immunology*, 3rd ed., 1993, p. 877.]

with the hapten-carrier complex DNP-BGG. The affinity of the serum anti-DNP antibodies produced in response to the antigen was then measured at 2, 5, and 8 weeks after immunization. The average affinity of the anti-DNP antibodies increased about 140-fold from 2 weeks to 8 weeks. Subsequent work has shown that affinity maturation is mainly the result of somatic hypermutation.

THE ROLE OF SOMATIC HYPERMUTATION

Monitoring of antibody genes during an immune response shows that extensive mutation of the Ig genes that respond to the infection takes place in B cells within germinal centers. A direct demonstration that germinal centers are the sites of somatic hypermutation comes from the work of G. Kelsoe and his colleagues. These workers compared the mutation frequencies in B cells isolated from germinal centers with those from areas of intense B-cell activation outside the germinal centers. To do so, they prepared thin sections of spleen tissue from animals immunized with the hapten 4-hydroxy-3-nitrophenylacetyl (NP) conjugated with chicken gamma globulin as a carrier. This system is convenient because the initial response to this hapten is dominated by a particular heavy-chain gene rearrangement and the use of a γ light chain (in mice, >95% of antibodies bear κ light chains). Consequently, antibodies against the idiotype of this antibody can be used to readily distinguish responding B cells. Using antibodies to the idiotype and immunohistological staining

techniques, these workers identified B cells bearing anti-NP antibody in germinal centers and nongerminal-center foci of B-cell activation present in thin sections cut from the spleens of recently immunized mice. They isolated these B cells by microdissection, used PCR to amplify the immunoglobulin genes of each individual cell, and then cloned and sequenced the immunoglobulin genes. Many mutations were found in the immunoglobulin genes obtained from B cells in germinal centers, few in the genes obtained from activated B cells in nongerminal-center foci. When the mutated sequences of the collection of B cells from germinal centers was examined, it was apparent that many of the cells had sequences that were sufficiently similar that they were likely to be related by common descent from the same precursor cell. Detailed analysis of the sequences allowed these workers to build genealogic trees in which one could clearly see the descent of progeny from progenitors by progressive somatic hypermutation.

The introduction of point mutations, deletions, and insertions into the rearranged immunoglobulin genes is strikingly focused. Figure 11-18 shows that the overwhelming majority of these mutations occur in a region that extends from about 0.5 kb 5' to about 1.5 kb 3' of the V(D)J segments of rearranged immunoglobulin genes. Although the hypermutation process delivers mutations throughout the V region, antigen-driven selection results in the eventual emergence of immunoglobulin genes in which the majority of the mutations lie within the three complementarity-determining regions (CDRs). It has been estimated that the mutation rate

during somatic mutation is approximately 10^{-3} /base pair/division, which is a millionfold greater than the normal mutation rate for other genes of humans or mice cells. Since the heavy- and light-chain V(D)J segments total about 700 base pairs, this rate of mutation means that, for every two cell divisions it undergoes, a centroblast will acquire a mutation in either the heavy- or light-chain variable regions. The extremely high rates and precise targeting of somatic hypermutation are remarkable features that are unique to the immune system. Determining the molecular basis of this extraordinary process remains a challenge in immunology.

Because somatic mutation occurs randomly, it will generate a few cells with receptors of higher affinity and many cells with receptors of unchanged or lower affinity for a particular antigen. Therefore, selection is needed to derive a population of cells that has increased affinity. The germinal center is the site of selection. B cells that have high-affinity receptors for the antigen are likely to be positively selected and leave the germinal center; those with low affinity are likely to undergo negative selection and die in the germinal center.

THE ROLE OF SELECTION

Somatic hypermutation of heavy- and light-chain variable-region genes occurs when centroblasts proliferate in the dark zone of the germinal center. Selection takes place in the light zone, among the nondividing centrocyte population. The most important factor influencing selection is the ability of the membrane Ig molecules on the centrocyte to recognize and bind antigen displayed by follicular dendritic cells (FDCs). Because the surfaces of FDCs are richly endowed with both Fc receptors and complement receptors, antigen complexed with antibody or antigen that has been bound by C3 fragments generated during complement activation (see Chapter 13) can bind to FDCs by antibody or C3 bridges. A centrocyte whose membrane Ig binds and undergoes cross-linking by FDC-bound antigen receives a signal that is essential for its survival. Those that fail to receive such signals die. However, centrocytes must compete for the small amounts of antigen present on FDCs. Because the amount of antigen is limited, centrocytes with receptors of high affinity are more likely to be successful in binding antigen than those of lower affinity (see Figure 11-17).

While antigen binding is necessary for centrocyte survival, it is not sufficient. A centrocyte must also receive signals generated by interaction with a $CD4^+$ T_H cell to survive. An indispensable feature of this interaction is the engagement of CD40 on the B cell (centrocyte) by CD40L on the helper T cell. It is also necessary that processed antigen on class II MHC molecules of the B cell interact with the TCR of the collaborating T_H -cell. Centrocytes that fail to receive either the T_H -cell or the antigen-membrane Ig signal undergo apoptosis in the germinal center. Indeed, one of the striking characteristics of the germinal center is the extensive cell death by apoptosis that takes place there. This is clearly evident in the presence of condensed chromatin fragments, indicative of apoptosis, in tingible-body macrophages, an

unusual type of macrophage that removes cells by phagocytosis from lymphoid tissues.

CLASS SWITCHING

Antibodies perform two important activities: the specific binding to an antigen, which is determined by the V_H and V_L domains; and participation in various biological effector functions, which is determined by the isotype of the heavy-chain constant domain. As described in Chapter 5, class switching allows any given V_H domain to associate with the constant region of any isotype. This enables antibody specificity to remain constant while the biological effector activities of the molecule vary. A number of cytokines affect the decision of what Ig class is chosen when an IgM-bearing cell undergoes the class switch (Figure 11-19). The role of cytokines in class switching is explored further in Chapter 12.

As noted earlier, the humoral response to thymus-dependent antigens is marked by extensive class switching to isotypes other than IgM, whereas the antibody response to thymus-independent antigens is dominated by IgM. In the case of thymus-dependent antigens, membrane interaction between CD40 on the B cell and CD40L on the T_H cell is essential for the induction of class switching. The importance of the CD40/CD40L interaction is illustrated by the **X-linked hyper-IgM syndrome**, an immunodeficiency disorder in which T_H cells fail to express CD40L. Patients with this disorder produce IgM but not other isotypes. Such patients fail to generate memory-cell populations, fail to form germinal centers, and their antibodies fail to undergo somatic hypermutation.

Memory B Cells and Plasma Cells are Generated in Germinal Centers

After B cells are selected in the germinal center for those bearing high-affinity mIg for antigen displayed on follicular dendritic cells, some B cells differentiate into plasma cells and others become memory B cells (see Figure 11-17). While germinal centers are important sites of plasma-cell generation, these Ig-secreting cells are formed in other sites as well. Plasma cells generally lack detectable membrane-bound immunoglobulin and instead synthesize high levels of secreted antibody (at rates as high as 1000 molecules of Ig per cell per second). Differentiation of mature B cells into plasma cells requires a change in RNA processing so that the secreted form of the heavy chain rather than the membrane form is synthesized. In addition, the rate of transcription of heavy- and light-chain genes is significantly greater in plasma cells than in less-differentiated B cells. Several authors have suggested that the increased transcription by plasma cells might be explained by the synthesis of higher levels of transcription factors that bind to immunoglobulin enhancers. Some mechanism also must coordinate the increase in transcription of heavy-chain and light-chain genes, even though these genes are on different chromosomes.

As indicated above, B cells that survive selection in the light zone of germinal centers also differentiate into memory

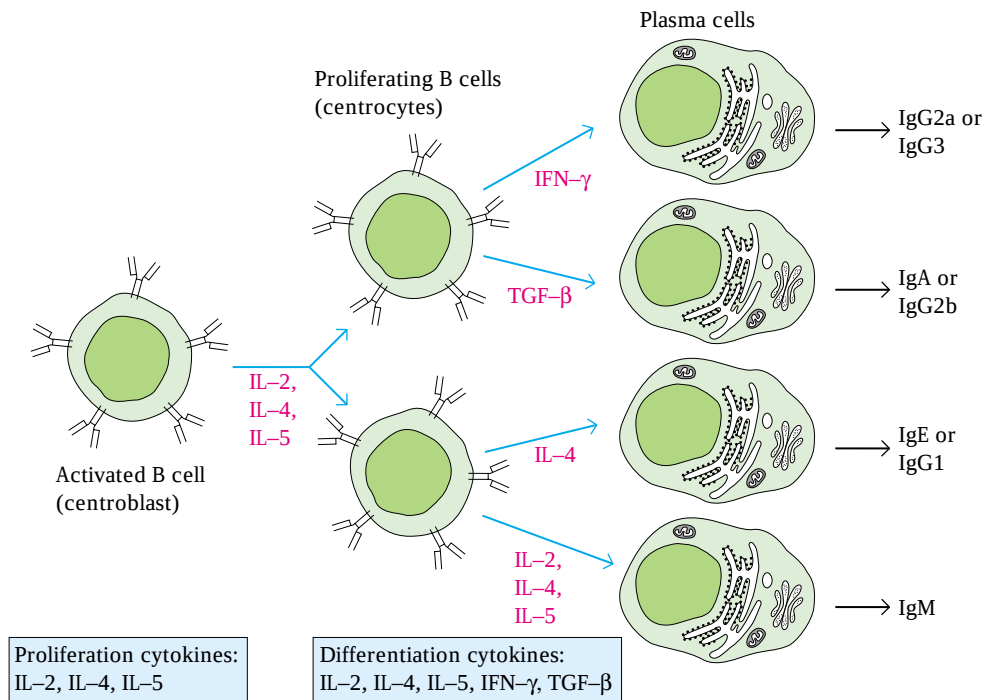


FIGURE 11-19 The interactions of numerous cytokines with B cells generate signals required for proliferation and class switching during the differentiation of B cells into plasma cells. Binding of the proliferation cytokines, which are released by activated T_H cells, provides the

progression signal needed for proliferation of activated B cells. Similar or identical effects may be mediated by cytokines beyond the ones shown. Class switching in the response to thymus-dependent antigens also requires the CD40/CD40L interaction, which is not shown here.

cells. Some properties of naive and memory B cells are summarized in Table 11-6. Except for membrane-bound immunoglobulins, few membrane molecules have been identified that distinguish naive B cells from memory B cells. Naive B cells express only IgM and IgD; as a consequence of class switching, however, memory B cells express additional isotypes, including IgG, IgA, and IgE.

Regulation of B-Cell Development

A number of transcription factors that regulate expression of various gene products at different stages of B-cell development have been identified. Among these are NF-6B, BSAP, Ets-1, c-Jun, Ikaros, Oct-2, Pu.1, EBF, BCF, and E2A. Like all

TABLE 11-6 Comparison of naive and memory B cells

Property	Naive B cell	Memory B cell
Membrane markers		
Immunoglobulin	IgM, IgD	IgM, IgD(?), IgG, IgA, IgE
Complement receptor	Low	High
Anatomic location	Spleen	Bone marrow, lymph node, spleen
Life span	Short-lived	May be long-lived
Recirculation	Yes	Yes
Receptor affinity	Lower average affinity	Higher average affinity due to affinity maturation*
Adhesion molecules	Low ICAM-1	High ICAM-1

* Affinity maturation results from somatic mutation during proliferation of centroblasts and subsequent antigen selection of centrocytes bearing high-affinity mIg.

transcription factors, these DNA-binding proteins interact with promoter or enhancer sequences, thereby either stimulating or inhibiting transcription of the associated gene. Analyses of the effects of knocking out the gene that encodes a particular transcription factor have provided clues about the role of some factors in B-cell development. For example, in knockout mice that carry a disrupted *Ikaros* gene, there is a general failure of lymphocyte development, and pro-B cells fail to develop in the bone marrow.

One of the most critical B-cell transcription factors, *B-cell-specific activator protein (BSAP)*, which has been previously mentioned (see Figure 11-3), appears to function as a master B-cell regulator. It is expressed only by B-lineage cells and influences all the cell stages during B-cell maturation. Moreover, recent evidence indicates that BSAP also influences the final differentiation events leading to the formation of memory B cells and plasma cells. The latter are the only B-lineage cells that do not express BSAP. BSAP binds to promoter or enhancer sequences of various B-cell-specific genes, including the $\lambda 5$ and Vpre-B genes of the surrogate light chain, the J-chain gene of polymeric IgM, and the 3' α heavy-chain enhancer region, one of the two enhancers that lie 3' of the α gene in heavy-chain germ-line DNA. In addition, BSAP binds to various immunoglobulin heavy-chain switch sites and to several genes involved in B-cell activation.

The heavy-chain 3' α enhancer ($E_{3'\alpha}$) contains binding sites for several transcription factors in addition to BSAP. Binding of BSAP to $E_{3'\alpha}$ appears to influence B-cell development by preventing the binding of other transcription factors. For example, when BSAP levels are high, this factor appears to block binding of NF- α P to the 3' α enhancer, thereby blocking transcription of the heavy-chain gene and promoting formation of memory B cells. When BSAP levels are low, NF- α P can bind to $E_{3'\alpha}$. As a result, transcription of the immunoglobulin heavy-chain gene is increased, leading to formation of plasma cells.

Regulation of the Immune Effector Response

Upon encountering an antigen, the immune system can either develop an immune response or enter a state of unresponsiveness called **tolerance**. The development of immunity or tolerance, both of which involve specific recognition of antigen by antigen-reactive T or B cells, must be carefully regulated since an inappropriate response—whether it be immunity to self-antigens or tolerance to a potential pathogen—can have serious and possibly life-threatening consequences.

Regulation of the immune response takes place in both the humoral and the cell-mediated branch. Every time an antigen is introduced, important regulatory decisions determine the branch of the immune system to be activated, the intensity of the response, and its duration. Chapter 12 describes the im-

portance of the cytokines to the orchestration of appropriate immune responses. In addition to cytokines, other regulatory mechanisms may also play important immunoregulatory roles. Greater knowledge about these regulatory events, which are still not well understood, may allow the deliberate manipulation of immune responses, selectively up-regulating desirable responses and down-regulating undesirable ones.

Different Antigens Can Compete with Each Other

The immunologic history of an animal influences the quality and quantity of its immune response. A naive animal responds to antigen challenges very differently from a previously primed animal. Previous encounter with an antigen may have rendered the animal tolerant to the antigen or may have resulted in the formation of memory cells. In some cases, the presence of a competing antigen can regulate the immune response to an unrelated antigen. This **antigenic competition** is illustrated by injecting mice with a competing antigen a day or two before immunization with a test antigen. For example, the response to horse red blood cells (HRBCs) is severely reduced by prior immunization with sheep red blood cells (SRBCs) and vice versa (Table 11-7). Although antigenic competition is a well-established phenomenon, its molecular and cellular basis is not understood.

The Presence of Antibody Can Suppress the Response to Antigen

Like many biochemical reactants, antibody exerts feedback inhibition on its own production. Because of antibody-mediated suppression, certain vaccines (e.g., those for measles and mumps) are not administered to infants before the age of 1 year. The level of naturally acquired maternal IgG, which the fetus acquires by transplacental transfer, remains high for about 6 months after birth. If an infant is immunized with

TABLE 11-7 Antigenic competition between SRBCs and HRBCs

IMMUNIZING ANTIGEN		HEMOLYTIC PLAQUE ASSAY (DAY 8)*	
Ag1 (day 0)	Ag2 (day 3)	Test Ag	PPC/10 ⁶ spleen cells
None	HRBC	HRBC	205
SRBC	HRBC	HRBC	13
None	SRBC	SRBC	626
HRBC	SRBC	SRBC	78

* See Figure 23-1 for a description of the plaque assay.

measles or mumps vaccine while this maternal antibody is still present, the humoral response is low and the production of memory cells is inadequate to confer long-lasting immunity. If an animal is immunized with a specific antigen and is injected with preformed antibody to that same antigen just before or within a few days after antigen priming, the immune response to the antigen is reduced as much as 100-fold.

There are two explanations for antibody-mediated suppression. One is that the circulating antibody competes with antigen-reactive B cells for antigen inhibiting the clonal expansion of the B cells. The second explanation is that binding of antigen-antibody complexes by Fc receptors on B cells reduces signalling by the B-cell-receptor complex.

As the antibody response proceeds, antibody feedback produces inhibition of the response. As more secreted IgG molecules become involved in antigen-antibody complexes, the Ig portions of these complexes become bound to Fc γ receptors present on the B cell membrane and the antigen of the complex binds the Ig of B-cell receptors. This crosslinking brings Fc γ receptors into close association with activated B-cell-receptor complexes, allowing phosphatases bound to the cytoplasmic tails of the Fc receptor to dephosphorylate sites in the BCR complex that are necessary to maintain B-cell activation. As a consequence, the activity of the B cell is progressively down-regulated as the amount of IgG bound to antigen increases. Evidence for such competition between passively administered antibody and antigen-reactive B cells comes from studies in which it took over 10 times more low-affinity anti-DNP antibody than high-affinity anti-DNP antibody to induce comparable suppression. Furthermore, the competition for antigen between passively administered antibody and antigen-reactive B cells drives the B-cell response toward higher-affinity antibody. Only the high-affinity antigen-reactive cells can compete successfully with the passively administered antibody for the available antigen.

SUMMARY

- B cells develop in bone marrow and undergo antigen-induced activation and differentiation in the periphery. Activated B cells can give rise to antibody-secreting plasma cells or memory B cells.
- During B-cell development, sequential Ig-gene rearrangements transform a pro-B cell into an immature B cell expressing mIgM with a single antigenic specificity. Further development yields mature naive B cells expressing both mIgM and mIgD.
- When a self-reactive BCR is expressed in the bone marrow, negative selection of the self-reactive immature B cells occurs. The selected cells are deleted by apoptosis or undergo receptor editing to produce non-self-reactive mIg. B cells reactive with self-antigens encountered in the periphery are rendered anergic.
- In the periphery, the antigen-induced activation and differentiation of mature B cells generates an antibody response. The antibody response to proteins and most other antigens requires T_H cells. These are thymus-dependent or simply T-dependent (TD) responses. Responses to some antigens, such as certain bacterial cell-wall products (e.g., LPS) and polymeric molecules with repeating epitopes, do not require T_H cells and are independent (TI) antigens. The vast majority of antigens are dependent.
- B-cell activation is the consequence of signal-transduction process triggered by engagement of the B-cell receptor that ultimately leads to many changes in the cell, including changes in the expression of specific genes.
- B- and T-cell activation share many parallels, including: compartmentalization of function within receptor subunits; activation by membrane-associated protein tyrosine kinases; assembly of large signaling complexes with protein-tyrosine-kinase activity; and recruitment of several signal-transduction pathways.
- The B-cell coreceptor can intensify the activating signal resulting from crosslinkage of mIg. This may be particularly important during the primary response to low concentrations of antigen.
- Activation induced by TD antigens requires contact-dependent help delivered by the interaction between CD40 on B cells and CD40L on activated T_H cells. The CD40/CD40L interaction is essential for B-cell survival, the formation of germinal centers, the generation of memory-cell populations, and somatic hypermutation.
- The properties of the primary and secondary antibody responses differ. The primary response has a long lag period, a logarithmic rise in antibody formation, a short plateau, and then a decline. IgM is the first antibody class produced, followed by a gradual switch to other classes, such as IgG. The secondary response has a shorter lag time, a more rapid logarithmic phase, a longer plateau phase, and a slower decline than the primary response. Mostly IgG and other isotypes are produced in the secondary response rather than IgM, and the average affinity of antibody produced is higher.
- Within a week or so of exposure to a TD antigen, germinal centers form. Germinal centers are sites of somatic hypermutation of rearranged immunoglobulin genes. Germinal centers are the sites of affinity maturation, formation of memory B cells, class switching, and plasma-cell formation.

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USEFUL WEB SITES

<http://www.ncbi.nlm.nih.gov/Omim/>

<http://www.ncbi.nlm.nih.gov/htbinpost/Omim/getmim>

The Online Mendelian Inheritance in Man Web site contains a subsite that lists more than a dozen different inherited diseases associated with B-cell defects.

<http://www.bioscience.org/knockout/knohome.htm>

The Frontiers in Bioscience Database of Gene Knockouts features information on the effects of knockouts of many genes important to the development and function of B cells.

Study Questions

CLINICAL FOCUS QUESTION Patients with X-linked agammaglobulinemia are subject to infection by a broad variety of pathogens. Suppose you have three sources of highly purified human immunoglobulin (HuIg) for the treatment of patients with X-linked agammaglobulinemia. The human Ig from all three sources is equally free of disease-causing agents and is equally well tolerated by recipients, but the number of donors whose blood was pooled for the preparation of each source differs widely: 100 individuals for source A, 1000 for source B, and 60,000 for source C. Which would you choose and what is the basis of your choice?

1. Indicate whether each of the following statements concerning B-cell maturation is true or false. If you think a statement is false, explain why.
 - a. Heavy chain V_H - D_H - J_H rearrangement begins in the pre-B-cell stage.
 - b. Immature B cells express membrane IgM and IgD.
 - c. The enzyme terminal deoxyribonucleotidyl transferase (TdT) is active in the pre-B-cell stage.

- d. The surrogate light chain is expressed by pre-B cells.
- e. Self-reactive B cells can be rescued from negative selection by the expression of a different light chain.
- f. In order to develop into immature B cells, pre-B cells must interact directly with bone-marrow stromal cells.
- g. Most of the B cells generated every day never leave the bone marrow as mature B cells.

2. You have fluorescein (Fl)-labeled antibody to the μ heavy chain and a rhodamine (Rh)-labeled antibody to the δ heavy chain. Describe the fluorescent-antibody staining pattern of the following B-cell maturational stages assuming that you can visualize both membrane and cytoplasmic staining: (a) progenitor B cell (pro-B cell); (b) precursor B cell (pre-B cell); (c) immature B cell; (d) mature B cell; and (e) plasma cell before any class switching has occurred.
3. Describe the general structure and probable function of the B-cell-coreceptor complex.
4. In the Goodnow experiment demonstrating clonal anergy of B cells, two types of transgenic mice were compared: single transgenics carrying a transgene-encoded antibody against hen egg-white lysozyme (HEL) and double transgenics carrying the anti-HEL transgene and a HEL transgene linked to the zinc-activated metallothioneine promoter.
 - a. In both the single and double transgenics, 60%–90% of the B cells expressed anti-HEL membrane-bound antibody. Explain why.
 - b. How could you show that the membrane antibody on these B cells is specific for HEL and how could you determine its isotype?
 - c. Why was the metallothioneine promoter used in constructing the HEL transgene?
 - d. Design an experiment to prove that the B cells, not the T_H cells, from the double transgenics were anergic.
5. Discuss the origin of the competence and progression signals required for activation and proliferation of B cells induced by (a) soluble protein antigens and (b) bacterial lipopolysaccharide (LPS).
6. Fill in the blank(s) in each statement below (a–i) with the most appropriate term(s) from the following list. Terms may be used more than once or not at all.

dark zone	centroblasts	memory B cells
light zone	centrocytes	plasmablasts
paracortex	follicular dendritic cells	T_H cells
cortex	medulla	

- a. Most centrocytes die by apoptosis in the _____.
- b. Initial activation of naive B cells induced by thymus-dependent antigens occurs within the _____ of lymph nodes.
- c. _____ are rapidly dividing B cells located in the _____ of germinal centers.
- d. _____ expressing high-affinity mIg interact with antigen captured by _____ in the light zone.
- e. Class switching occurs in the _____ and requires direct contact between B cells and _____.
- f. Centrocytes expressing mIg specific for a self-antigen present in the bone marrow are subjected to negative selection in the _____.

- g. Within lymph nodes, plasma cells are found primarily in the _____ of secondary follicles.
- h. Generation of _____ in the _____ of germinal centers is induced by interaction of centrocytes with IL-1 and CD3.
- i. Somatic hypermutation, which occurs in proliferating _____, is critical to affinity maturation.
7. Activation and differentiation of B cells in response to thymus-dependent (TD) antigens requires T_H cells, whereas the B-cell response to thymus-independent (TI) antigens does not.
- a. Discuss the differences in the structure of TD, TI-1, and TI-2 antigens and the characteristics of the humoral responses induced by them.
- b. Binding of which classes of antigen to mIg provides an effective competence signal for B-cell activation?
8. B-cell-activating signals must be transduced to the cell interior to influence developmental processes. Yet the cytoplasmic tails of all isotypes of mIg on B cells are too short to function in signal transduction.
- a. How do naive B cells transduce the signal induced by crosslinkage of mIg by antigen?
- b. Describe the general result of signal transduction in B cells during antigen-induced activation and differentiation.
9. In some of their experiments, Nemazee and Burki mated mice carrying a transgene encoding K^b, a class I MHC molecule, linked to a liver-specific promoter with mice carrying a transgene encoding antibody against K^b. In the resulting double transgenics, K^b-binding B cells were found in the bone marrow but not in lymph nodes. In contrast, the anti-K^b single transgenics had K^b-binding B cells in both the bone marrow and lymph nodes.
- a. Was the haplotype of the mice that received the transgenes H-2^b or some other haplotype?
- b. Why was the K^b transgene linked to a liver-specific promoter in these experiments?
- c. What do these results suggest about the induction of B-cell tolerance to self-antigens?
10. Indicate whether each of the following statements is true or false. If you believe a statement is false, explain why.
- a. Cytokines can regulate which branch of the immune system is activated.
- b. Immunization with a hapten-carrier conjugate results in production of antibodies to both hapten and carrier epitopes.
- c. All the antibodies secreted by a single plasma cell have the same idiotype and isotype.
- d. If mice are immunized with HRBCs and then are immunized a day later with SRBCs, the antibody response to the SRBCs will be much higher than that achieved in control mice immunized only with SRBCs.
11. Four mice are immunized with antigen under the conditions listed below (a–d). In each case, indicate whether the induced serum antibodies will have high affinity or low affinity and whether they will be largely IgM or IgG.
- a. A primary response to a low antigen dose
- b. A secondary response to a low antigen dose
- c. A primary response to a high antigen dose
- d. A secondary response to a high antigen dose
12. DNA was isolated from three sources: liver cells, pre-B lymphoma cells, and IgM-secreting myeloma cells. Each DNA sample was digested separately with the restriction enzymes *Bam*HI and *Eco*RI, which cleave germ-line heavy-chain and κ light-chain DNA as indicated in part (a) of the figure below. The digested samples were analyzed by Southern blotting using a radiolabeled C μ 1 probe with the *Bam*HI digests (blot #1) and a radiolabeled C κ probe with the *Eco*RI digests (blot #2). The blot patterns are illustrated in part (b) of the figure. Based on this information, which DNA sample (designated A, B, or C) was isolated from the (a) liver cells, (b) pre-B lymphoma cells, and (c) IgM-secreting plasma cells? Explain your assignments.

For use with Question 12 (a)

